

CHAPTER ONE

MONOPOLISTIC COMPETITION

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1.1 INTRODUCTION

Dear students, in pervious course on microeconomics we have seen the basic features of monopoly and prefect competitive market. As you know prefect competitive market assumes free entry and large number of small firms producing homogenous product. In such market environment, firms are worry only about the amount of output produced and take market price as given. In case of monopoly, however only single firm producing a product for which there is no close substitute dominates the industry. As a result, the monopolistic firm has power to set price and output level, which maximizes its profit.

Until 1920s, these two extreme forms of market model are the only model that used to explain the behavior of firms. However, by 1930s some economists began to question the capacity of these models to explain the modern market economics. They cannot explain several empirical facts. The assumption of homogenous product of prefect competitive market did not fit to the real world. Advertising and other selling activities, which are widely practiced by businesspersons, could not explained by both market models. Moreover, under the existence of legal monopoly on its trademarks and brand names for a firm, it is possible for other firms to produce similar product, which is not exactly the same. Such product viewed as a substitute by a consumer to some degree unlike non-existence of close substitute assumption of monopoly market model. From the viewpoint of a single firm therefore, production decision of its competitor will be a very important consideration in deciding how much it can produce and what price it can charge. This implies perfect competition and monopoly market models are useful tools for shedding light on how market works but they rarely represent the real market situation.

Edward Chamberlin (1933) from Harvard University in his theory of monopolistic competition and Joan Robinson from Cambridge University in her economics of imperfect competition working independently, comes up with a new model, which falls between the two extreme models. This model, which is the subject of this chapter, is termed as monopolistic competition market structure. Relatively it is a market model that can better explain the existing market situations. For example, the way retail trade, fast food and cosmetics market works better explained by monopolistically competitive market model than monopolistic or perfect competitive market model.

If we take retail traders, they sell goods in many different stores by competing with one other through differentiating their product based on location, availability and expertise of sales peoples and the provision of credit terms. In addition, entry is relatively easy. Therefore, if the business is profitable new store will be established to supply slightly differentiated products. Such feature of retail trade better fits to monopolistic competition than to perfectly competitive and monopoly market. Thus, monopolistic competition said to exists when there are many firms, as in perfect competition and each firm produce a product that is slightly different from that of the other.

Learning Objective

At the end of this chapter, the students will be able to:

Identify the features of monopolistic competitive market structure.

Define concepts like product differentiation, product group and industry.

Describe the demand and cost structure of monopolistic competitive firm.

Explain how short and long run equilibrium of the firm attained.

Define the excess capacity of monopolistic competitive firm.

Describe the welfare effect of monopolistically competitive firm resource allocation.

1.2 MONOPOLISTICALLY COMPETITIVE MARKET STRUCTURE

Dear students, can you guess what do we mean by monopolistically competitive market means? Is there any character shared by monopoly and monopolistic competitive as well as perfect competition and monopolistic competitive market structure? What is there in monopolistic competition but not is perfect competition and monopoly? What are the features of demand; cost and products of firms operate in monopolistically competitive market? Is there any market for a product that approximates monopolistically competitive market in your surrounding? Well students, we are going to answer these questions gradually while we deal with monopolistically competitive market in this chapter.

We began our discussion by stating the basic assumption of Chamberlin's large group model of monopolistic competition market. These are:

There are large numbers of sellers in the product group. This assumption implies each firm in the product group contributes small proportion of the total supply. As a result, there is no

single firm dominance in setting market price and collusion between firms to gain market power is impossible.

Sellers found in the same product group produce differentiated product and yet they are close substitutes of one another. This is to mean that each firm makes its product different from other firm by changing their inherent characteristics of their product or through different sales promotion activities. As a result, firms get certain monopoly power to set price. However, their monopoly power usually small relative to monopolistic firm because of the existence of stiff competition from other firms producing similar product.

It is relatively easy for new firms to enter the market with their own brands and for existing firms to leave the product group if they are unprofitable. Since there is no barrier to entry, a firm cannot make positive profit in the long run.

The objective of the firms is profit maximization in both the short run and long run.

Firms assumed to behave as if it knew its demand and cost curves with certainty.

Both demand and cost curves for all products are uniform through out the group. This assumption made in order to present the equilibrium of the firm and product group on the same diagram.

As indicated in the assumption, a monopolistically competitive market structure has some element of both monopoly and perfect competition. The fact that there are many firms in a monopolistically competitive situation is a perfect competitive element that restricts the ability of firms to control price through product differentiation. A significant difference in price between firms quickly leads customers to abandon the loyalties built by product differentiation. This will cause shift in preference of consumer from high price product to lower price close substitutes. Another perfect competitive element is easy entry of firms in a monopolistically competitive market even though product differentiation, which takes the form of brand name or firm's loyalty, can act as barrier to entry. Therefore, we cannot say that there is complete freedom of entry as in pure competition. Rather there is reasonable freedom of entry.

On the other hand, the product of each firm in monopolistic competition is differentiated. That is each firm sells a brand or a version of the product that differs in quality, appearance or reputation and it is a sole producer of its own brand. Such product differentiation gives monopolistically competitive firm some market power to set different price from their

competitors rather than passively accepting the market price. Usually the greater the differentiation, the greater the discretionary market power of a given firm to set its product price. When a product is highly differentiated, the product of each seller is similar but not identical. This may give them a monopoly power over the specific product it sells.

Activity 1.1

Describe the feature of monopolistic competition that resembles perfect competitive and then monopolistic market structure.

Determine whether the following firms operate in a monopolistic, perfect competitive or monopolistically competitive market structure.

barber shop

d) laundry soap producers

restaurants

e) tooth paste producers

Ice cream producers.

From the above discussion, it is possible to conclude that monopolistic competition as a market structure in which:

A large number of independent firms compete.

Each firm produces a differentiated product.

Firms compete on product quality, price and marketing activities.

Easy enter and exit of firms to the market.

1.3 Product differentiation and the demand curve

Chamberlin develops his theory of monopolistic competition based on some empirical facts. There are very few monopolists because there are very few commodities for which close substitutes do not exist. Similarly, very few commodities are entirely homogeneous to make perfect competition assumption realistic. For instances there are no homogeneous automobiles, soaps, suits, television sets, grocery stores, magazines and others. They are differentiated. In most cases, each producer tries to differentiate his product to make it unique and reduces the number of its close substitutes. The process of making a product unique from other product is called product differentiation. Chamberlin uses the concept of product differentiation to develop the theory of monopolistic competition.

There are two types of product differentiation: real and fancied (spurious) product differentiation. Product differentiation is said to be real, if the products found in the same product group differ in terms of their inheritance characteristics. They may be different in the type of input used to produce the product; specification and location of the firm in terms of convince to be accessed by the consumer. For example, shampoos with conditioner and without conditioner are differentiated in their content. Grocery stores found near to the house of customer and far from the house of a customer are differentiated in terms of their location. Fancied (spurious) product differentiation is a case where the products are the same but the producers that its product differs from other close substitutes convince consumers. Such differentiation occurs through advertisement, difference in packing, design, brand name and other sales promotion activities. Whatever the type, product differentiation determines the nature of demand curve facing a give firm.

The demand curve facing a firm will depend on output decisions and prices charged by other firms that produce similar product. That is the slope of demand curve facing the firm will depend on how similar the firm's products are. If large number of firms produces identical or homogenous products, then the demand curve facing the firms is flat. Each firm must sell its product at market price. Any firm that tries to raise its price above market price would loss all of its customers. On the other hand, if a firm has exclusive right to sell a particular product, it may raise its price without losing all of its customers.

Firm can gain certain monopoly power through product differentiation by making a given product unique to the mind of the consumer. This will create brand loyalty of consumer for a product and given some discretion power for the firms to set the price of their product different from their competitor price. As a result the demand facing individual firm becomes down ward sloping. The firm did not loss its entire customer through price rise even though some of them switch to its competitors product. However given the competitive element of monopolistically competitive product group (large number of firm and easy entry), small rise in price results in large fall in quantity demanded. For instance if price increases from P_1 to P_2 in figure1.1, quantity demanded decreases from Q_1 to Q_2 . Because increase in price leads the firm to lose some of its customers.

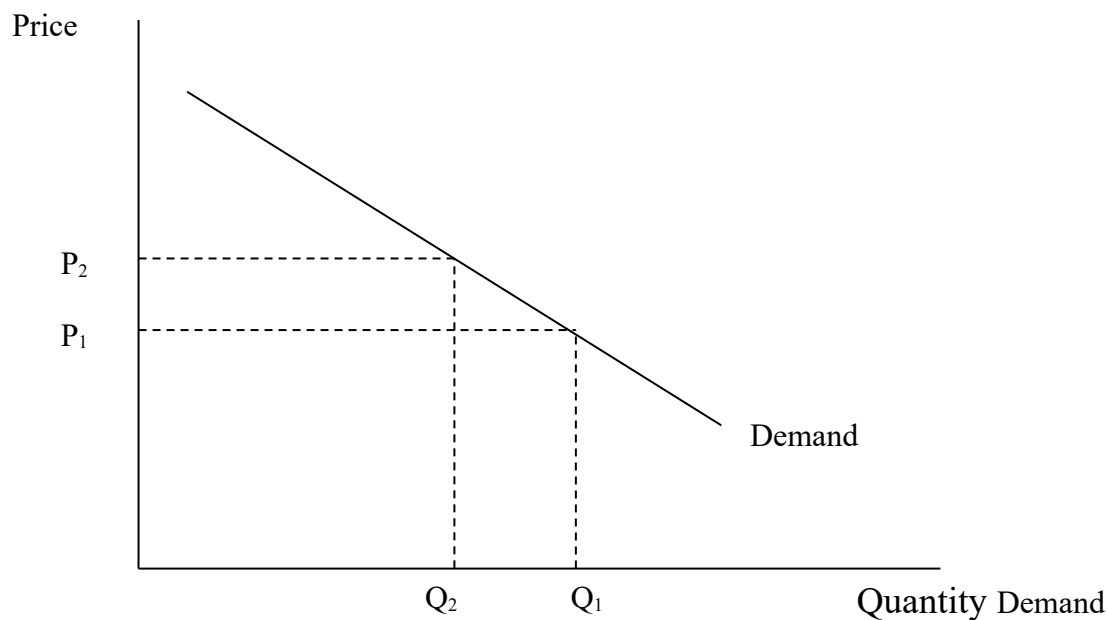


Figure1.1: Demand curve of monopolistically competitive firm

In summary product differentiation, which is the basis of non-price competition, give monopolistically competitive firm certain monopoly power and then down ward sloping demand curve. In other word in monopolistically competitive market structure, firms have a certain capacity to change the demand for their product through product differentiation. Even though firms in monopolistically competitive market faces downward sloping demand curve, their demand curve is highly elastic because of the existence of large number of firms producing closely substitute products.

Activity 1.2

How does a firm under take product differentiation?

Does product differentiation have any implication on the demand for the product of a firm and its market power?

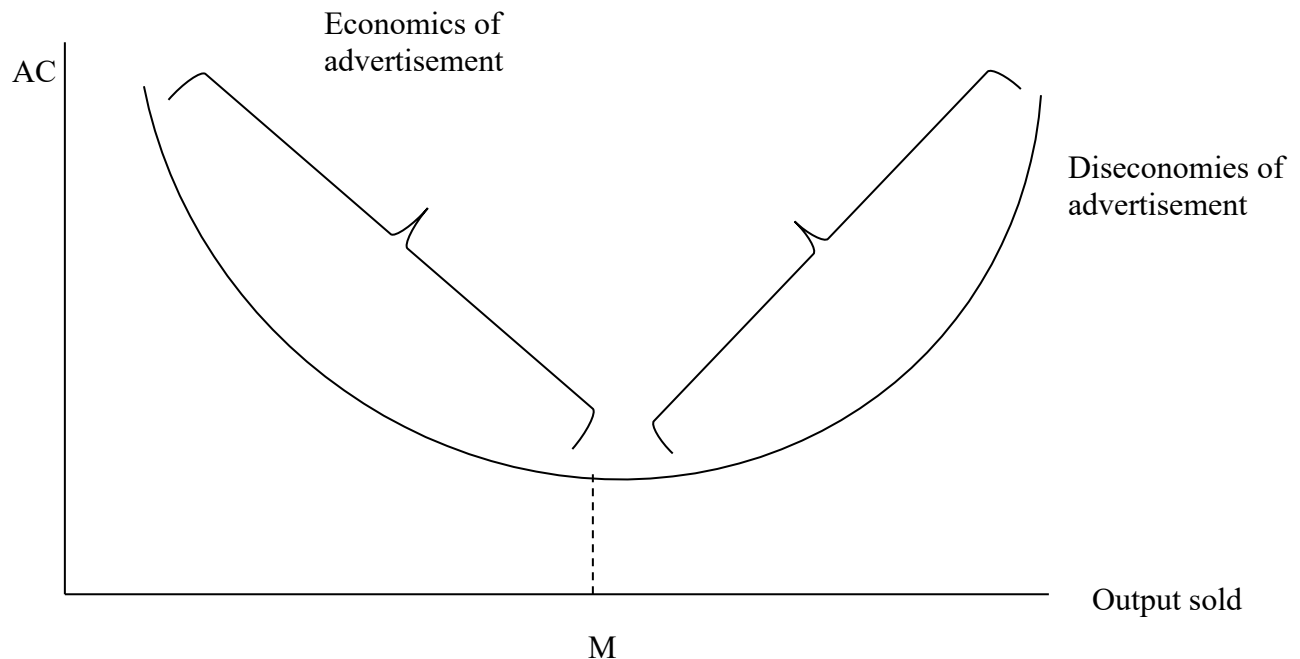
From demand curve of monopolistic competition and monopolistic firm, which one is more price elastic and Why?

1.4 COST OF MONOLOSTIC COMPETITION

Dear students, we have seen that monopolistically competitive firm undertake different activities and changes compositions of goods to differentiate their product from their competitors. For example, firms use intensive advertisements to increase demand and even to inform potential buyers about the availability of the product. This causes strong attachment of consumer to the product so that price elasticity of demand for the product becomes very low. In other words, firms often devote considerable resources to differentiate their product from their competitors through such devices as quality and style variations, warranties and guarantees, special services features, and product advertisement. All of these activities require firms to employ an additional resource that incurs additional cost to them. This cost is known as selling cost or cost of product differentiation. Therefore, firms have to consider this cost during pricing and output decision in addition to traditional production costs.

Like perfect competition and monopolistic firm, traditional production cost of monopolistically competitive firm's average costs and marginal costs are all U-shaped. Such shape indicates the operation of law of diminishing marginal return in the short run and law of return to scale in the long run during production process. Moreover, Chamberlin tries to show the average and marginal selling costs curve of monopolistically competitive firm have U-shape. If we take advertisement, cost that comprises the major proportion of selling cost shows economies and diseconomies of scale.

As indicated in figure 1.2, initially the amount of sales expansion through advertisement will not result in equally increase in selling cost. It is characterized by falling of the average selling cost because of increase return to scale of advertisement (economies of scale). However, when large quantity of output sold, after a certain level, firms have to spend more per unit in order to increase the amount of output sold. That is diseconomies of advertisement start to operate and average selling cost increase after its minimum point. Such feature of average selling cost give rise to U-shaped average selling cost. Therefore, since both production and selling costs have U-shaped, the total average and marginal costs of monopolistically competitive firm is U-shaped similar to cost curves of monopoly and perfect competitive firm. However, the average and marginal cost of monopolistically competitive firm is greater than average and marginal cost of perfect competitive and monopolistic firm because of the additional costs incurred to during product differentiation.



1.5 THE CONCEPT OF PRODUCT GROUP AND INDUSTRY

Figure 1.2:

It is defined in perfect competition that an industry is a group of firms producing a homogenous product. However, when products are differentiated like in the case of monopolistic competition, one cannot define an industry in this narrow sense. There is no automobile, soap or furniture industry. Each firm in the automobile, soap or furniture industry produces its own distinct product. This implies product differentiation create difficulties in defining the boundaries of an industry. Heterogeneous group cannot add together to form an industry and to their market demand and supply schedule.

To overcome this problem, Chamberlin introduces the concept of product group. It is formed by lumping together firm's producing similar products, which are close substitutes. They are group of product with higher price and cross elasticities of demand. The consumer preference for each product shifts to other product in the group when its price increases. For instance, we can form a product group by putting together different model of automobiles. Automobiles are differentiated product. They are close substitute (used for the same purpose)

and their price and cross elasticity of each automobile model are high. If the price of one model increase consumers shift their preference to other model.

Dear students, how does chamberlain's theory work with the idea of product group? During the determination of equilibrium market price and output, industry demand and supply should be considered. This can be possible by summation of individual demand and supply which need common price. With product differentiation however, we cannot derive industry demand and supply curve as we did for perfect competitive market. We do not have a single equilibrium price for differentiated product, but a cluster of prices.

For this reason, in order to analyze his model of monopolistic competition, Chamberlin made assumption about demand and cost of firms in the product group. That is every firm in the product group faces the same demand curve with identical cost even though it is not the case in real situation. Such assumption enables us to get unique equilibrium price and to treat firms and market demand on the same graph. However, if firms supply different product to market then why should their demand and costs are identical? This is one point up on which Chamberlin model of monopolistic competition was criticized.

Activity 1.3

Does cost structure of monopolistically competitive firm have any difference from firms found in other market structure? If yes how?

What is the difference between product group and industry?

State the assumption made by Chamberlin in order to get unique equilibrium price and market demand and supply for differentiated product.

1.6 EQUILIBRIUM OF MONOPOLOSTICALLY COMPETITIVE FIRM IN SHORT RUN

By now we have get familiar with the different component of profit of monopolistically competitive firm, i.e. their cost and demand structure. Each firm's product in the Chamberlin world is somewhat different from the product of its rivals. This situation gives rise to downward sloping demand curve for the firm. This implies a decrease in the price of individual product results in increase in sales volume of the firm by attracting customers of

other rivalry firms. On the other hand, increase in price will result in decrease in the sales volume of the firm. This is because some customers shift their preference to other product because of increase in price. Such relationship between price and sells volume of firms can be indicated by individual demand curve as a result individual demand curve also known as planned sale curve.

As in the case of monopoly, since the demand curve of monopolistically competitive firm is negatively sloped, the corresponding marginal revenue sloped downward. This is indicated by figure1.3. Q_1 and P_1 represent short run equilibrium level of output and price respectively. . Both are characterized by the intersection of marginal revenue (derived from effective demand curve) and the firms marginal cost curve. Having achieved this point, the firm would have no incentive to change its price from p_1 . The equality of marginal revenue and marginal cost at output Q_1 means that firm believes that it would maximize its profit by maintaining its price at p_1 . At the indicated point monopolistically competitive firm earn a positive profit equals to the area of shaded region.

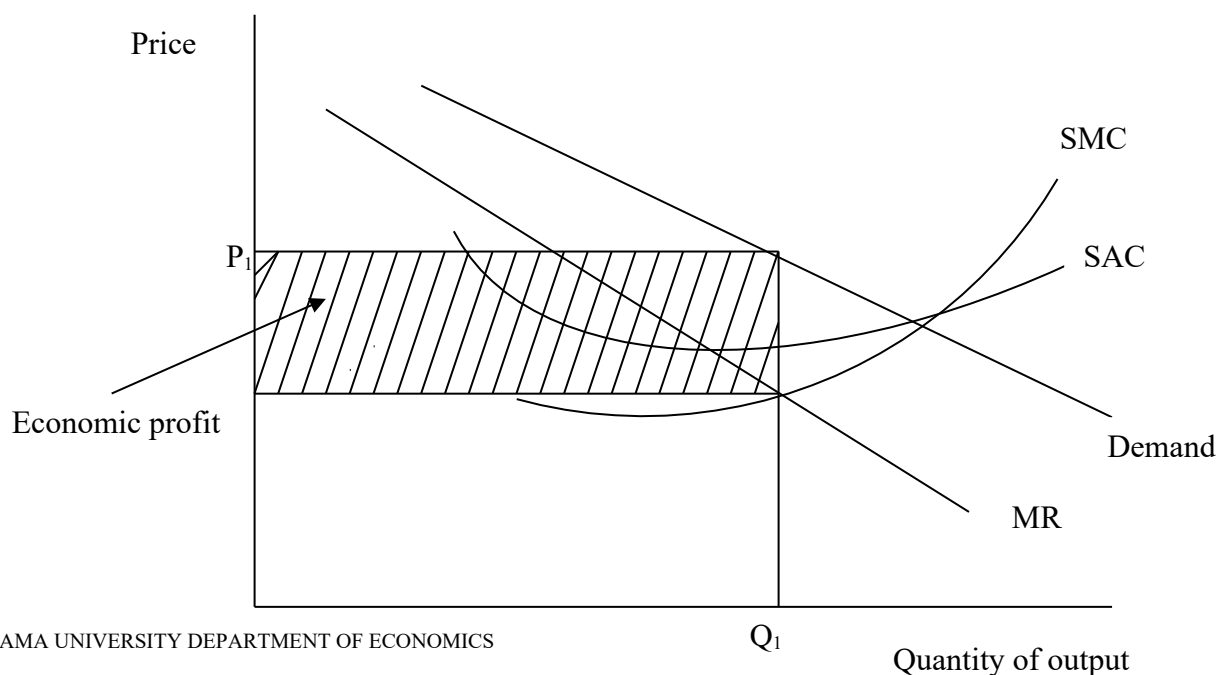
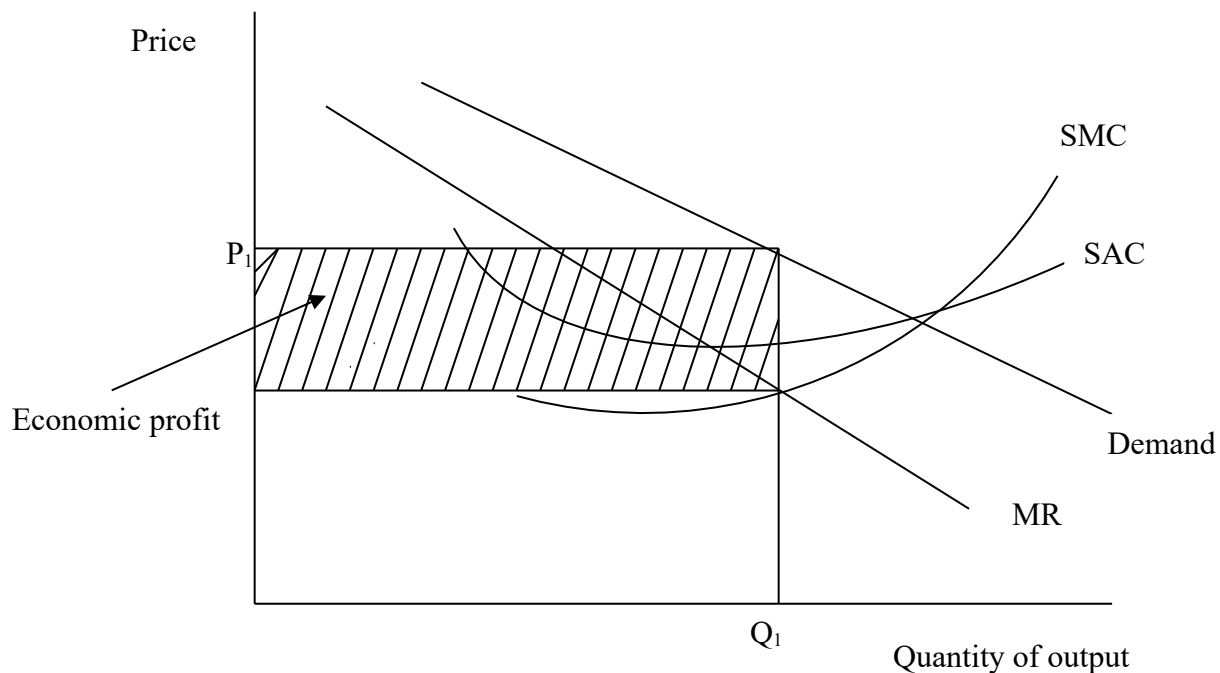


Figure 1.3(a) Short run equilibrium under monopolistic competition.



However, t **Figure 1.3(a) Short run equilibrium under monopolistic competition.**

firms. This is totally depending on a point at which marginal revenue equals to marginal cost. As indicated in figure 1.3(b) below, if the point of intersection $MR=MC$ (marginal cost equals marginal revenue) takes places where average cost equals to the demand curve, the firm earns zero (normal) profit since their cost equal to their revenue at this point

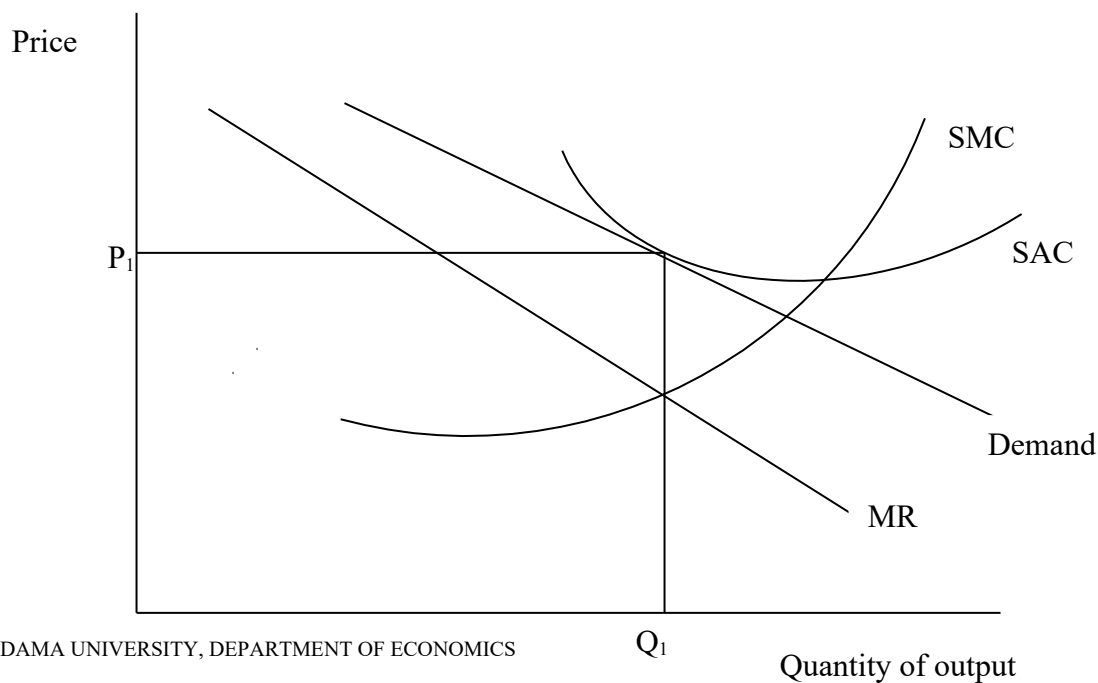
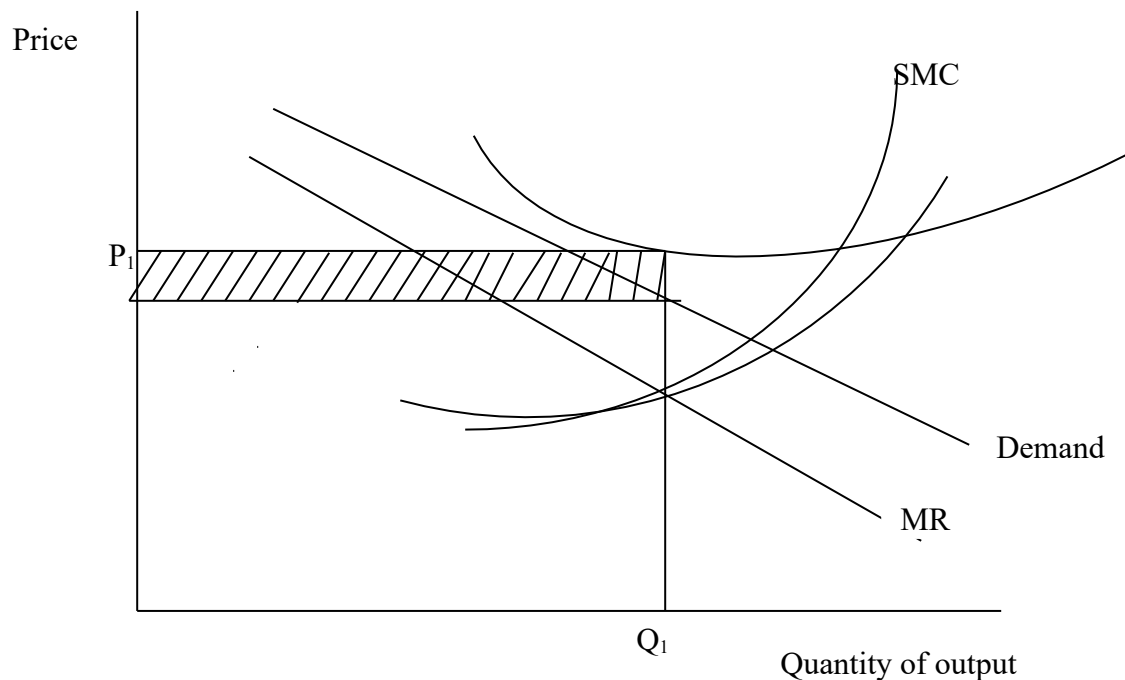


Figure 1.3(b) Short run equilibrium under monopolistic competition.

If the point of intersection takes place, where demand curve is below the average cost curve, the firm incurs a loss equal to the shaded region indicated by figure 1.3(c). This is because their total cost greater than total revenue.



Therefore, for monopolistically competitive firm to earn positive abnormal profit at equilibrium in the short run, the equality between MR and MC should take place when demand curve is greater than short run average cost curve.

1.7 Long run equilibrium

In the long run just like perfect competition, monopolistically competitive firm make change to the scale of their plant to minimize cost of production. They also decides to leave industry if they see the prospect to earning negative profit and enter in to the industry if there is a prospect of earning positive profit. Such adjustment process leads to long run equilibrium level of output and price, which is characterized by the tangency of long run average cost

curve and demand curve. It is also characterized by the equality of long run marginal cost and marginal revenue. At this point total revenue equals to total cost resulting in zero economic profit. This will create disincentive to enter the market. The adjustments to such point takes place through change in the position of demand curve resulted from enter and exit of firms or price adjustment by the existing firm or a combination of the two.

In order to analyze how long run equilibrium level of output and price for a firm and an industry on the same diagram, Chamberlin made two heroic assumptions. **Firms have identical cost and consumer preferences are evenly distributed among different product.** That is even though the products are differentiated; all firms assumed to have identical cost and demand curves. Under these assumptions, it is possible to get a unique market equilibrium price. Given the above assumption, Chamberlin developed three distinct model of long run equilibrium.

Equilibrium with new firm entering the industry

Equilibrium with price competition

Equilibrium with price competition and free entry of new firms

1.7.1 Model 1 Equilibrium with new firms entering the product group

In this model, the existing firms are assumed to be in short run equilibrium realizing abnormal profit, i.e., existing firm do not have any incentive to adjust their price. Therefore, how is long run equilibrium run achieved? According to the equilibrium with new enter model, equilibrium position can be attained through entry of new firms who are attracted by the short run positive economic profit. The entry of new firms and exit of old firms can causes shift in demand curve facing any single firm. That is increase in the number of firms through entry in the product group shifts the individual demand curve of the firm inward to the left. This is because market demand (which is relatively fixed) divided among more firms. The market share of individual firms decreases causing inward shift of individual demand curve of a firm. Therefore, entry and exit push demand curve facing any single firm toward equilibrium position where it is tangent to long run average cost curve.

As indicated in figure 1.4 below firm with long run average cost of LAC, long run marginal cost of LMC and facing a demand curve dd_1 , will set price at a point where marginal revenue equals to marginal cost in the short run. At this point firms in the product group earn abnormal

profit of area $ABCP_1$. Therefore, there is no incentive for these firms to change their price P_1 and output Q_1 . However, the abnormal profits obtained by existing firms attract other new firms to enter in to the market in the long run.

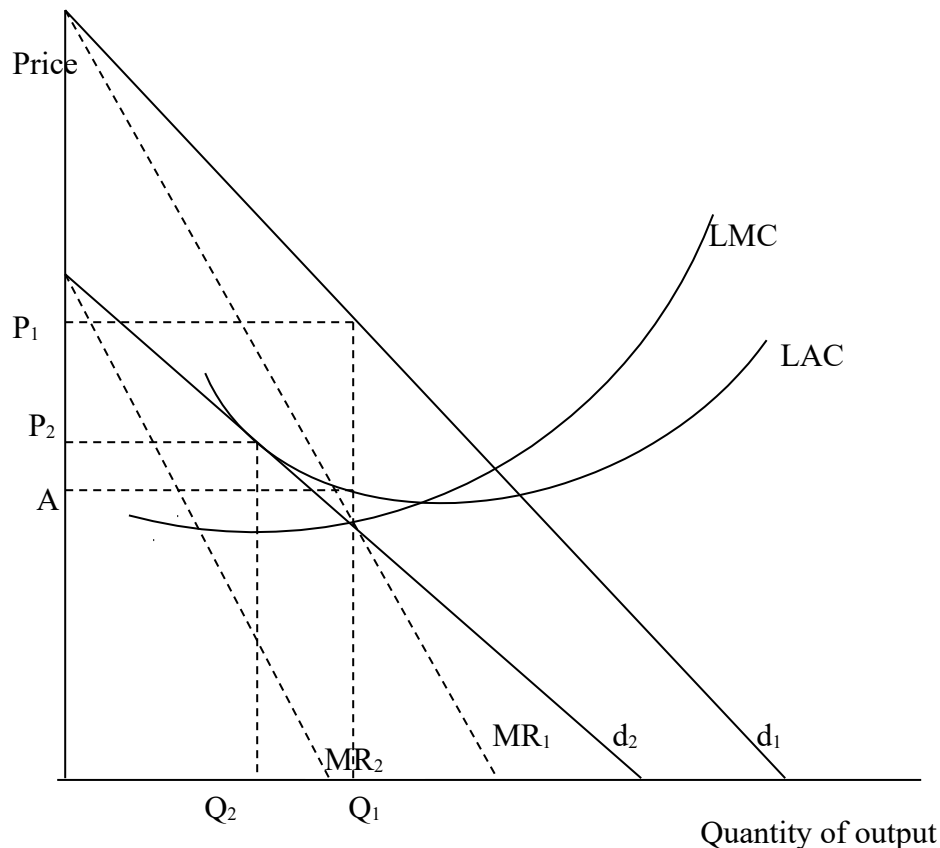


Figure 1.4: long run equilibrium with new entry of firms

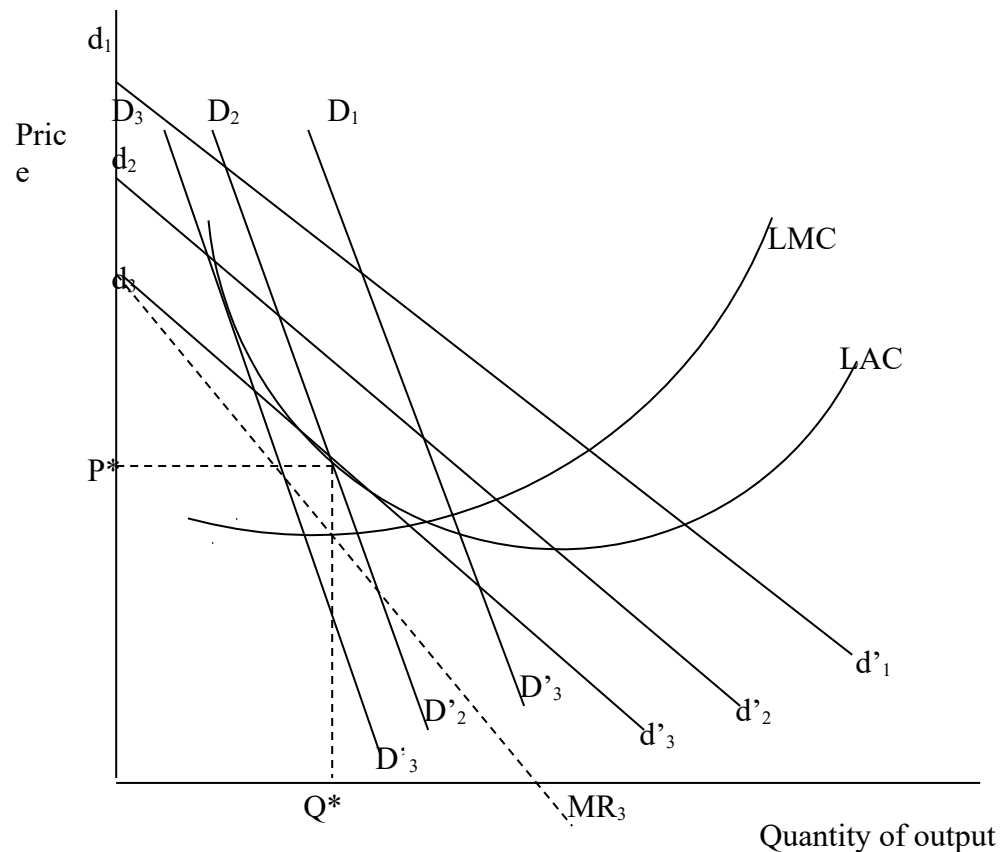
As shown in figure 1-4, firms found in the product group reach short run equilibrium at price level p_1 and they do not have any incentive to change its price from p_1 . However, the abnormal profit earned by existing firms attracts other new firms in to the market. When new firms enter in to the market, the market share of individual firm decreases. This will causes inward shift in individual demand curve. Assuming that cost curves will not shift as entry occurs, each shift in demand curve result in establishment of new equilibrium at a point where new marginal revenue intersect marginal cost curve. That is, as more and more new firms entry in to the product group, there is continuous shift in the original demand curve, dd_1 . Such adjustment process continues until the shifted demand curve dd_2 tangent to long run average cost curve

(LAC) at point E. At point E, a firm charges p_2 and produces Q_2 level of output and earns zero economic profit. When firms reach such tangency point, there is no further entry since further entry makes firms to earn negative profit (loss). Thus, the long run equilibrium becomes stable when the shifted demand curve is tangent to the long run average cost curve.

1.7.2 Model 2 Equilibrium with price competition

Equilibrium with price competition model assumes the number of firms in the industry is compatible with long run equilibrium. Therefore, no entry or exit that will take place in the long run. However, since short run equilibrium price is higher than long run equilibrium price, long run equilibrium level of price attained through price competition among firms in the product group.

To conduct the analysis of how long run equilibrium achieved under the assumption of price competition model, the second demand curve represented DD' is introduced as shown in figure 1.5. It is a curve that shows the actual sale of a firm at each price level after adjustment of price to capture more share of the market. This demand curve sometimes called actual sales or share of the market curve since it indicate the **share of each firm in the product group. When competitors change their price, the market share of each firms changes and individual demand curve shifts accordingly.** Connecting points that show market share of a firm when price adjustment made by competitor firm independently gives actual sales curve (market demand). A movement along DD' shows change in the actual sales of the existing firms as all of them adjusts their price simultaneously and have similar market share as before the price adjustment made.



Dear **Figure 1.4: long run equilibrium with price competition** ved

through price competition? Let us began from short run equilibrium position like P_3 and see how long run equilibrium achieved through price adjustment of existing firm. As an attempt to maximize its profit a given firm reduce its price to increases its market share. However, the attempt is not realized, because all other firms having the same demand and cost condition have the incentive to act in the same way simultaneously. Each firm attempt to maximize their profit, ignoring the reaction of other competitor on the assumption that the effect of other firms on demand of its product is not significant.

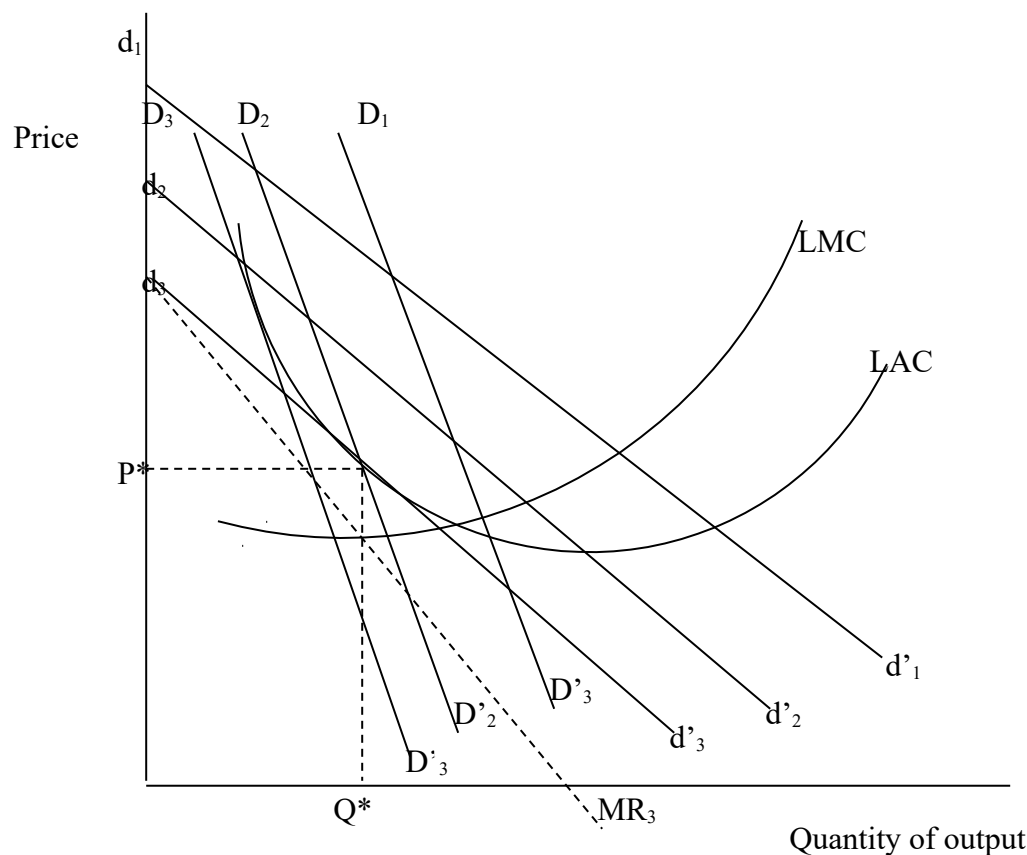
Such action of firms shifts the individual demand curve inward resulting in small amount of sale than expected amount on the shifted demand curve dd_2 along the share curve. In the future if firms learn from their post experience, they will not do the same since it reduce their profit. However according to the model firms suffer from myopia and cannot learn form their experiences. It continues to behave on the assumption that its new demand dd_2 will not shift further upon lowering their price. Thus the firm lower it price again in an attempt to get

maximum profit, still other firms also do the same thing. This will cause an inward shift in demand curve to dd_3 . The process will continue until the LAC is tangent to the intersection point of the shifted individual demand curve and actual sales curve at point E. This point represents long run equilibrium of monopolistically competitive firm competing through price adjustment.

1.7.3 Model 3 Equilibrium through price competition and free entry

Dear students, is the assumption of equilibrium with price competition model and new entry of firms to reach long run equilibrium holds in real world? That means, is the number of firms in the product group in the short run optimal as in the second model and is equilibrium achieved through free entry or exit only as in the first model. As Chamberlin suggests in actual life long run equilibrium is achieved both through price adjustment of existing firms and by new entry. This is the main assumption of the third model that we consider next.

As it is indicated in Figure 1.6, price adjustment is made along the dd curve while entry (exit) causes shifts in DD curve. Equilibrium of the firm is stable if DD curve is tangent to LAC at the intersection point between dd and DD curve (actual sales equals to planned sales). Let us see how the long run equilibrium of tangency solution is achieved through price adjustment, entry and exit of firms. At short run equilibrium point, firms in the industry get abnormal economic profit. New firms being attracted to such abnormal profit, enter into the market. This will cause a shift in DD_1 curve inward stepwise as more new firms enter to take place until they start to earn normal profit. At the same time, existing firms independently reduce their price on myopic assumption to maximize their profit through increasing market share emanated from price reduction. Reduction in price causes planned sales curve, dd to shift inward along DD curve and assumed to stop at a point where dd curve is tangent to LAC curve. However, if they continue price reduction below tangency point, each firm realizes a loss instead of positive or normal profit.



For example, **Figure 1.6: Equilibrium with price competition and free entry and exits** shows that as new firms enter the industry, the demand curve for each firm shifts down from D_1 to D_2 to D_3 . Firms will eventually leave the industry first, so that surviving firms will have a large share. Exit will continue until the demand curve (planned sales curve) is tangent to the average cost curve at the cutting point of the demand curve at point E as indicated in figure 1.6. This point is the long run equilibrium of a monopolistically competitive firm obtained through price competition and entry of new firms and exit of old firms.

Activity 1.4

State short run profit maximization condition of a monopolistically competitive firm.

Describe how long run equilibrium of a monopolistically competitive firm is achieved according to:

Equilibrium with price competition model

b) Equilibrium with entry and exit of firms model

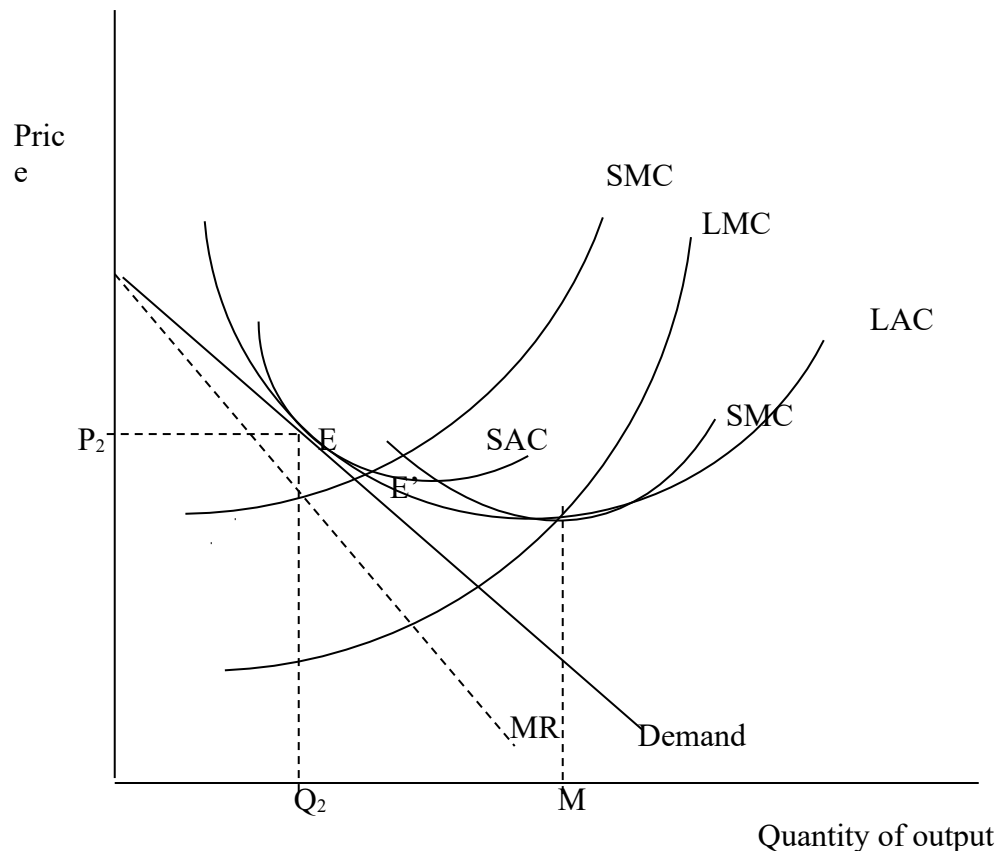
3. What condition should satisfy for a monopolistically competitive firm to maximize long run profit? How much profit would the firm earn at this long run equilibrium point?

1.8 Excess capacity and welfare loss in monopolist competition

As you remember from microeconomics I, firm in perfect competitive market structure operate at a minimum point of long run average cost curve in the long run. That means, it used plants at their full capacity and produced output at a minimum feasible cost. This implies resources allocated efficiently by firm operating in perfect competition market structure compared to firms in other market models. In addition, output production decision take place at a point where $P=LMC = \min LAC$ in the long run. Thus, prices and output set at equilibrium maximize the total consumers and producer surplus compared to other firms operating in imperfect market structure. On this ground, we can use perfect competition firm decision as a benchmark to analyze the efficiency and welfare implication of firms operate in other market environment.

Dear students, do firms in monopolistically competitive firm will tend to operate with excess capacity? In other words, is monopolistically competitive firm expanding production in the long run to a point at which its cost minimized? To answer this question let us revisit Chamberlin's long run equilibrium of monopolistically competitive firm.

Long run equilibrium of the firm under monopolistic competition attained at a point where the perceived demand curve is tangent to LAC curve. Since the demand curve is downward sloping, the LAC is also downward sloping at the point of tangency. Thus, unlike perfect competition, the firm's equilibrium will not achieved at the minimum point of average cost curve. Instead, the tangency between the LAC curve and demand curve take places at downward sloping portion of the LAC curve. That is, firms in monopolistic competition construct a plant smaller than the minimum cost size and operate it at less than the minimum cost output. Therefore monopolistically competitive firm in the long run produce output below their full capacity (below the optimal point). Their cost of production is higher than that of pure competition. This is because monopolistic competition firm incurs additional cost, which is known as selling cost.



It is clear from fig

Figure 1.7 Excess capacity of monopolistic competition firm

behave in Chamberlin way building smaller plant size than a minimum cost plant and operate at point E. In addition, long run average cost curve is tangent to demand curve at point E before it reach minimum point. Therefore, firm in monopolistic competition would not even be producing at the minimum of it short run average cost, point E.

This implies monopolistically competitive firm working at suboptimal scales without exhaustively using the advantage of economies of scale. They are operating with excess capacity defined as the difference between the long run equilibrium level of output and the output level at a minimum point of LAC (equilibrium point of perfect competitive firms). It is the difference between M and Q_2

From social benefit point of View, Monopolistic competition decision did not maximize social welfare since equilibrium price is higher than MC and output is below social desired level. If attempt is made to equalize price and marginal cost, the firm incurs a loss in the long run. This

is because long run marginal cost curve intersect DD_2 (shifted actual sales curve) below long run average cost curve. This implies the social desired point, $P=MC$ cannot achieve with the assumption of monopolistically competitive market structure.

However, Chamberlin argued that the crisis of excess capacity and misallocation of resource is valid only if one assumes that demand curve of individual firm is horizontal. If demand curve is down ward sloping and firms enter in to price competition while entry is free, then point M cannot be considered as social optimal level of output. Consumer wants to have varieties of products. Product differentiation and downward sloping demand curve reflect the desires of consumer to pay higher price in order to have choice among differentiated products. If such products produced at higher cost than minimum amount, it is socially acceptable rather than considered as social cost. Therefore, the difference between M and Q_2 would not be a measure of excess capacity for Chamberlin. Rather it represents social cost of producing and offering consumers a large diversity of products. For Chamberlin, excess capacity is the result of non-price competition coupled with free entry. In this case firm ignores its dd curve (since on price adjustment are made) and concern itself with its market share. In other words, actual sales curve (DD) becomes the relevant demand curve of firms. Thus for Chamberlin excess capacity is the difference between Q_2 and Q_1 indicated in figure 1.

Recently, different types of research suggest that the argument of excess capacity is somewhat myopic. Avinash Dixit, Michael Spence, Joseph Stiglitz and others have suggested that product diversity offered to consumers by monopolistically competitive firms support the idea of social benefit. Consumers have got more chance to select from wide variety of products when products are differentiated (diversified) and so the society benefits. Consumers have got more chance to select from wide variety of products when products are differentiated (diversified) . The society benefits more than its cost and social welfare is maximized

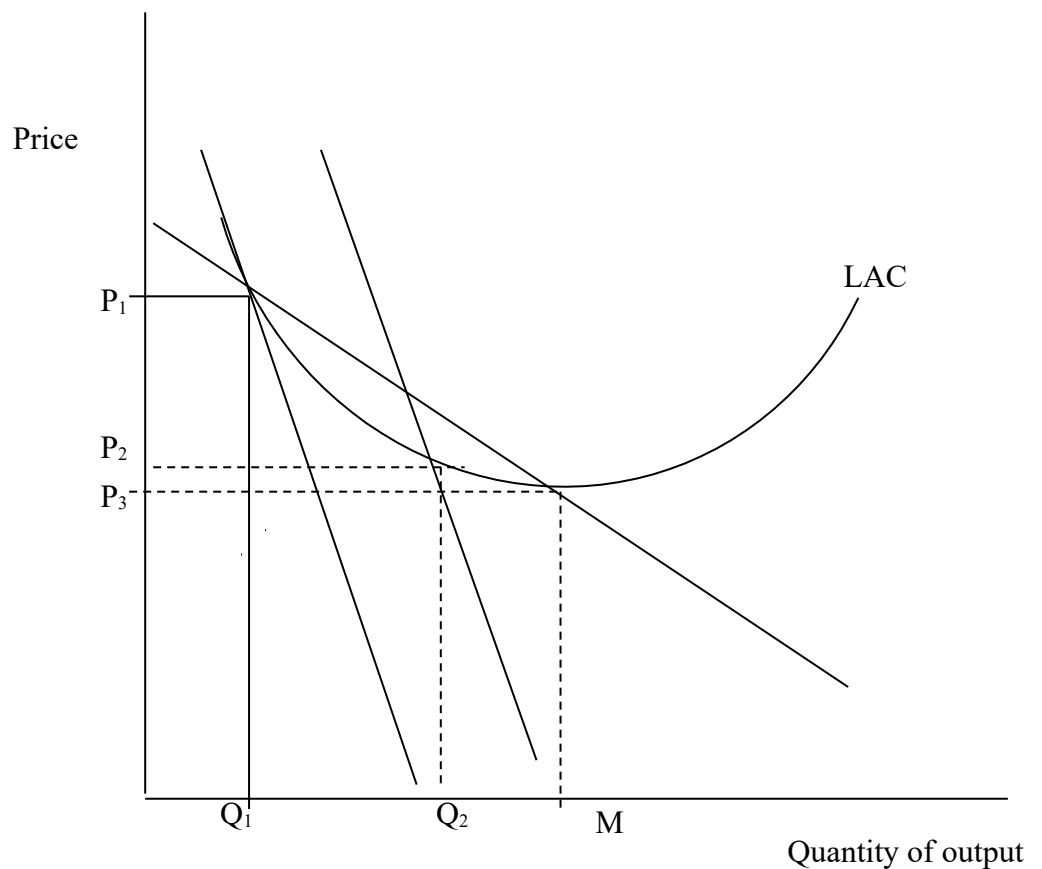


Figure 1.8 Chamberlin excess capacity

In general, compared to firm in a perfect competitive market, firm under monopolistic competition probably produces less and set a higher price. The demand curve confronting the monopolistic competitor would not be perfectly elastic as in perfect competitive firm. Its marginal revenue therefore, is lower than price compared to the equality of marginal revenue with price in perfect competitive market. On the other hand, monopolistically competitive firms are likely to have lower profit greater out put, and lower price compared to monopolistic firm.

1.9 Critiques of Chamberlin large group monopolistic competition market model

This model could be attacked on the following points:-

The assumption of product differentiation is incompatible with the assumption of independent action and free entry of firms. It assumes that the firm actions are independent but firms are dependent on each other. There are a\

Also entry barrier for new firms.

It is difficult to accept the myopic behavior of business firm implied in the model. Normally, business firms learn from their experience.

The concept of industry is destroyed by recognition of product differentiation. Heterogeneous products cannot be added together to form industries demand curve in order to come up with a unique Equilibrium price.

The model assumes a large number of firms with higher price elasticity of product sold in monopolistic competitive market. But, they do not objectively define the number of firms and the level of elasticity required to have a monopolistic competitive market structure.

Contribution of the model

Introduce the concept of product differentiation and selling strategy as two additional policy variables in optimal decision-making process of the firm. These factors are the basis of non-price competition, which is a typical form of competition in real world.

It introduces the concept of share of the market demand curve as a tool of analysis.

Activity 1.5

Define monopolistic competition excess capacity.

State the condition at which output and price set to maximize social welfare and achieve production efficiency.

Do the long run equilibrium level output and price maximize social welfare? Why?

Describe the contributions of monopolistically competitive market model to microeconomics theory.

Summary

Monopolistic competition is a market structure in which large number of firms competes by selling differentiated products. Even though the products are differentiated, they are still close substitute.

Product differentiation can be fancied or real. Fancied product differentiation is a case when the products are the same but the consumer is persuaded through different selling promotion activity such as advertisement that its product is unique from its close substitute.

Real differentiation is case when the inherent character of the product is different. Each close substitute differs in terms of the input used, quality and location.

Products differentiation gives monopolistic competition firm a certain monopolistic power to influence the price of its product. Compared to monopolistic firm the monopoly power is restricted by the existence of close substitute.

Product differentiation give rise to downward sloping demand curve. Firms can increase price without losing all its customers or has to reduce price to attract more customers.

Monopolistically competitive market has U-shaped cost curves like the case in a perfectly competitive market. However, cost of monopolistic competition firm includes selling cost in addition to traditional production cost.

In monopolistic competition, there is a need for advertising to convince consumers that the product is indeed unique and better.

In the short run, monopolistic competition firms acts as a monopolist and optimal decision by equating marginal revenue with marginal cost. At a point where marginal revenue equals to marginal cost, the firm earn positive or abnormal profit provided that it takes place where demand curve greater than the minimum point of short run AC curve.

Long run equilibrium is of monopolistic competitive firm achieved at a point where demand curve is tangent to the falling portion of long run average cost curve. The tangency point arrive through adjustment caused by shift in demand curves due:

☞ Price competition

entry of new firms attracted by short run abnormal profit

Both through entry of new firms and price competition at the same time

Monopolistic competition firm in the long run make output decision at higher cost. It is produced at a point before the minimum long run average cost. As a result firms in monopolistic competition industry said to be inefficient. They could not use the full capacity.

From social point of view monopolistic competition firm would not maximizes social welfare in the long run. This because they set prices above the marginal cost.

Key concepts and terms

Product differentiation	Industry
Product group	Chamberlin excess capacity
Selling cost	real and fancied differentiation
Heroic chamberlin assumption	equilibrium with new entry of firms

equilibrium with price competition

Suggested readings

Nicholson, w: **Microeconomic theory; basic principle and extension**. 5th edition

Salvatore, D(2003):**Microeconomics; theory and applications**.4th edition

Mansfield, E.and Yohe,G (2000):**Microeconomics, theory and applications**.10th edition

Varian, R.Hall (1999): **Intermediate microeconomics** 3rd edition

Pindyck, R.S. and Rubinfeld, D.L. (2003): **Microeconomics**. 6th edition

CHAPTER TWO

OLIGOPOLY MARKET STRUCTURE

2.1 Introduction

Learning objective

2.2 Non collusive oligopoly

2.2.1 The kinked demand curve model

2.2.2 Cournot model

2.2.2.1 Reaction curve approach

2.2.2.2 Mathematical version of cournot duopoly model

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2.3.1 Cartel

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2.3.1.2 Market Sharing Cartel

2.3.2 Price leadership

2.3.2.1 Low cost price leadership

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Summary

Key concepts and terms

Exercise

Suggested readings

2.1 Introduction

Dear students, in theory of Microeconomics I, we have studied perfect competition and monopoly market structure. Both of them in common assume that the decision taken by any particular firm has no diffused effect on the environment in which other firms operate. Individual firm therefore safely neglect the reaction or behavior of its competitor in making its decision. Especially, the issue of strategic interaction between firms is irrelevant in perfect competition because the prevailing market price convey all the external information that was relevant to the firm. Is the above feature of firms can be applied to firms in oligopoly market structure? No, because when small number of large firms dominating a particular industry producing homogenous or differentiated product, any change in a firm's price or output influence the sales and profit of its competitors. As a result, each firm formulates its policies and strategies with an eye on its competitor decision. A market model, which considers such interdependence of few firms when making optimal decision, is called oligopoly market model/structure, which is the subject of this chapter.

The chapter begins with discussion about the characteristics and source of oligopoly market model; then, examine various collusive and non-collusive duopoly model optimal price and output determination under different behavioral assumption about the firms operating in different oligopoly industry.

Learning objective

At the end of the chapter, you are expected to:

Differentiate oligopoly from other forms of market

Identify how interdependence between firms affect their optimal decision

Explain the difference between different duopoly models

Explain how equilibrium price and output determined in collusive and non-collusive oligopoly market structure.

Oligopoly

As it is mentioned above, oligopoly is a market structure dominated by few sellers of homogenous or differentiated product. As a result, the action of each firm affects the other firm's decision in the industry. A beer industry in Ethiopia is a good example of such type of

industry. Each of the major beer producers takes in to account the reaction of others when they formulate their price and output policies. Bedel or Dashen in this case know that its own action will have significant impact on the rest of the beer producers. Therefore, Bedel or any other producer considers the possible reaction of its competitor in deciding prices, degree of product differentiation to be introduced, the level of advertisement to be undertaken and the amount of service to be provided and so on.

Automobile industry, which produces different cars and aerospace industry producing different airplanes are also an example of oligopoly industry. Note also that, all oligopolists are not necessary large firms as given in the above example. Two grocery stores that exist in isolated community can be oligopolists given that their decisions are interdependent and they are the sole supplier of a specific product. This implies that the distinguishing feature of oligopoly is the interdependence of decision making by rivalry firms in an industry. Such interdependence between firms is the natural result of the existence of few numbers of firms in an industry.

Activity 2.1

List the key feature of oligopoly market structure

Identify industry you know that approximate oligopoly market other than the one given in the example

What are the similarity and difference between oligopoly and monopolistically competitive market structure?

In summary, oligopoly market structure is a market structure which contain firms characterized by the following features

Few number of firms in a given industry

Interdependence of firms in decision making

Firms produce homogenous or differentiated product

Firms have some power to set price

Well students by now we have get familiar to the key features of the four different types market structure (perfect competition, monopolistic competition, oligopoly and monopoly)

To recap the main points, their comparison in terms of market power, entry condition of new firms and strategic behaviors of the firms are summarized in table 2.1.

Table 2.1 properties of monopoly, oligopoly, monopolistic competition and perfect competition

	Characteristics	Monopoly	Oligopoly	Monopolistic competition	Perfect competition
1	Number of firms in an industry	One	Few	Few or Many	Many
2	Type of product	Single (unique)	Homogenous or differentiated	Differentiated	Homogenous
3	Ability to set price	Price setter	Price setter but have very limited power	Price setter with limited power	Price taker
4	Profit maximization condition	MR=MC	MR=MC	MR=MC	MR= MC =P
5	Entry condition	No entry	Limited entry	easy entry	Free entry
6	Strategic dependence firms on its rivals	Has no rival	Yes	Yes	No
7	Example	Local natural gas utility	Automobile manufacturer	Retail trades	Apple farmers

Causes of Oligopoly

There are many cause of oligopoly market. Some of them are

Economies of scale: low costs cannot achieved in some industries unless a few firms are producing output that account for substantial percentages of the total market demand .That means the average cost of production reach minimum only when the output produced in large amount by a few firms. As result, the number of firms in such type of industry should be reduced in order to make use of the advantage of economies of scale in production. Economies of scale in sales promotion and advertising may also promote oligopoly.

Barriers to entry: - there are varieties of barrier that did not allow the entry of some firms in to the industry. This barrier may be technological, skill, cost, and size of the market in relation to economies of scale, patent right and different activities of government such as licensing and marketing quota

Collusion (merger of small firms): Small firms collide to get market power and overcome their competitor's pressure. If they gain market power, firms set higher price and restricts output supply that maximizes their profit. Such firms develop to oligopoly while other removed from the market.

Dear students, from our pervious section discussion you have some idea about the general features of oligopoly market structure and its cause. Now let us identify different types of oligopoly model and see how equilibrium level of output and prices are determined in each model given their underlying assumption. For simplicity, we consider only the case of two firms that is termed as duopoly. In addition, we limit our self to the study of firms producing homogenous product. This will allow us to avoid the problem related to analysis product differentiation and focus on the study of strategic interaction between firms.

Based on the reaction pattern of firms in the industry, it is possible to classify oligopoly in to two groups. These are collusive oligopoly and non-collusive oligopoly. Non-collusive oligopoly is a condition in which firms operate independently to determine the optimal level of price and output. That is firms in the industry will not go in to contractual agreement to cooperate in making optimal decision. Under such cases, negotiation and enforcement of binding agreement is not possible even though each firms make some expectation (assumption) about the reaction of its rivalry in response to its action or observe the decision of its rivalry while setting profit maximizing level of output and price. For example, if a firm in non-collusive oligopoly wants to increase output or price to maximize its profit, it has to assume something about the possible reaction of its rivalry and the effect of the reaction on profit maximization process.

Dear students do you think that the assumption firms make about their rival firm is the same in all different type of non-collusive oligopoly model. The answer to this question will obtain after discussing the different non-collusive oligopoly in subsequent subsections. Here are some of non-collusive oligopoly model that we will consider right now after a moment

Kinked demand model

Cournot duopoly model

Bertrand duopoly model

Stackelberg duopoly model

In collusive oligopoly however, firms get together to make open and formal agreement in setting prices and output so as to maximize the total profit of the industry as in the case of cartel. It can be also implicit cooperation of firms in the industry without actually making explicitly agreement with one another as in the case of price leader

2.2 Non collusive oligopoly

2.2.1 The kinked demand curve model

Dear students as you know from your previous microeconomic course, Price in perfect competition, monopoly and monopolistic competition markets adjusts rapidly to changing cost or demand conditions. Is the adjustment of price possible to change in cost and demand conditions in the case of oligopoly market like in other market? For instance, is the price of beer or soft drink changes frequently as their demand and cost condition changes? You might answer these questions based up on the assumption how oligopolist react to each other's decisions according to kinked demand curve model.

The kinked demand curve model, developed by Paul Sweezy in 1939, explains why prices are rigid in some oligopoly market. The model assume that oligopolist often have strong desire to keep stable price. Even under the condition when cost and demand changes, firms are reluctant to change their prices. If costs fall or market demand declines, they fear that the lower prices send the wrong message to their competitors and initiate price war among them. If costs and demand rises, they do not increase price because they are afraid that their competitors may not raise their price.

According to this model therefore, demand curve facing each firm in oligopoly market is kinked at prevailing market price (Chamberlin's intersection of individual and market demand) reflecting the following behavioral pattern of oligopolists. Rivalry firms expected to follow decrease in price but ignore price increase. That means if a firm increases its price, due to increase cost of production or increase demand for its product, it would loss most of its customer and cause total revenue to decrease. This is because other firms in the industry would not follow the increase in price. As result, given that the product produced in the industry is homogenous, consumer preference shifts from the other hand an oligopolist could not increase its market share by lowerintend to maintain prices constant even under the

condition where their demand and cost changes as shown in figure 2.1. The demand curve is much more elastic above point E (kinked point) than below the kink on the assumption that the competitors will not follow price increase but quickly follow price decrease.

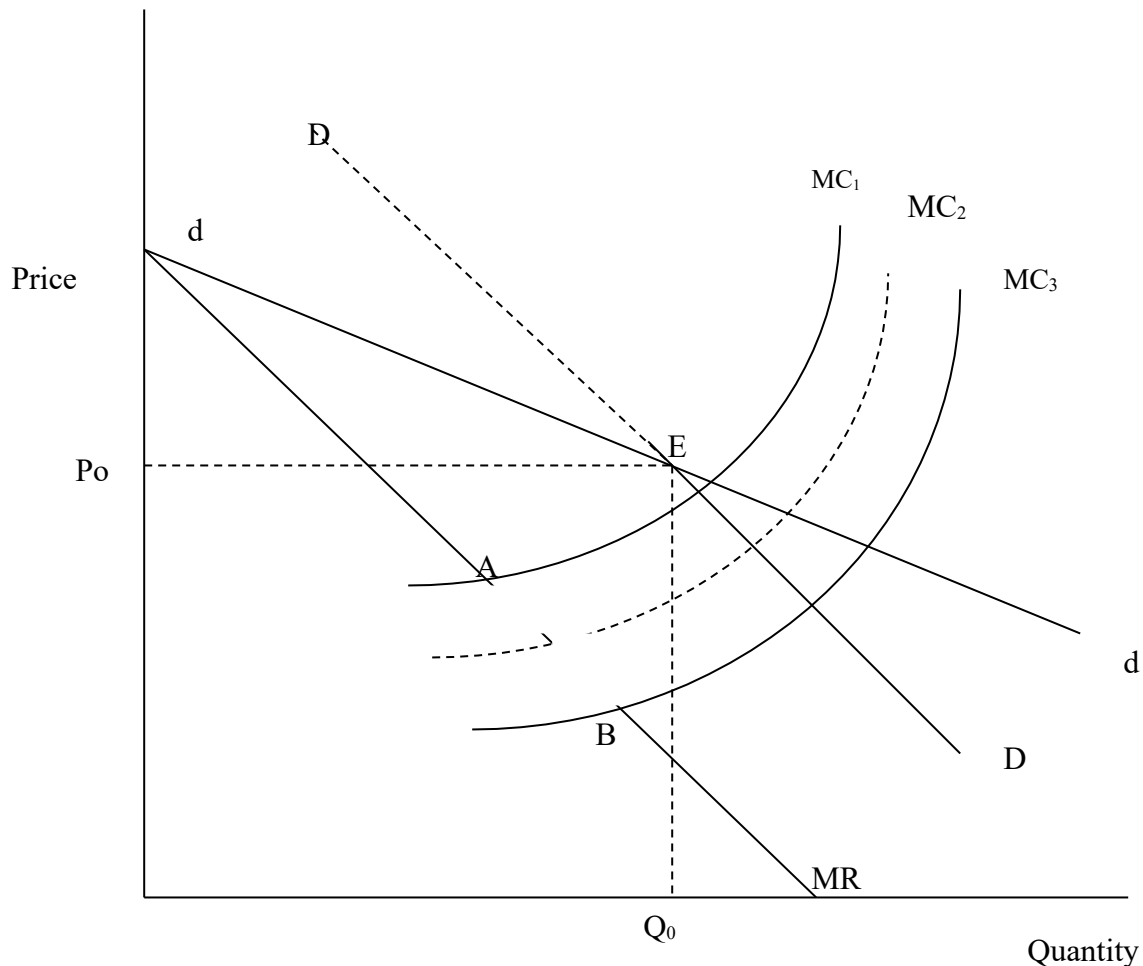


Figure 2.1: The kinked demand curve

As shown in the above figure 2.1, an oligopolist firm faces two demand curves for different ranges of prices. Above P_0 the relevant demand curve for the firm is dE . Because if the firm increases its price, it would lose some of its customer to firm that maintained their previous price. The firm will then face a demand curve given by dE , which is very elastic. On the other hand, if the firm decreases its price below P_0 from the intention of increasing their market share, they will not able to increase their market share, since other firms also decrease their price in order to keep up their customers. Therefore, below P_0 the relevant demand curve of

the firm is ED. This implies the demand curve facing oligopolies is not straight line, rather kinked at a certain price.

Well students, we have seen that oligopolistic is reluctant to price change in response to change in their cost and demand condition. Thus, their demand curve is kinked at the intersection point of market and individual demand curve to reflect the rigidity of prices. Now let us see how their marginal revenue curve derived and how they undertake optimal decision. Normally, marginal revenue curve derived from demand curve. So the marginal revenue curve associated to kinked demand curve is discontinuous at the level of output corresponds to the kinked point. It has two segments as indicated in figure 2.1, dA and BMR. Segment dA corresponds to the upper part of demand curve, while the segment BMR corresponds to the lower part of demand curve and the kink point on demand curve corresponds to the discontinuous portion of marginal revenue curve.

The point of kink defines equilibrium of the firm since at any point to the left of the kink, MC is below MR, while to the right of the kink; MC is larger than MR. Thus profit of the firm maximized at the point of the kink. However, this equilibrium is not necessarily defined by the intersection of MC and MR curve. So long as MC passes through segment AB, the firm maximizes its profit by producing Q_0 and charging p_0 level of price. This level of price and output is compatible with wide range of cost.

To sum up, the kinked demand curve model give us some insight about why price and output will not change despite changes in cost and demand in oligopoly market structure. **The only case where a rise in cost results in increase in price is when the rise in cost equally affects all firms in the industry. It is important to note here that a kinked demand curve model does not explain how equilibrium price and output determined like other models rather, explain why price once set remain fixed.** Lastly some economists have been very critical of the model's assumption .It would certainly be a mistake to conclude that oligopolies in general are unresponsive to change in cost as well as unresponsive to change in demand in the world of technological advancement and change in taste of consumer preference. Perhaps we observe the opposite in many oligopoly industries.

Activity 2.2

Why price in oligopoly industry becomes unresponsive to change in demand and cost conditions

Describe the nature of demand curve facing an oligopolist according to kinked demand curve model.

kinked demand curve model;

Explain how an equilibrium price is established.

Explain how firms change their price when cost of production and demand changes

Explain how equilibrium price determined

It explains prices are rigid in oligopoly market.

2.2.2 Cournot model

A French mathematician, Augustine Cournot in 1838 using two firms producing homogeneous products, illustrated Cournot model for the first time. He illustrates the model assuming two firms having identical cost facing linear market demand. Given these assumptions, each firm knows that the market price will depend on the total output of both firms. Thus to maximize their profit each of the Cournot duopolists simultaneously decides, how much to produce by taking its rivals' output constant at existing level regardless of what output it decides to produce. Thus, each firm takes the other firm's output level as given and chooses its own output level to maximize profit. **The level of output that it chooses will, of course depend on how much it thinks its rival will produce.** In other words, each firm recognizes that its own decision about output y_i will affect its revenue through affecting market price but any one firm's output decisions do not affect those of any other firm. That is, each firm recognizes that

$\frac{\partial p}{\partial y_i} \neq 0$ but assume that $\frac{\partial y_j}{\partial y_i} = 0$, for $j \neq i$.

Dear students now let us see Cournot's simplified illustration of the model. He started his illustration by assuming that there are two firms (firm-A and firm-B), each producing mineral water at zero cost and face linear demand curve DD as shown in figure 2.2. Each firm also acts on the naïve assumption that its competitor will not change its output when deciding its profit maximizing level of output.

Assume that firm A is the first to start producing and selling mineral water assuming that firm B produce nothing. Firm-A therefore thinks that its effective demand curve is the market demand because the firm thinks that it will be the sole producer of mineral water. So to find the profit maximizing level of output and price, we use the marginal principle ($MR=MC$). Since it is assumed that cost of production equals to zero MC is also zero. Therefore profit maximization condition of the firm reduced to $MR=0$. This point corresponds to Q_1 level of output that is half of the total market demand.

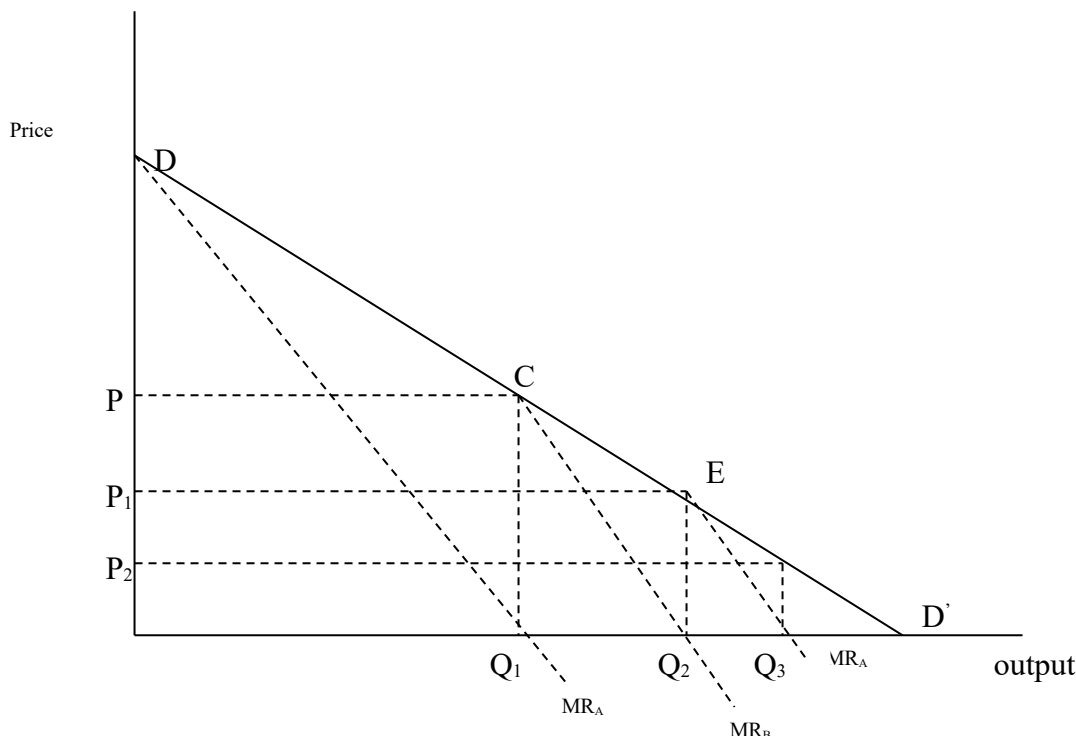


Figure 2.2.: Cournot's equilibrium

Now firm-B assume that firm-A will keep its output fixed at Q_1 and define CD' as its relevant demand curve to determine its profit maximizing level of output. Applying the same marginal rule to the perceived demand section, firm-B produce half of the market which is not supplied by firm-A. That is B's profit maximizing level of output is $\frac{1}{4} = (\frac{1}{2} \cdot \frac{1}{2})$ of the total market demand. This point represented by Q_2 .

In the next round, firm-A assumes B will produce fixed level of output as before ($\frac{1}{4}$ of the total market), it will produce half of what is not supplied by B, $\frac{1}{2} \left(1 - \frac{1}{4}\right) = \frac{3}{8}$ of the total

market demand firm-B. Also reacts in the same fashion and produces $\frac{1}{2}\left(1 - \frac{3}{8}\right) = \frac{5}{16}$ of the total market. This action and reaction pattern continues, since the firms are acting on the naive behavioral assumption (firms cannot learn from their experience). Eventually, the equilibrium level of output obtained, when each firm produces one-third of the total market. This equilibrium level can be obtained as follows

Firm A's equilibrium level of output at different successive period;

$$\text{Period-1: } \frac{1}{2}$$

$$\text{Period-2: } \frac{1}{2}\left(1 - \frac{1}{4}\right) = \frac{3}{8} = \frac{1}{2} - \frac{1}{8}$$

$$\text{Period-3: } \frac{1}{2}\left(1 - \frac{5}{16}\right) = \frac{1}{2} - \frac{1}{8} - \frac{1}{32}$$

$$\text{Period-4: } \frac{1}{2}\left(1 - \frac{43}{128}\right) = \frac{43}{128} = \frac{1}{2} - \frac{1}{8} - \frac{1}{32} - \frac{1}{128}$$

We can get equilibrium level of output for firm-by adding up the different level of output firm-A produces by assuming a given level of output supply by firm-b's:

$$\begin{aligned} &= \frac{1}{2} - \frac{1}{8} - \frac{1}{28} - \frac{1}{512} \dots\dots \\ &= \frac{1}{2} - \left[\frac{1}{8} + \frac{1}{8} * \frac{1}{4} + \frac{1}{8} * \left(\frac{1}{4}\right)^2 + \frac{1}{8} * \left(\frac{1}{4}\right)^3 + \dots \right] \end{aligned}$$

The expression in parenthesis is infinite geometric progression with common ratio $r = \frac{1}{4}$.

Applying the summation formula for an infinite geometric series, $G_n = \frac{G_1}{1-r}$ (G_n - the sum of N geometries infinite series, G_1 - the first term of geometric infinite series and r - Common ratio)

$$= \frac{1}{2} - \frac{\frac{1}{8}}{1 - \frac{1}{4}} = \frac{1}{3} \quad \text{-Of total market demand}$$

For firm -B:

$$\text{Period-1: } \frac{1}{4} = \frac{1}{2} \left(1 - \frac{1}{2} \right)$$

$$\text{Period-2: } \frac{5}{16} = \frac{1}{2} \left(1 - \frac{3}{8} \right) = \frac{5}{16} = \frac{1}{4} + \frac{1}{16}$$

$$\text{Period-3: } \frac{21}{64} = \frac{1}{2} \left(1 - \frac{11}{32} \right) = \frac{1}{4} + \frac{1}{16} + \frac{1}{64}$$

$$\text{Period 4: } \frac{85}{256} = \frac{1}{2} \left(1 - \frac{43}{128} \right) = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256}$$

We can write B's equilibrium level of output as $\frac{1}{4} + \frac{1}{4} \left(\frac{1}{4} \right) + \frac{1}{4} \left(\frac{1}{4} \right)^2 + \frac{1}{4} \left(\frac{1}{4} \right)^3, \dots$

$$= \frac{G_1}{1-r} = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{1}{3} \text{ Of the total market demand.}$$

Example: If the market demand for cournot's mineral water in the example above is, equals to 120, what level of output maximize the profit of each firm.

Solution: we said under the above assumption (zero cost and cournot behavioral assumption) the two firms maximize its profit by producing $\frac{1}{3}$ of the total market demand each, that is, $\frac{1}{3}(120) = 40$

Therefore, under cournot costless production assumption, the output level that maximizes the profit of a duopolist is equal to 1/3 of the total market. By similar analogy, it is also possible to generalize for N-firm cases. That is for an industry with n-firms the output, which maximizes the profit of a firm, is $\frac{1}{n+1}$ of the total market demand. For example if there are four firms in an industry act under cournot naïve assumption, the output level that maximizes the profit of each firm is $\frac{1}{5} = \left[\frac{1}{1+4} \right]$ of the total market demand.

Dear students, is the costless production (identical cost and demand) assumption of cournot in the above analysis is realistic? No, because there are different type of cost involved in production in real world and definitely, the cost of production in most cases different from one firm to the other. Hence, let us reconsider the model after relaxing identical cost and demand assumptions. In other words how much should each firm

produce to maximize their profit if they undertake production of their output at different cost and demand condition. This can be determined using another approach called a reaction curve approach that we consider next.

2.2.2.1 Reaction curve approach

Under this approach, we try to see how Cournot duopolist choice output level that maximizes their profit after relaxing the identical cost and demand assumption. Here we use the reaction curves of the two firms to determine the optimal level of output of the duopolist under the basic Cournot behavioral assumption. So before using the reaction curves as a tool for determination of the optimal choice of the firms, it is helpful first if we understand what a reaction curve is and how one can derive a reaction curve.

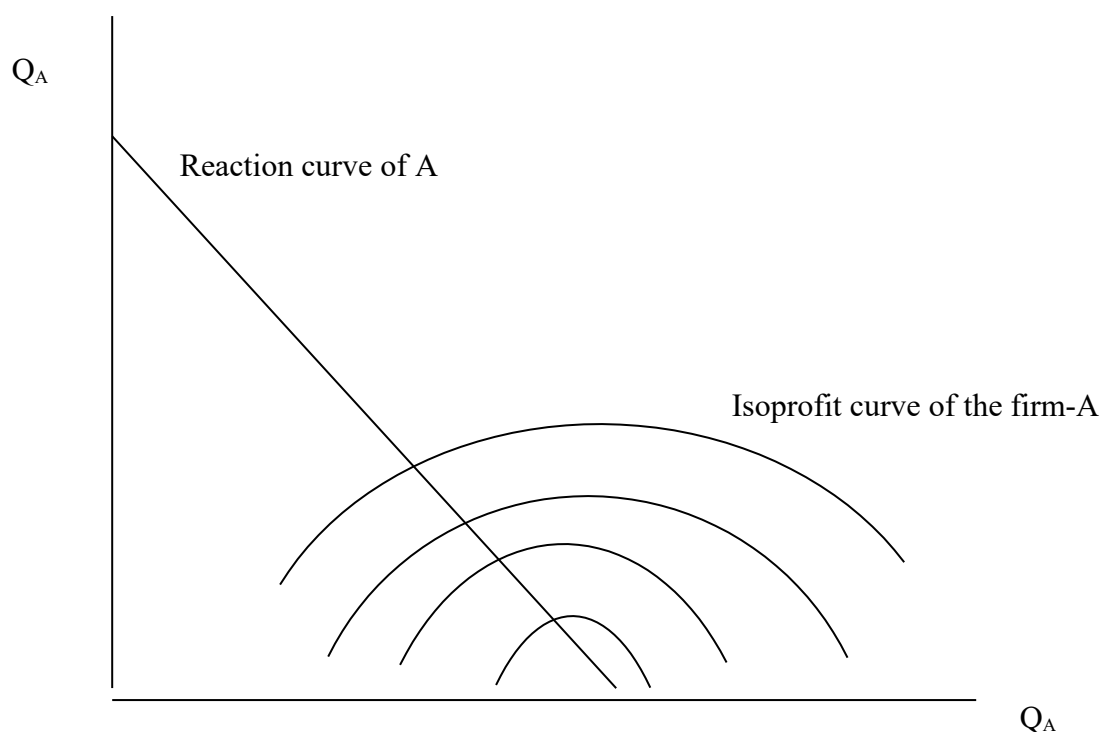


Figure 2.3.: (a): Isoprofit of firm-A

A reaction curve is a graphic representation of reaction function. A reaction function is a function that shows the functional relationship between the optimal output level of one firm and its beliefs about other firm optimal choice. For instance, reaction function of firm-A shows, how firm-A will react in making its output decision to its perception of how much it thinks firm-B will produce and sell. **That is for firm-A to decide to produce let say**

Y_1 level of output, it has to forecast first the amount of output firm-B produce such as Y_2 . A function which shows the amount of Y_1 as a function of Y_2 gives the reaction of firm-A. Similarly, reaction function of firm-B defined in the same fashion.

Graphically the reaction curve of cournot duopolists derived from their respective isoprofit curve. A curve contains a locus of point that yields the same level of profit for the firm. Isoprofit of firm-A is the locus of points defined by different level of output of firm-A and its rival B which yields the same level of profit to firm-A. Similarly, isoprofit curve of firm-B is the locus of point defined by different level of output of firm-B and its rival firm-A which yields the same level of profit to firm B. The isoprofit curves of the two firms given in figure 2.3 (a) & (b). From the definition, it should be clear that the isoprofit curves are a type of indifference curve

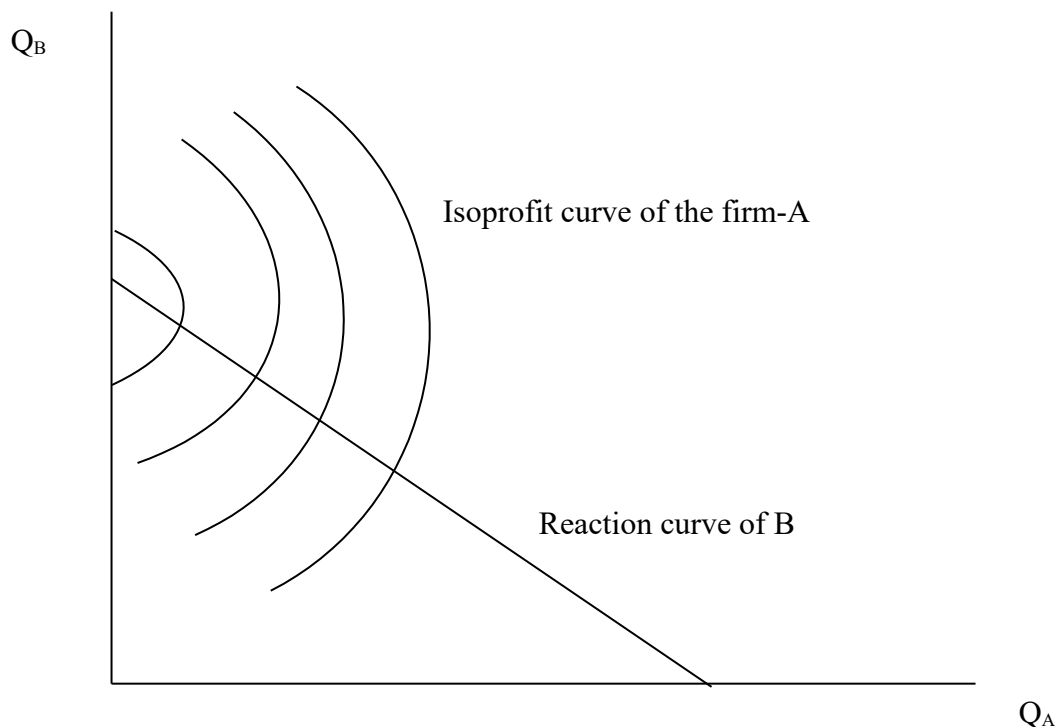


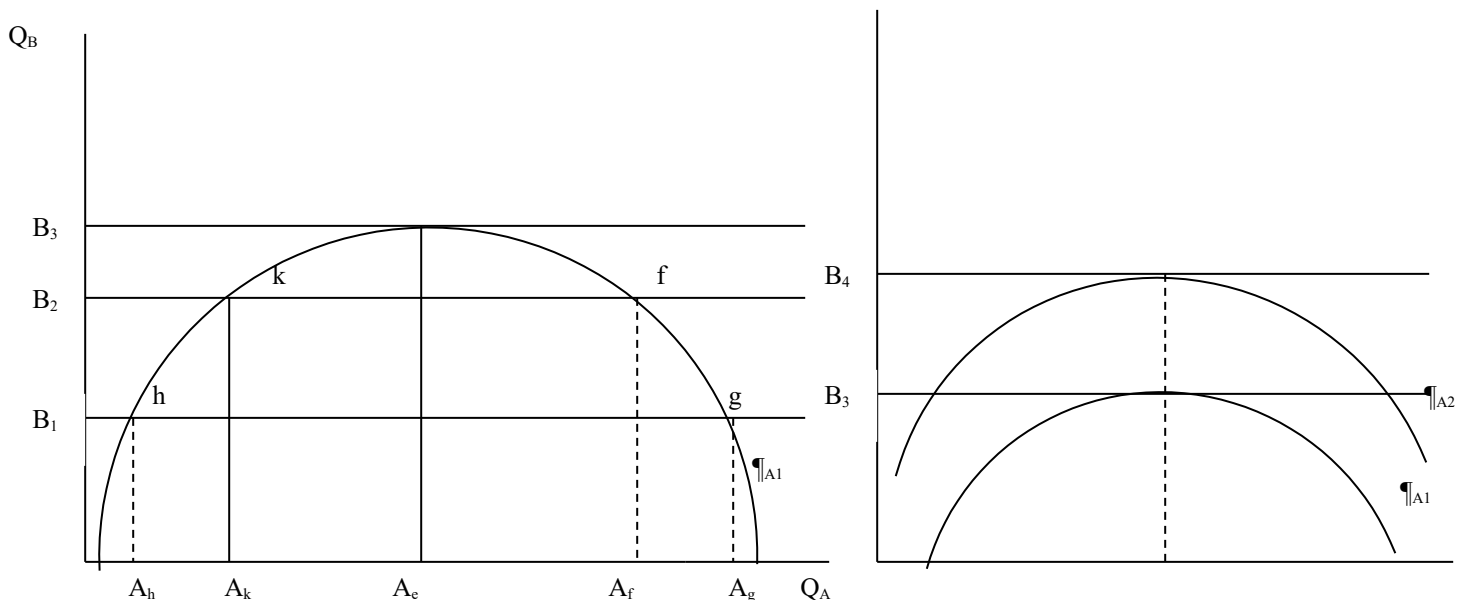
Figure 2.3.: (b): isoprofit of firm- B

Properties of isoprofit curve

Isoprofit curves for substitute commodities are concave to the axis along which we measure the output of the rival as indicated in figure 2.3.a and b. Therefore, isoprofit of firm-A is concave to axis that measures the output of A and isoprofit curve of firm-B is concave to the

quantity axis of firm-B. The shape of isoprofit curve of firm-A shows how A react to B's output decision so as to retain a given level of profit. For example, consider the isoprofit curve of firm-A, π_A and firm-B, π_B as in the figure- 2.4.below. It is concave to A's quantity axis

Let us suppose that B decide to produces B_1 amount of output, firm-A will realizes the same level of profit by producing A_h or A_g amount of output. Assume firm-A decide to produce A_g amount of output. If firm-B increases its output to B_2 , firm-A must decreases its output to A_f to maintain the same level of profit. If firm-A continue producing A_g amount while firm-B increase its output level to B_2 the total amount of output supplied to the market increase and result in decrease in prices and hence profit of the firms. If B further increases its output to point B_3 , firm-A should decrease its output to A_e , other wise it result in reduction of price and then profit .Beyond point E, increase in output by firm-B result in fall in market price and increase cost of production which result in decrease in profit of the firm. This is indicated by out ward shift in isoprofit function to π_{A2} , representing lower level of profit .This implies the isoprofit line which is found far from quantity axis represent lower level of profit while the isoprofit curve found nearer to quantity axis represent higher level of profit

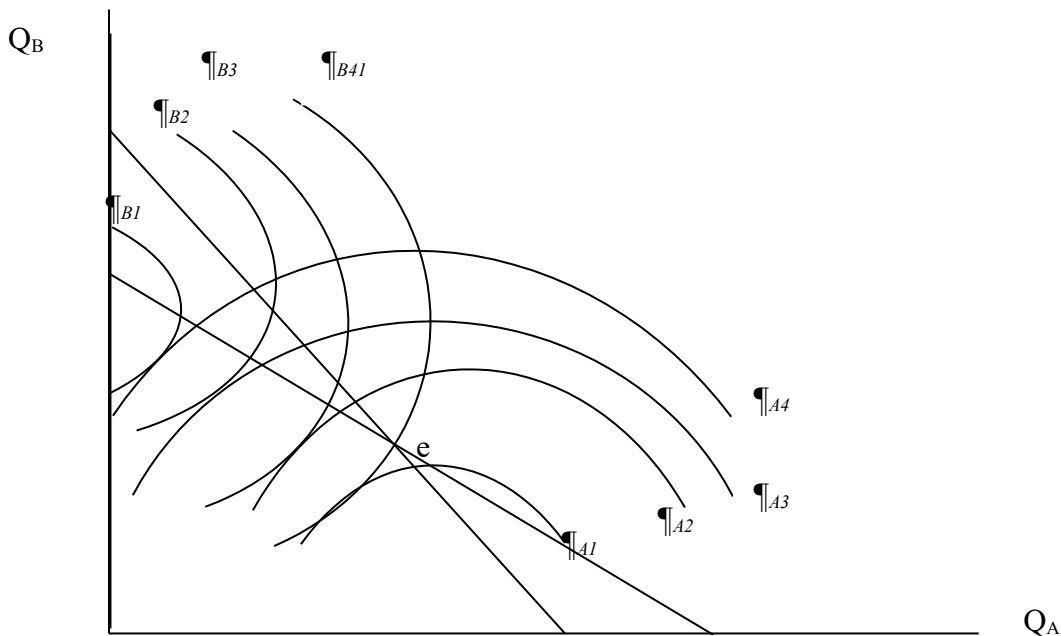


For firm-A the highest points of successive isoprofit curves lies to the left of each as we go
Figure 2.4.: isoprofit curves each other as we go towards to

quantity axis. On the other hand the highest point of the isoprofit curves of firm-B lies to the right of each other as we move further away from the quantity axis) Q_A and lies to the left as we move toward the quantity axis. Joining these respective highest points of isoprofit curve of the two firms gives their reaction curve. Hence, the reaction curve of firm-A is a locus of

highest point of Isoprofit that firm-A can attain, given the level of output of rival B. The reaction curve of firm-B also represent the locus of highest isoprofit point that firm-B can attain as indicated in figure 2.3. (a) and (b).

Dear students we have seen what a reaction curve is and how to derive it from isoprofit curve. Now let us see how reaction curves used to determine the equilibrium level of output for Cournot duopolist. Each firm's reaction curve tells us how much to produce given the output of its competitor. Therefore, at equilibrium the belief that each firm makes about other's firm output level during setting their output is equal to the actual amount of output produced by its rival. This point corresponds to the intersection of two reaction curves, point e as shown in figure 2.4.



Note that at point e, each firm maximizes its own profit but the industry's (joint profit) profit is not maximized. *Figure 2.4.: Cournot's duopoly model equilibrium* curve. It is the curve formed by joining the tangency point between the two firms' isoprofit curves. Points on this curve represent optimal point (maximize industry's profit). Points off contract curve however represent lower level profit to one firm or both profits. Therefore, since point e is found off the contract curve it cannot maximize industry's profit and represents suboptimal level of output. Such suboptimal resource allocation is the result of naïve Cournot behavioral assumption of

firms cannot learn from their experience. Had they learned from experience, they would show a different strategic behavior. For instance, they would have come together and agreed to share the market in such a way that would raise the profit of the two firms or the industry as a whole.

Activity 2.3

Define the following terms.

Isoprofit curve

c) contract curve

Reaction curve

d) Cournot's equilibrium

State the basic assumption made about firms in the Cournot duopoly model while they are setting output level that maximizes their profit.

Is Cournot equilibrium level of output maximizes industry's profit? If not, why?

Find the level of output produced that maximizes the profit of a firm that acts under costless production assumption of Cournot, if the total market demand facing the firm is 3000.

2.2.2.2 Mathematical version of Cournot duopoly model

So far, we have seen graphical presentation of Cournot model. Now let us see mathematical treatment of the model. Determining equilibrium output level of the Cournot duopolists mathematically given their cost and demand function. Here we begin from the assumption of the model, each firm decides how much to produce by treating its competitor's level of output fixed at existing level. For example, if the units of output that firm-2 produce currently is Y_2 , firm 1 assumes firm 2 continues producing Y_2 amount of output while deciding to produce Y_1 unit of output. So the total output supplied to the market equals $Y = Y_2 + Y_1$ and the market price of the product, $P(y) = P(Y_1 + Y_2)$. Given the cost function of firm 1, $C_1(Y_1)$ and firm 2, $C_2(Y_2)$, we can define the profit maximization problem of the firms as follows.

For firm 1: $\text{Max } P(Y_1 + Y_2) Y_1 - C_1(Y_1)$

For firm 2: $\text{Max } P(Y_1 + Y_2) Y_2 - C_2(Y_2)$

Following the usually optimization procedure, take the first order condition of the profit function of each firm with respect to choice variable (their output level). The result of first order condition gives the reaction function of each firm. Then solve the two-reaction curves to gather simultaneously to get the equilibrium output level of the two firms.

Example: Suppose that two firms producing identical product and supply to a market with a linear demand function, $p(y) = 60 - 2y$. Where $y = y_1 + y_2$ and y_1 represent unit of output produce by firms-1 and y_2 represent unit of output produce by firm two. The cost of production of firm 1 and firm-2 is given by $c_1(y_1) = y_1^2$ and $c_2(y_2) = 4y_2$ respectively. Find the output level that the two firms should produce to maximizes their profit under cournot behavioral assumption.

Solution: starts solving the problem by defining the profit maximization problem of the two firms.

$$\text{Firm-1 } \max \pi_1 = \max [p(y_1 + y_2)]Y_1 - c_1(y_1)$$

$$\Pi_1 = [60 - 2(Y_1 + Y_2)]Y_1 - Y_1^2 = 60Y_1 - 2Y_1^2 - 2Y_1Y_2 - Y_1^2$$

$$\Pi_1 = 60Y_1 - 3Y_1^2 - 2Y_1Y_2$$

$$\text{F.O.C: } \frac{\partial \pi_1}{\partial y_1} = 0$$

$$\Rightarrow 60 - 6y_1 - 2y_2 = 0$$

$$\Rightarrow y_1 = 10 - \frac{1}{3}y_2 \dots \dots \dots (1)$$

Equation (1) represents the reaction curve of firm –1

Firm 2

$$\max \pi_2 = \max [p(y_1 + y_2)]Y_2 - c_2(y_2)$$

$$\Pi_2 = [60 - 2(Y_1 + Y_2)]Y_2 - 4Y_2$$

$$\Pi_2 = 60Y_2 - 2Y_1Y_2 - 2Y_2^2 - 4Y_2$$

$$\text{F.O.C: } \frac{\partial \Pi_2}{\partial y_2} = 60 - 2Y_1 - 4Y_2 - 4 = 0$$

$$\Rightarrow 28 - 2Y_1 - 4Y_2 = 0$$

Equation (2) represents the reaction function of firm 2. Solve simultaneously the two reaction curves together to get the cournot equilibrium level of output.

$$y_1 = 10 - \frac{1}{3} y_2 \dots\dots (1) - \text{Reaction curve of firms -1}$$

$$y_2 = 14 - 0.5 y_1 \dots\dots (2) - \text{Reaction curve of firms -2}$$

Substituting equation (1) in equation (2)

$$y_2 = 14 - 0.5(10 - \frac{1}{3} y_2) = 14 - 5 + \frac{1}{6} y_2$$

$$y_2 = 9 + \frac{1}{6} y_2 \rightarrow \frac{5}{6} y_2 = 9$$

$$y_2 = \frac{54}{5} = 10.8 \text{ (Equilibrium level of output of firm -2)}$$

$$y_1 = 10 - \frac{1}{3} y_2 = 10 - \frac{1}{3}(10.8) = 10 - 3.6$$

$$y_1^* = 6.4$$

The total output of the industry at equilibrium is the sum of Y_1 and

$$Y_2, (y = 6.4 + 10.8 = 17.2)$$

The above result can be represented graphically as shown in figure 2.5 below

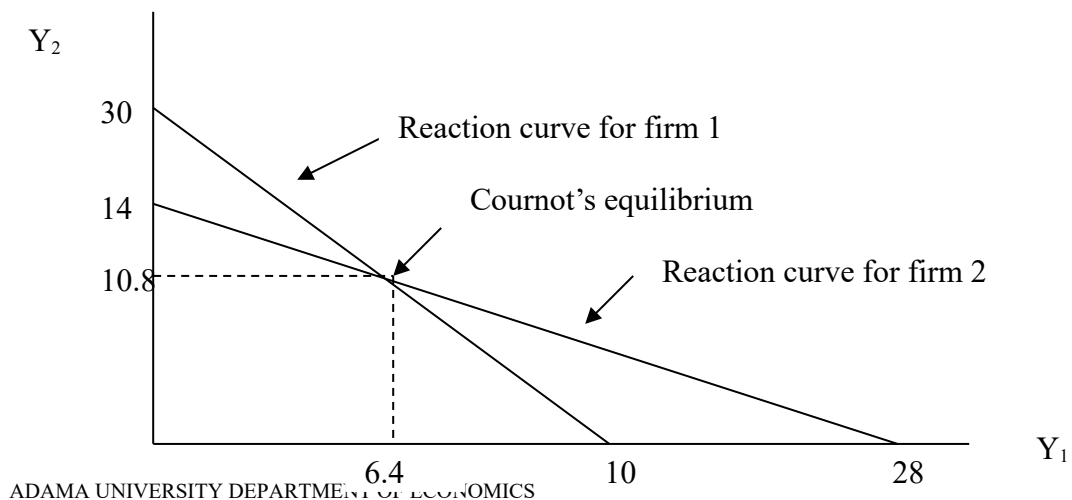


Figure 2.5 Cournot equilibrium

The market price of the product

$$p(y^*) = 60 - 2(y^*)$$

$$= 60 - 2(17.2)$$

$$= 60 - 34.4$$

$$P(Y^*) = 25.6$$

The profit of each duopolistic is

$$\Pi_1 = p(y^*)y_1 - c(y_1)$$

$$\Pi_2 = p(y^*)y_2 - c(y_2)$$

..... Profit of firm-1

$$\Pi_2 = p y_2^* - c(y_2^*)$$

$$= (25.6)(10.8) - 4(10.8)$$

$$= 276.48 - 43.2$$

..... Profit of firm-2

Total industry's profit, $\Pi = \Pi_1 + \Pi_2$

$$= 163.84 + 233.28$$

$$\Pi = 397.12$$

Activity 2.4

1. Given the following demand and cost function of cournot duopolist

$$Y = 40 - 0.2p \text{ where } Y = Y_1 + Y_2$$

$$C_1 = 50 + 2y_1 \quad C_2 = 100 + 10y_2$$

Find the profit function of each firm and their reaction function

Find cournot equilibrium level of output and price

Calculate the profit of each firm

Show graphically the equilibrium level of output of the two firms.

2. Suppose that we have two firms that face a linear demand curve $y = a - bp$ and have constant marginal costs, c , for each firm. Find the equilibrium level of output for the two firms in terms of a and b assuming that they act according to Cournot model.

2.2.3 The Bertrand duopoly model

In Cournot model, we have seen that firms are choosing quantities of output produced and letting the market to determine price. However, in case of Bertrand model, firms set their price and letting the market to determine the amount of output sold. This implies the strategic variable upon which firms are competing to maximize their profit is price rather than output for Bertrand duopolist. Similar to Cournot model however, each firm makes decision about the level of price that maximizes their profit simultaneously by assuming their competitor's price fixed at existing level.

The model also assumes that firms operating in the industry produce homogenous product with identical cost. This implies that each firm faces the same demand curve and consumer will prefer to purchase from lower price seller or firm. Thus if the two firms charge different price, lower price firm will supply the entire market while the firm which charges higher price sell nothing. If both firms charge the same price, the consumers are indifferent between the two firms' product. This leads to a return to firm-1 of the form:

$$\pi_1(p_1, p_2) = \begin{cases} [(p_1 - c)y(p_1)] & \dots\dots\dots (1) \\ [(p_1 - c)y(p_1)]\left(\frac{1}{2}\right) & \dots\dots\dots (2) \\ 0 & \dots\dots\dots (3) \end{cases}$$

Equation (1) to (3) represent the profit earned by firm-1, when it set price less than, equal to and greater than firm-2 respectively.

Therefore, Bertrand model is an oligopoly model in which firms producing homogenous product set price simultaneously that maximize their profit by assuming their competitors price fixed at a certain level.

Activity 2.5

Given the above underline features of the Bertrand model, what level of prices should each firm charge to maximize their profit?

It is known that price cannot be set less than marginal cost since each firm has the incentive to decrease their price to increase its profit by reducing production level. So let us see how Bertrand duopolist set price under the case where price is greater than marginal cost for the two competing firms.

Suppose that both firms sell their product at some price greater than marginal cost. If firm-1 lower its price by small amount while the other firm keeps its price fixed, the entire consumer will prefer to purchase from firm-1 and the other firm sell nothing as stated above. However, the other firm also acts in the same way (have the incentive to reduce its product) if its price is greater than marginal cost. Therefore, price greater than marginal cost cannot be stable equilibrium because each firm has an incentive to cut price as long as production remain profitable. The only equilibrium point where firms have no incentive to change their decision is where price equal to marginal cost. This is the same as competitive market equilibrium condition. Both firms set price equals to marginal cost in the short run and both will set price equal to average cost in the long run because of constant cost assumption, which leads to earning of zero profit.

The equilibrium price level of Bertrand model can be also determined through reaction curve approach. As we have seen under Cournot model, reaction curves are derived from Isoprofit maps. However, Bertrand Isoprofit curve represent different thing from Cournot's isoprofit curve. Bertrand model isoprofit curves contain locus of point that represents a combination of prices of a firm and its competitor that yields the same level of profit to the firm. Unlike Cournot isoprofit curve, the shape of Bertrand duopolist isoprofit curve is convex to the price axis of the firms.

Dear students, what does the shape of Bertrand duopolist isoprofit imply? The shape of Isoprofit of Bertrand duopolist show how a firm reacts to price cut by its competitor. For example if the competitor of a firm cut price, the firm also adjust its price to maintain its profit at the same level. Such process continues up to the minimum point of the isoprofit curve, which represents lower level of profit. If the competitor cut price beyond the minimum point, the firm cannot adjust its price to keep the same level of profit. The profit of the firm decreases due to fall in price and increase in output level. This can be indicated by moving to the lower level of Isoprofit curve.

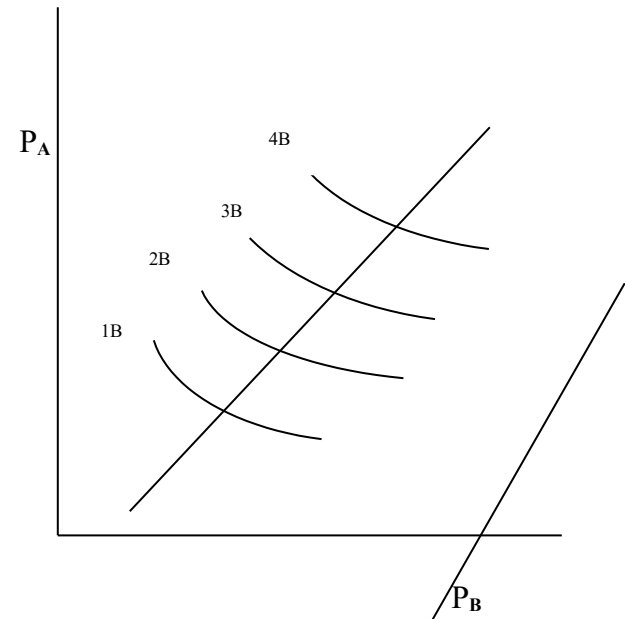
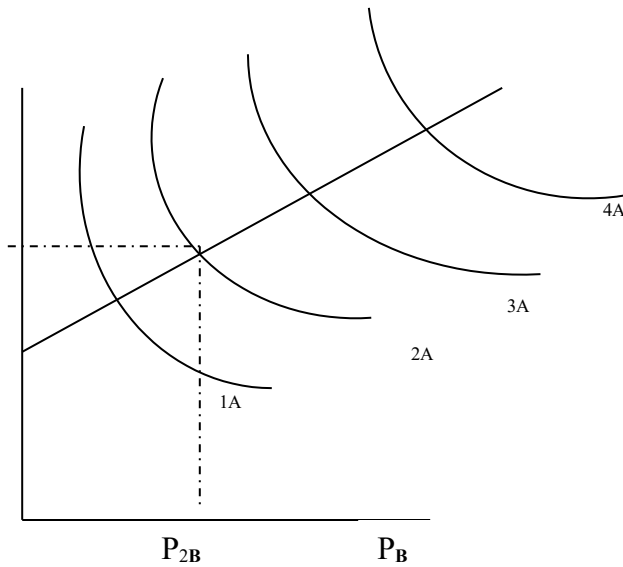
P_{2A} 

Figure: 2.6. Reaction and Isoprofit curves of Bertrand duopolist

For example, firm-A in figure 2.6 moves from Isoprofit curve π_{2A} to Isoprofit curve π_{4A} when firm-B cut its price below P_{2B} . This implies Isoprofit curve found nearer to the price axis of the duopolist represent lower level of profit. If we join the minimum point of successive isoprofit curves of firm-A, it result in reaction curve of A. They are locus of point firm-A can attain by charging a certain price, given the price of its rival. The reaction curve for firm-B also derived in the similar way by joining the lowest point of isoprofit curves given the price level of firm-A. Given the two reaction curves, Bertrand equilibrium defined by the intersection of the reaction curves of the firms as indicated by figure 2.7.

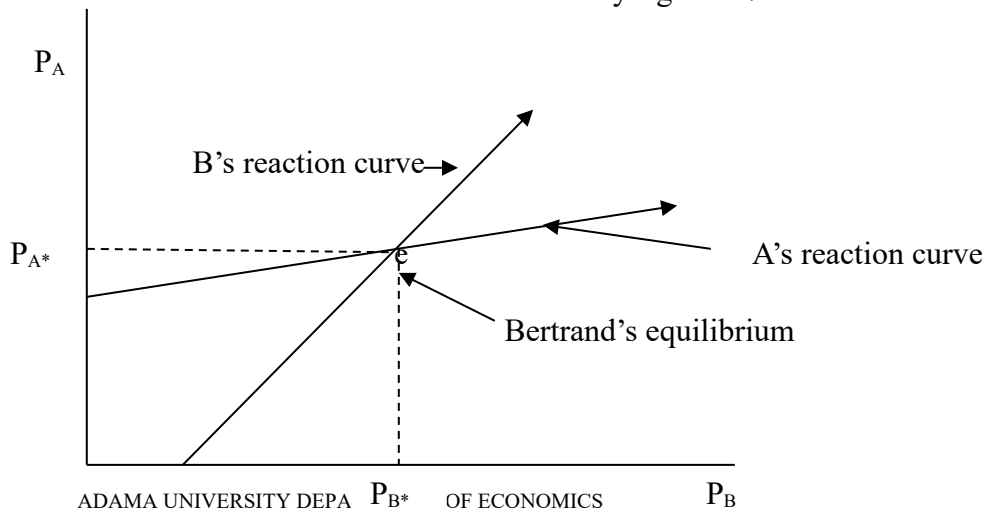


Figure: 2.7. Bertrand equilibrium

Price level (P_A^* , P_B^*) is the equilibrium level of Bertrand duopoly model. Also like Cournot model, Bertrand equilibrium does not lead to maximization of industry (joint) profit, due to the fact that firms behave naively (firms set their price by assuming its rival keeps its price fixed and they never learn from past experience) to decide the price that maximizes their profit. The industry profit could be increased if firms recognized their past mistakes and abandoned the Bertrand pattern of behavior.

Although the Bertrand model is used to understand the strategic interaction of oligopolists on price setting, it has plenty of shortcomings for various reasons. For one thing, firms, which produce exactly the same product, seem to compete more by focusing on non-price competition than on price competition. Moreover, if they focus on price competition, they set the same price in accordance with the model; there is no real assurance that they split the market equally. Also like Cournot model, Bertrand model is criticized for its naïve assumption of firms.

Activity 2.6

What are the difference and similarity between Bertrand and Cournot model.

On which variable do firms in Bertrand model make strategic interaction (output, price)

What does the concave shape of the isoprofit curve of Bertrand duopoly imply?

In what aspect is the Bertrand model similar with perfect competition market model

Stackelberg Model

Dear students, in the previous duopoly model we have seen that duopolists make simultaneous optimal decision about level of output produced and price charged. However, according to Stackelberg model firms make decision about output that maximizes their profit sequentially.

That is, there is a firm known as a dominant (stackelberg leader) which knows the other firm behaves naively in cournot fashion (i.e. known the reaction function of naïve firm). The firm, which behaves in cournot, fashion (take competitor's output as given) and make decision after observing leader's decision is called the follower. The leader has extra information and potential than the follower firm to make decision before the follower firm. In choosing its own output, therefore, the leader could account the effect of its output on follower behavior while the follower naively took leader's output as given.

For instance, IBM is often considered as a dominant firm in the computer industry. Small firms in the industry wait for the decision made by IBM in order to make decision how much to produce and the type of product they supply to the market. In general, the leader firm (the first mover) decides to produces certain amount of output, which maximizes its profit; it will take into consideration the impact of follower firm. The follower firm, after identifying the level of output produced by the leader firm, responds by producing certain amount of output to maximize its profit. Each firm act in the stated way while making its output decision because each of them know that the total output produced determined the market price and then profit they earn.

Given the structure of the model, what output should the leader choose to maximize its profits? Leader firm recognizes its influence on the action of followers firm and total level of output when making decision on the level of output to maximize profit. This relation ship between follower and leader optimal choice can be summarized with the help of reaction function of follower firm. So the leaders first determine the reaction function of its follower and then incorporate it to its own profit function. Then it maximizes the newly formed profit function like monopoly firm by setting marginal revenue equals to marginal cost. On the other hand follower firm react to the optimal choice of the leader according to its reaction function to come up with its profit maximizing level of output.

The stackleberg solution can also be illustrated graphically using isoprofit curves and reaction curves. Both firms have the same shaped isoprofit curve as in the case of cournot. The highest

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profit level is represented by the isoprofit curve found near to the quantity axis of each firm and the lowest level of profit is represented by isoprofit curve, which found away from the quantity axis. Given such properties of isoprofit curves of the firms, they are acting in the following ways to determine their optimal output level.

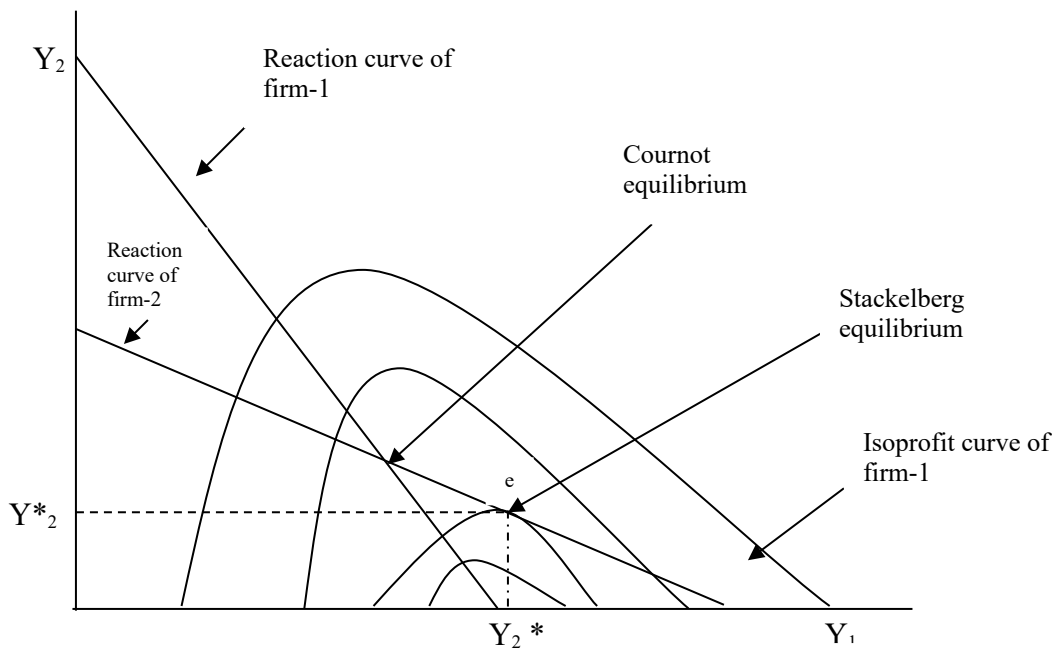


Figure: 2.8. Stackelberg equilibrium

Firm 2 as a follower will choose an output along its reaction curve, while firm-1 (the leader) choose the output level on the reaction curve of firm-2 that gives him the highest maximum profit. As a result, the equilibrium point of the stackelberg duopolists is not defined by the intersection of their reaction curve. This is because the leader firms no more take the follower's output as given. It knows the follower output would depend on its own output level in accordance with its reaction function. This implies the equilibrium point of the Stackelberg model defined by the tangency of the isoprofit of the leader with the reaction curve of the follower as indicated by figure 2.8 below.

Since the leader, firm-1 makes choose on the reaction curve of follower, firm 2, point e represents stable stackleberg's equilibrium. At point-e, the leader get higher profit and the follower firm get lower profit compared to cournot equilibrium.

In short if one firm is sophisticated, it will emerge as a leader and stable equilibrium will be established since the naïve firm act as a follower. However if both firms are sophisticated, both want to be a leader to get higher profit. In this case, market situation becomes unstable. Such situation is known as Stackleberg disequilibrium. The effect of such situation will be either a price war until one of them surrenders and agrees to act as follower or collusion of both firms abandoning their reaction function and move to a point close to edge worth contract curve where both of them get higher profit

Numerical illustration of Stackleberg model

Consider the demand and cost function used for Cournot model

$$P = 60 - 2(y_1 + y_2)$$

$$C_1 = y_1^2, \quad C_2 = 4y_2$$

Find

- a) Stackleberg equilibrium assuming firm – 1 is a leader
- b) Stackleberg equilibrium assuming firm – 2 is a leader.

Solution

If firm-1 is a leader, it is the one which chooses output level that maximizes its profit first by incorporating reaction function of its follower into its own isoprofit function as follows:

Follower's profit function,

$$\pi_2 = [60 - 2(y_1 + y_2)] y_2 - 4y_2$$

$$\pi_2 = 60y_2 - 2y_1 y_2 - 2y_2^2 - 4y_2$$

$$\pi_2 = 56y_2 - 2y_1 y_2 - 2y_2^2$$

Take first order partial derivative of the profit function with respect to y_2 to determine the reaction function of follower firm

$$\frac{\partial \pi_2}{\partial y_2} = \frac{\partial (56y_2 - 2y_1 y_2 - 2y_2^2)}{\partial y_2} \quad \text{The reaction function of follower firm}$$

$$56 - y_1 - 4y_2 = 0$$

$$\Rightarrow y_2 = 14 - 0.5y_1$$

Leader's profit function can be defined as:

$$\pi_1 = [60 - 2(y_1 + y_2)] y_1 - y_1^2 \dots (2)$$

$$\pi_1 = 60y_1 - 3y_1^2 - 2y_1 y_2$$

To determine its optimal level of output the leader incorporate follower's reaction function into its own profit function:

Then take first order condition to determine the level of output that maximizes the profit of the leader.

$$\frac{\partial \pi_1}{\partial y_1} = 32 - 4y_1 = 0$$

$$\Rightarrow 4y_1 = 32$$

$$y_1 = \frac{32}{4} = 8$$

$$y_1^* = 8$$

$\Rightarrow$ The leader produces 8 unit of output to maximize its profit.

To determine follower's optimal choice substitute leader's output level in the reaction function of the follower firm.

$$y_2 = 14 - 0.5y_1$$

$$y_2 = 14 - 0.5(8) = 10$$

$$y^* = 10$$

$$p = 60 - 2[y_1 + y_2]$$

$$= 60 - 36 = 24$$

$$p^* = 24$$

Profit of the leader

Profit of follower

$$\pi_1 = 60y_1 - 3y_1^2 - 2y_1y_2$$

$$= 60(8) - 3(64) - 2(8 \times 10)$$

$$= 480 - 352$$

$$\pi_1 = 128$$

$$\pi_2 = 56y_2 - 2(y_1y_2 - 2y_2^2)$$

$$= 56(10) - 2(80) - 2(100)$$

$$= 300$$

$$\pi_2 = 300$$

b) Assume firm-2 is a leader and firm-1 is a follower

Solve backward starting from the follower profit maximization

$$\pi_1 = [p(y_1 + y_2)]y_1 - C_1(y_1)$$

$$\pi_1 = 60y_1 - 3y_1^2 - 2y_1y_2$$

F.O.C

$$\frac{\partial \pi_1}{\partial y_1} = 60 - 6y_1 - 2y_2 = 0$$

$$y_1 = 10 - \frac{1}{3}y_2 \dots (1)$$

Equation one above represent the reaction function of firm-1, the follower and then

Firm-2(the leader) substitutes the reaction function of the follower, firm-1 into its profit function to determine profit maximizing level of output as follows:

$$\pi_2 = 56y_2 - 2y_1y_2 - 3y_2^2$$

$$\pi_2 = 56y_2 - 3y_2^2 - 2y_2(10 - \frac{1}{3}y_2)$$

$$= 56y_2 - 2y_2^2 - 20y_2 + \frac{2}{3}y_2^2$$

$$\pi_2 = 36y_2 - \frac{4}{3}y_2^2$$

F.O.C.

$$\frac{\partial \pi_2}{\partial y_2} = 36 - \frac{8}{3}y_2 = 0$$

$$y_2 = 36(\frac{3}{8}) = 13.5$$

To determine the optimal level of output for the follower, substitute the optimal output level of the leader (firm-2) in the reaction function of the follower:

copy
paste
here

Leader profit's

follower's profit

calc
25.5 42
27.6
17.5

$$\pi_1 = R_1 - C_2 = 22(5.5) - (5.5)^2$$

$$= 121 - 30.25$$

$$\pi_2 = 90.75$$

Self-test exercise

Suppose that the stackelberg duopolist face the following market demand curve:

Where y is the total output supplied to the market (111). Also suppose that the two firms have zero marginal cost, 111. Assuming firm-1 is stackelberg leader find:

- a). The profit maximizing level output of the leader firm and the amount of output Produced by the follower at the profit maximizing level of the leader
- b). The maximum level of profit earned by the two firms.

- 2) In Stackelberg model, which firm set optimal output level first? (The leader or the follower).
 - 3) What do we mean by stackelberg disequilibrium? Does it has any effect?
-

2.3 Collusive oligopoly

All oligopoly models described up until now are an example of non-collusive oligopoly. In such type of industry each firm make independent decision to maximize its profit even though each of them take other's likely behavior in to account during the process of decision making. Now we consider other possibility in which firms found in a given industry make collusive agreement implicitly or explicitly to some degree in setting price and output. Firm enter in to such collusive agreement in order to cultivate the advantage of increasing profit, decreasing uncertainties and to create better opportunity to prevent other's entry to the industry. To see how firms operating in such arrangement make decision we will consider cartel and price leader ship in this sub section.

2.3.1 Cartel

A cartel is a formal organization of firms producing the same product. It is formed to coordinate the policies of member firm so as to increase their joint profit by limiting the scope of competition among them. It also reduces uncertainty arising from their mutual interdependence. Here we will study two typical forms of cartels.

- a) Cartels aiming at joint profit maximization
- b) Market sharing cartel

2.3.1.1 Cartel aiming at joint profit maximization.

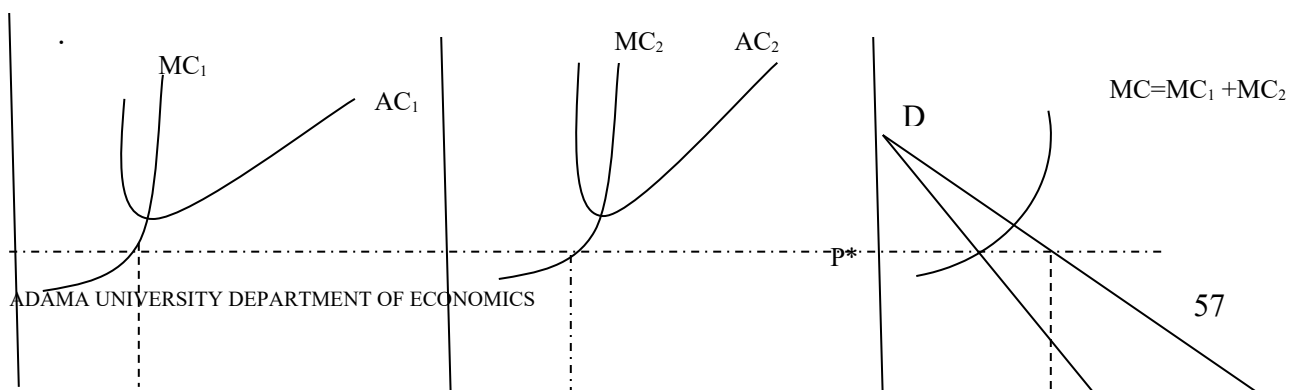
Cartel aiming at joint profit maximization is formed as its name indicates with the objective of maximizing the total industry's profit. In order to achieve this objective member firm appoint a central agency to which they delegate the authority to decide not only the total quantity and price at which the industry's profit maximized but also allocation of output and profit among the member of cartel.

To make such decisions, it is assumed that central agency have access to information about the cost of individual firms and know market demand. Given the information about cost and demand, central agency determine price and output that maximizes industry's profit defined by the intersection of marginal revenue and marginal cost curve. Marginal revenue curve can be derived from market demand and the marginal cost used for decision can be obtained by summing up individual marginal cost. In this case, the cartel act as multiplant monopoly and chooses Q to maximize the total industry profits:

Take first order condition of industry profit function to determine profit maximization level of output and price:

Since total revenue depends on the sum of all cartel members' output levels, marginal revenue is the same no matter whose output level is sold. At the profit maximization point therefore, this common marginal revenue will be equated with each firm's marginal production cost.

For simplicity let us assumes there are only two firms in the cartel with marginal cost MC_1 and MC_2 as indicated in figure 2.9. So the industry's marginal cost (MC) obtained by summing up marginal costs of the two firms. From the market demand DD , we can derive industry's marginal revenue. The intersection point of MC and MR gives profit-maximizing level of output and price.



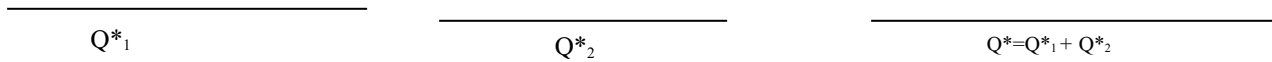


Figure: 2.9 Joint profit maximizing cartel

At the same time central agency, allocate production between the two firms similar to Multiplant monopoly by equating the MR to individual marginal cost. Mathematically the profit maximization of cartel can be derived as

$$\text{Max } \pi = R_1 + R_2 - C_1 - C_2$$

$$Y_1, Y_2$$

$$\pi = p(y_1 + y_2)(y_1 + y_2) - C_1(y_1) - C_2(y_2)$$

Taking F.O.C for joint profit maximization

$$1. \frac{\partial \pi_1}{\partial y_1} = \frac{\partial [(y_1 + y_2)(y_1 + y_2) - c_1(y_1) - c_2(y_2)]}{\partial y_1} = 0$$

$$2. \frac{\partial \pi_1}{\partial y_1} = \left[\frac{\partial (y_1 + y_2)(y_1 + y_2) - C_2(y_1) - C_2(y_2)}{\partial y_2} \right] = 0$$

Rearranging equation (1) and (2) we can get

$$MR = MC_1 = MC_2 \text{ Since } \frac{\partial R_1}{\partial y_1} = \frac{\partial R_1}{\partial y_2} = \frac{\partial R}{\partial y}$$

Cartels aiming at joint profit maximization determine the amount of output produced by each member firm using the above condition ($MR = MC_1 = MC_2$)

Numerical illustration

Assume that the Market demand facing the members of a cartel is $P = 60 - 2(y_1 + y_2)$ and their cost of production is given by $C_1 = Y_1^2$, $C_2 = 4y_2$. Find the level of output and price that maximizes the industry's profit.

Solution

We derive profit maximization condition for cartel aiming at joint profit maximization as $MR = MC_1 = MC_2$

$$MR = \frac{\partial TR}{\partial y} = \frac{\partial(60 - 2y).y}{\partial y} = 60 - 4y$$

$$MC_1 = \frac{\partial c_1}{\partial y_1} = \frac{\partial y_1^2}{\partial y_1} = 2y_1$$

$$MC_2 = \frac{\partial c_1}{\partial y_2} = \frac{\partial 4y_2}{\partial y_2} = 4$$

$$MR = MC_1$$

$$MR = MC_2$$

$$60 - 4y = 2y_1 \text{ Where } y = y_1 + y_2$$

$$60 - 4(y_1 + y_2) = 2y_1$$

$$60 - 4y_1 - 4y_2 = 2y_1$$

$$60 - 6y_1 - 4y_2 = 0$$

From Equation (3)

$$60 - 4y = 4$$

$$60 - 4y_1 - 4y_2 = 4$$

$$56 - 4y_1 - 4y_2 = 0 \dots (4)$$

Solve simultaneously equation 3 and 4

$$56 - 4y_1 - 4y_2 = 0$$

$$\underline{60 - 6y_1 - 4y_2 = 0}$$

$$4 - 2y_1 = 0$$

$$\underline{Y_1 = 2}$$

Substitute in equation (1)

$$60 - 6y_1 - 4y_2 = 0$$

$$60 - 12 - 4y_2 = 0$$

$$48 - 4y_2 = 0$$

$$Y_2 = 2$$

Total output that maximizes industry's profit is the sum of individual firm's output

$$Y = y_1 + y_2 = 12 + 2 = 14$$

$$\underline{Y = 14}$$

$$P = 60 - 2(y)$$

$$= 60 - 2(14) = 22$$

$$\underline{\underline{P = 22}}$$

Joint profit of the industry $\pi(y_1 + y_2) = [p(y_1 + y_2)] [y_1 + y_2] - c_1(y_1) - c_2(y_2)$

$$\begin{aligned}
 \pi(y_1 + y_2) &= 60y_1 + 56y_2 - 3y_1^2 - 2y_2^2 - 4y_1y_2 \\
 \pi(y_1 + y_2) &= 60y_1 + 56y_2 - 3y_1^2 - 2y_2^2 - 4y_1y_2 \\
 &= 60(2) + 56(12) - 3(2)^2 - 2(12)^2 - 4(2)(12) \\
 &= 120 + 672 - 12 - 288 - 96 \\
 &= 792 - 396 = \underline{\underline{396}}
 \end{aligned}$$

In Practice however, cartels rarely achieve maximum joint profit. There are several reasons why cartel's industry profit cannot be maximized. Some of them are

Mistakes in the estimation of market demand. Mistake in estimation of market demand leads to mistake in the derivation of the MR and hence price and output which maximizes profit.

Mistakes in estimation of MC due to incomplete knowledge of the individual firm's costs at different level of output. Such error leads to equilibrium level of output and price, different from the expected cartel solution.

Slow process of cartel negotiation. Since member firms have different cost and market share, to bring them to common level of price it takes time. When the time taken for making decision is longer, the price of goods and services changes for what it is expected to be. So the information used by central agency for decision making becomes out dated and correct level of price that maximizes the industry profit can not be set.

Stickiness of the negotiated price. Even when market condition changes, once price set, it need long time for negotiation and set profit maximizing price.

Cheating by member firm during bargaining process through giving biased information in order to achieve large market share.

Existence of high cost firms. If a firm operates with higher than equilibrium MC, it reduces the total joint profit. As a result, it has to close down to maximize joint industry's profit.

Nevertheless, there is no such thing with firm aiming of joining industry maximization.

Fear of government interference to change the level of price which maximizes cartels' industry profit

The wish to have good public image cartel agency will not increase price to profit maximization level of the industry.

Self-test exercise

Suppose that the market demand facing two firms producing the same product is given by $p = 300 - 3y$ where $y = y_1 + y_2$, total output. If the production cost of the two firms is equal and given by $C(y) = 30y + 1.5y^2$ and then the agree to form cartel aiming at joint profit maximization, find:

- a) The amount of output that maximizes their profit.
 - b) The industry's price.
 - c) The output and the profit of each firm.
-

2.3.1.2 Market Sharing Cartel

In market sharing cartel the member firms agree on how to share the market but they keep a considerable degree of freedom concerning the style of their output, their selling activities and other decisions. Each firm therefore operates in one area or region agreed upon without encroaching on others' territories. This form of cartel is more common in real world and more popular than cartel aiming at maximization of joint industry profit. There are two basic methods for sharing market by the member firm of cartel: non-price competition and quota.

1. Non- price competition

In such form of cartel the member firm agree on a common price at which they sell their output. The price up on which they agreed set by the process of bargaining. In the bargaining process, low cost firm pressing for lower price and high cost firm press for higher price. At the end common price which allow all members certain amount of profit, is set. At such price firms compete to maximizing their profit by increasing their sell volume through different way of non-price competition like quality, style, selling activities and advertising. Such form of cartel is unstable most of the time compared to cartel aiming at joint profit maximization. Because, whenever there is cost and liquidity difference, low cost firms have the incentive to cheat by lowering their price and initiates price war among member firms, which result in instability.

2. Sharing of the market by agreement on quotas

In this case, member firms agree to supply a certain quantity of output at agreed price. The quota allocation made based on the cost structure of the firms. If they have identical costs, the

monopoly solution will emerge by sharing the market equally. However if costs are different, the quota of market share differs between the firms. Under such cases quota sharing depends on the bargaining power of the firms. During bargaining process past levels sales and or the basis for productive capacity of the firms are considered for decision.

The other form of sharing market is made through defining geographical boundaries to which each member firm supply their product. Like non-pricing competition, market sharing cartels or regional sharing cartels agreement are also unstable. The agreements are violated intentionally or by mistake by low cost firm who have the incentive to expand their output by cutting price.

Most of the above forms of cartels are unstable. The firms will not act according to output and price level upon which they agreed due to a number of reasons. Some of the reasons includes: the member firms have different costs, different assessments of the market demand and even have different objective. So most of the time they want to set price at different level. Furthermore, each member of the cartel tempted to cheat by lowering price slightly to capture larger share of market than allocated for the firm. This implies there should be some sort of enforcement mechanism for a cartel to be successful. Threat in the long run for the member who breaks the agreement should be there. Also gaining certain monopoly power to set price by cartel is other factor for the success of cartel i.e., if the potential gain from cooperation are large, cartel member have more incentive to solve their organizational problem and hence profit for cartelization is large enough to give incentive for the members to act according to the contractual agreement.

2.3.2 Price leadership

Dear students, unlike firms in cartel, which agree explicitly to cooperate in setting price and output, firms in price leadership collusive oligopoly, agree to cooperate implicitly in making decisions about price and output without any formal discussion. They enter in to such agreement voluntarily to avoid any uncertainty about the competitor reaction. In such type of model, one firm implicitly recognized as the leader and set price. The other remaining firm, the follower take price as given and adopt the price set by the leader firm even though its profit did not maximized.

This follows from the assumption that the two firms sell identical products. If one charged a different price from the other, all of the consumers would prefer the producers with the lower price. In addition, if two firms do not agree implicitly on a common price, there is a possibility to enter into a price war. Price war through reduction of prices causes low profit level and even causes destruction of the firm. These conditions may force oligopoly firms to cooperate without actually making explicit agreements with one another.

There are various forms of price leadership, the most common ones are:

1. Price leadership by low cost firm
2. Price leadership by large (dominant) firm
3. Barometric price leadership

2.3.2.1 Low cost price leadership

This model assumes that there are two firms in the industry producing homogeneous products at different costs. One firm produces at low cost compared to its competitor. Moreover, firms may have equal or unequal market share. Given firms with the stated features, the low cost firm becomes the leader and sets price, which maximizes its profit. The follower firm, by sacrificing some of its profit, takes the price set by the leader. This is to avoid a price war, which would eliminate the firm from the industry if the price set lower than its LAC.

However, the price set by the leader firm using marginal principle ($MC=MR$) would remain at the stated position through maintaining output constant. Deviation of output from the point where $MR=MC$ due to over or under supply of output by the follower firm will change the price and then the profit of the leader will not be maximized. This implies that the follower must supply a quantity sufficient to maintain the price set by the leader. So at the optimal price level, the firms must also enter into an agreement on the share of the market formally or informally. Otherwise even though the follower adopts the leader's price, producing higher or lower level of output required to maintain the price (set by the leader) in the market pushes the leader to a non-profit maximizing position. In this respect, the follower is not completely passive; it can affect the market price and then the profit of the leader unless they enter into a formal or informal agreement to supply certain proportion of the total output.

Numerical illustration

Recall that the examples we use for others duopoly model with:

$$\text{Market demand: } P = 60 - 2(y_1 + y_2)$$

$$\text{Cost } C(y_1) = y_1^2 \quad C(y_2) = 4Y_2$$

When we see their cost structure, firm two is a low cost firm so that it can be taken as the leader. So, it is the one who set price using the marginal principle but the two firms go into agreement about the share of the market at leader's profit maximizing level of price. Let assume they agree to share equally. With this assumption, the demand curve relevant to leader's decision would be modified as:

$$P = 60 - 2(Y_1 + Y_2) \quad , \quad Y_1 = Y_2$$

$$P = 60 - 2(Y_2 + Y_2) = P = 60 - 4Y_2$$

$$\Pi_2 = PY_2 - C_2$$

$$\Pi_2 = (60 - 4Y_2)Y_2 - 4Y_2^2 = 56Y_2 - 4Y_2^2$$

Take F.C.O $\frac{d\Pi_2}{dY_2} = 0$ to determine equilibrium level of output that maximizes leader's

profit.

$$\frac{d\Pi_2}{dY_2} = \frac{d(56Y_2 - 4Y_2^2)}{dY_2} = 0$$

$$56 - 8Y_2 = 0 \rightarrow Y_2 = Y_1 = 7$$

$$P = 60 - 4Y_2$$

$$= 60 - 4(7) = 60 - 28 = 32$$

$$\mathbf{P^* = 32}$$

However, $p = 32$ and output level $Y_1 = 7$ would not maximizes the profit of the follower firm. The price that maximizes the profit of the follower firm can be determined through maximizing the its profit function as follows;

$$\Pi_1 = R_1 - C_1 = (60 - 4Y_1)Y_1 - Y_1^2$$

$$\Pi_1 = 60Y_1 - 5Y_1^2$$

Take F.O.C $\frac{d\Pi_1}{dY_1} = 60 - 10Y_1 = 0$

$$Y_1 = 6$$

$$P = 60 - 4Y_1$$

$$P = 60 - 4(6) = 60 - 24 = 36$$

$$P^*=36$$

The output level which maximizes the profit of firm one (follower) would be $Y_1 = 6$ units and charge a price $P^*=36$ if it did not recognize the leadership of firm two.

Activity 2.7

In low cost price leadership model what is the variable upon which firms strategically interact during decision-making.

State the unique feature of the firm, which are classified as leader firm in low cost price leadership.

Why firms in low cost price leadership enter explicit or implicit agreement on the level of output to be supplied when the leader sets price?

Suppose that the market demand facing two firms producing the same product is given by $p = 105 - 25y$ where $y = y_1 + y_2$ and their costs of production is :

$C_1 = 5y_1, C_2 = 15y_2$. If the two firms enter into an implicit agreement to produce the same level of output according to low price leadership oligopoly model,

- a) Which firm is low cost price leadership?
- b) Find the price that maximizes the profit of the leader.
- c) If the follower firm wants to maximize its profit what price and output level should it set?

2.3.2.2 Dominant firm price leadership

In such oligopoly industry there is one large firm having large share of the total market with a group of smaller firms supply the rest. The larger firm, which is also known as, the dominant firm becomes the leader and sets price that maximizes its profit. The other smaller firms, which could have little influence over price, would be a price taker like firms in perfect competitive firm. That is each of the small firms produces an amount determined by setting marginal cost equals to price announced by the leader and the leader supplies the residual.

Given the structure of the model, what price would the dominant firm set to maximize its profit? Is the leader firm considering the impact of other small firms while setting price?

By assumption, the dominant firm knows the market demand and the marginal cost of small firms. This assumption enables the dominant (the leader) firm to derive its own demand function from the market demand and marginal costs of smaller firms. That is since small firms in the industry behave like perfect competitive firm, the dominant firm equate price with the sum of their marginal costs to get their supply function ($p = \sum SMC_i$). Then the leader firm's demand function, which is also known as, the residual demand curve, can be obtained by subtracting the supply function of the smaller firms from the market demand.

Once the leader identifies its demand function, it acts like a monopoly to set price and output level which maximizes its profit by equating marginal revenue with its marginal cost. Here the marginal revenue used for decision is derived from residual demand that actually measures how much output it will be able to sell at each given price.

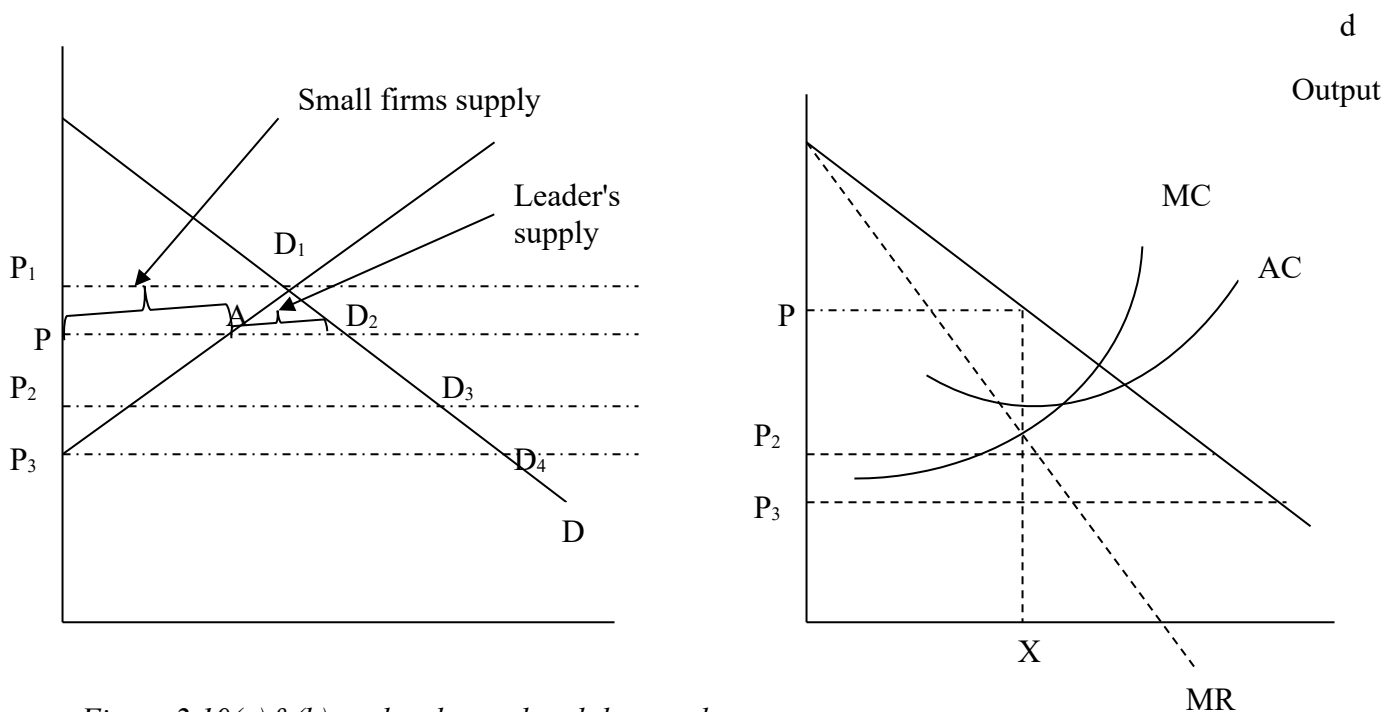


Figure 2.10(a)&(b) market demand and the supply curves

As indicated in figure 2.10. for example at P_1 the demand for the product of the leader will be zero, because the total quantity D_1 is supplied by smaller firm. As price fall below p_1 the demand for the leader's product increases. At p the total demand is D_2 out of which PA is supplied by small firms and the remaining AD_2 is supplied by the leader.

Having derived the demand curve of the leader as in figure 2.10(b) and given its MC, the dominant firm will set price at which his $MR=MC$. However smaller firms may or may not maximizes it profit depending up on its cost structure at the set level of price.

Numerical illustration

Suppose that the market demand facing the industry's product is $D(P) = 50 - 0.3P$ and the supply function of smaller firms as a function of price is $S = 0.2P$

So the dominant firm's demand function can be derived as $Y = D - S$

$$= 50 - 0.3p = 50 - 0.5P$$

$$P = 100 - 2Y$$

If dominant firm's cost function is $C = 2Y$ then its profit function, $\Pi = R - C$

$$\Pi = PY - 2Y$$

$$\Pi = (100 - 2)Y - 2Y$$

$$\frac{d\Pi}{dY} = \frac{d[(100 - 2Y)Y - 2Y]}{dY} = 0$$

$$98 - 4Y = 0$$

$$\rightarrow Y^* = \frac{98}{4} = 24.5$$

Leader will set price $P = 100 - 2X = 100 - 2(24.5)$

$$P = 51$$

At this price the total quantity demanded

$$D = 50 - 0.3(51) = 34.7$$

$$D = 34.7$$

The amount supplied by smaller firms is $= 0.2P$

$$S = 0.2(51) = 10.2 \quad \Rightarrow S = 10.2$$

Activity 2.8.

1) Describe the structure dominant firm price leadership.

- 2) What is the difference between low cost price leadership and the dominant firm price leadership?
- 3) Suppose that the market demand for a product by one larger firm and many small firms is given as $p = 300 - y$. The total amount of output supplied by the smaller firms is given by $y = 49p$. If the marginal cost of the larger firm is equal to $C(y) = 2.96y$, what price should the firm charge to maximize its profit? How much the industry produces as whole at this price?
-

2.3.2.3 Barometric price leadership

In this type of price leadership oligopoly, there is a firm which has good knowledge of the prevailing condition in the market and can forecast better than others about the future development in the market. Such type of firm is used as the barometer for other firms to reflect the changes in economic environment. Such type of firm acts as a leader and when its price changes, other firms follow the change in price. Usually a barometric firm may or may not be a firm with low cost or larger firm. However, it is a firm which establishes good reputation in the past in forecasting economic changes. Even a firm belonging to another industry having good reputation may also be used as a barometric leader to affect the decision of other firms. For example, a firm in the steel industry might be taken as a barometric price leader for the motor car industry.

Summary

- Oligopoly is a market structure with only a few firms dominating the whole industry. Economic decision in such type of market involves strategic considerations; each firm must consider how its action affects its rivals and how they likely react.
- There are two different forms of oligopoly. These are non-collusive models and collusive models. The classification is based upon whether there exist some forms of cooperation between firms or not. It is also possible to classify non-collusive oligopoly into four types depending upon the way they make decision and the strategic variable upon which they interact: kinked demand, Cournot duopoly, Bertrand duopoly, and Stackelberg duopoly models.

- The kinked demand curve model explains why prices often remain stable in oligopoly markets, even when costs rise. The other three models try to predict the behaviours of oligopoly firms based on different kinds of assumption about the rival firm.
- In Cournot duopoly model of oligopoly, firms make their output decisions at the same time by assuming others firm keep its output fixed at existing level. At equilibrium therefore each firm maximizes their profit given the output of its competitor.
- In Bertrand firm make strategic decision on price as the same time by assuming the price of their competitor remain fixed. In this model the equilibrium outcome is the same as perfectly competitive market even though the number of firm in the industry is two.
- In Stackelberg's duopoly model it is assumed that one duopolist with better information becomes sophisticated to recognize his competitor act on the cournot assumption. Thus the sophisticated firm will determine the reaction curve of his rival and incorporates it in his own profit function, to determine price that maximizes its profit and taken as given by the follower firm.
- The other form of oligopoly is collusive oligopoly in which firms make cooperative agreement implicitly or explicitly to set optimal level of output produced and price charged.
- Cartel is a collusive form of oligopoly in which firm producing the same product enter in to explicit contractual agreements to set price and the level of output produced. This will avoid the uncertainty created between for decision making. There are two different forms of cartel. These are Cartel aiming at joint profit maximizing of the industry and market sharing cartel
- Joint profit maximizing cartel has the objective to maximizing the profit of the industry. Decision for such type of cartel is made by the central agency assigned by the member firm. The central agency set price which maximizes the industry's profit and allocate the profit among the member firms. In practice the industry profit can not maximized due to various reason like cheating by low cost firm, mistake in estimation of costs and market demand by central agency while setting price and output levels.
- In Market sharing cartel, the firm agree to share the market, but keep a considerable degree of freedom concerning the style of their output, their selling activities and other decisions. This can occur when firms agreed not to compete on price.
- Because of the difficult of forming effective cartel, oligopolists may attempt to cooperate implicitly without making explicit agreements with one other. Such type of oligopoly is

called price leader oligopoly. There are three forms of price leadership; price leadership by a low cost firm, price leadership by a large (dominant) firm and Barometric price leadership. In all the three forms the leader firm set price and the follower firm take price set by the leader as given

Key concepts and terms

Kinked demand	cournot equilibrium
Reaction function	Bertrand equilibrium
Isoprofit curve	stackelberg equilibrium
Naïve behavioural assumption	cartel
Collusive oligopoly	Barometric firm
Non-collusive oligopoly	sticky price
Leader firm	stackelberg disequilibrium
Follower firm	residual demand
Contract curve	price leader oligopoly

Exercise

I. Answer the following question briefly

- 1). Discusses the behavioral assumptions made about firms operating in the following duopoly model
 - a) Cournot duopoly model
 - b) kinked demand curve model
 - c) Stackelberg duopoly
 - d) dominant firm price leadership.
- 2). According to kinked demand curve model why does price rigidity occur in oligopolistic market? What are the limitations of this model?
- 3) explain how different types of price leader oligopoly determine profit maximization level of price; a) low cost price leadership b) dominant firm price leadership
- 4) In most cases, the profit maximization level of price cannot maximize the profit of the industry. Why?

II. Choose the best answer for the following questions.

- 1). All of the following oligopoly model have something in common except

- a). cournot's duopoly model
 - b). Bertrand duopoly model
 - c). Kinked demand curve model
 - d). Stackelberg duopoly model
- 2). In which of the following model optimal decision made by firms simultaneously
- a). low cost price leadership
 - b). Bertrand duopoly model
 - c). Stackelberg duopoly model
 - d). Dominant firm price leadership
- 3). In which one of the following duopoly model the strategic interaction between firms made on quantity of output produced.
- a). low cost price leadership
 - b). Bertrand duopoly model
 - c). Stackelberg duopoly model
 - d). Dominant firm price leadership
- 4) One of the following is true about duopoly models;
- a) In Bertrand duopoly model firms are making decision at the same time about output that maximizes their profit.
 - b) The isoprofit curve found near to the price axis in case of Bertrand model represent higher level of profit like the of cournot model.
 - c) Firms in cournot model get higher profit if they decided to operate on contract curve rather deciding to produce at the intersection between the reaction curves.
 - d). In low cost price leadership follower firm act passively.

III work out questions;

- 1). Suppose the market for TV set have one dominant firm and five smaller firms. The market demand facing firms is $Y = 400 - 2P$. The dominant firm has constant marginal cost of 20. The small firms each have a marginal cost of $MC = 20 + 5Y$.
- a). verify that the total supply curve for the five small firms is $Y_s = P - 20$

- b). Find the dominant firms demand curve
- c). Find the profit maximizing quantity produced and price charged by the dominant firm, and the quantity produced and price charged by each of the small firms.
- d). suppose there are ten small firms instead of five how does this change your result?
- e). suppose the marginal cost of each of the five firms changed to $MC = 20 + 2Y$. How does this change in result.

2). Assume that four firms producing identical product in an oligopolistic industry form a cartel aiming at maximization of joint profit. The total market demand facing the cartel is $Y = 20 - 2P$

and each firms marginal cost function is given by $Mc = \frac{1}{4}Y$;

- a). find the level of output and price that maximize the profit of cartel industry.
- b). how much should each firms produce at cartel's profit maximizing level of price.
- c). how much profit can the cartel will earn.

3). given the demand function of two duopolist as

$$p = 40 - 0.4y \quad \text{where} \quad y = y_1 + y_2, \text{ output produced by two firms}$$

y_1 - output produced by firm – 1

y_2 – out put produced by firm - 2

If the cost function of the two firms given as

$$Tc_1 = 0.6 y_1^2 \quad Tc_2 = 10.4 y_2$$

Find the equilibrium price and output level of each firms assuming both firms are operating under cournot's assumption

Find the equilibrium price and output level of each firms assuming the two firms are Stackleberg duopolist and firm 2 is the leader firm.

Assuming these two firms are merged to form one monopoly firm, calculate the equilibrium and output of the firm.

If the two firms agreed to share the market in such a way that low cost firm produce three fourth of the total output determine the price and quantity that will maximize the profit of low cost firm

4). suppose that the market of a give product has one dominant firm and few small firms. The market demand of the firms is $Q=80-3P$. The cost of dominant firm is $C_1 = 20Q_d$ and the total cost of the small firms is $C_2=0.5Q_s^2+20Q_s$ the supply function of small firms is equal to $Q_s = 20-P$. Assuming the market structure is dominant firm price leadership type:

Find: a) dominant's firm demand function.

b) Profit maximization level of output and price for both dominant and smaller firms.

c) If the dominant firm is a low cost firm and set price for the market in such way that the quantity of output produce by the dominant and small firms is equal, find the profit maximizing level of price and output of the dominant and small firms.

d) If the follower (smaller) firm wants to maximize their profit by setting price, what level of price and output maximizes their profit?

Suggested readings

Nicholson, w: **Microeconomic theory; basic principle and extension**. 5th edition

Salvatore, D(2003):**Microeconomics; theory and applications**.4th edition

Mansfield, E.and Yohe,G (2000):**Microeconomics, theory and applications**.10th edition

Varian, R.Hall (1999): **Intermediate microeconomics** 3rd edition

Pindyck, R.S. and Rubinfeld, D.L. (2003): **Microeconomics**. 6th edition

CHAPTER THREE

GAME THEORY

3.1 Introduction

Learning objectives

3.2 Game theory and Strategic Behavior

Basic Concepts

3.3 Dominant Strategy and Nash Equilibrium

3.3.1 Elimination of dominated strategies

3.4 The Prisoners' Dilemma

3.5 Mixed Strategies Nash equilibrium

3.6 Repeated Game

3.7 Sequential Game

Summary

Key concepts and terms

Exercise

Suggested readings

3.1 Introduction

Dear students in previous chapter we have seen that how each firm in oligopoly market take in to account its rival's reaction to make decision on level of price and output that maximizes their profit using classical economic theory. Such strategic interaction between firms can be also studied by a tool known as game theory. It was developed in 1920 and grew rapidly during World War II in response to the need to develop formal way of thinking about military strategy. In this chapter, we will see this theory to give you some flavor of how it works and how it can be used to study economic behavior in oligopolistic market.

The chapter begins with explanation of the basic concepts of game theory. It then defines a dominant strategy and Nash equilibrium and their usefulness in the analysis of oligopolistic behavior. Next, it describes the prisoner's dilemma and its applicability to the analysis of price and non-price competition. Finally, we conclude the chapter by analyzing repeated and sequential game

Learning objectives

At the end of this chapter, students able to

- Define a game, payoff, strategy

Identify optimal strategy of players

Differentiate between pure and mixed strategy

Describe prisoner dilemma

Determine equilibrium of simultaneous and sequential game

Differentiate between Nash and dominate strategy equilibrium

Describe the behavior of firms using game theory

3.2 Game theory and Strategic Behavior

Basic Concepts

It is known that oligopolistic firms interact strategically to make decision in a variety of ways. Such interaction of economic agents and the expected reward of their decision can be studied

using game theory. It helps us to show how an oligopolistic firm can make strategic decision to gain a competitive advantage over its rivals or how it can minimize the potential harm from a strategic move by a rival. However, what do we mean by the term game? A game is any situation in which players (participants) make decision by taking in to account each other's action and response and the outcome of the game depends on the decision of each player. For example, Cournot duopolist set price by taking in to account the reaction of its competitor and hence the level of profit obtained depends on decision made by the two firms. Such decision-making represents a game.

All games have three basic elements: players, strategies and payoffs. Each decision maker in a game is called a **player**. These players may be individuals, firms (as in oligopoly markets), entire nation (as in military conflict) or organization. Players are assumed rational in the sense that each player chooses the course of action that yields the most favorable outcome. Moreover, players start to play the game by assuming that their competitors are just rational and as smart as they are.

A strategy is a course of action a player will take under each contingency in playing of the game. In the price setting, a strategy might be increase price or decreasing price depending up on the action of its rival

Usually the number of strategies available of each player will be finite and many aspect of game theory can be illustrated for situation in which each player has only two or three strategies. A strategy is said to be optimal strategy if it maximizes expected return of the strategic choice. In the above example if increasing price maximizes the profit of the firm it is said to be the optimal strategy of the firm who take the action.

The final return to the players of a game to chosen combination of strategy is termed as payoffs. In other words, a payoff is the outcome or the consequence of each combination of strategy played by the two players. Payoffs are usually measured in the level of utility obtained by the player although frequently monetary payoffs (say, profits for firms) are used instead. In general it is assumed that players can rank the payoffs of a game from most preferred to least preferred and choose the highest ranked payoff attainable. In the above price setting game example, the return to strategy (Payoff) of the game to increase or decrease price

is different level of profit for each firms for their respective action. Therefore, by ranking the profit gained under different strategy, each firm plays strategy that gives it the highest possible level of profit.

A table, which summarizes the payoffs for all the strategies open to the firm and rivals response, is called the payoff matrix.

Dear students, we can illustrate the basic elements of a game using a simple advertising game played by two firms as summarized in Table 3.1.

		Firm- A's Strategies	
		A	D
Firm B's Strategies	A	(7,5)	(5,4)
	D	(6,4)	(6,3)

Table 3.1 Payoff matrix for advertising game.

The game contain two players, firm-A and firm-B and each firm has two strategies; to advertise (A) and do not advertise (D).The actual level of profit firms earn depends on the decision they made to advertise or not to advertise as well as its competitor decision. Thus, each strategy by firm- A (advertise or do not advertise) can be associated with each of firm B's strategies (advertise and do not advertise). The four possible outcomes from this simple game are illustrated by the payoff matrix in Table-3.1.

In the payoff matrix, the first number in each of the four cells represents payoff to the player that found to the left of the table associated with the strategy it chooses. Where as the second number represents the payoff of player found across the top of the table associated with its strategy. In the above example the first number of each cell represents the payoff firm-B obtain by choosing specific strategy and the second number belongs to firms A's payoff decides to advertise, firm-B earns a profit equals to 7 and firm-A earns a profit equals to 5. On the other hand, if firm-B advertise while firm-A decides not to advertise, firm-B get a profit of 5 and firm-A gets a profit of 4. The others payoff found in the rest of the cells for others combination of strategies interpreted in the same way.

3.3 Dominant Strategy and Nash Equilibrium

Well students, in the preceding section we have seen that how a game is represented by a payoff matrix and identify the payoff associated with different combinations of strategies. Now let us see how firms choose optimal strategy among different strategies when playing a game. We will also study how equilibrium of a game is determined.

Consider two firms, firm-A and firm-B producing homogenous product. Both have two strategic options to maximize their profit; decrease price or do not decrease price. What strategy should each firm choose if the expected payoff to the game is summarized in Table-3.2

		Firm-A	
		decrease	Doesn't decrease
Firm-B	decrease	(10,5)	(15,0)
	Doesn't decrease	(6,8)	(10,2)

Table 3.2 Pay off matrix for pricing game.

Let us consider firm-A first. If firm-B decreases its price, firm-A will earn a profit of 5 by decreasing price and 0 if it does not. Thus the optimal strategy for firm-A is to decrease its price while firm B decreases its price. On the other hand, if firm-B does not decrease its price, firm-A will earn a profit of 8 by decreasing its price and 2 if it does not decrease. Thus, firm-A is better off by decreasing its price also while firm- B does not decrease its price. This means firm-A will always get large profit if it decreases its price regardless of what firm-B does. Thus we can say decreasing price is the dominant strategy for firm-A. Dominant strategy is an optimal strategy of a player no matter the strategy chosen by other firm. In the example above decrease price strategy, provide greater profits to firm- A, no matter what B does.

The same is true for firm B. whatever firm-A does (decrease price or does not decrease price) firm-B is better off (earn large profit) by decreasing its price. That is if firm-A decrease price, firm-B's profit is equal to 10 by decreasing price and 5 if it does not. Similarly, if firm-A does not decrease its price firm B's profit would be 15 by decreasing its price and 10 if it does not. Thus, the dominant strategy for firm- B is to decrease price.

Therefore, the optimal strategy for both firms is decreasing of price. Whenever player has dominant strategy, we call the outcome of the game is an equilibrium in dominant strategy. In the above pricing game strategy; (decrease, decrease) with the outcome represented by left top cell payoff represents equilibrium in the dominant strategy of the game. In the case of advertising game summarized by table 3.1, advertising is a dominant strategy for firm-A. No matter what firm-B does, advertisement is the strategy that give maximum profit to firm-A than does not advertise strategy. Similarly, advertising strategy gives maximum profit than does not advertise strategy. So advertising is a dominant strategy for firm B. this implies the strategy (advertise, advertise) represent equilibrium in dominant strategy.

One additional point that we should notice here is that the advertising game and price setting game indicated above represent a game in which player move simultaneously. Each player selects a strategy before observing any action or strategy chosen by the other player. Not all games are of this type. As we will see later on, sometimes one player most goes first. In such type of game, the order play can make big difference in the outcome.

Activity 3.1

Describe the following concepts

- a) Game, strategy, player, payoff and payoff matrix
 - b) Dominant strategy and optimal strategy.
2. The following payoff matrix represents the utility that husband and wife get from buying microeconomics and macroeconomics textbook.

		Husband	
		microeconomics	Macroeconomics
Wife	microeconomics	(4,4)	(2,3)
	macroeconomics	(3,1)	(1,5)

- Determine: a). the dominant strategy of the husband if any
- b). the dominant strategy of the wife if any.
- c). the optimal strategy of each firms.

d). equilibrium in the dominant strategy of the game if any.

Nash equilibrium

Dominant strategy equilibrium does not exist always. For example, the game shown in table 3.2 does not have dominant strategy equilibrium. Here when B chooses left the payoffs to A is 2 or 0. When B chooses right, the payoffs to A is 0 or 1. This means that when B chooses left, A would want to choose top and when B choose right, A would want to choose bottom. Thus A's optimal choice depends on what he thinks B will do.

Player B's optimal choice also depends on what he thinks player-A will do. That is the optimal strategy of firm-B is to play right when A play bottom and left when A play top. So the strategies (Top, left) and (bottom, right) are the equilibrium of the game. Such type of equilibrium is said to be Nash equilibrium named after John Nash, the Princeton university mathematician and 1994 Nobel Prize winner who first formalized the notion in 1951.

		Player-B	
		Left	Right
Player-B	Top	(2,1)	(0,0)
	bottom	(0,0)	(1,2)

Table 3.2 Nash equilibrium

The Nash equilibrium is a situation in which each player chooses an optimal strategy, given the strategy chosen by the other player. In words a Nash equilibrium is a set of strategies such that each player beliefs (accurately) that it is doing the best it can given that the other player play its equilibrium strategy. Note that neither of the players knows what other person will do when he has to make choice of strategy. But each person has some expectation what the person choose will be. There Nash equilibrium is a pair of expectation, such that, when other choice is revealed, neither player wants to change its behavior. In our example, (Top, left), (bottom, right) are a Nash equilibrium.

This is because if A choose top, then the best strategy for B is to choose left since the payoff to B from choosing left is 1 and from choosing right is 0. In addition, if B chooses left, then the best thing for A is to choose top since A will get a payoff 2 rather than 0. Thus, if A chooses top, the optimal choice for B is to choose left and if B choose left, then optimal choice for A is top. This indicates each player optimal choice will be made after making certain expectation about other player's strategic choice.

Nash equilibrium is a generalization of cournot equilibrium that we have seen in chapter two. In cournot duopoly model each firm choose its output level that maximizes its profit by assuming the other firm continuing to produce the output level it had chosen, i.e. by taking the strategy of its rival firm as if it keep constant level of output. Under this assumption cournot, equilibrium occurs when each firm is maximizing profits given the other firm's behavior, which is similar to Nash equilibrium.

Dear students, have you notice any difference between Nash equilibrium solution and a solution of a game in which each player has a dominant strategy? Yes, if each player has a dominant strategy then that strategy is its best choice regardless of what other players do. In Nash equilibrium, on the other hand each player adopts a strategy that is its best choice given what the other players do at equilibrium. However, dominant strategy equilibrium is Nash equilibrium, but all Nash equilibria are not a dominant strategy.

Nash equilibrium has some problems. The first problem with the concept of Nash equilibrium is that there are games that have no Nash equilibrium as in the case of matching pennies game. In matching pennies game, there are two players, Row and Column. Each player has a coin, which he can arrange so that either the head side or the tail side faces up .thus each player has two strategies; head and tail. Once the strategies are chosen, there are payoffs to each player which depend on the choice that both players make. These choices are made independently and neither player knows others choice. If both player shows heads or tails ,then row wins a dollar and a column loses a dollar .If on there hand ,one player exhibit heads and the other shows tail, then column wins a dollar and row loses a dollar. The payoff matrix for the game is shown in table 3.3

The game has no Nash equilibrium because there is no combination of heads or tails leaves both players satisfied. Row's payoff is the negative of Column's payoff. Such type of game is a zero sum game, the interest of the players are opposite. As a result, one player or the other will always want to change their strategy.

		Column	
		Head	Tail
Row	Head	(1,-1)	(-1,1)
	Tail	(-1,1)	(1,-1)

Table 3.3 matching pennies game

Second problem with Nash equilibrium of a game is that it does not necessarily lead to paretoefficient outcomes. This will be illustrated using prisoner's dilemma game in the next section. Another, problem with Nash equilibrium is that the game may have more than one Nash equilibrium as in the case of the example above. Under such cases, it is possible to identify a unique equilibrium by a process known as elimination of dominated strategy.

3.3.1 Elimination of dominated strategies

When there is no dominant strategy equilibrium, we have to restore to the idea of Nash equilibrium. Our problem then is to try to eliminate some of the Nash equilibrium as being unreasonable. One sensible belief to have about players' behavior is that it would be unreasonable for them to play strategies that are dominated by other strategies. This suggests that when given a game, we should first eliminate all strategies that are dominated and then calculate the Nash equilibrium of the remaining game. This procedure is called elimination of dominated strategies; it can sometimes result in a significant reduction in the number of Nash equilibrium. For example, consider the following game.

		Column	
		Left	Right
Row	Top	(2,2)	(0,2)
	Bottom	(2,0)	(1,1)

Table 3.4 a game with dominated strategy

Note that there are two pure strategy Nash equilibria, (top, left) and (bottom, right). Therefore, to get a unique Nash equilibrium applies the iterative elimination of dominated strategy in favor of its dominant strategy. Here if we start from Row player, the strategy bottom dominates the strategy top. This implies eliminate the strategy top to find the equilibrium of the game and left Row player with bottom strategy. The payoff matrix after elimination of Row's dominated strategy represented by table 3.5.

		Column	
		Left	Right
Row	Bottom	(2,0)	(1,1)

Table 3.5 elimination of dominated strategy.

In this simplified after elimination of the dominated strategy, bottom continue determining the dominated strategy for column player. That is left is a dominated strategy by right .So eliminate a strategy left and end up with unique Nash equilibrium (Bottom, right)

Dear student, iterative elimination of dominated strategies enables us to solve games in which each player has more than two strategies. Let us take a game involving two players, each of whom has three strategies to choose from. All that we have to do is to eliminate dominated strategies from a player, then go to the other player and eliminate his dominated strategies from the remaining strategies and payoffs after the first elimination and repeat the process until we end up at the equilibrium. This three by three model is shown in table 3.6.

		Column		
		Left	Center	right
Row	Top	(-5,-1)	(2,2)	(3,3)
	middle	(1,-3)	(1,2)	(1,1)
	Bottom	(0,10)	(0,0)	(0,-10)

Table 3.6 a three by three game

The process of iterated elimination of dominated strategies is illustrated by actually eliminating the Row and Column's dominated strategies turn by turn as follows.

For the Row player, top and middle strategies dominate bottom strategy. Therefore, we can eliminate the bottom as a dominated strategy. This is because if the Row player expects the column player to play left, she would choose middle, if she assumes the column player would play center, she would go for top; and if she assumes the column player will choose right, she will choose top again. That is, no matter what the column player does, the Row player will not choose bottom as her strategy. This implies the Bottom strategy is dominated by Top and middle and can be eliminated.

		Column		
		Left	Center	right
Row	Top	(-5,-1)	(2,2)	(3,3)
	middle	(1,-3)	(1,2)	(1,1)

Table 3.7 eliminating dominated strategy for Row player

After bottom strategy is eliminated, let us identify dominated strategy for Column player. When we see into the payoff of Column player strategy left is dominated by center and Right. This is because if column think that Row play top, column play right. If column thinks that Row play middle, column play center. So left strategy is strictly dominated by left and right and should be eliminated.

		Column	
		Center	right
Row	Top	(2,2)	(3,3)
	middle	(1,2)	(1,1)

Table 3.7b eliminating dominated strategy for Column player

At this round, the Row player's dominated strategy is the middle, so we can eliminate the middle strategy.

		Column	
Row		Center	right
	Top	(2,2)	(3,3)

Table 3.7c eliminating dominated strategy for row player

Finally, column chooses Right over left, yielding a unique outcome after the iterated elimination of dominated strategies, which is (Top, Right).

The iterated elimination of dominated strategies is a useful concept, and when it applies, the predicted outcome is usually quite reasonable. Certainly, it has the property that no player has an incentive to change their behavior given the behavior of others. However, there are games where it doesn't apply.

Activity 3.2

- 1) Differentiate between dominant strategy equilibrium and Nash equilibrium
- 2) Why Nash equilibrium is not paretoefficient?
- 3) Describe elimination of dominated strategy method of finding unique Nash equilibrium.
- 4) Two firms producing the same product have a choice to produces high quality (H) and low quality (L). The resulting profit is given by the following payoff matrix.

		Firm-2	
		Low	High
Firm-1	Low	(-10, -30)	(100 ,160)
	High	(90, 80)	(50,50)

Determine the Nash equilibrium of the game

Suppose that the firms coordinate the action, what would be the equilibrium outcome of the game.

- 5) Two competing firms are each planning to introduce a new product. Each will decide whether to produce product A, product B, or product C. They will make their choice at the same time. The resulting payoffs are as shown below.

		Firm-2		
		Product-A	Product-B	Product-C
Firm-1	Product-A	(-10,-10)	(0,10)	(10, 20)
	Product-B	(10,20)	(-20,-20)	(-5, 15)
	Product-C	(20, 10)	(15,-5)	(-30,-30)

Find the Nash Equilibria of the game if any

Determine a unique Nash equilibrium of the game by using elimination of dominated strategy method.

3.4 The Prisoners' Dilemma

Dear students, having described the basic features of game theory, we now turn our attention to an important type of game known as the prisoner's dilemma. It has been the source of some insight with which we approach modern oligopoly theory as well as many other areas of economics, behavioral and political science.

The game consider a situation where two prisoners who are partners in a crime where questioned in two separate rooms and communication is not allowed. Each prisoner had a choice of confessing for the crime and there by implicating the other or denying that he had participated in the crime.

If only one suspect confessed, then he would go free while the other suspect (who does not confess) will receive 6 year sentence. If both suspects denied being involved in the crime, then both would be held for 1 year. If both confessed, they would be held for 3 years in prison. The payoff matrix for the game is given in table 3.8.

		Suspect (player)-B	
		Confess	Deny
Suspect (player)-A	Confess	(-3,-3)	(0,-6)
	Deny	(-6,0)	(-1,-1)

Table 3.8 Prisoners' dilemma game

Which strategy is the optimal strategy for the two suspects? What is the equilibrium outcome of the game? Well students, to answer these questions let us compare the payoff that the player receives by choosing a specific combination of strategies. Let us start from player A.

If player-B decides to deny, player-A is better off confessing since it will be left free. If player B decides to confess, still player-A is better off by confessing since it receives a 3-year sentence rather than a 6-year sentence. Thus whatever player B does, player A is better off by confessing. That is the dominant strategy for player A is confessing. Also the dominant strategy for player B is confessing. The reason is that if player A confesses, player B gets a three-year jail sentence and a six-year jail sentence if he does not. This implies the unique Nash equilibrium of the game is both players confessing. This equilibrium is also a dominant strategy equilibrium, since each player has the same optimal choice independent of the other player.

But the equilibrium outcome (confess, Confess) is not a Pareto efficient outcome. Because if they coordinate their action and choose a strategy (deny, deny) both suspects are better off. That is the two suspects receive only one-year imprisonment by denying. The problem is however there is no way for the two suspects to communicate with one another to make sure that the two suspects cooperate. If each could trust the other, then they could be better off by choosing a strategy (deny, deny).

Prisoners' dilemma game is useful in analyzing the behavior of oligopoly. It can be used to indicate the situation under which firms would be tempted to cheat on a cartel agreement by cutting price from agreed level or increasing the output produced more than its quota.

Suppose that there are two firms' production oil, firm- A and firm- B form cartel. Each firm has two possible strategies: to stick with the cartel agreement or to cheat. There are four possible outcomes, depending up on which strategy each firm pursues. The payoff matrix is given in table 3.9.

		Firm-B	
		Stick to agreement	cheat
Firm-A	Stick to agreement	(4,4)	(1,5)
	cheat	(5,1)	(3,3)

Table 3.9 cartel cheating game.

As indicate in the payoff matrix, cheating is a dominant strategy for both firms. That is firm-A get a profit of 5 million by cheating and 4 million by stick to the agreement while firm-B stick to the agreement. When firm-B decide to cheat firm-A get a profit of 1 million by sticking to the agreement and 3 million by cheating. This indicates firm-A get higher return by cheating whatever strategy chosen by firm B. Similarly firm-B gets higher return by cheating whatever strategy chosen by firm-A. So both firms would cheat even though they would be better off if they would both stick to the agreement and sustain cooperative solution. This example is the same as prisoners' dilemma game.

Activity 3.3

Think of U.S. and Japanese trade policies as prison's dilemma. The two countries are considering policies to open or close their import markets. The payoff matrix to their decision is shown below.

		Japan	
		Open	Close
U.S.	Open	(10, 10)	(5,5)
	Close	(-100, 5)	(1, 1)

U.S.

Assume that the two countries know the payoff matrix and believe that the other country act in its own interest. Does either country have a dominant strategy? What will be the equilibrium policies if each country acts rationally to maximize its welfare?

3.5 Mixed Strategies Nash equilibrium

In all of the games that we have seen so far, we consider a strategy in which players once make a specific choice, it will stick to it. Such type of strategy is said to be a pure strategy. It is a condition in which players choose a strategy with a probability equals to one. There is another form of strategy that we consider in this section. It is a situation in which agents (players) assign a probability to each choice (strategy) and play their choices according to those probabilities. For example firm-A might choose to play top 50 percent of the time and bottom 50 percent of the time, while B might choose to play left 50 percent of the time and right 50 percent of the time. This kind of strategies are said to be a mixed strategies. Strategies in which a player makes a random choice among two or more possible actions, based on a set of chosen probabilities.

If firm-A and firm-B follow the mixed strategies given above, then they will have the probability of $\frac{1}{4}$ of ending up in each of the four cells of the payoff matrix. Thus the average payoff to A will be zero, $[= (\frac{1}{4} \times 0) + (\frac{1}{4} \times 0) + (\frac{1}{4} \times (-1)) + \frac{1}{4} \times (1)]$ and the average payoff for B will be $\frac{1}{2}$, $[= (\frac{1}{4} \times 0) + (\frac{1}{4} \times (-1)) + 0 \times \frac{1}{4}) + (3 \times \frac{1}{4})]$.

		Firm-B	
		Left	Right
Firm-a	Left	(0,0)	(0, -1)
	Right	(1,0)	(-1,-3)

Table 3.10 Game with no Nash equilibrium in pure strategies.

Students, one advantage of mixed strategies over pure strategies is that, it allows us to determine Nash equilibrium to some game which do not have Nash equilibrium in pure strategies like the game represented by table 3.10. But how can Nash equilibrium in mixed strategies determined? Is it differs from Nash equilibrium in pure strategy?

Nash equilibrium in mixed strategies refers to an equilibrium in which each agent chooses the optimal frequency with which to play his strategies given the frequency choices of the other agent. In other words Nash equilibrium in mixed strategies is a pair of mixed strategy (P_r, P_c) such that each agent's choice a probability that maximizes his expect payoff, given the strategy of the other agent. Where P_r represents the probability that Row plays his strategies and P_c represents the probability that column play its strategies.

Let us consider the matching pennies game once again to illustrate how to determine equilibrium in a mixed strategies.

		Column	
		Head	Tail
Row	Head	(1, -1)	(-1, 1)
	Tail	(-1,1)	(1, -1)

Table 3.11 matching the pennies game.

Suppose that Row believes column plays head with probability P . Then if Row plays head, Row gets q with probability P and -1 with probability $(1-P)$. So the expected payoff of player Row while player column play head is equal to $1P + (-1)(1-P) = 2P-1$. Similarly if Row plays Tails, Row gets -1 with Row's expected payoff by playing tail while column play head is equal to $-1P + 1(1-P) = 1-2P$.

The expected payoffs for both Row and column player are summarized Table3.12 below

Column

Row		Head	Tail	Row's expected payoff
	Head	(1, -1)	(-1, 1)	$1p+(-1)(1-p)$ $=2p-1$
	Tail	(-1, 1)	(1, -1)	$-1p+1(1-p)$ $=1-2p$
	Column's expected payoff	$-1q+1(1-q)$ $=1-2q$	$1q+(-1)(1-q)$ $=2q-1$	

Table 3.12 Expected pay off both players

Now let q be the probability that column would play Head then, if column plays head, column gets -1 with probability q and 1 with probability $(1-q)$. So the expected payoff of column by playing head, while Row play head is equals to $-1q + 1(1-q) = 1-2q$. We can find the expected payoff for Row player using the same fashion to be $2p-1$ as indicated in the above-expected payoff Table.

Dear students, given the expected payoff for the two player how can we determine the optimal strategy of them? To determine their optimal strategy we have to compare their expected payoff. If $2P-1 > 1-2P$, then Row is better off on average playing Head than Tail. Similarly if $2P-1 < 1-2P$, Row is better off playing Tail than Head. On the other hand, if $2p-1 = 1-2P$, then Row get the same expected payoff no matter what it does. In this case, Row can play Head, Tail or could flip a coin and randomize its play. For column, also we can determine the optimal strategy following the same fashion.

As we define earlier, a mixed strategy Nash equilibrium involves at least one player playing a randomized strategy, and no player able to increase their expected payoff by playing a pure strategy. Randomized strategy however, requires equality of expected payoffs. If a player is supposed to randomize over strategies, then both of these strategies must produces the same expected payoff. Otherwise, the player would prefer one of them and would not play the other. So to find the probability with which player chooses a given strategy can be obtained by equating the expected payoff of the two strategies of a given player.

That is:

For Row, $2P - 1 = 1 - 2P$

$$4P = 2 \rightarrow P = \frac{1}{2}$$

For Column, $1 - 2q = 2q - 1$

$$4q = 2 \rightarrow q = \frac{1}{2}$$

So the Nash equilibrium in mixed strategies for the matching pennies game is defined by the probability which maximizes the expected payoff for Row and column. That is at equilibrium Row play 50 percent of the time head and 50 percent of the time tail. Also column play 50 percent of the time head and 50 percent of the time tail. So the unique Nash equilibrium in mixed strategies of the game becomes $(\frac{1}{2}, \frac{1}{2})$

Activity 3.4

- 1). differentiate between pure strategy and mixed strategy.
- 2). describe a Nash equilibrium in mixed strategy.
- 3). given the following payoff matrix,

		Column	
		Left	Right
Row	Left	(-3,-3)	(2, 0)
	Right	(0, 2)	(1,1)

- Find:
- a). pure strategy equilibrium of the game
 - b). mixed strategy equilibrium of the game
 - c). the expected payoff of Column and Row player
 - d). the probability that the strategic choice of the player fall in a (bottom, right)
-

3.6 Repeated Game

Dear Students, up to now we have been assuming that players, play the game only one (one shot game) and there is no communication between players. Now let us consider the case where there is a possibility to repeat the game. Do you think that the equilibrium outcome of the game changed if the game is to be played repeatedly?

In repeated game if one player chooses to defect on one round, then the other player can choose to defect on the next round. That means the player who defects in the first round can be punished by the other player for his bad behaviors in the next round. So each player has the opportunity to establish a reputation for cooperation and thereby encourage the other player to do the same in a repeated game. However, such type of result is expected depending upon whether the game is played for a fixed number of times or an indefinite number of times.

In the case of a fixed number of repetitions, the equilibrium outcome of the game is similar to a one-shot game if there is no way to enforce cooperation in the last round. Basically, players cooperate because they hope that cooperation in the present will induce further cooperation in the future. But this requires that there will always be a possibility of future play. Since there is no possibility of future play in the case of fixed (finite) numbers of repeated games, players do not cooperate in the last round. Thus, the outcome (result) of playing for a fixed number of games is the same as playing a one-shot game.

However, if the game is going to be repeated an indefinite number of times, and then players have influence on their opponent's behavior through the tit-for-tat strategy. This means to do to your opponent what he or she has just done to you. That is, beginning by cooperating and continuing to cooperate as long as your opponent cooperates. If one player refuses to cooperate this time, the other player refuses to cooperate next time. As long as both parties care enough about future payoffs, both parties cooperate for fear of retaliation in the next round. This will result in a Pareto efficient equilibrium outcome. Such a game might explain price setting strategy in cartels and other oligopolistic firms operating in the same industry.

For example, consider a price setting game played by duopolists producing a homogeneous product. If each firm charges a high price, then both get a larger profit. This is the situation where both firms cooperate to maintain a Pareto efficient outcome. But if one firm is charging a high price, and the other lowers its price, the one who cuts its price captures the other's market and thereby gets an even higher profit. However, if both firms cut prices, both firms end up making lower profits. Whatever price the other firm is charging, the dominant strategy of each firm is to cut price and the equilibrium outcome becomes a Nash equilibrium. The result of the game is summarized in the table below.

Firm-2

Firm-1		Lower price	High price
	Lower price	(5,5)	(0,20)
	High price	(20,0)	(10, 10)

Table 3.13 price setting game

However, if the game is repeated an infinite number of times, there may be other possible outcomes since firms use tit-for-tat strategy. If one firm cut price in this week and get higher profit at the expensive of the other, the other firm retaliate in the next round by cutting its price and get higher profit.

If each player knows that the other players playing tit for tat, then each player fear of cutting price which initiate price war, they would cooperate. So to avoid such price war both firms prefer to maintain higher price and get Pareto efficient level of profit. This outcome is represented by bottom right cell in the price setting game payoff matrix example.

For tit-for-tat strategy to become an optimal strategy, however there are certain conditions that must be holds. First, a reasonable stable set of players is required. If the numbers of players change frequently, there is little chance for cooperative behavior to develop. Second, there must be small number of players (otherwise, it becomes difficult to keep track of what each is doing). Third, it is assumed that each firm can quickly detect and willing to retaliate if other firm cheat. Cheating (defect) that goes undetected for a long time encourage cheating.

Lastly, it must be perceived by the player that the game is repeated infinitively for uncertain number of times. In most market however, the game is repeated for long and uncertain period of time. As a result, in some industries, particularly those in which only a few firms compete for long period under stable demand and cost condition, cooperation prevails even though there are no contractual agreements are made.

3.7 Sequential Game

As we have seen in most of the games that we have discussed so far, both players move at the same time and it is known as simultaneous game. In this section, we are going to consider a case where on player move first and the other player play its strategy after observing the strategy played by its opponent. Such type of game is called sequential game. An example of

such type of game includes the Stackleberg model of oligopoly- One firm (the leader) first set its output before the other does; an advertising decision by one firm and the response of its competitor; entry- deterring investment by an incumbent firm and the decision whether to enter the market by a potential competitor; or a new government regulatory policy and the investment and output response of the regulated firm. Is there any difference in the equilibrium outcome of a game when played simultaneously and sequentially? How can we go about it to determine the equilibrium of sequential game? We will try to give answer to such type of question in this section

Consider a game given in table 3-14. In the first round, player A has a strategy to move top or bottom. Player B after observing the optimal strategy plays a strategy left or right. The return to each combination of strategies chosen as summarized in table 3.14.

		Player B	
		left	Right
Player A	Top	(1,9)	(1,9)
	bottom	(0,0)	(2,1)

Table 3.14 the payoff matrix of a sequential game

When the game is present using a payoff matrix, it has two Nash equilibria (top, left) and (bottom, right). However, we will show below that one of this equilibrium is not reasonable equilibrium. The payoff matrix hides the fact that one player gets to know what the other player has chosen before she makes her choice. As a result sequential games are easier to visualize if we present the possible move in the form of decision tree. Such representation of a game is called extensive form of a game. It shows how player sequential play the game one after the other. The above game, therefore should be present is the extensive form as in figure 3.1 to easily identify the equilibrium of the game

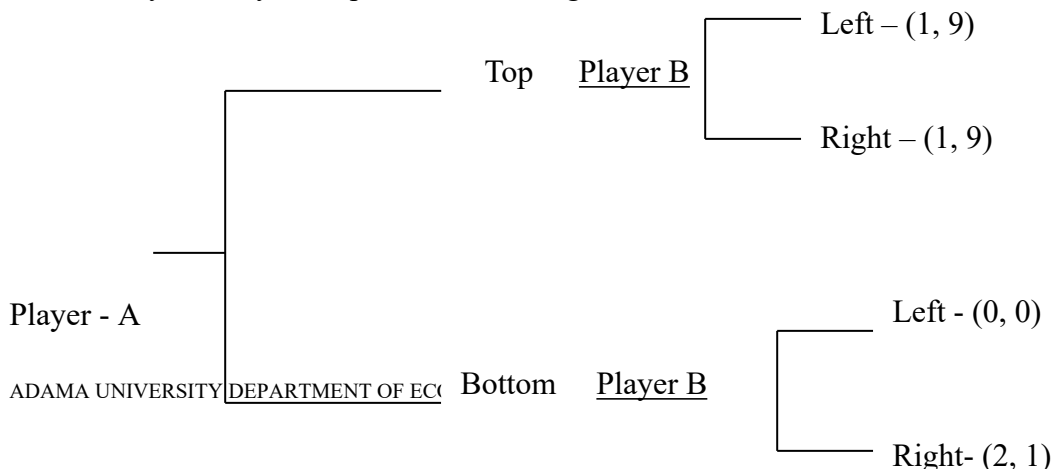


Figure 3.1 a game in extensive form.

In this game, first player-A has to choose from strategy top or bottom, and player-B after detecting the strategy chosen by player-A, choose a strategy left or right. With this basic rule of the game, the equilibrium of the game can be determined by playing the game backward starting from the player, which moves second. Suppose that player-A has already made her choice and we are identify the branch that corresponds to A's choice. For example if player-A chosen top, then it does not matter what player B does, the payoff faced by the player is (1, 9). If player-A choice to play bottom ,then the reasonable choice for player-B is to choose right and the payoff corresponds to the indicated combination of strategy is (2, 1).

Now think about player A's initial choice. If she choose top, the outcome will be (1, 9) and thus she will get a payoff of 1. But if she chooses bottom, she gets a payoff of 2. So the reasonable thing for her is to choose bottom. Thus the equilibrium outcome of the game will be (bottom, right), so that the payoff to player A will be 2 and to player B will be 1.

The strategies (top, left) are not a reasonable equilibrium in this sequential game. That is they are not an equilibrium given the order in which the players actually get to make their choices. It is true that if player-A choose top, player-B could choose left but it would be silly for player-A to ever choose top.

For player B's point of view this is rather unfortunate, since she ends up with a payoff of 1 rather than 9. What might she do about it? She can threaten to, for example, to play left if player A plays bottom. If player B would actually carry out this threat, she would be well advised to play top. For top gives her 1, while bottom- if player B carries out her threat will only give her 0.

Activity 3.5

What are the difference between sequential game and simultaneous game?

2) Two Oligopolist firms A and B are planning to maximize their profit through investing in new technology that improves their product quality. Market research indicates that the following table gives the resulting profits to each firm for the alternative strategies.

		Firm-B	
		Invest	Not invest
Firm-A	Invest	(25,20)	(40,30)
	Not invest	(30,25)	(5,0)

If the two firms play simultaneous game to decide in making the investment, determine the equilibrium of the game. Is the equilibrium dominant or Nash equilibrium?

Suppose that firm-A move first and firm-B play (decides) after observing the strategy chosen by firm A. What is the equilibrium of the game assuming that the threat from firm B is not credible? Use extensive form of game representation.

Summary

Game theory is a tool that is used for examining strategic behavior in economic circumstances. A game is any situation in which players (participants) make strategic decisions i.e. decisions that take into account each other's actions and responses.

A game is cooperative, if the player can communicate and arrange binding contracts otherwise, it is non-cooperative.

Strategic decisions result in payoffs to the players and the optimal strategy for a player is the one that maximizes her expected payoff.

A Nash equilibrium is a set of strategies such that all players are doing their best given the strategies of the other players. We will say that a pair of strategies is a Nash equilibrium if A's choice is optimal given B's choice, and B's choice is optimal given A's choice. Equilibrium in dominant strategy is a special case of Nash equilibrium.

A dominant strategy is a strategy that will maximize the expected payoff of a player no matter what the other player does.

The prisoner's dilemma is a type of game that is very much help fully in understanding oligopoly behavior. In particular, it helps explain why firms have a tendency to cheat on cartel agreement. However, if the game is repeated indefinitely, then firms may not cheat. When the game is repeated a fixed known number of times, on the other hand cheating solution is return. The dominant strategy equilibrium can be found by eliminating dominated strategies of each player alternatively mostly in the case of more than three strategies case.

A pure strategy is strategy in which a player makes a specific choice or takes a specific action with probability equal to one. Where as, mixed strategies are those in which a player makes a random choice among two or more possible actions, based on a set of chosen probabilities.

Nash equilibrium in mixed strategies refers to an equilibrium in which each agent chooses the optimal frequency with which to play his strategies given the frequency choices of the other agent.

A game said to be sequential game if player move in turn. While simultaneous game a case where a player move at the same time by assuming a given strategic choose for the other player.

Key concepts and terms

- | | | |
|-------------------------------------|---------------------|--------------------|
| ▶ Game | ▶ dominant strategy | ▶ pure strategy |
| ▶ Player | ▶ optimal strategy | ▶ pare optimality |
| ▶ Payoff | ▶ Nash equilibrium | ▶ prisoner dilemma |
| ▶ Strategy | ▶ mixed strategy | ▶ sequential game |
| ▶ Simultaneous game | ▶ repeated game | ▶ payoff matrix |
| ▶ Expected payoff | | |
| ▶ Elimination of dominated strategy | | ▶ extensive form |

Review questions.

Answer the following question briefly

We know that a single-shot prisoner's dilemma game results in a dominant Nash equilibrium strategy that is Pareto inefficient. Suppose we allow the prisoners to retaliate after their respective prison terms .formally what aspect of the game would affect? Could a Pareto efficient outcome result?

What is the dominant Nash equilibrium strategy for the repeated prisoner's dilemma game when both players know that the game will end after one thousand-time repetition?

Letu and Lata are playing a number matching game. Each chooses 1, 2 or 3. If the numbers match, Letu pays Lata 3 birr. if they differ Lata pays Letu 1 birr .

Describe the payoff matrix for this game, and show it does not have a Nash equilibrium strategies pair.

Show that with mixed strategies this game does have Nash equilibrium if each player plays each number with a probability $\frac{1}{3}$. Find the expected payoff of the two players.

Problems.

Two firms, A and B exist in a particular market. Each has two strategies. The payoff matrix for each combination is shown below.

		Firm-A	
		Left	right
Firm-B	Top	(-5,-6)	(5,10)
	Bottom	(10,5)	(-6, -5)

If firm A move first, what strategy would it choose? What strategy would firm-B then move?

If firm B moves first what strategy, would he choose? What strategy firm-A then choose to move.

Represent the game under question (a) and (b) with extensive form or game tree.

consider the game shown below :

		Column		
		Left	Middle	Right
Row	Top	(1,0)	(1,2)	(2,-1)
	Middle	(1,1)	(1,0)	(0,-1)
	Bottom	(-3,-3)	(-3,-3)	(-3,-3)

- Determine the dominant strategy of Row what ever Column does
- Which strategy of Row is dominated strategy?
- Which strategy of column is dominated strategy?

d) Determine the equilibrium of the game using elimination of dominated strategy method.

3) From the following payoff matrix, where the payoffs are the profits or losses of two firms,

		Firm-B	
		Low price	High price
Firm-A	Low price	(1,1)	(3,-1)
	High price	(-1,3)	(4,2)

Determine

Whether firm-A has a dominant strategy

Whether firm-B has a dominant strategy

The optimal strategy for each firm.

The Nash equilibrium if there is

Suggested readings

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Salvatore, D(2003):**Microeconomics; theory and applications**.4th edition

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CHAPTER FOUR

PRICING OF FACTORS OF PRODUCTION AND INCOME DISTRIBUTION

4.1. INTRODUCTION

LEARNING OBJECTIVES

BASIC CONCEPTS OF FACTOR MARKET

FACTOR PRICING IN PERFECTLY COMPETITIVE MARKET

4.3.1. DEMAND OF A FIRM FOR A SINGLE VARIABLE FACTOR IN SHORTRUN: LABOR

CHAPTER FOUR

PRICING OF FACTORS OF PRODUCTION AND INCOME DISTRIBUTION

4.1 INTRODUCTION

Why do major executives seem to earn more than worker for example wages in the neighborhood of \$2.5 million per year?

Why do secretaries earn about \$15 to \$20 pr hour?

Why do football players earn many million dollars per year?

To answer these and others questions we have to understand the determination of inputs prices.

In the preceding chapters, have completed our examination of the product market, where the prices of goods and services are determined. These products are produced by processing one or more factors of production: land, labor, capital, and entrepreneurship. As they involve cost of production of certain product their prices (rent for land, wage for labor, interest for capital, profit for entrepreneurship) determines what to produces and how to produces. The theory of input pricing is, therefore, a theory of the income distribution among landowners, wage earners and capitalist. It is equally important to see factors market that takes us in to a battleground of economics where factors of productions are determined.

The mechanism of determination of factors prices does not differ fundamentally from the pricing of commodities. The difference lies in the determination of the demand and supply of productive resources and in the reversal of the role played by firms and consumers. Factors price are determined in market under the forces of demand and supply. Firms supply of commodities to the market, but they demand inputs; consumers demand commodities from the market places, but they supply some very important inputs. This is what circular flow of economic activities for simplified economy in which there are firms and households with their respective objectives all about.

It is therefore important for us to understand the general principles that govern the determination of factors prices. Although there are many common factors of production underlying the determination of the prices of inputs, a general framework will be developed to analyzing the pricing mechanism of any productive resources in this module. Hence, in the coming topics we select labor as representative factor of production and see how price of

labor and capital is determined in their respective market through the interaction of demand and supply.

LEARNING OBJECTIVES

The general objective of this chapter is to show how the prices of productive factors are determined and how this in turn determines the level of income enjoyed by their owners.

Therefore, after completing this chapter, you will be able to:

Understand the difference between price makers and price takers in the factor markets

Calculate the marginal contribution of a factor to a firm's revenue

Explain the profit-maximizing rule in the factor market under different market structure.

Describe how the functional distribution of income is determined.

Measure the substitutability of factors.

Explain the implication of technological progress on income distribution.

4.2. BASIC CONCEPTS OF FACTOR MARKET

Production function is the relationship between inputs and outputs. The firm's demand for a factor input depends upon the input's physical productivity and the demand for the good the factor is being used to produce. A business firm participates in both the product market, where it sells the goods and services it produces, and in the factor market, where it buys factors such as the land, labor and capital it needs to produce the output. Factor markets differ from product market in three important aspects.

First, factor market tends to be more competitive than product market. In product market, firms compete against other sellers/firms on same or similar products. In many cases producers can sufficiently differentiate their product and can exercise some control over prices. In factor market, all business firms tend to compete for the same factor resources. For instance, General Motors, Microsoft television stations ... etc compete for the general pool of labor, capital and land though producing quite different goods and services. They all need word processing specialists, computers, trucks drivers and managers. It is, therefore, rare that one firm can exercise sufficient control over a particular factors market to be able to influence the prices of those factors. Most firms are price takers in factor market in a sense that business firms simply must pay the going rate for the factors of production.

Second, the demand for factors of production is a derived demand in a sense that its demand comes from the demand for the goods and services produced by the factors of production. The firm buys factors of production because they produce goods and services that generate revenue for the firm. The garment industry, for instances, buys sewing machines because it helps to produce suits, shirts, and dresses that consumers will buy or need. The principle of derived demand is essential to understanding the workings of factor markets. If consumers reduce their demand for lettuce, the demand for workers employed in lettuce decline, the demand for farmland used for lettuce and even the demand for water used in farm irrigation will also fall.

Third, the production of a good requires the cooperation of different factors of production. Farm workers can produce no corn without farmland and farmland without farm labor is useless. Both farmland and farm workers require farm implements (ranging from hand tools to sophisticated farm machinery) to produce corn. Thus, it is joint determination of factors demand.

The determination of prices of factors of production is something related the supply and demand of products as it is a derived demand. The theory of prices is related to the behavior of production function. All production functions exhibits the law of diminishing returns as larger quantities of variable factors are combined with fixed amounts of the firm's other factors, the marginal physical product (the amount of extra output of an extra works) of the variable factor will eventually decline. The interdependence of the marginal physical product of land, labor, and capital does not make the problem of factors pricing in a market setting difficult.

The determination of factor prices differs depending on the type of market structures (perfect market and different ranges of imperfect market). In this regard, we will first examine the pricing system of these inputs under perfectly competitive market. Subsequently we relax our assumption and see how the theory woks under imperfect competition. In these sections, we will examine the economic forces that govern wages and other conditions of employment.

4.1. Activity

1. How do factor markets differ from the product market?

2. It has been argued that the demand for factor is derived demand. Explain this concept with the help of the demand for labor by Wanji Sugar Factory assuming the factory is privately owned?

FACTOR PRICING IN PERFECTLY COMPETITIVE MARKET

We have seen how firm's make factor employment decisions based on the objective function under production chapter specifically in module –1. We have developed the optimization conditions of production and input employment; how much of each input the firms would demand under various sets of circumstances based on its decision of the level of output. This is an important beginning for constructing a model of input pricing and utilization under perfect competitions.

For our better understanding lets, we review some of the main points. We showed there that a firm minimizes its cost by picking the combination of inputs for which the ratios of their marginal products to their prices are all equal. That is, a cost-minimizing firm should set

$$\frac{MP_x}{P_x} = \frac{MP_y}{P_y} = \dots = \frac{MP_z}{P_z}$$

Where MP_x is the marginal production of input x, P_x is the prices of input x, MP_y is marginal product of input y, P_y is the prices of input y, and so on. If it did not hold, firm always make certain adjustments in such a way that it reduce costs and hence optimize his objective function. Going a step further, it can also be shown that the above cost minimizing firm's ratios equals to the reciprocal of the firm marginal cost. That is

$$\frac{P_x}{MP_x} = \frac{P_y}{MP_y} = \dots = \frac{P_z}{MP_z} = MC$$

Where MC is its marginal cost

More over, we have seen that the profit maximizing firm must operate at the point for which marginal cost equals marginal revenue. It follows, therefore, that

$$\frac{P_x}{MP_x} = \frac{P_y}{MP_y} = \dots = \frac{P_z}{MP_z} = MC = MR$$

Where MR is the firm's marginal revenue

Rearranging the terms in equation above we get

$$MP_x \times MR_x = P, MP_y \times MR_y = P_y, \text{-----} MP_z \times MR_z = Z$$

We concluded that the profit maximizing firm should employ each input up to the point where its marginal product multiplied by the firm's marginal revenue equals in the inputs per unit price. In this competitive product and input markets, factors are paid the value of their marginal physical product, which is determined by the force of total demand and supply.

A firm has a demand for factors (labor and capital) is related the contribution of the factor to the objective function of the firm. The contribution of labor, for an instance, to the firm's profit function is determined by two things: the quantity of additional output a unit labor input can produce and the price that the firm can sell its product. The analysis of the demand for the labor differs when labor is the only variable factor of production and when there are several variable factors. Accordingly, we will develop first the demand for labor by a single variable factor for the firm and market demand. Then we take a look at the derivation demand curve for input when there are several variable factors. By the same analogy we will develop the supply of labor by individual consumer and then market supply.

4.3.1.DEMAND OF A FIRM FOR A SINGLE VARIABLE FACTOR IN SHORTRUN: LABOR

Our first step in analyzing the demand for an input is to consider the demand curve of an individual firm for some input (call labor) under the assumption that labor is only variable factor in the firm's production process. Our analysis of demand labor underlines the following assumptions:

A single commodity x is produced in perfectly competitive market. For example a firm is producing wheat and its price P is given for all firms in the market

The goal of the firm is profit maximization

There is a single variable input in the firm's production process: Labor and all other inputs are fixed. Workers who are assumed to be homogeneous in our example of wheat, for instances, varies in their ability to produce, but the farm, machinery, fertilizer, seeds and so on are all fixed.

Perfectly competitive market.

The price of labor services, w , is given for all firms as the firm is price taker.

This implies that the supply of labor to the individual firm is perfectly elastic.

This can be denoted by a straight line parallel to the horizontal axis (Figure 4.1). At the going market wage rate the firm can employ (hire) any amount of labor it wants.

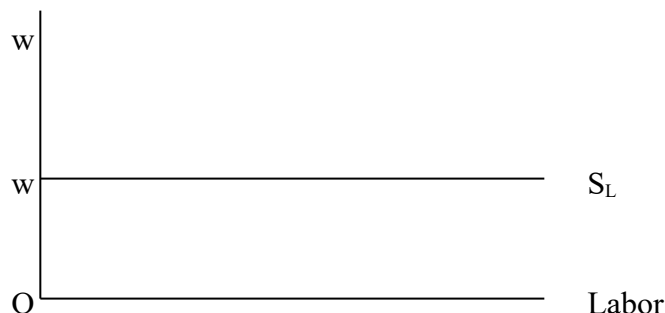


Figure 4.1: Labor supply for an individual firm

Technology is given. The available technology enables the firm to produce certain level of output. This is shown by the production function of the following figure 4.1.

Given the above assumption, the demand curve for labor by the firm is the quantity of input that the firm would demand at each possible price (wage). The firm will demand certain amount of input for which the value of the extra output produced by the employment of the last unit of input is equal to the prices of the product. To make these meaning more concert, we use the production function based on our assumption and deduce the concept from marginal product of input at each level of employment. Look at the following production function. O measured in units depends on the number of workers hired.

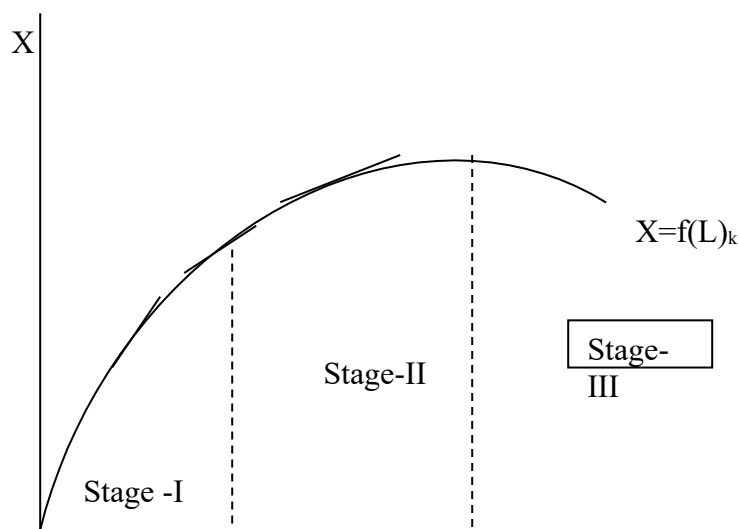


Figure 4.2. The production function.

The more workers, the more output. However, as the workforce increases, the increase in output is unlikely to change at the same proportion as the increase in the number of workers employed add lesser and lesser even nothing to the production process because of diminishing marginal product of labor. As more and more worker is added (employed) the marginal physical product of additional labor decline though the total product increases. The slope of the production function measures the marginal physical product of labor (MPP_L). That is

$$\frac{dX}{dL} = MPP_L$$

From production, theory we know that a given production function has three stages of production. The first stage (Stage-I production) is the range over which the marginal physical product of labor increase at increasing rate (the extra output gained from extra input is higher as compared to the previous input). The firm hires more factors over this range. The second stage is where the MPP_L decline at higher levels of employment, given the law of variable proportions. Over third stage of production, the extra input (labor) has negative effect or contributions on total output.

We are converging to the second stage, where MPP_L is falling but positive because these profits maximizing firms operate in this range. If we multiply the MPP_L at each level of employment by the given prices of the output P_x, we obtain the Value of Marginal Product or Marginal Revenue Product in monetary term-VMP (MRP). In short:

$$VMP = MPP_L \cdot P_x$$

A profit maximizing firm will hire a factor as long as it adds more to total revenue than to the cost. This is so, as firm is guided by profit maximization objective in the factor market which is basically the same as profit maximizing decisions in the product market as deciding on the quantity of inputs determines the level of output. Hence, the higher the

prices of an input, the lesser the firm wants to buy. It is a down-ward sloping curve because the greater the amount of factor used, the lower is the marginal revenue product as the law of diminishing returns holds. In addition, as the firm is a price taker in the product market to some extent, higher levels of output will result in lower marginal revenue. From these one can see with the quantity of factor increases, both marginal physical product and hence the marginal revenue tend to decline, so that MRP (which $MPP \times MR$) declines.

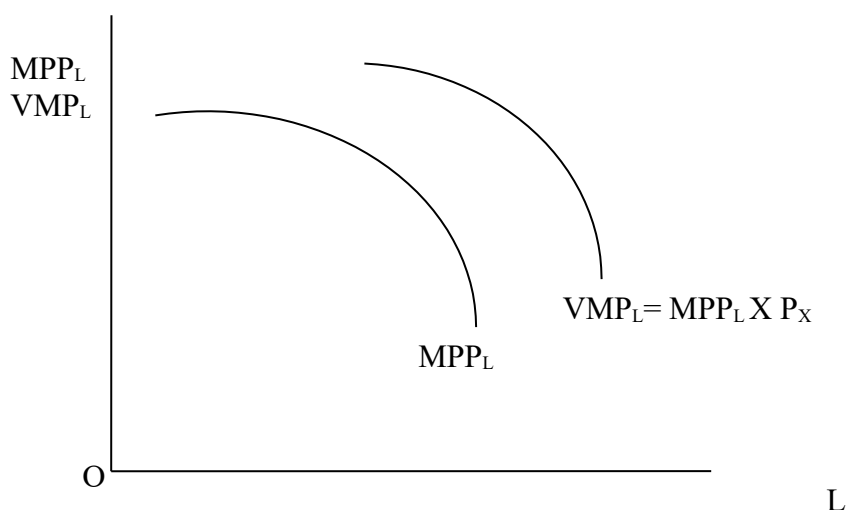


Figure 4.3. Demand curve of the firm

VMP_L curve is the firm's demand curve for a factor as it determines the level of extra output produced by additional labor and hence demands for labor. Hence, how do equilibrium level of employment is determined given the labor demand and supply?

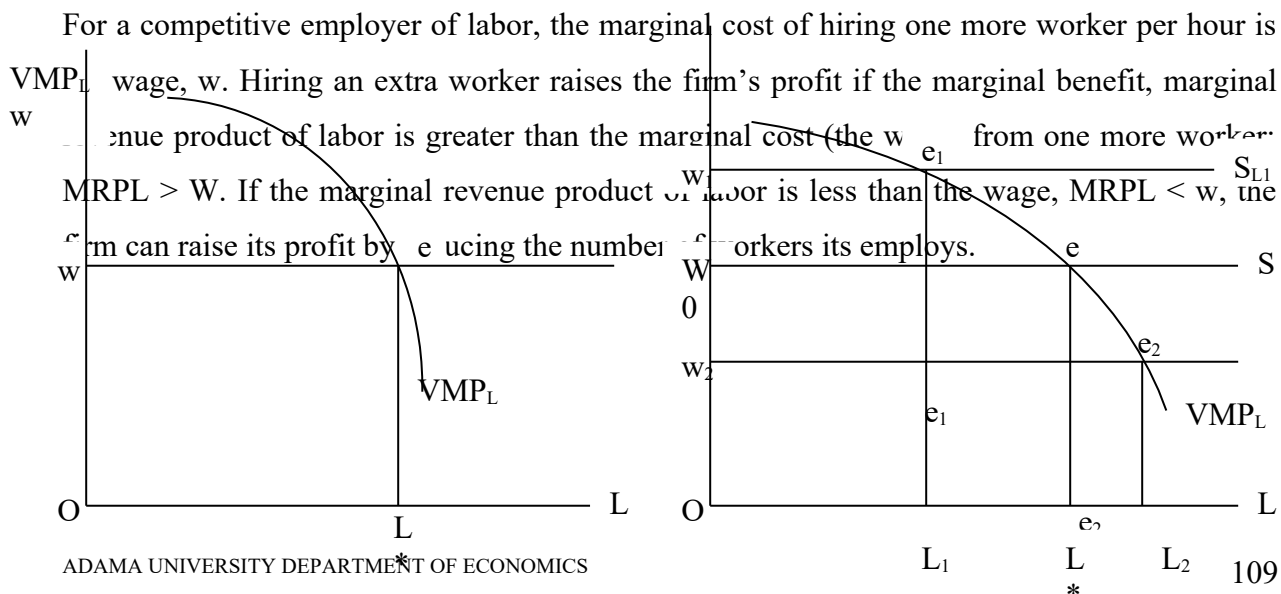


Figure 4.4-A Demand curve of the firm Figure 4.4-B Demand curve when wage changes

Thus, the firm maximizes its profit by reducing the number of workers until the marginal revenue product of the last labor is exactly equal to the marginal cost of employing that work, which is the wage; $MRP_L = w$. This is indicated by point 'e' on figure 4.4-B.

As the wage rate changes, the firm will move along its VMP_L curve to determine the profit maximization level of employment. If the market wage is raised to w_1 from original point, the firm will reduce its demand for labor to L_1 in figure 4.4 -B in order to maximize its profit (at e, where $W_1 = VMP_L$). Similarly, if the wage falls to W_2 , the firm will maximize its profit by increasing its employment to L_2 . It follows from the above analysis that the demand curve of a firm for a single variable factor, labor, is value of marginal product curve and is downward sloping.

Formal derivation of the equilibrium of the firm is as follows:

The production function is specified as $X = f(L)_k$

The total cost consists of the variable cost $W.L$ and the fixed cost F can be stated as

$$C = W.L + F$$

The revenue of the firm is $R = P_x . X$ which is $R = P_x . [f(L)]$.

The firm wants to maximize its profit

$$\pi = R - C$$

$$\pi = P_x . [f(L)] - (W.L + F)$$

The first order derivative of the profit functions with respect to labor equal to zero we obtain

$$\frac{d\Pi}{dL} = \hat{P}_x \cdot \frac{dX}{dL} - \bar{w} = 0$$

Rearranging

$$P_x \cdot (MPP_L) = W \quad \text{given} \quad \frac{dx}{dL} = MPP_L, \quad VMP_L = W$$

The above discussion can be illustrated with numerical example by assume a production process, which involves a fixed amount of machinery giving rise to a total fixed cost of Birr 50, and the labor which is the only variable factor. The wage rate is 40 and the prices of the commodity produced are 10. The production function is specified by the information of the first four columns of table 4.1. Column 6 shows total revenue ($= X \cdot P_x$), column 10 indicates total variable cost ($= L \cdot W$). Finally column 12 shows the profit of the firm ($\pi = R - TVC - FC$)

Table 4.1: Data for the derivation of the demand curve for the firm

The production function				Costs, revenues, profit							
Unit of fixed capital K	Unit of labor L	Total output (unit) X	Marginal physical product of labor MPPL	Price of product P_x £	Total revenue $R = X \cdot P_x$ £	Value of marginal physical product of labor VMP_L (£)	Total fixed cost £	Wage rate w £	Total variable Cost $TVC = W \cdot L$ £	Total Cost $TC = TVC + FC$	Profit $TI = R - TC$ £
10	0	0	—	10	0	0	50	40	0	50	-50

10	1	20	20	10	200	200	50	40	40	90	110
10	2	38	18	10	380	180	50	40	80	130	250
10	3	54	16	10	540	160	50	40	120	170	370
10	4	68	14	10	680	140	50	40	160	210	420
10	5	80	12	10	800	120	50	40	200	250	550
10	6	90	10	10	900	100	50	40	240	290	610
10	7	98	8	10	980	80	50	40	280	330	650
10	8	104	6	10	1040	60	50	40	320	370	670
10	9	108	4	10	1080	40	50	40	360	410	670
10	10	110	2	10	1100	20	50	40	400	450	650

Source: Kotisyanos , 1984

There are two ways of looking at profit maximization of the firm based on the Table 4.1 .

A. Total revenue or total cost approach

Profit is at a maximum where the differences between total cost and revenue are greatest. The maximum deviation is at point where the distances between two curve is the largest. This is at point where $MR = MC$, which is $MRP = W = MC_L$ at labor utilization is equal to eight (see table above).

The VMPL approach

The supply of labor to the individual firm is the straight lines equal to given wage rate for competitive firm. The equilibrium is where the VMP_L intersect the supply $S_L = w$. According to our data, this occurs at wage rate, \$40 and labor employment equals to 8. (See Table xx). Under both approach profit is maximized at the 8th level of employment where $W = VMP_L$.

Example

Suppose certain hypothetical firm has a Cobb – Douglas production function for Canadian thread mill is specified $Q = L^{0.6} k^{0.2}$. Suppose further that mill's capital, K , is fixed at 32 units, labor is only variable input. Determine, MPP_L , VMP_L , and profit maximizing level of output?

Solution:

$F = L^{0.6} k^{0.2} = L^{0.6} (32)^{0.2} = 2L^{0.6}$, then extra level of output or MPP_L can be determined by looking the extra output from the last worker when firm's goes from 31 to 32 workers is

$F = (2 \times 32^{0.6}) - (2 \times 31^{0.6})^{0.3}$. The VMPL at this point is $VMPL = MPP_L \times W = 0.3 \times 5 = 1.5$.

The firm can sell its output at \$50 per unit, the firm's $MRP_1 = P_x MP_L = \$50 \times 0.3 = \15

Thus, when the price is \$50 and wage is 15, at this level the firm hires 32 workers. More generally, the marginal product of labor function, when we hold capital fixed at $k = 32$, is $MPL = 1.2 L^{-0.4}$. This is obtained by holding capital fixed at $k = 32$ and differentiating the short run production function, $f = 2L^{0.6}$ with respect to labor:

$$MPL = \frac{dq}{dL} = 0.6 \times 2 \times 2^{0.6-L} = 1.2L^{-0.4}$$

The above calculator method for discrete change from 31 to 32 workers can best be estimated by infinitesimally small change in labor. We can find MPL at $L = 32$ is exactly $0.3 = 1.2 (32)^{-0.4}$

4.2. Activity

1. Given the above example of Canadian mill, what will be the VMP_L and MPP_1 when capital is fixed at $K=35$?

4.3.2. DEMAND OF A FIRM FOR SEVERAL VARIABLE FACTORS

Suppose the now that the firm uses more than one variable. What is the demand curve for factor of production? Do you think VMP_L is demand curve for labor? Unlike the case of one Variable, the VMP_L is not longer demand curve for the several variable factors of production. Because when the many varied factors are used together in production change in price of one factor lead to change in employment of the other factor of production. Change in employments of other factor in turn shifts the MPP curve of inputs whose price initially changed as the productivity of a given labor is not the same when it is associated with different levels of factors such as capital.

Consider the case of two variable inputs: labor and capital. The firm will employ labor and capital to the point where the value of the marginal product of each factor is equal to the prices of the factors. If the inputs are independent, a change in the quantity of one input has no effect on the marginal product of the other; the demand curves for each factor can be derived separately as it is done for labor.

Thus: Value of marginal product of labor $(VMP)_L = (MP)_L \cdot P_o = P_L$
 Value of marginal product of labor $(VMP)_K = (MP)_K \cdot P_o = P_K$

However, the demand for one factor depends on the nature of the relationship between the inputs. Because most of the times inputs are complementary so that the marginal product of one input increases with higher employment of the other inputs. The marginal product of labor for each unit of labor, for instances, increases with a larger quantity employed of the complementary inputs such more tools and machinery.

To see this point clearly, consider the impact of a decline in the wage rate that makes the firm hire more labor. The lower wage causes the marginal cost to falls so that the firm expands output. As the firm tries to expand output by increasing employment of labor, it will probably necessary to hire more capital. With capital usage increases, the VMP curve of capital and the demand for labor curve shift to the right, including a further increase in the quantity of labor employed. The resulting demand curve for labor looks like the one shown by figure xx with dotted line. Point A also lies on curve VMP_L - initial marginal revenue product curve for input combination associated with its price. Its very construction assumes that the employment of other inputs is fixed, and it would be demand curves for labor if none of other factors were variable. Nevertheless, what would happen when the price of labor decline to \$6 and other inputs were variable? The marginal revenue product of labor now exceeds its new prices and the firm expands labor employment. However, increasing employment of labor would shift the marginal revenue product of other factor from $K=10$ to $K=15$.

These change further increase the labor and the system goes on. After all of these changes have worked their way through the firm's cost minimization decision system, the firm would be on a different marginal revenue product of labor- $(VMP_L)^*$. Now the employment of labor would be determined by the point where its new marginal revenue product curve intersects. The firm now demands about 200 labor with wage rate \$6. This indicated by Point C lies on curve $(VMP_L)^*$ -

the marginal revenue product curve for an input combination associated with higher prices. As a result, we can be sure that points A and C would both lie on the firm's demand curve for input (labor). Other points for other prices could be determined in a similar fashion. The result of the determination is the demand curve for labor. This demand curve is represented in figure 4.5 by curve D_L , as would be expected it can be shown that all demand curves of this type slope down and to the right.

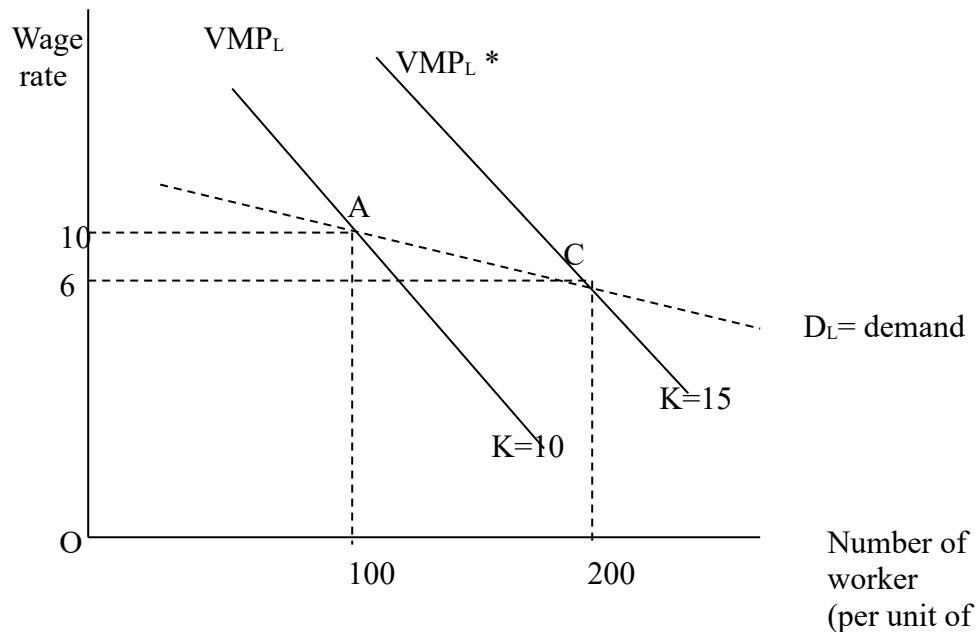


Figure 4.5: demand for labor allowing for changes in the capital

Alternatively, When there are various factors are used together in production; the demand curve for labor is derived using Iso-quant analysis. This has been discussed in one of the previous chapters. The change in the wage rate has in general three effects:

Substitution effects

Output effects (income effect or sales effects)

Profit maximizing effect

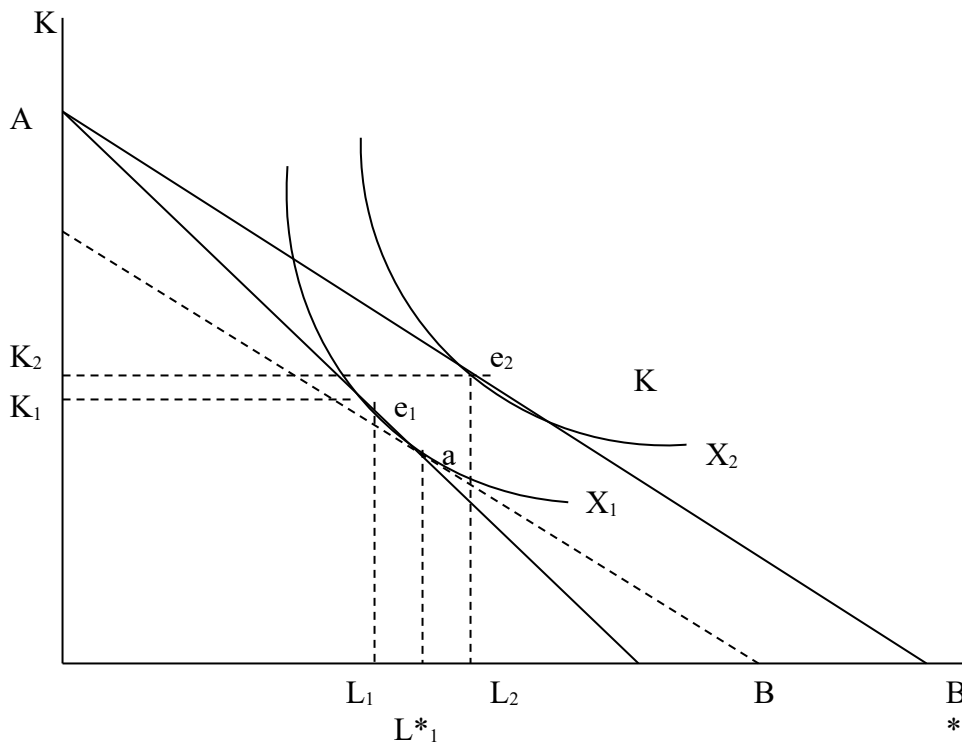
This can be presented using simplified diagram 4.6. Suppose that initially the firm produces the profit maximizing output x , with the combination of factor K , L , given the (initial) factors prices w_1 and r_1 , whose ratio defines, the slope of the Iso-cost line AB .

Suppose wage rate falls to w_2 , so that the Iso-cost line AB (the prices of capital remains the same) shifted to AB^* as the ratio changes. The firm using the same expenditure can now

produce ten higher output denoted by the Iso-quant X_2 , using K_2 , L_2 amount of capital and labor respectively. The equilibrium moves from point e_1 to e_2 . The movement can be split in to two separate effects:

substitution effect

Output effect



To Figure 4.6: Derivation of demand curve for variable input L is AB^* so that it reflects the new prices ratio, but tangent to the old Iso-quant X_1 . The tangency occurs at point 'a'. The movement from point 'e', to 'a' along old Iso-quant is pure substitution effect that show the cheaper labor substituted for capital. Thus, the employment of labor increases from L_1 to L_1^* . But, as the firm can buy more labor and capital with the same expenditure, it can produce the higher output X_2 , employing K_2 of capital and L_2 of labor. The increase of employment from L_1^* to L_2 , corresponding the movement from 'a' to 'e₂', is the output effect.

Point 'e₂' is not the final equilibrium of the firm. It would be it the firm were to spend the same amount of money as initially. However, keeping the total cost expenditure constant does not maximize the profit of the firm. The firm will increase its expenditure and its output in order to maximize its profit. To understand this let us assume that the initial equilibrium of the firm is

denoted by point H in figure xx where the firm's MC is equal to the prices of X. The fall in the wage rate shifts the MC curve downwards to the right, and the profit maximizing output of the perfectly competitive firm increases to X_3 . This requires an increase in expenditure equal to the shaded area X_1HGX_3 .

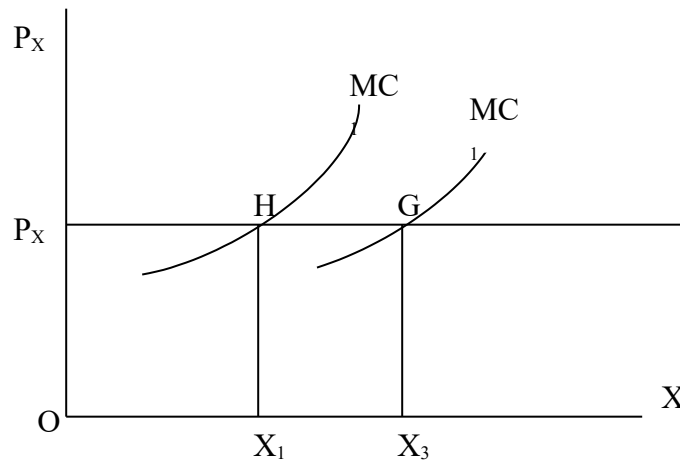


Figure 4.7: Effect of increase in expenditure

This, Iso-cost line AB^* must shift out wards, parallel to itself at a distance corresponding to the increase in the firms outlays(see figure 4.8 below). Actually the new Iso-cost can be determined by dividing the increase (addition) in total cost by price of capital, r , and adding the result to the distance OA . The new point 'A', on the vertical axis is the vertical intercept of the required Iso-cost and parallel to AB' a new Iso cost line A^*B^* . The final equilibrium of the firm will be determined by the point of tangency of the new Iso-cost A^*B^* with the Iso-quant denoting the profit-maximizing output X^X at e_3 of figure 4.8.

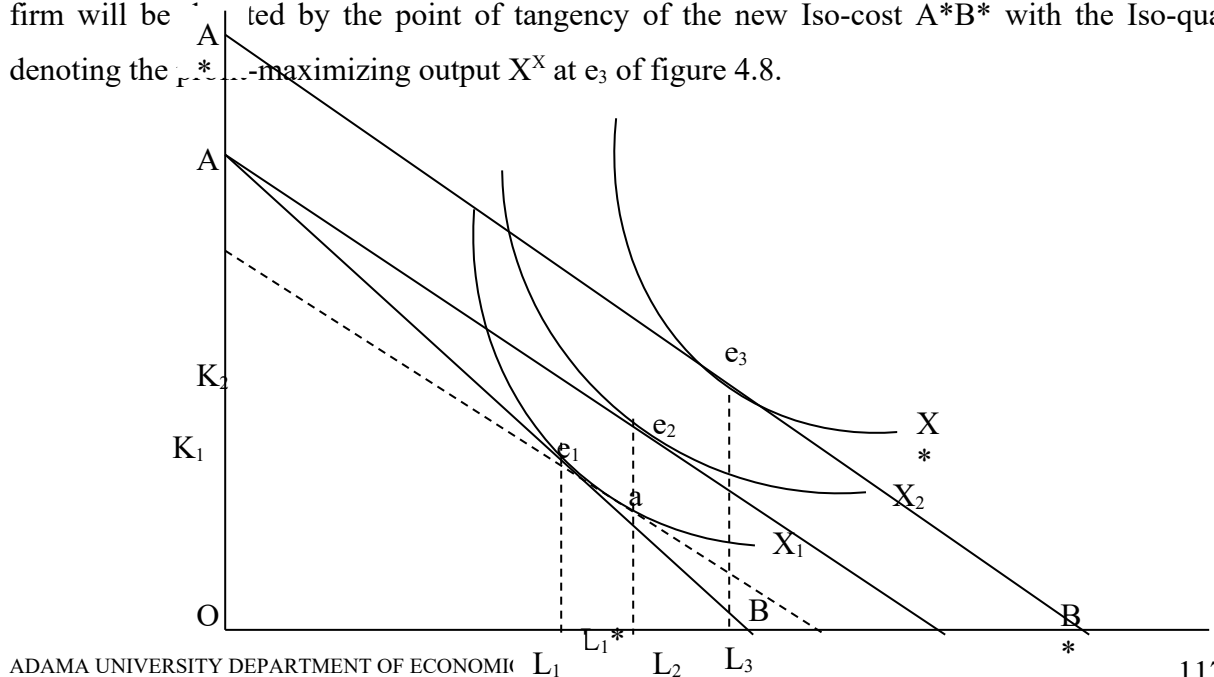


Figure 4.8: Derivation of demand curve for variable input

In sum, the substitution effect of a decrease in the wage rate causes a decrease in the MPP, even though, the output effect and profit maximizing effect result in an increased employment of both inputs. Last two effects exceeds the former one hence the MPP_L curve shift outward.

Given the prices of the final commodity, P_x , the VMP_L shift to the right when several variable factors are used in the production processes. The new equilibrium demand for labor is then denoted by point B on VMP_{L2} . By repeating the above analysis with different wage rates we can generate a series of point such as A, B and C. The locus of these points is the demand for labor by the firm when several factors are variable. (See figure 4.9 below)

This long run demand for a factor is negatively sloped as the three effect of an input -prices change must cause quantity demanded of the factor to vary inversely with prices.

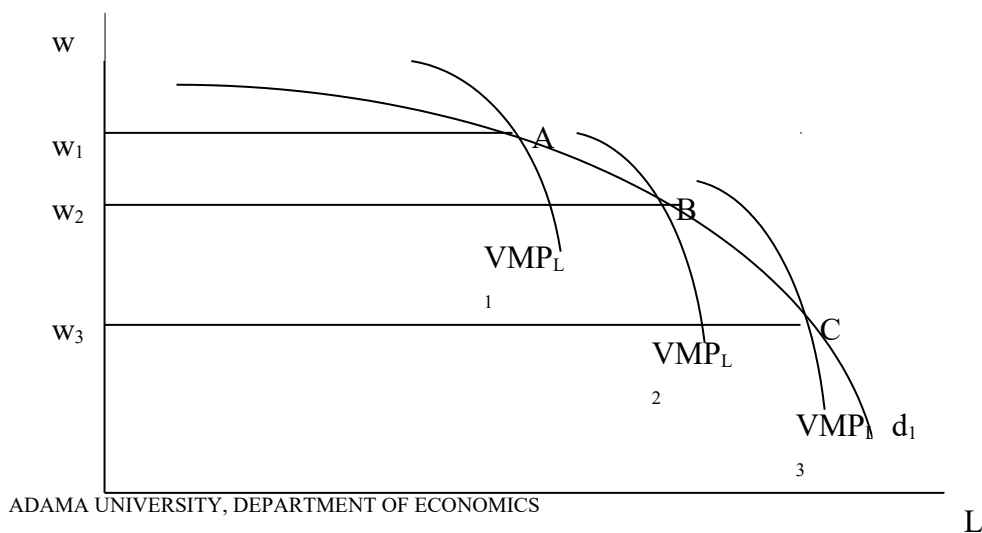


Figure 4.9 : Demand curve for the firm under more than variable

This is somewhat the demand for factors in the long run when all factors can vary because of adjustments of the structure of production.

4.3. Activity:

Assume Coca-Cola Company uses labor, capital and land as factors of production: labor and capital are relatively variable factors where as land is fixed factor of production. Further more, the demand for factors are based on the marginal product of the factor (MP_L and MPP_K) and the price of the product (P) and inputs (w and r). Based on the above information derive the demand curve for labor using the Is-cost line theory discussed above?

4.3.3. THE MARKET DEMAND CURVE FOR A FACTOR

Having derived the individual firms demand for inputs, the next step is to derive market demand curve for an input. Recall how we derived a market demand curve for a commodity in previous section. We summed horizontally over the demand curves of individual consumers of the commodity. Do you think we can apply the same technique to drive market input demand?

Although these provide a first approximation, it would not yield the correct result because it would neglect the effect of changes in the input prices on the product price. For instances, as the prices of the input falls all firms will seek to employ more of this factor and expand their output. Thus, the supply of the commodity shifts downwards to the rights, leading to a fall in

the price of commodity, P_x . Since these prices is one of the components of the demand curves of the individual firms for the factor, these curves shift down ward to the left.

As inputs such as labor are used in many output markets, thus, to derive the market demand curve for labor, we first determine the labor demand curve for each output market and then sum across output markets to obtain the factor market demand curve.

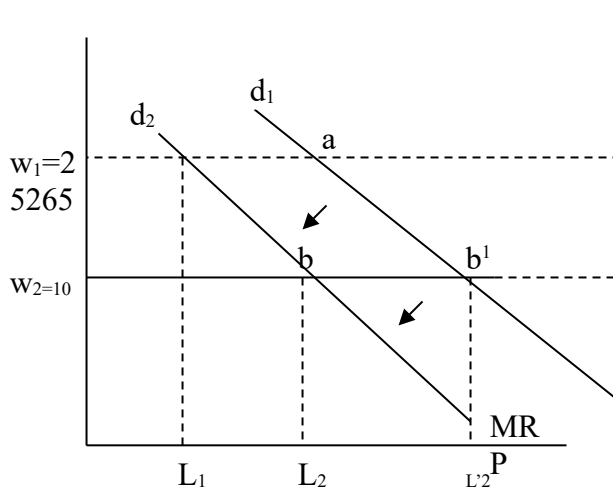


Figure 4.10A: Demand curve of a single firm

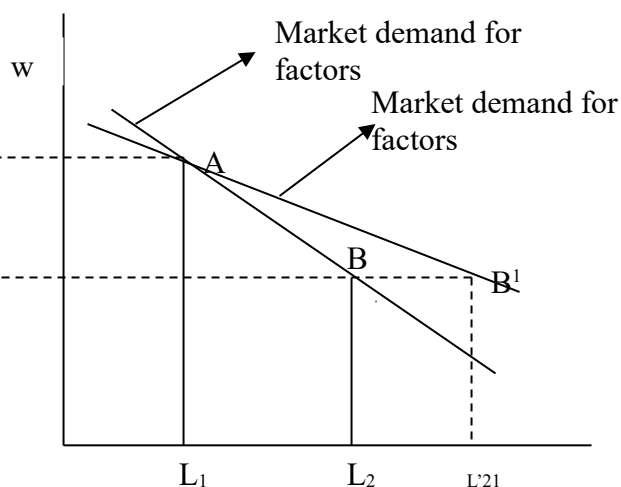


Figure 4.10B: Market demand curve for labor

Earlier we derived the factor demand of a competitive firm that the output market price depends on the factor's prices. As the factor's price falls, each firm, taking the original market prices as given, uses more of the factors to produce more output. This extra production by all the firms in the market causes the market prices to fall. As the market price falls, each firm reduces its output and hence its demand for the input. Thus a fall in input price causes less of an increase in factor demand than would occur if the market price remained constant. This can be illustrated with figure 4.10. Initially suppose the wage rate is w_1 (e.g. $w_1 = 25$) and output market prices P_x (e.g. $P_{x1} = 9$ Birr). The firm is at point 'a' on its demand curve and employs L_1 units of labor.

Summing over all employing firms, we obtain the total demand for the input at wage rate w_1 . Assume next that the wage rate declines to w_2 ($w_2 = 10$). Other thing being equal, the firm

would move along its demand curve d_1 , to point b^1 , increasing the employed labor to L'_2 because cost decline that motivate firm to produce more. As a result of more output, the price of product decline, that in turn leads to decline in employment and less demand for inputs. The marginal revenue product of labor decline and have inward shift of demand curve for labor.

The equilibrium under new wage rate is not at point b^1 , but at point b on new demand curve d_2 . Summing horizontally over all firms we obtain point B of market demand curve. If the fall of the commodity-price was not taken in to account, we would be led to an overestimation of the demand for labor following a decline in wage rate. This indicated by point B' of the figure 4.10 - (b). Thus the market demand curve for labor is shown by the line AB rather than AB' .

This long run demand for a factor is negatively sloped as the three effects of an input-prices change must cause quantity demanded of the factor to vary inversely with prices.

4.4. Activity

Discuss what do the demand for labor in the long run look like when wage rate rises?

In summary, the determinants of the demand for a variable factor by individual firm depend on:

The prices of input

The marginal physical product of the factor

The prices of the commodity produced by the factor as $VMP_L = P_x \cdot MP_L$

The amount of other factors which are combine with labor (e.g. capital)

Technological progress as it changes the MPP of all inputs and hence their demand,

We have derived the demand curve for the individuals and the market. The next thing to be done is deriving the corresponding individual as well as market supply of under competitive market structure in such away that we see the market equilibrium.

4.4. SUPPLY OF LABOR IN PERFECTLY COMPETITIVE MARKET

The supply of labor to the market depends on different factors. Some of these are:-

The prices of labor (wage rate)

The tastes of consumers (that defines trade-off between leisure and work)

The size of the population

The labor - force participation rate

The occupational, educational and geographic distribution of the labor forces

The relation between the supply of labor and the wage rate defines the supply curve. The other factors can be considered as shift factors of the supply curve. As we are interested in the supply curve, we assume all other factors as given. Just like for the supply of commodity in product market, the market supply for factors is the supply of labor by individuals. Thus, we begin by the derivation of supply of labor by a single individual and then that of market.

4.4.1.THE SUPPLY OF LABOR BY AN INDIVIDUAL

For ordinary rational economic agent, higher the price of input is required to increase the quantity of input willingly supplied. But an individual labor face with certain constraint especially related to the time available for the agent. This take us to the theory of consumer preference covered in module one of the course. An individual economic agent has only 24 hour with a day and tend to allocate his/her time for work and for leisure which in turn affect the supply of labor for work. We derive the supply curve using this theory.

Hence, the supply of labor by an individual can be derived by indifference curves analysis. A simple consumer choice problem is between two goods “income from work” and “leisure” in the standard consumer choices problem. The individual is assumed to have preferences over the two goods that can be summarized in the form of an indifference map. The supply of labor directly related to the amount of time an individual is willing to work.

The derivation can be illustrated with Figure 4.11 on which the horizontal axis measures the hours available for leisure (and work) over a given period of time. For example, there are OZ maximum hours in a day, which an individual can use for leisure or for work. On the vertical axis we measure money income. The slope of a line from Z to any point to the vertical axis represents the wage per hour. If the individual were to work all the OZ hours and earn a total

income of OY_0 , the wage rate would be $w = \frac{OY_0}{OZ} = [\text{slope of } ZY_0 \text{ line}]$. The steeper the line or indifference curves the higher the hourly wage rate.

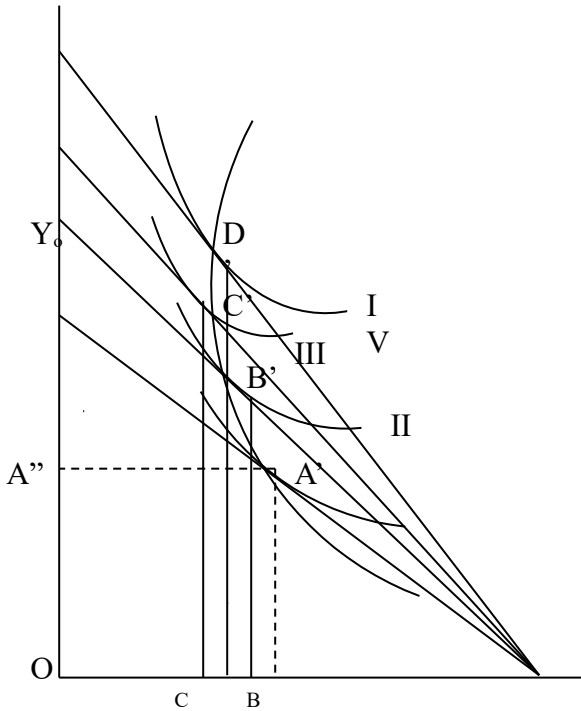


Figure 4.11-1; The optimal choice of Leisure and income

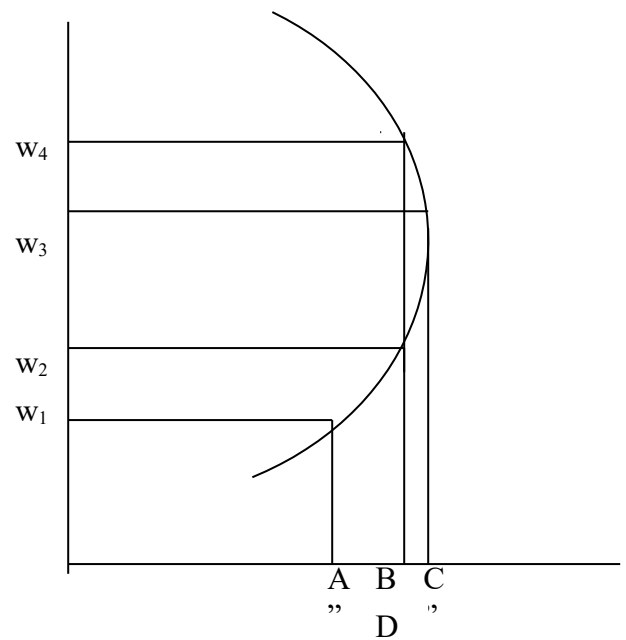


Figure 4.11-2 labor supply curve of the workers

For example on indifference curve I, the individual is in equilibrium by working AZ hours (OA^{11}) and get wage rate W_1 . If the wage rate increases to W_2 the individual will work more hours ($BZ > AZ$) and will earn a higher income (BB^1) and will have less hours (OB) for leisure. The equilibrium at this time is at point B^1 . Similarly when wage rate rise to W_3 , the equilibrium become C^1 .

As we know from consumer behavior, individual tend to work more hours as wage increase up to certain point. At some higher wage rate the hours offered for work may decline because of some other reasons. Higher wage creates disincentives for longer hours of work after a certain point. When rates of pay reaches high levels, the incomes of such individuals may become so high that they may choose to reduce the amount of work and they do rather than increase leisure time. Because of these the supply of labor is backward bending curve.

As can be seen from above diagram, if wage rate is increased to W_4 , the individual will work BZ hours, the same as wage rate W_2 . If the wage rate increases still further (to W_5) the hours supplied for work decline even more: They drop back to AZ. This is so for the fact that longer working hours implies less leisure hours. This pattern of response to higher wage rates produces a backward – bending supply curve for labor as. Thus, the supply of labor can be obtained from the locus of the equilibrium point A', B', C', D', etc. This is shown by figure 4.11-2. As wage rate increases, the individual's income rise, and this enable the worker to have shorter leisure activities. However, the time for such activities is less. Beyond a certain level of wage rate, the supply of labor decreases as worker prefers to use his income on more insures activities.

This nature of supply curve of labor is the result of income and substitution effects that accompany increases in the prices paid to labor, the wage rate. The substitution effect suggests that as wage increase people tend to substitute their leisure time by working more hours and increases the amount of time they work. On the other hand, the income effect shows that as the people are getting higher income because of the increases in wage rate, the person feel wealthier and starts to enjoy more leisure time by cutting his labor supply. These means substitution and income effect works together. At lower level of employment the substitution effect tends to be greater than the negative income effect so labor supply tend to increases with an increases in wage rate. Gradually , however, the negative income effect tends to dominate the positive substitution effect and hence, individual reduces their labor supply if wage rate keeps on increasing at this high wage rate.

4.5. Activity

How do the substitution and income effect of time affect the supply of labor individual under perfectly competitive market?

4.4.2. THE MARKET SUPPLY OF LABOR

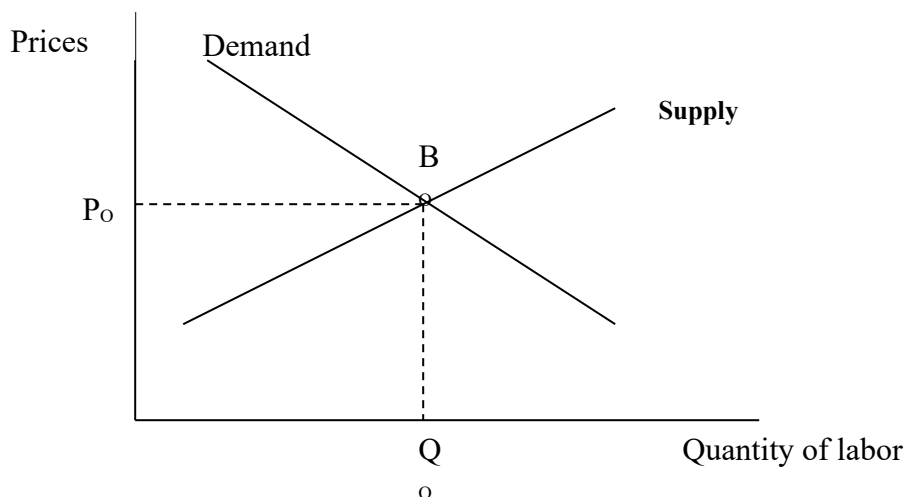
Although there is disagreement on individual backward bending labor supply pattern among economist, in short run market supply of labor may have segments with positive and segments with negative slope. But, in long run the market supply must have a positively sloped.

Because, more wage is an incentives for young workers, and change the jobs. The others maintain back word-bending supply curve as people tend to supply more labor until the income reach the level required for a comfortable standard of living. It seems that a positive aggregate supply of labor work in general case.

Just as the market supply curve in product markets could be determined by summing the quantities individual firms would supply at various prices/wage rate. This is mainly because any negative effect of an increase in wage can be off sated, as the all individual do not behave in the some way. That is when someone is tend to decrease the supply of labor the others may increase the willingness to supply that makes the supply curve upward sloping.

4.5. EQUILIBRIUM PRICE AND EMPLOYMENT OF AN INPUT

The market demand and supply curves of an input determine the inputs equilibrium price. The price of input will tend in equilibrium to settle to the level at which the quantity of the input demanded equals the quantity of the input supplied. This is indicated by using figure4.12. This point B is the equilibrium position for input market with price of the input is P_o and quantity of input Q_o . If the prices were higher than P_o , then the quantity supplied would exceed the quantity demanded and there would be down word pressure on the prices and vice versa. Hence, the market model is valid for the determination of the equilibrium prices of a commodity or productive resources.



4.6. F Figure 4.12: Determination of equilibrium prices and quantity of inputs
ADAM/

We have seen how prices of input are determined under perfectly competitive market. The next step is the determination of input pricing under imperfect competition. But what would the theory say about a case in which there was perfect competition in the market for input but imperfect competition (monopoly, oligopoly, monopolistic, competition) in the relevant product market? The input price under imperfections in the commodity and the factor markets is determined in the same manner as the case of perfectly competitive markets. The demand and supply determines the prices of factor and the level of its employment though the determinants differ from perfect market. Further more, there are different structures of market based on the power of economic agent in two markets (product and input market). There are four known simplified models are under these. These are:

Model	Product market	Factor market
A	Monopoly	Perfectly competitive
B	Monopoly	Monopsony
C	Monopoly	Monopsony +unionized labor supply
D	No monopoly	No monopsony but unionized labor

The above models are extension of the marginal productivity theory of factor pricing and income distribution. We will see each of them in dome detail.

4.6.1.MODEL–A: MONOPOLISTIC POWER IN THE PRODUCT MARKET

In this model, we assume that the firm has monopolistic power in the product market, while the factor market is perfectly competitive. As we did for factor pricing under perfectly competitive product and factor markets we will analyze first how demand and supply of labor is determined. Then we will see determination equilibrium factor price and employment. The analysis of firm's demand for labor differs when it is a single variable factor and when there are many variable factors. We will see each one detail independently.

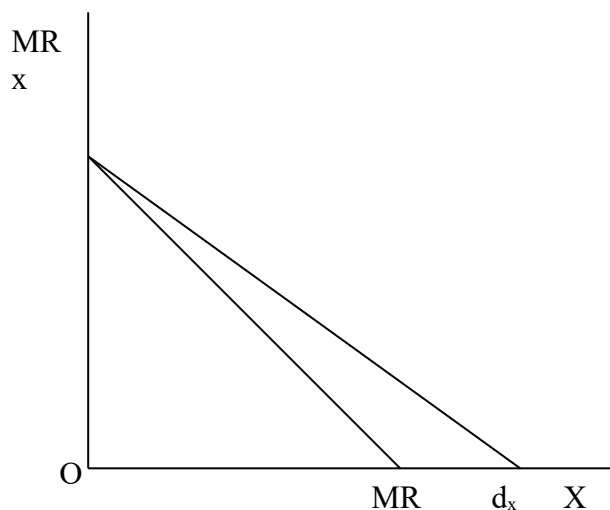
4.6.1.1Demand for a single variable factor by a monopolistic firm

To illustrate how the factor prices are determined under this condition, we make some assumptions. Assume:

- A. A firm with monopolistic power in a product market. Recall how the market demand curve for output of a firm with monopoly power behaves in product market. We have said that the demand for the product of the firm is down-ward sloping and the marginal revenue is smaller than the price at all levels of output.
- B. Firm in the market that the firm uses a single variable factor, labor in perfect market. The supply of labor to the individual firm is perfectly elastic and the wage rate is given or determined by the market.

Under these conditions the demand for labor by an individual firm is not the VMP_L curve but marginal revenue product curve. We may illustrate the derivation of the Marginal Revenue Product of labor curve, graphically using the example of the previous section, with the differences that the prices of the commodity produced declines as output increases.

Because of the theory underlying the monopolist the price and marginal revenue differs. There is difference between $VMP_L (= MPP_L \cdot P_x)$ and $MRP_L (= MPP_L \cdot MR_x)$. The few differs, due to the fact that $P_x > MR_x$ at all levels of output and employment, hence $VMP_L > MRP_L$. Except that both the VMP_L and MRP_L have negatively sloped because their components (MPP_L, P_x, MR_x) decline as output expands and the prices of the product falls.



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Figure 4.13A: Imperfect product market

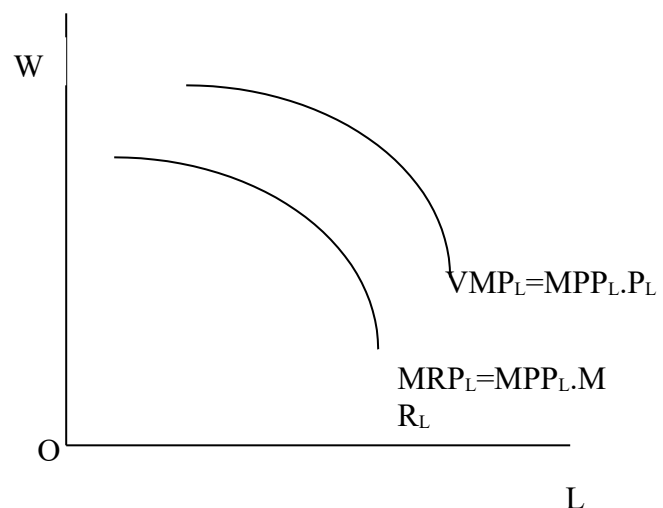


Figure 4.13B: perfect factor market

This can be derived algebraically as follows by first deriving parts MR_x and MPP_L , and then MRP_L as well as MR_x from total revenue function.

1. Let the demand function for the product in generalized form be $P_x = f_1(Q_x)$.

Total revenue of the firm's is $TR = P_x \cdot Q_x$ then,

The marginal revenue is obtained by taking the first order derivative with respect to Q_x

$$\frac{d(TR)}{dQ} = P_x \cdot \frac{dQ_x}{dQ_x} + Q_x \cdot \frac{dP_x}{dQ_x}$$

Or

$$MR_x = P_x + Q_x \cdot \frac{dP_x}{dQ_x}$$

2. The production function with labor as the only variable factor is $Q_x = f_2(L)$

By taking first order derivative we get $MPP_L = \frac{dQ_x}{dL} \equiv MPP_L$

3. The marginal revenue product of labor is the change in total revenue attribute to a unit

$$\text{change in labor } MRP_L = \frac{d(TR)}{dL}$$

Given $TR = P_x \cdot Q_x$, the derivative of total revenue with respect to L is

$$\frac{d(TR)}{dL} = P_x \cdot \frac{dQ_x}{dL} + Q_x \left[\frac{dQ_x}{dQ_x} \frac{dQ_x}{dL} \right]$$

$$\text{Or } MRP_L = \frac{dQ_x}{dL} \left[P_x + Q_x \cdot \frac{dQ_x}{dQ_x} \right]$$

$$\text{But from (3) } \frac{dQ_x}{dL} = MPP_L \text{ and from (2) } MR_x \left[P_x + Q_x \cdot \frac{dP_x}{dQ_x} \right]$$

Therefore: $MRP_L = MPP_L \cdot MR_x$

Thus, the profit maximizing firm's the demand for labor is marginal revenue product curve as the decision to hire extra labor is based on the additional revenue it generate. . The supply of labor to the individual firm is perfectly elastic shown by horizontal line S_L . Then, the

equilibrium is determined by the interaction of the two (demand and supply) like that of competitive market. The equilibrium is at point 'e' when $MRP_L = MC_L = W$ with 'L' level of employment. To the left of 'e' a unit of labor adds more to the revenue of the firm than the amount of its cost: hence it pays the firm to increase its employment. Conversely at any point to the right of 'e' an additional unit of labor adds more to total cost than to total revenue. Thus, firm employ labor up to the point of equality.

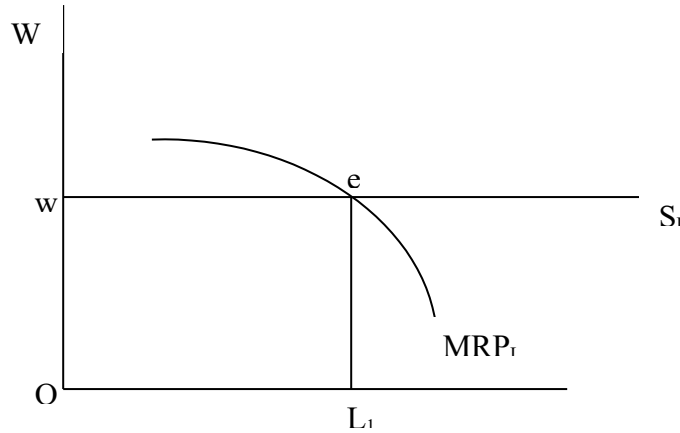


Figure 4.14: Determination of equilibrium

Formal derivation of the equilibrium of the monopolistic firm is as follows;

The firm wants to maximize its profit

$$\Pi = R - C = P_x Q_x - (WL + F)$$

$$\text{Given } P_x: f_1(Q_x) \quad \text{and } Q_x = f_2(L)$$

The first order condition for Π max is that the derivative of Π w.r.t is equal to zero

$$\frac{d\Pi}{dL} = p_x + Q_x \left[\frac{dp_x}{dQ_x} \frac{dQ_x}{dL} \right] - W = 0$$

Rearranging we obtain
$$\frac{dQ_x}{dL} \left[p_x + Q_x \frac{dp_x}{dQ_x} \right] = W$$

We have shown that
$$\frac{dQ_x}{dL} = MPP_L \quad \text{and} \quad \left[p_x + Q_x \frac{dp_x}{dQ_x} \right] = MR_x$$

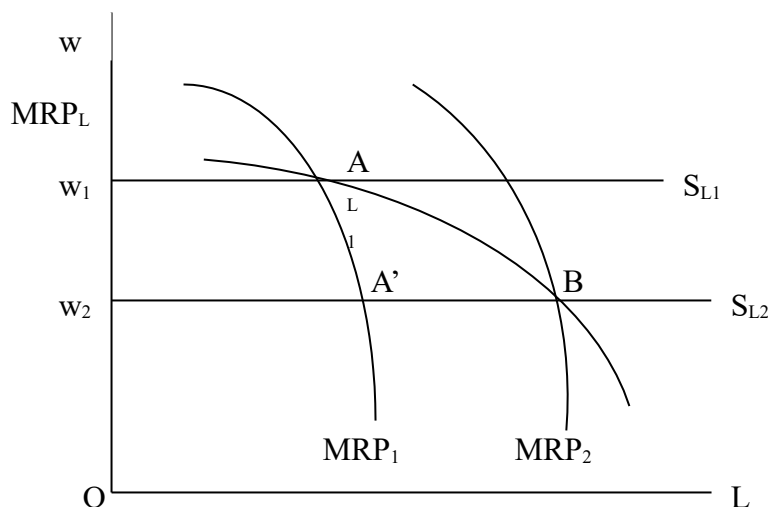
Therefore

$$(MPP_L), (P_x) \left[1 - \frac{1}{e_p} \right] = W \text{ where } MR_x, P_x \left[1 - \frac{1}{e_p} \right]$$

This shows the relations among commodity price, factor price, elasticity of demand and the production functions.

4.6.1.2. DEMAND OF VARIABLE FACTOR BY A MONOPOLISTIC FIRM UNDER SEVERAL FACTORS

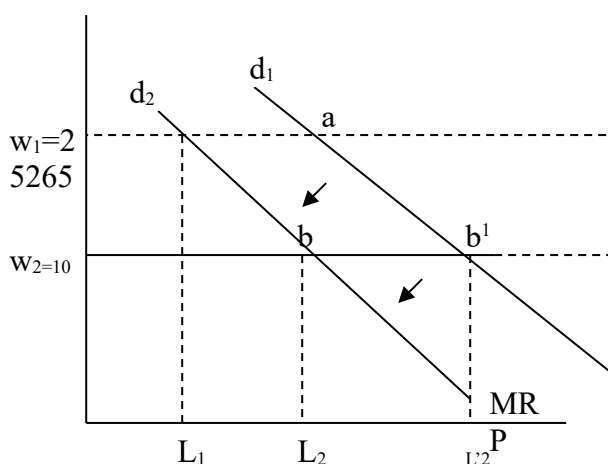
Unlike the case of one variable, marginal-revenue product curve is not demand curve, for several variable case, rather the demand curve is formed from points on shifting MRP curves. The analysis is similar to that of the previous sections the case of perfectly competitive market. Suppose that the market price of labor is w_1 with MRP_1 (see figure 4.15). The monopolistic firm is in equilibrium at point A, employing L_1 units of labor. If the wage rate fall to w_2 the firm would move along its MRP_2 curve to point A', if other things remained constant. However, the fall in wage rate has three effects (substitution, output and profit –maximizing effects) that result in shift of the marginal revenue produce curve to the right which also lead equilibrium point to B. By similar analogy one can derive other point of equilibriums. Connecting points such as A and B at various level of 'w' we obtain the demand curve of labor. From our analysis one can conclude that the demand curves of inputs are negatively sloped irrespective of the condition of competition in the product market.



4.6.1.3 The market demand for and supply of labor

The market demand for a factor is derived in the same way as in the case of perfectly competitive market from the summation of the demand curves of the individual monopolistic firms. In aggregating these curves, we must take into account their shift as the price of the factors falls: as all monopolistic firms expand their output, the market price falls. The individual demand curve and the marginal revenue As opposed to competitive market curve for the commodity produced shift to the left. Graphically, the derivation of demand curve is exactly the same as perfectly competitive market.

The market price for the factor is determined by the market demand and supply, just like that of perfectly competitive market. The difference was that the figure is based on the MRP_L and not on VMP_L . This means that when the firm has a monopolistic power, the factors are paid the MRP_L which is smaller than VMP_L as the MRP_L is the market demand for imperfect market and $MRP_L < VMP_L$. This effect has been called monopolistic exploitation by Robinson. This is indicated by figure 4.16 (a). Monopolistic exploitation says a productive factor is exploited if it is paid a price less than the value of its marginal product (VMP).



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Figure 4.16A: Demand curve of a single firm

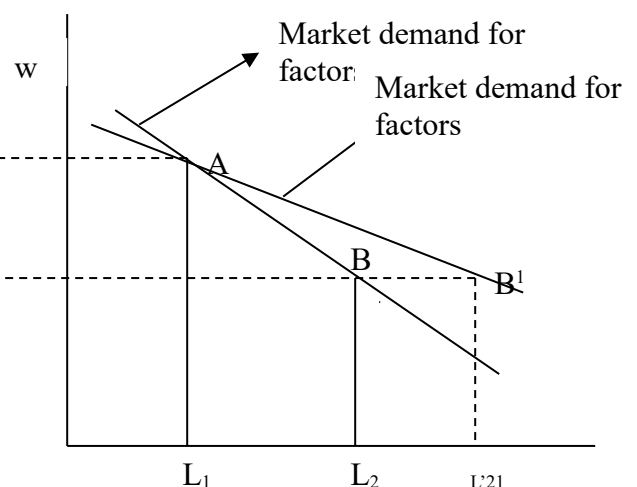


Figure 4.16B: Market demand curve for labor

But the market supply is not affected by the fact that firms have monopolistic power. Thus, the market supply of labor is the summation of the supply curve of individuals as derived earlier. The equilibrium is determined by the interaction of supply and demand. Thus, the analysis is the same, but there is an important difference is that the market demand is based on the marginal revenue product of the factors not on value on marginal product of factors.

This can be derived algebraically.

For a perfectly competitive firm

$$VMP_L = MRP_L$$

Because $VMP_L = MPP_L \cdot P_x$ and $MRP_L = MPP_L \cdot MR$,

$P_x = MR_x$ so that $VMP_L = MRP_L$

But for monopolist $P_x > MR_x$ so that $VMP_L < MRP_L$

Thus, when the commodity market is competitive the profit maximizing level show $VMP_L = MRP_L$. But for the monopolistic firm in product market, the payments of factors are paid less than value of the marginal product, hence inefficiencies. The amount $W_c - W_m$ shown by the figure above indicated that profit maximizing behavior of imperfectly competitive firms causes the factors prices less than its VMP_L . Furthermore, the level of employment is lower in industries which are not perfectly competitive $L_c > L_m$.

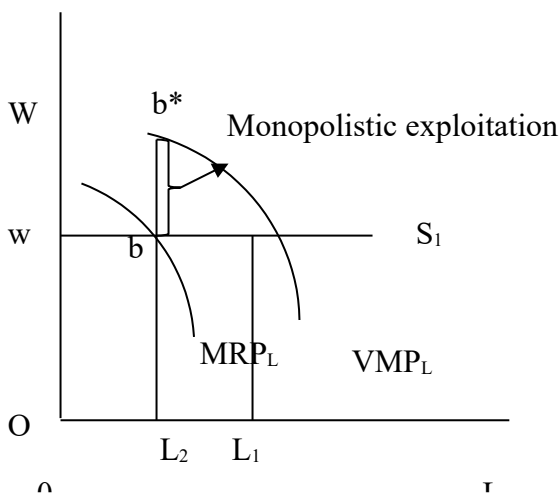


Figure 4.17: Monopolistic exploitation (a)

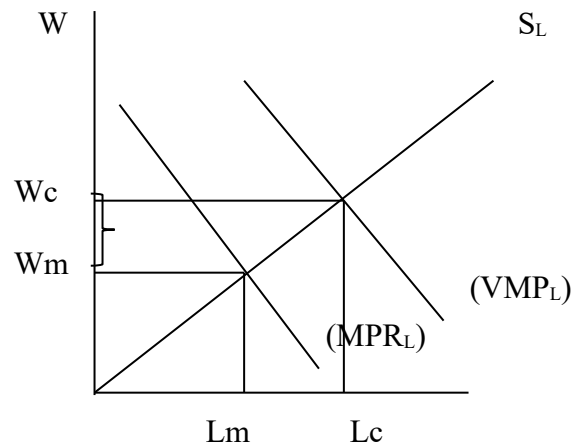


Figure 4.17: demand curve for labor (b)

4.6.2. MODEL-B: THE FIRM WITH MONOPOLISTIC POWER IN COMMODITY MARKET AND MONOPSONISTIC POWER IN THE FACTOR MARKET

Assume:

- The monopoly in products market, a single seller of product
- Monopsony power in factor market- a single buyer of factor of production: labor

Both parties want to maximize profit of their activity. We will look at the equilibrium level under this model when labor is only variable factor and for several factors as we did for model A. Does this ever happen?

It is difficult to tell the story that fits the bill. Group small firms maybe set up to provide tools, supplies, or material for a single large manufacturing firm. This firm may be the only of its type in the area, and its requirements might be very specialized.

4.6.2.1. EQUILIBRIUM OF MONOPSONIST, WHO USES A SINGLE VARIABLE FACTOR

Suppose that a firm is the sole employer in a town in local market- monopsony. The firm uses only one factor, Labor (L), to produce final good. What do the demand and supply curve look like? How equilibrium is determined? The labor demand for the firm is MRP_L as for the case of model –A, and perfectly competitive markets. The firm values the last extra worker it hires by marginal revenue product of that worker. Hence, the firm has a downward sloping demand curve as shown in figure 4.18.

The supply curve for labor to individual firm is not horizontal and perfectly elastic because the firm is large- a monopsonist firm in the labor market. The supply of labor is positively sloped

(upward-sloping) as the monopsonist expands the use of labor by pay a higher wage. The higher its daily wage, w_1 , the more people want to work for the firm. Because the firm pay all workers the same wage, the monopsony must pay more to each worker it was already employed. When we multiply the input prices by the level of employment we find total expenditure of monopsonist. Hence, the supply of labor curve is shown by the average expenditure, at different level of employment.

However, what is relevant for the determination of level of employment is marginal expenditure that can be derived from change in total expenditure at all level of employment. Hiring an additional unit of input increases the total expenditure on the factor by more than the price of this unit because all previous units employed are paid the new higher price. The marginal expenditure (ME) curve lies above the supply curve (average expense curve). Because the prices per unit rise as employment increases, the marginal expense of the inputs is greater than its prices at all levels of employment. Hence, the ME curve has a positively sloped and lies above and to the left of the supply of inputs curves.

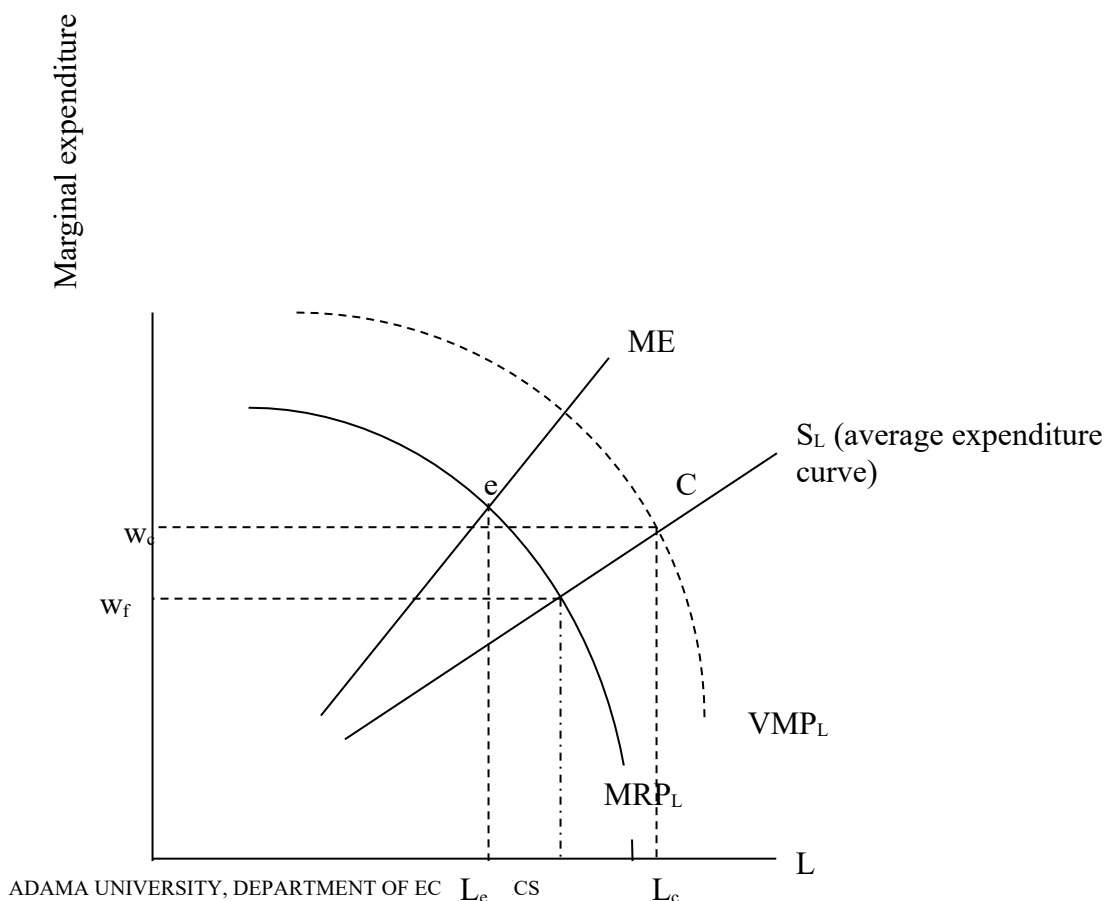


Figure 4.18: Demand and Supply curve of monopsonist

As can be seen from the table since the price per unit of input rises as employment increases, the marginal expenses (ME) of the input is greater than its prices at all level of employment. Under these conditions, the firm is in equilibrium when its marginal expenditure on the factor is equal to MRP_L or it buys labor services up to that point of equality. That is, at point 'e' on figure 4.18. Where $ME = MRP_L$. This is shown in figure xx above. If the last unit is worth more to the buyer than its marginal expenditure, the buyer purchases another unit. This refers to area to the right of 'e'. On the other hand, if the last unit is less valuable than its marginal expenditure, the buyer purchases one less unit. This is shown by area to the left of 'e'.

Thus at equilibrium point, 'e', the wage rate that the firm will pay for the L_e units of labor is W_F . However, if the market is competitive, the intersection of demand and supply curve determines the equilibrium at point C with the level of employment L_c and corresponding payment W_c . Thus, the monopsonist hires less labor and pay lower wage as compared to the perfectly competitive market.

The monopsony power is the ability of a single buyer to pay less than the competitive price profitably. The size of the Gap between the value the monopsony places on the last worker (the height of its demand curves and what it pays depend on the elasticity of supply at the monopsony optimum.

By creating a wage differences between the value on the monopsony and the value to the supplies, the monopsony causes a welfare loss in monopsonist exploitation. The exploitation arises from monopsonistic power of firms, and is something in addition to monopolistic exploitation. The difference between wage paid by competitive firm and monopsonistic firm is monopsonistic exploitation. Each units of labor receives its MRP which is less than the VMP.

Activity:

1. Derive ME curve and the equilibrium level of employment using the following data.

Table 4.2 Total and marginal expense on labor

Units of labor	Price of labor	Total expenditure on labor	Marginal expense on labor	TRP _L (P _x L)
1	4	4	-	4
2	5	10	6	10
3	6	18	8	18
4	8	28	10	32
5	9	40	12	45
6	10	54	14	60
7	11	70	16	77
8	12	88	18	96
9	13	108	20	117
10	14	130	22	140

4.6.2.2. EQUILIBRUM OF A MONOPSONIST WHO USES SEVERAL VARIABLE FACTORS

Recall that under perfectly competitive market, the firm hire labor up to the point where

$$\frac{MPP_L}{MPP_W} = \frac{w}{r} \quad \text{Or} \quad \frac{MPP_L}{W} = \frac{MPP_K}{r}$$

How do monopsonistic determine equilibrium level of employment?

If the factor markets are monopsonistic, changes in the amount of factors employed causes changes in the prices of factors. Thus 'w' and 'r' are not given. The monopsonist must look at the marginal expense of the factors. A monopsonist who uses several variable factors will use the input combinations at which the ratio of the MPP_L to the ME is equal for all variable inputs. The least combination is obtained when the marginal rate of technical substitution ($MRTS_{L,K}$) equals the marginal expenses of input ratio. For the two input case the equilibrium condition of the monopsonist may be stated as follows

$$MRTS_{L,K} = \frac{MPP_L}{MPP_K} = \frac{ME_L}{ME_K}$$

$$\frac{MPP_L}{ME_L} = \frac{MPP_K}{ME_K}$$

It is apparent that the least cost condition for perfect input markets. Namely

$$\frac{MPP_L}{MPP_K} = \frac{w}{r}$$

is a special case of expression since in perfect factor markets $ME_L=w$ and $ME_K=r$.

4.6.3. Model–C: Bilateral monopoly

Bilateral monopoly arise when a single seller (Monopolist) face a single buyer (Monopsonist) in the labor market. Assume that all the firms are organized in a single body which acts as like monopsonist. Thus, we also assume labors are organized in a labor union which acts like a monopolists. Hence, we have two monopolies participating in a market one on supply side and the other on the demand side. How do the two party bargains? Is there a unique equilibrium for the market like that of the above two models?

These can be better presented through graphical approach. The figure 4.19 below shows how the two parties interact in the market. The monopsonists (single buyer's) demand curve is D_b which is MRP_L of input being demand. From the point of view of monopolist (labor union) D_b represent average revenue curve, hence, $AR=D_b$. The seller's (Union) MRs curve can be derived by usual graphic techniques and it is below the ARs and will cut any horizontal line at

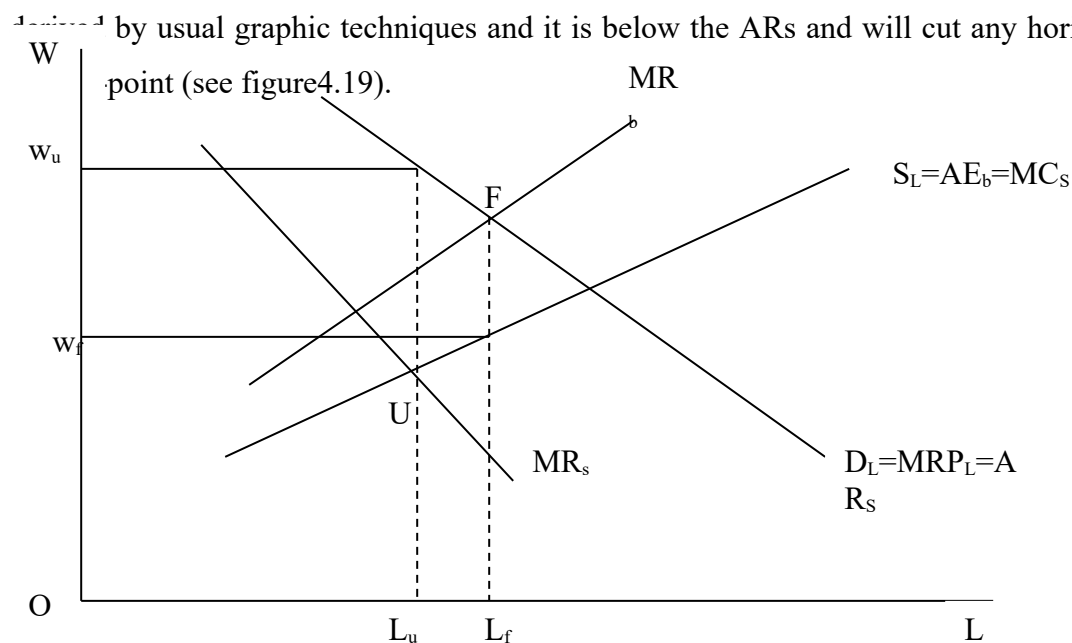


Figure 4.19 : The determination of equilibrium under bilateral monopoly

The supply of labor facing monopsonist is the upward sloping curves S_L which similar technique we have seen under monopsony. From point of view of the monopolist (labor union) the S_L is its marginal cost. Assume that monopolist's behave as if his prices were determined by outside force just like perfectly competitive market sellers, and hence MC_s is supply curve. Given the above cost and revenue curve we can find the equilibrium position of each participant in the market.

The monopsonist maximizes his profit at point F where marginal expense of labor (ME_L) equals to the marginal revenue product of labor. The monopsonist desire L_F units of labour and pay a wage rate equal to W_F . The monopolist (labor union), on the other hand, maximizes Wu profit (gains) at point U, where his marginal cost is equal to Wu marginal revenue. This, he wants to supply L_u units of labor and receives a wage equal to W_u .

As you can see from the diagram the point of equilibrium for the two parties differs. The price desired by the monopsonist (firm's management) is the lower limit of price, W_f , while the price desired by the monopolist seller, W_u is upper limit prices, since the price goals of two monopolists can not be realized, the price and quantity in the bilateral monopoly market are indeterminate in economic sense.

The equilibrium situation in bilateral monopoly situation is indeterminate. The model gives only the upper and lower limit within which the wage will be determined by bargaining. The level at which the price will be settled depend on the bargaining skills and power of the participants. The power of each participant is determined by his ability to inflict losses to the opposite party and his ability to withstand losses inflicted by the opponent. Hence, the outcome of the bargaining cannot be known with certainty. It depends on bargaining skills, political and economic power of the labor union and the firms, and on many other factors.

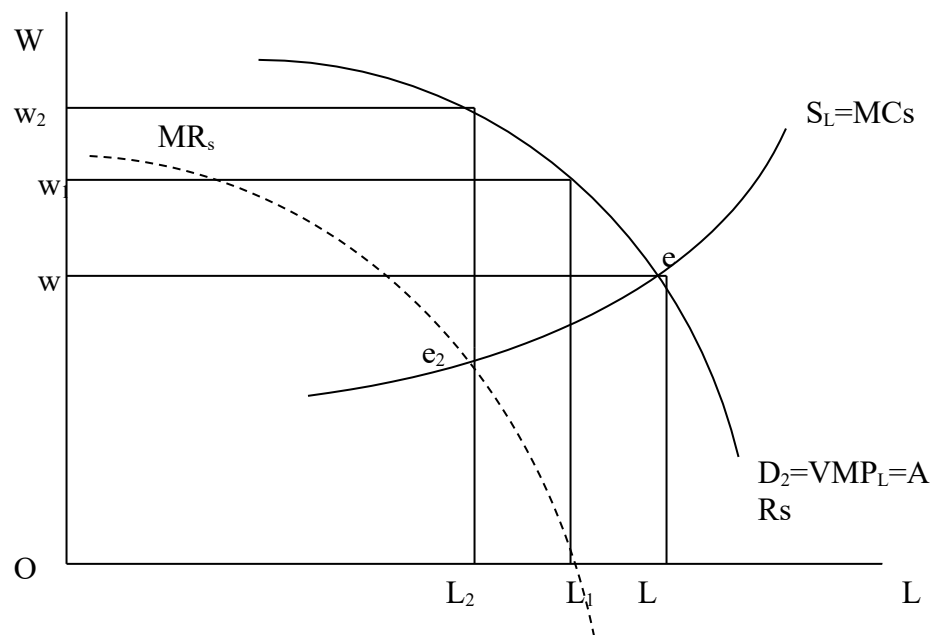
4.6.3. MODEL-D: COMPETITIVE BUYER-FIRM VERSUS MONOPOLY UNION

In this case we assume that firms have no monopolistic or monoposonistic power, but the labor force is unionized and behave like monopolist. The situation is presented in figure xx. In this case labor supply curve, S_L , is shown by marginal cost of the union as in the case of bilateral monopoly. The market demand for labor is D_L derives from VMPL through summation of individual firm's demand. It also represents MRs. How do wage is determined under these condition?

The determination of wage and other factor's payment is depending on the goals of the union. There are three common goals of labor union. These are:

The maximization of employment

The employment is determined by intersection of demand and supply curve through the market force. Hence the labor union will demand a wage rate equal to W . The firms, being price takers, will maximize their profit by equating W to the VMPL. The OL is total employment.



The maximization of the total wage bill.

If the goal of union is total wage bill maximization, they set wage at point where MRs is zero. Hence, the equilibrium of the union is at point e_z . The corresponding wage rate is w_1 and level of employment is at L_{s1} (see figure xx)

The maximization of the total gains to the union as whole

The goal is attained by setting wage at point where $MC=MR$ of the union. Thus, the equilibrium under this condition is at point e_2 . This points identified by w_2 level of wage that correspond with L_2 level of employments (see figure xx)

In sum, if the firms do not have monopsonistic power, the wage rate and the level of employment are determined by the good of the union.

4.7. THE ELASTICITY OF INPUT SUBSTITUTION, TECHNOLOGICAL PROGRESS AND INCOME DISTRIBUTION

In the previous section, we examined the determinants of factor price. As factor price change, the firms' structure of employment will change as it affects the profit maximizing behavior and income distribution. Under microeconomic theory of production we have seen the elasticity, technological progress and other factors as major shifting factors that are necessary for policy purpose. We will relate that theoretical concept with the input market and income distribution especially substitutability of factors. In the section we will examine how the substitutability of productive factors and technological progress affect income distribution.

Objectives:

At the end of this section, you will be able to:

Explain how substitutability of factors of production affects income distribution.

Analyze how technological changes or progress affect income distribution
 Identify how changes in factors prices affect the income shares of factors.

4.7.1.THE ELASTICITY OF INPUT SUBSTITUTION AND THE SHARES OF FACTORS OF PRODUCTION

How do changes in factors prices affect the shares of factors and income distribution?
 What is its consequence?

From our micro economic theory, rational economic agent tends to substitute a cheaper input for a relatively more expensive one. The ability to substitute one input for another is reflected in the elasticity of substitution between the two inputs. A large elasticity indicates that the two inputs are close substitutes in production. Now if there is a close substitute available, then when the price of an input rises, the firm can simply substitute the other input. Therefore, if labor and capital are close substitutes, then when the wage rate rises firms will substitute capital for labor, and the decline in employment will be greater. Hence, the demand for an input will be more elastic when it has close substitutes are available.

This will result in a change of the K/L ratio, and the size of this effect depends on the responsiveness of the change of the K/L ratio to the factor price change which we call elasticity of substitution. We discussed this in first module of microeconomics in connection with the production function. Recall that the elasticity of substitution is defined as the ratio of the percentage change in the K/L ratio to the percentage change of the $MRTS_{L,K}$.

$$\sigma = \frac{\frac{\Delta(K/L)}{(K/L)}}{\frac{\Delta(MRTS_{L,K})}{(MRTS_{L,K})}}$$

In the perfect input markets the firm is in equilibrium when it chooses the input combination at which the MRTS is equal to the ratio of factor prices.

$$MRTS_{L,K} = \frac{W}{r}$$

Thus, in equilibrium with perfect factor markets the elasticity of substitution may be

$$\sigma = \frac{\frac{\Delta(K/L)}{(K/L)}}{\frac{\Delta(W/r)}{(W/r)}}$$

The sign of the elasticity of substitution is always positive (unless $\sigma=0$) because the numerator and denominator change in the same direction. When wage to interest ratio

increase labor is relatively more expensive than capital and this will induce the firm to substitute capital for labor, so that in K/L ratio would decline.

The value of elasticity ranges from zero to infinity. If $\sigma=0$ it is impossible to substitute one factor for another; K and L are used in fixed proportions (as in the input –output analysis) and the isoquants have the shape of right angles. If $\sigma=\infty$ the two factors are perfect substitutes: the isoquants become straight lines with a negative slope. If $0<\sigma<\infty$ factors can substitute each other to a certain extent: the isoquant are convex to the origin.

The Cobb –Douglas production function has $\sigma=1$. In general the larger the value of σ , the greater the substitutability between K and L

We may classify σ in three categories:

$\sigma<1$: inelastic substitutability

$\sigma=1$: Unitary substitutability

$\sigma>1$: elastic substitutability

There is an important relationship between the above values of σ and the distributive shares of factors. By definition

$$\left[\frac{\text{Share of}}{\text{labour}} \right] = \frac{w \cdot L}{X} \quad \text{and} \quad \left[\frac{\text{Share of}}{\text{capital}} \right] = \frac{r \cdot K}{X}$$

The concept of the elasticity of substitution of factors has been introduced by Sir John R.

$$\frac{\left[\frac{\text{Share of}}{\text{labour}} \right]}{\left[\frac{\text{Share of}}{\text{capital}} \right]} = \frac{w'/r}{K/L}$$

We can easily find the effect of a change in the w/r ratio on the relative shares of the two factors. Assume that $\sigma<1$. This implies that a given percentage change in the w/r ratio results in a smaller percentage change in the K/L ratio, so that relative–share expression increases. Thus, if $\sigma=1$, an increase in the w/r ratio increases the distributive share of labor. For example assume that $\sigma=0.5$. Then a 10 percent increase in w/r results in a 5 percent increase in the K/L ratio. The new relative shares are

$$\left(\frac{wL}{rK} \right) = \frac{(w/r)}{(K/L)} \frac{(1+0.10)}{(1+0.05)} = \frac{1.10}{1.05} \cdot \left(\frac{w/r}{rK} \right)$$

Clearly, new relative share ratio $>$ initial relative share ratio

If $\sigma > 1$ change in w/r leads to a smaller percentage change in K/L so that the relative share of labor decreases: For example assume that $\sigma=2$. A 20 percent increase in w/r leads to a 40 percent increase in K/L . The new share ratio is

$$\frac{K}{L}$$

Clearly if $\sigma > 1$ the relative share of labor decreases following an increase in the w/r ratio.

With a similar reasoning it can be shown that if $\sigma=1$ the relative shares of K and L remain unchanged.

In summary, an increase in the w/r ratio will cause labor's share, relative to capital's share. It should be noted that there is a two-way causation between w/r and K/L . Changes in the capital-labor ratio result in changes in the relative factor prices, and hence in changes in the shares of the factors to total output. The effect of changes of the K/L ratio on factor shares can be explored with the use of the elasticity of substitution. The analysis is analogous to the one we used above for changes in the w/r ratio.

From the above discussions, it is clear that the concept of the elasticity of substitution is very important in the neoclassical theory of income distribution. It is extremely useful in examining the way in which changing input prices or input ratios affect income shares. Given that changes in w/r lead to changes in K/L and vice versa, it follows that prediction about factor shares following changes in one of these ratios are bound to be misleading if they do not take into account the possibility of the associated changes in the other ratio.

4.7.2. TECHNOLOGICAL PROGRESS AND INCOME DISTRIBUTION

Up until now we have assumed a given production function. However, technological change takes place continuously, and this shifts the production function, leading to changes in the K/L ratio and the elasticity of substitution. Thus, it is important to consider the effects of

technological progress on factor shares. Technological progress shifts the iso quant's down words, given that the same level of output that can be produced with smaller quantities of factor inputs as technological progress occurs. The progress can be classified into three types, neutral, capital-deepening and labor-deepening. These are:

Technological progress is neutral if at a constant K/L ratio the $MRTS_{L,K}$ remains unchanged. Since in equilibrium $MRTS_{L,K} = w/r$, it follow that when technological progress is neutral both the K/L ratio and the w/r ratio, are unchanged. Consequently, the relative shares of factors remain unchanged.

Technological progress is capital-deepening if at a constant K/L ratio the $MRTS_{L,K}$ declines. This implies that at equilibrium the w/r ratio declines, as r increase relative to w , while K/L remains constant. Consequently, the ratio of factors shares declines, i.e., n the share of labor decrease and the share of capital increases.

Technological progress is Labor –deepening if at a constant K/L ratio the $MRTS_{L,K}$ increases. Then, at equilibrium, the w/r ratio increase as technological progress Takes places. This implies that the shares of labor will increase and the share of capital will decrease.

In short, the relative share of labor increases if technological Progress is labor –deepening , remain unchanged if technological progress is neutral , and decreases if technological changes is capital- deepening.

SUMMARY

Production function is the relationship between inputs and outputs. The firm's demand for a factor input depends upon the input's physical productivity and the demand for the good the factor is being used to produces.

Factor demand is derived demand is a sense that it's comes form the demand for the product.

Factor markets differ from product market in three important aspects. First, factor market tends to be more competitive than product market. Second, the demand for factors of production is a derived demand in a sense that its demand comes from the demand for the goods and services produced by the factors of production. Third, the production of a good requires the cooperation of different factors of production.

A firm has a demand for factors (labor and capital) is related the contribution of the factor to the objective function of the firm. The contribution of labor to the firm's profit function is

determined by two things: the quantity of additional output a unit labor input can produce and the price that the firm can sell its product.

Under perfectly competitive market the firm demand for only one variable input –labor is value of marginal product of labor for which the value of the extra output produced by the employment of the last unit of input is equal to the price of the product multiplied by price. However, this is no longer true when the firm uses more than one variable inputs which is the marginal revenue product of labor. In both cases the demand for labor is downward sloping.

Even though the supply of labor is determined by different factors and evolves behavioral aspect for most cases is backward bending curve. Just like that of perfectly competitive market, the equilibrium is determined by the forces of supply and demand for labor.

Determination of factor prices and employment are under imperfect competition slightly differing from that of perfect competition depending upon the model under consideration. The major difference lies in the marginal revenue product of labor and value of marginal product of labor in which case the former is the demand curve of factor of production as opposed to the perfect model. As a result of these there is exploitation of labor by monopolist.

Factors substitutability and technological progress are the basic issue for the firm's factors employment decisions. Furthermore, it shows income distribution and area of intervention for government.

KEYTERMS

Marginal physical product	Value of marginal physical product
Output effect	Substitution effect
Income effect	Profit maximizing firm
Backward bending supply curve	Monopsonist
Bilateral monopoly	Elasticity of factor substitution
Labor union	Marginal revenue product of labor
Labor-deepening technological progress	Capital –deepening technological progress
Demand for labor	

Review Questions

Given the following in Table 4.2 below derive the demand curve for labor assuming it is the only variable factor of production. Assume further, fixed output prices equal to 3 and wage rate equal to $w=4$

Table 4.2: Employment of factor and its marginal productivity

Quantity of labor used	3	4	5	6	7	8	9
Marginal product of labor	8	7	6	5	4	3	2

Suggested readings

Nicholson, w: **Microeconomic theory; basic principle and extension**. 5th edition

Salvatore, D(2003):**Microeconomics; theory and applications**.4th edition

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UNIT CONTENTS

- 5.1. Introduction
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- 5.3. General equilibrium in pure exchange
- 5.4. General equilibrium in production, Two-input Two- output case
- 5.5. The production possibilities curve
- 5.6. Production and exchange
- 5.7. General equilibrium in two-consumer economy (2X2X2 model)
- 5.8. There conditions for economic efficiency
- 5.9. Unit Summary

CHAPTER FIVE

GENERAL EQUILIBRIUM ANALYSIS

5.1. INTRODUCTION

Dear readers, until this point we have examined the behavior of individual decision making units. To recall, we have seen individuals as consumers of commodities and suppliers of inputs, and firms as employers of inputs and producers of commodities. Further more, we have discussed the working of individual market for commodities and inputs under various market structures and their respective equilibrium. These have been done by isolating other factors that affect the relationships-each unit or market being considered separately. Do think such situation exists? Rather the economic agents are highly interrelated interdependent. For instance, the activities of consumers affect the producers and the vice versa.

In this chapter, we take up the topic of interdependence or relationship among the various decision making units and markets. This allows us to trace both the effects of a change in any part of the economic system on every other part of the system, and the repercussions from the latter on the former. In other word we examine how the various individual pieces fit together to form an integrated economic system. The main objective is just like that of others (individual market) to analyze the equilibrium of the system that maximizes the social welfare. We begin the chapter by distinguishing between partial equilibrium and general equilibrium analysis and examining the conditions under which each type of analysis is appropriate. Then, we will discuss the conditions required for the economy to be in general equilibrium of exchange, production and production and exchange simultaneously. Lastly the welfare implication of general equilibrium analysis will be analyzed.

OBJECTIVE

After completing this chapter, students should be able to:

Understand the difference between partial and general equilibrium

Explain the general equilibrium in exchange
Explain the general equilibrium in production
Show the implication of general equilibrium on welfare

5.2 GENERAL EQUILIBRIUM ANALYSIS

5.2.1 PARTIAL VERSUS GENERAL EQUILIBRIUM ANALYSIS

Dear readers, in the previous chapter we studied the behavior of individual decision making units and individual markets in isolation. We examined how an individual maximizes satisfaction subject to his or her income constraint, how a firm minimizes its costs of production and maximizes profit under various market structures, and how the prices and employment of each type of input is determined. We have shown how demand and supply in each market determine the equilibrium price and quantity in that market independently of other markets. The model that we have used has taught us that the prices and quantity in each such market is determined by the supply and demand. Each market has been regarded as independent and self-contained units, for practical purpose. In doing so, we have abstracted from all the interconnections that exist between the market under study and the rest of the economy. The analyses that assume that changes in prices can occur without causing significant changes in prices in other markets are called Partial Equilibrium

However, the assumption may be in reality be seriously flawed. No market can adjust to changes in its condition without causing changes in other market, and the change in the other market can be substantial and significant. A change in any market has spillover effects on other markets and the change in the other markets will, in turn, have repercussions or feedback effects on the original market. For example, suppose demand for housing increases, supply remaining the same, at some point of time in an economy. As a result of this (increase in housing demand) home rents will go up and profitability of the housing industry increases which in turn increases the number of firms in the industry who are attracted by the increase in profits.

Furthermore, the demand for intermediate goods steel cements and bricks and construction labor increases. The number of firms producing intermediate goods increases and their

demand for factor inputs such as land, labor and capital. If these factors are in short supply, factor payment increases. Consequently, larger factor incomes flow to the households. When household income increases, demand for other consumer goods and services increases and creates another chain of action and reaction between household and firms in product and factor markets. The complexity is further increased as change in relative prices of products and factors lead to substitution effects and other effects. In this way the whole economic system functions in a complex manner. These effects are studied by general equilibrium analysis.

General equilibrium analysis studies the interdependence or interconnections that exist among all markets and prices in the economy and attempt to give a complete, explicit, and simultaneous answer to the questions of what, how and for whom to produce.

A general equilibrium is, thus, defined as a state in which all economic units optimize their respective objective function and all prices are simultaneously in equilibrium, and all markets are cleared.

Both are very useful. Partial equilibrium analyses are perfectly adequate in the case in which the effect on a change in the market condition in one market has little or no effect on prices in other market. The studying the effect of small excise tax on the production of an insignificant commodity may work perfectly well under the assumption that the price of all other goods are fixed. Those prices might not change, and so it would not be worth the effort to model feedbacks and interactions that are not at large; a partial equilibrium of sort presented above would then be appropriate. On the other hand, studying the effects of the large of excise tax on commodity whose purchases absorb significant proportion consumers income could be inappropriate if it repercussions on other prices were not considered. In this case, a general equilibrium analysis could be required to make certain that the feedbacks and interaction did not seriously undermine the validity of the conclusions drawn from a partial equilibrium approach.

5.2.2. THE EXISTENCES OF GENERAL EQUILIBRIUM

The general equilibrium analysis, like partial equilibrium analysis, can be used to solve problems of many kinds. It is based on the notion that general equilibrium for all the economic agent and market can be reached. The general equilibrium is a state of in which the following conditions hold:

Every consumer chooses his or her preferred market basket subjected to his or her budget line.

Every consumer supplies whatever amount of inputs he or she chooses, given the input and product prices that prevail.

Every firm maximizes its profit subjected to the constraints imposed by the available technology , the demand for its product, and the supply of inputs

Long-run economics profits are zero for every firm.

The quantity demanded equals the quantity supplied at the prevailing prices in all product and input market.

It is evident from the definition that a great many conditions must be satisfied simultaneously if the equilibrium is to be achieved. These include equilibrium in
exchange

production

Consumption... etc.

Given the above conditions for equilibrium; can we be sure that it can be achieved? If so, what are these conditions?

The first and simplest general equilibrium model was introduced in 1874 by Leon Walras, thought he was never able to prove the existence of a general equilibrium. In the Walrasian system, all prices and quantities in all markets are determined simultaneously through their interaction with one another (by the solution of the system of equations). Thus the first task in establishing the existence of a general equilibrium is to describe the economy by means of a system of equations in which the number of unknowns should be equal to the number of equations. His system one of the equations has a redundant (not an independent) equation and deprives the system of a solution since the number of independent equations is less than the number of unknowns. Hence, in this model, the absolute level of prices cannot be determined.

To solve these problem general equilibrium theorists have adopted the device of choosing arbitrarily the price of one commodity as a numeraire (or unit of account) and express all other

prices interims of the price of the numeraire. With this device each price is given relative to the price of the numeraire. Even if there is equality of independent equations and unknowns there is no guarantee that a general equilibrium solution exists.

In 1954 Arrow and Debreu provided a proof of the existence of a general equilibrium in perfectly competitive markets, in which there are no indivisibilities and no increasing returns to scale. In 1971 Debreu and Hahn proved the existence of a general equilibrium for an economy with limited increasing returns and monopolistic competition, with out indivisibilities. There is no proof so far that shows the existence of a general equilibrium in oligopoly and monopoly markets.

Apart from the existence problem, two other problems are associated with equilibrium: the problem of its stability and the problem of its uniqueness. That is related to the price and output that make up the general equilibrium. Are the price and output that make up the general equilibrium be unique? Or is there one set of prices and outputs that for which the quantity supplied equals quantity demanded in all market? Clearly the answer is no. This is beyond the scope of the course.

Though it have all these drawbacks, it is important to know the general equilibrium can be achieved under perfectly competitive market. Because the circumstances under perfectly competitive economy has a variety of desirable characteristics as the market is a benchmark for comparison with other market. Hence, we look at general equilibrium form under competitive market from those major activities: exchange, production, consumption and their interdependences.

5.1.Activity:

1. What are the differences and similarities of partial and general equilibrium?
 2. Which one do you think is appropriate for decision making and policy making?
-

5.3. GENERAL EQUILIBRIUM IN PERFECTLY COMPETITIVE MARKET

5.3.1. SIMPLE MODEL OF GENERAL EQUILIBRIUM

To illustrate the concept of general equilibrium in simple model we make assumptions about the economic agent and the market the economy. Suppose that our simplified economy have only two individuals with certain level of resource endowments isolated from the rest of the world. More specifically we create a world where;

There are two commodities “X: and “Y”; and these commodities are available in a fixed amount.

There are two individuals say, A and B, each possess a given amount of both resources as their initial endowment.

The preference and taste of agents for the two products differs. All assumption for the rational individual we discussed under consumer preference theory. For instances, their goal is maximization of utility, non satiation, usual U-shaped indifference curve...etc

Suppose the total amount of goods X and Y are represented by $\bar{x}$ and $\bar{y}$ respectively. Both individuals own some of X and some of Y as their initial endowment. Thus, $\bar{x} = \bar{x}_A + \bar{x}_B$ and $\bar{y} = \bar{y}_A + \bar{y}_B$.

This can be presented graphically by labeling the vertical axis the level of output Y consumed by each individual, whereas the horizontal axis the level output X consumed by each individual in such a way that each of them are on their respective indifference curve I_A and I_B . At any point in the diagram indicate a certain allocation of the commodity for consumption.

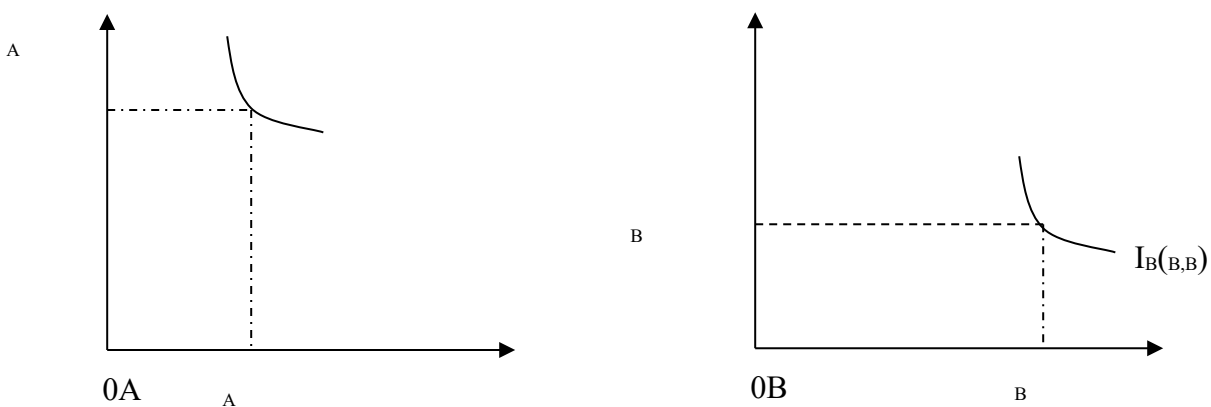


Figure 5.1. Edge worth Box of individual A and B

Suppose further that the two agents interact in such a way that each maximizes its satisfaction. In other words they voluntarily participate in trade assuming both individuals gain from the exchange. There is a convenient graphical tool known as the Edgeworth box that can be used to analyze the exchange of two commodities between two people. Simple models can make productive use of what economists have dubbed the “Edgeworth box diagram”. The box allows us to depict the endowment and preferences of two individuals in a convenient diagram, which can be used to study the various trading process.

An Edgeworth box diagram shows the interaction between two economic activities when the total quantities of commodities consumed or inputs are fixed. The length and width of an Edgeworth box diagram represent the quantities of two commodities that are available to both consumers. And each point in the box represents an allocation between two consumers of the total quantities of the two goods supplied. To show their interaction we put together these two graphs by making certain rearrangement on the presentations. This can be done if we rotate B's graph by 180° it appears to be upside-down and connect it with that of A. The Edgeworth Box diagram is formed by bringing the two graphs together as shown below.

The allocation can be read from the box. Point P for example, indicates that A consumes OAXA unit of X and OAYA unit of Y. But B's consumption is measured from its vertex at the top upper right corner of the box diagram -OB. The quantity of X consumed by B is measured by the horizontal distance to the left of a second origin OB. That is, OBXB unit of X. Vertical distances downward from the same origin measures the quantity of Y consumed by B – OBYB unit of Y.

Every point in the box therefore indicates a certain allocation of food to A and B. The dimension of the box provides even more information- they define the total quantity available for consumption for the two agents. So point P indicates that A consumes OAXA of X, thus the rest will be consumed by B. That is $X - OAXA = OBXB$. The same analogy will hold for Y.

As we have seen in module one the under consumer preference theory, the economic agents' preferences for good differs depending on the endowment and other factors. Hence, agent may have series of indifference curves. Suppose that individual A's preferences for good X and good Y are shown by indifference curves, I_A , II_A , III_A . On the other hand, individual B's indifference curves are shown by I_B , II_B and III_B in the following figure

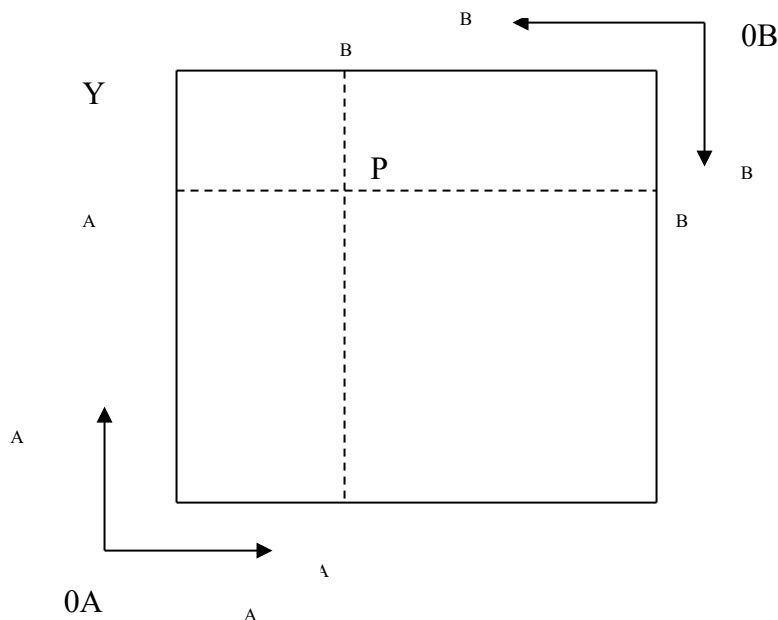


Figure: 5.2. Edge worth Box of both individuals in one.

If we rotate individual B's indifference curve by 180° we get the following Figure 5.3. We can now use the Edge worth box construction to anchor a discussion of the process of exchange. To that end, consider our simple economy of two consumers (A and B), and only two commodities (X and Y). There is no production. The only economic problem is the allocation of a given amount of X and Y between the two consumers. Not much of an economy, to be sure, but one with enough richness to assist in exploring the process of exchange. We need first to establish individual endowments of X and Y. Suppose that point P represents the initial distribution (Endowment) of commodity X and Y by individual A and B. Individual A has $\bar{x}_A$ units of X and $\bar{y}_A$ unit of Y. Similarly individual B has $\bar{x}_B$ units of X and $\bar{y}_B$ unit of Y. What sort of

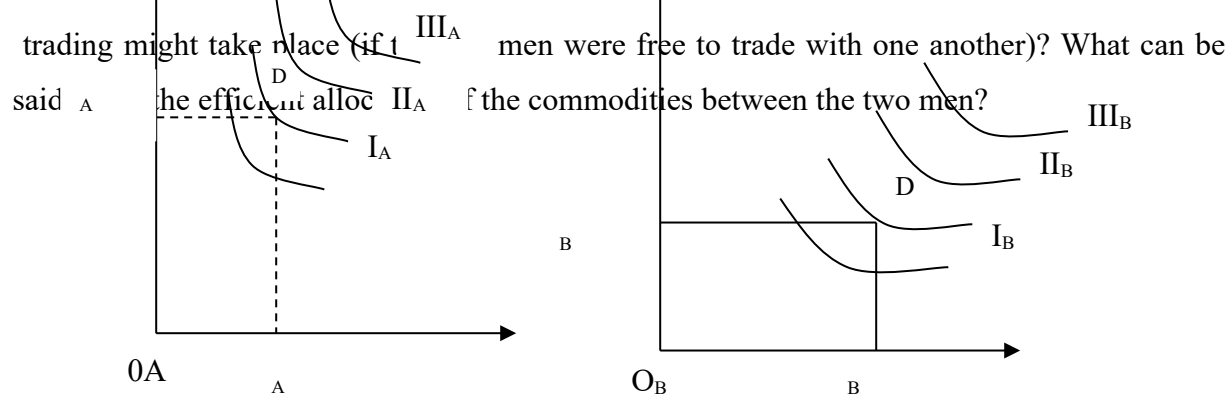


Figure:5.3. Individual A's (a) and B's (b) preference for good X and Y

(a)

(b)

To find out, we must insert indifference curves to our analysis and select the combination that maximize their satisfaction. Given the indifference curve of A, A's well-being is increased by moving towards the origin of B. Three of A's indifference curves are drawn there: I_A , II_A , and III_A . Of the three, the highest indifference curve is III_A ; the lowest is I_A . Similarly, B's well being will be increased by moving towards the origin of A. Three of B's indifference curves are also drawn there: I_B , II_B and III_B . The highest indifference curve of the three is III_B ; the lowest is I_B . A's satisfaction would generally improve if we moved his allocation from points close to his origin (point OA) to points closer to the upper-right corner of the box. Conversely, B's satisfaction would generally improve if we moved his allocation from points close to the upper-right corner of the box to points closer to the origin OA.

Given the initial allocation of food and medicine, we find the A is on indifference curve II_A and B is on indifference curve I_B at point P. At this point, A's marginal rate of substitution of X for Y is much higher than B's. How do we know that? Figure 5.4 shows that the slope of II_A at point P is larger than the slope of I_B . If both men were free to trade, therefore, A would be interested in trading some Y to B in exchange for some X. The exact point to which they might move cannot be predicted. If A were the more astute bargainer, then he might get B to accept the allocation at point E_4 where B would be no better off than before (since he would still be on indifference curve II_B). If, on the other hand, B were better negotiator, then he might get A to accept the allocation at point E_2 where A would be no better off than before (since he would still be on indifference curve II_A). If neither B nor A were so skilled in negotiating that he could extract all of the "surplus value" from the trade, then (as would be most likely), the ultimate point of equilibrium would lie somewhere between E_2 and E_4 .

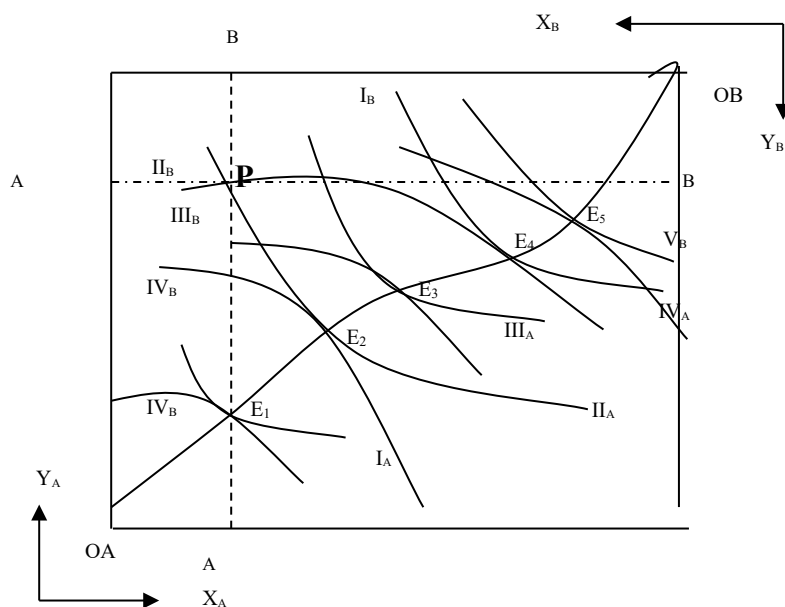


Figure.5.4. Simultaneous equilibrium of both individuals

And what would characterize this equilibrium? The absence of any trades that (1) would be acceptable to both B and A and (2) would improve the welfare of one or both of them. That is to say, the marginal rates of substitution of food for medicine would have to be the same for both i.e. their indifference curves would have to be the same for both i.e., their indifference curves would have to be tangent to one another at the point of equilibrium.

5.3.2. PARETO EFFICIENCY

The notion of equilibrium can be extended. If the object were to make the men as well-off possible (given their unfortunate circumstance to begin with), then an efficient allocation of the commodities would be one in which the marginal rate of substitution of X for Y were the same for both men. As we just argued from the other side, if that were not the case, one man could be made better off without making the other worse off. Tangency of indifference curves is therefore the key to characterizing equilibrium, but it does not yield a unique solution to the trading question. Indeed, there is a locus of points at which the two men's indifference curves are tangent; this locus is called the contract curve. It is shown in Figure 5.4 and it includes all of the points, like E₁, E₂, E₃, E₄ and E₅, for which the marginal rates of substitution are equal

for both consumers. The contract curve is a set of efficient points in a very special sense. Consumers like B and A who might be at a point that lies off the contract curve can always find a point on the contract curve that is preferable to both. Why? Because moving to such a point would not cost either any welfare even though it guarantees that the welfare of at least one of them would improve.

Pareto efficiency is defined, as a point (condition) where it is impossible to make any one better-off with out making some one worse-off. But, if it is possible to make any one better-off with out making the other worse- off, then the condition (that point) is not Pareto - efficient. A point is a Pareto optimal if and only if there is no change that can make some one better off, with out making any one worse- off.

Thus, the above condition (point P) is not Pareto efficient for both individuals. Because, it is possible to make better- off one of the two individuals with out making the other-worse-off by having voluntary exchange between these two individuals. The initial endowment places individual A on indifference curve II_A and individual B on II_B . A is willing to give up more units of Y in order to get additional unit of X (Because it will be on higher level of indifference curve). Similarly B is following to give up more unit of X to obtain additional unit of Y. Such situation will lead to the exchange between individuals

From the point P, A will trade some Y to B, receiving X in exchange

The exact bargain reached by the two parties.

If A is more skill-full negotiator, A may induce B to move along indifference curve II_B form point P to e_4 while A moves from indifference curve II_A to II_A .

Thus, individual A receives all the gain from the exchange while B gains or loses nothing (indifferent) That is, the Pareto optimality occurs, (the Pareto improvement). At point e_4 IV_A and II_B is tangent so that their slope of indifference curve (MRS_{XY}) are equal and there is no future base for exchange.

On the other hand, if B is more skill-full negotiator and induces A to move along its indifference curve II_A from point P to e_4 then B would get all benefits of exchange, and A would neither gain nor lose. AT E_2 the MRS_{XY} of A equals $MRTS_{XY}$ of B. And there is not base for further exchange.

Finally if A moves from point P to E_3 and B moves also from P to E_3 (i.e. A moves from II_A to III_A and B moves from II_B to III_B) both individuals gain from exchange.

Thus, starting from point P, both individuals gain through exchange by moving to a point on a line E_2E_4 or between E_2 and E_4 . The curves that join the locus of all tangency points of the indifference curves of the two individuals (OA, OB) is called the contract curve of exchange. Along this curve the MRS_{XY} is the same for individuals A and B, and the economy is in general equilibrium of exchange.

Thus, equilibrium implies $MRS_{XY}^A = MRS_{XY}^B$

Starting from any point outside the contract curve A or B or both can be benefited from exchange without making the other worse-off. Once on the contract curve the two individuals can not be better off without making the other worse-off. Thus, every point on the contract curve is Pareto optimal and the contract curve is a locus of Pareto optimality.

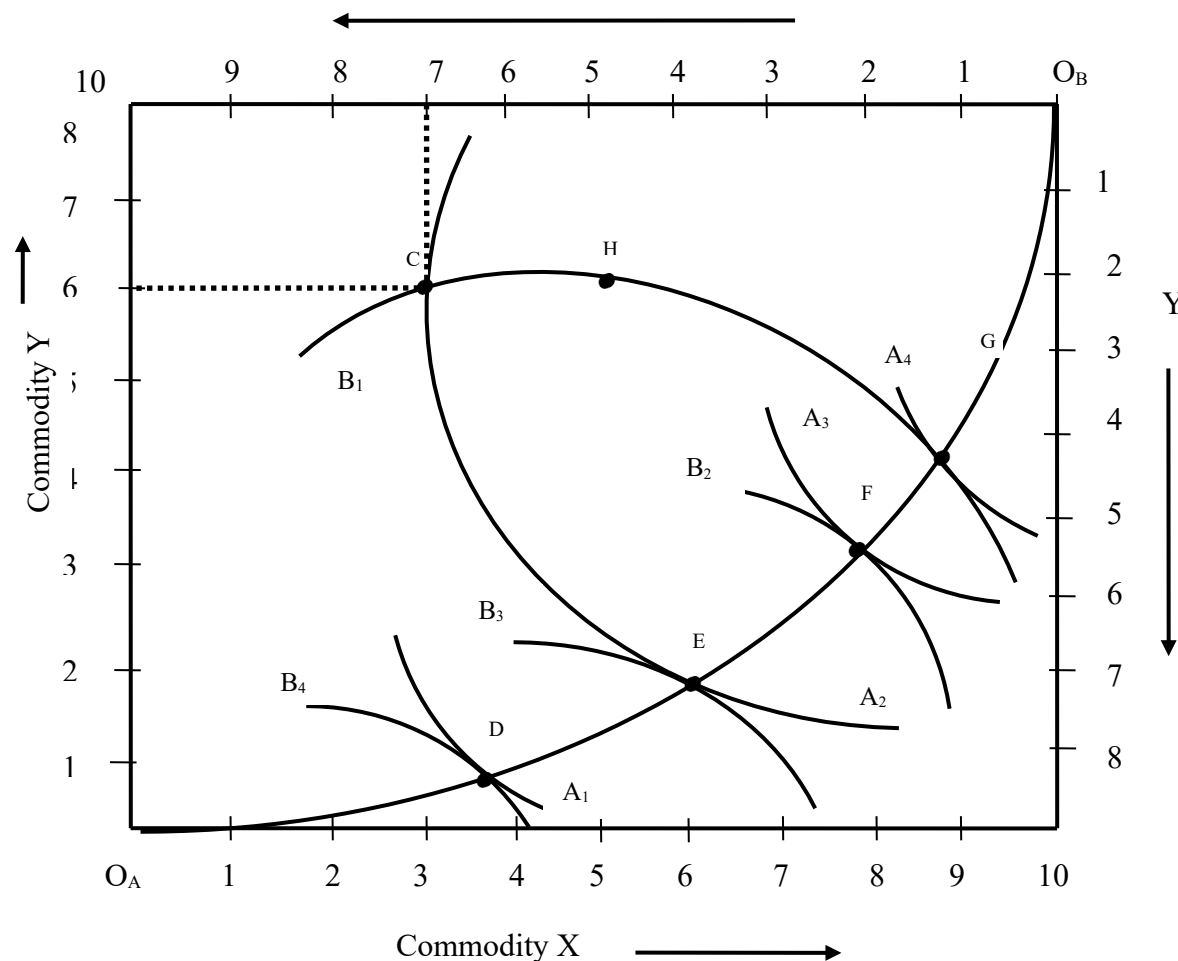
Consider the following Edgeworth¹ box diagram for exchange. The dimensions of the box are given by the total amount of the two commodities (10X and 8Y) owned by the two individuals together. In the box, curves denoted by $A_1, A_2 \dots A_4$ represent the indifference curves of individual A and the curves denoted by $B_1, B_2 \dots B_4$ represent the indifference curves of individual B.

Any point inside the box indicates how the total amount of the two commodities is distributed between two individuals. For example, point C indicates that individual A has 3X and 6Y for the combined total of 10X and 8Y (the dimensions of the box).

Suppose that point c does in fact represent the original distribution of commodities X and Y between individuals A and B. Since at point C, indifference curves A_2 and B_1 intersect, their slope or marginal rate of substitution of commodity X for commodity Y (MRS_{xy}) differs. Starting at point C, individual A is willing to give up 4Y to get one additional unit of X (and move to point E on A_2), while individual B is willing to accept 0.2Y in exchange for one unit of X and move to point H on B_1) i.e., $MRS_{xy} = 4$ for A and $MRS_{xy} = 0.2$ for B. Because A is willing to give up much more Y than necessary to induce B to give up 1X, there is a basis for

¹ The Edgeworth box is a convenient graphical tool that can be used to analyze the exchange of two goods between two people. It is named in honor of Francis Ysidro Edgeworth (1845-1926), an English economist who was one of the first to use this analytical tool.

exchange that will benefit either or both individuals. This is true whenever, as at point C, the MRS_{xy} for the two individuals differs.



For example, starting from point C, if individual A exchanges 4Y for 1X with individual B, A moves from point C to point D along the indifference curve A_1 while B moves from point C on B_1 to point D along A_1 while B moves from point C on B_1 to point D on B_3 . Thus, individual B receives all of the gains from exchange while individual A gains or loses nothing (since A remains on A_1). At point D, A_1 and B_3 are tangent, so that their slopes (MRS_{xy}) are equal, and there is no further basis for exchange. At point D, the amount of y that A is willing to give up for 1X is exactly equal to what B requires to give up 1X. Any further exchange would make either individual worse off than he/she is at point D. the same can be said about points E and F,. The locus of tangency points of the indifference curves of the two individuals (O_ADEFO_B) is the contract curve for exchange. That is, along the contract curve for exchange,

the marginal rate of substitution of commodity X for commodity Y is the same for individual A and B, and the economy is in general equilibrium of exchange. Thus, for equilibrium,

$$MRS_{xy}^A = MRS_{xy}^B$$

Starting from any point on the contract curve, both individuals can gain from exchange by getting to a point on the contract curve. For an economy composed of many consumers and many commodities, the general equilibrium of exchange occurs where the MRS between every pair of commodities is the same for all consumers consuming both commodities.

5.2. Activity

When do we say that a given economy is in a general equilibrium of exchange?

5.4. General Equilibrium in production; Two-Input- Two- Output Case

The preceding section focused our attention on the case in which consumers exchange quantities of commodities in the absence of production. In this section and the next, we take up the equivalent simple case in which there is production but no consumption. What do general equilibrium look like?

To examine general equilibrium in production, we make assume about the product produced and factor used. Let us consider

a very simple economy that produces only two goods, X and Y, with only two inputs (L and K)

The country has fixed endowment of $\bar{L}$ and $\bar{K}$ which will be used to produce both X and Y

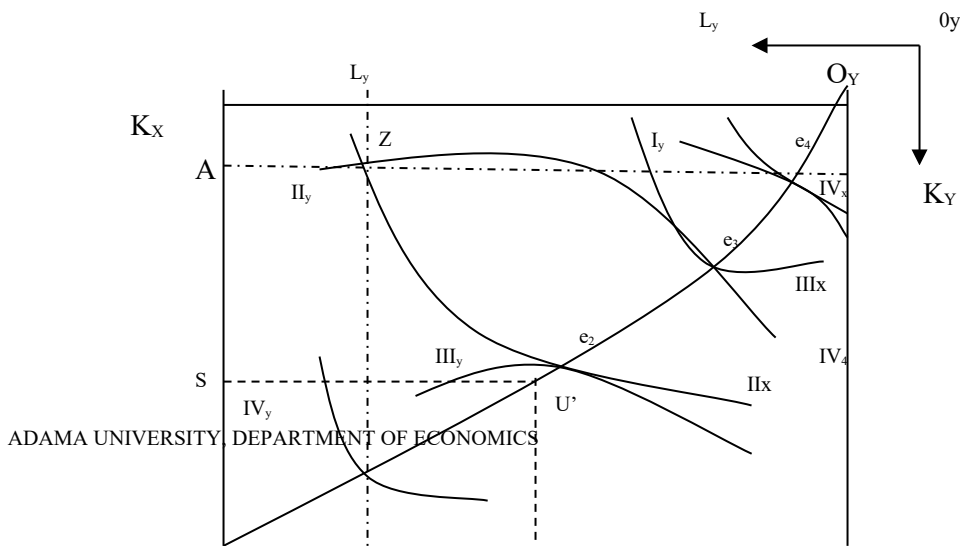
Suppose $\bar{L}_x$ unit of labor and $\bar{K}_x$ unit of capital are used in producing commodity X.

The remaining $\bar{L}_y$ unit of labor, and $\bar{K}_y$ unit of capital are used to produce commodity

Y. That is $L = L_x + L_y$ and $K = K_x + K_y$

To examine general equilibrium of production, we again start from and abstract an Edgeworth box diagram for production using the isoquants for commodities X and Y in a manner completely analogous to the Edgeworth box diagram for consumption above of Figure 5.4. The general equilibrium in production is presented by the figure 5.5. In the Edgeworth box diagram for production shown above in figure 5.5, the size of the box refers to the total amount of L and K available to the economy. While the curves inside the box indicate how the total amount of the two inputs is utilized in the production of the two commodities. Accordingly, let the total amount of labor to be allocated between the two sectors be O_xL and the total amount of capital to be allocated between the two sectors O_yL . Finally, suppose that the initial allocation of labor and capital between the two sectors were represented by point Z in the Edgeworth box diagram portrayed in Figure 5.5. Note that; the X industry starts with O_xA units of labor and O_xB units of capital, while the Y sector comes to the table with $O_yL - O_xA$ units of labor and $O_yK - O_xB$ units of capital.

On the basis of the production functions for X and Y, we can insert isoquants for both X production and Y production in Figure 5.5. There isoquants for X production are displayed there: I_x , II_x , and III_x . Of the three, the isoquant that reflects combinations of capital and labor with the highest output of food is III_x , and the isoquant for the lowest –output of foods is I_x . Three isoquants for medicine production are also displayed: I_y , II_y , and III_y . Again, of the three, the isoquants that reflects combinations of capital and labor with the highest output of medicine is III_y , and the isoquant for the lowest output of medicine is I_y .



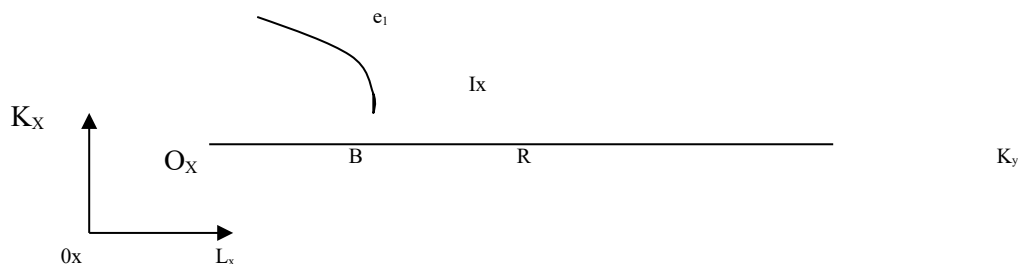


Figure 5.5 : General Equilibrium in production

What would be an efficient allocation of inputs between the two outputs? At the original allocation at point Z, the marginal rate of technical substitution of capital for labor in producing X is higher than it is in producing Y. This observation is based on the fact that the slope of I_X at point Z is greater than the slope of I_Y . The fact that the marginal rates of technical substitution are unequal at point Z means that the inputs are not being allocated efficiently.

To see why, suppose that the X industry operating at point Z could substitute 2 units of labor for 1 unit of capital without changing its output while the Y industry would have to substitute 1 unit of labor for 2 units of capital to maintain its output. In this case (where the Y industry uses more labor and less capital relative to the food industry), it would be possible for one industry to expand its output without any reduction in the other industry's output. Specifically, it would be possible to move to point 'e₂', where the output of X would be the same as at point Z even though the output of Y would climb to the level corresponding to I_{Y2} . It would be possible to move to point 'e₃', where the output of medicine would be the same as at point Z but the output of X would climb to the level corresponding to I_{X2} . And, of course it would be possible to move to a point between 'e₂' and 'e₃' for which the outputs of both sectors would expand relative to their levels at point Z.

Regardless of which point were chosen, the same idea described above should apply. Production should occur at a point at which the marginal rates of technical substitution between inputs are the same for all producers. Only then would the allocation of inputs be efficient in the sense that an increase in the output of one commodity could be achieved only by a reduction in the output of the other commodity. An efficient allocation of inputs must,

therefore, lie somewhere along the locus of points where the marginal rates of technical substitution are equal, and so it must lie at a point where a Y isoquant is tangent to a X isoquant. The locus of these points, like the analogous set in the preceding section, is also called the contract curve; it is shown in Figure xx. The curve $O_x e_1 e_2 e_3 O_y$ is the contract curve for production. It is the locus of tangency points of the isoquants for X and Y at which the Marginal Rate of Technical Substitution of labor for capital is the same in the production of X and Y. that is, the economy is in general equilibrium of production when

$$MRTS_{LK}^X = MRTS_{LK}^Y$$

Producers who find themselves at a point that lies off the contract curve can, if society is interested in producing as much as possible of each good, move to a point where the output of one industry can be increased without a reduction the other industry's output. Thus by simply transferring some of the given and fixed amounts of available L and k between the production of X and Y, this economy can move from a point not on the contract curve for production to a point on the curve and increase its output of either or both commodities. For an economy of many commodities and many inputs, the general equilibrium of production occurs where the marginal rate of technical substitution between any pair of inputs is the same for all commodities and producers using both inputs.

5.3. Activity

What do we mean by a contract curve for production?

What is the condition for the attainment of a general equilibrium of production?

5.4.1. THE PRODUCTION POSSIBILITIES CURVE

The contract curve in Figure 5.5 showed the various allocations of inputs that are efficient in the special sense describe in the preceding section. There is, of course, a level of output for X and Y corresponding to each point on the contract curve. Consider, for example, point e_2 in Figure xx. If the level of output of X corresponding to isoquant I_x were 100 and if the level of

output of Y corresponding to isoquant IIIy were 200, then an output of 100 units of X and 200 units of Y would correspond to the point e_2 . Similarly, if the level of output of X corresponding to isoquant IIx were 200 and if the level of output of Y corresponding to isoquant IIy were 100, then an output of 200 units of X and 100 units of medicine would correspond to the point e_3 .

Proceeding in this way, we can find the pair of outputs corresponding to each point on the contract curve. And we can then plot each such pair of points in a graph like that in Figure 5.6 below, where the amount of X produced is shown on the horizontal axis and the amount of Y produced is shown on the vertical axis. For example, the pair of outputs corresponding to point e_2 on the contract curves is plotted as point A in Figure 5.6, and the pair of outputs corresponding to point e_3 on the contract curve is plotted as point B in Figure 5.6. And the opposite origin in Figure 5.6, where X production is zero because Y employs all of the capital and labor, is portrayed by point P in figure 5.6. By joining points or the output pairs corresponding to each isoquant tangency points we can derive the certain curve. The curve PP' in Figure 5.6 is the result of plotting all these points i.e. the output combinations associated with all of the points on the contract curve. It is called the **production possibility curve**. You may remember this concept from your first lecture in introductory economics (but it was probably derived from a production function with only one input).

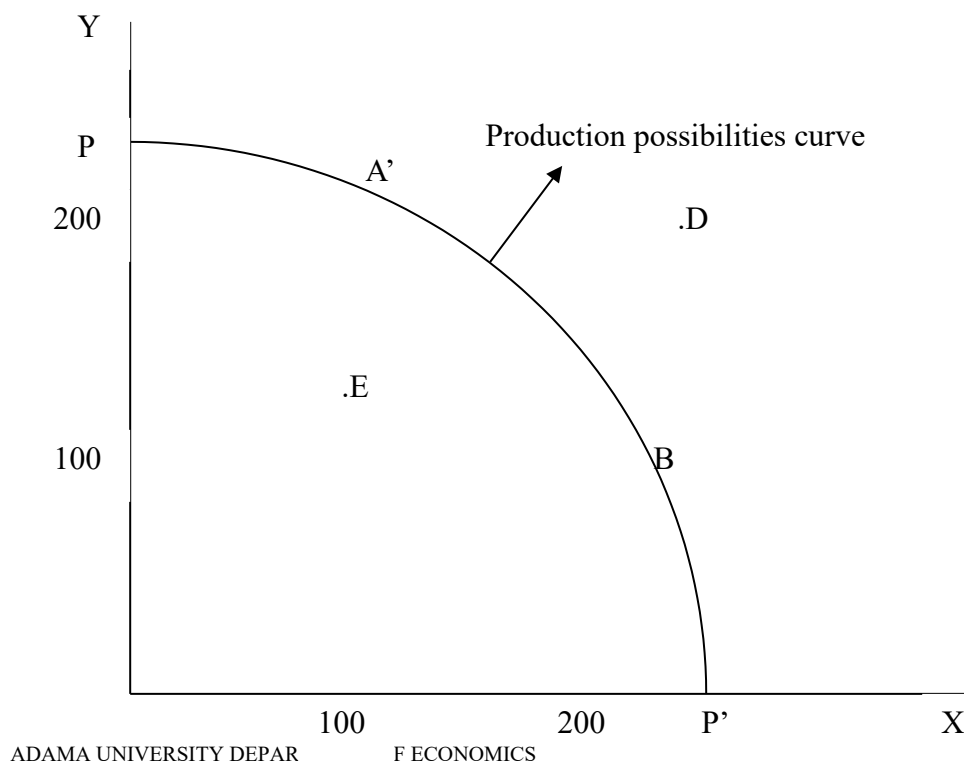


Figure 5.6: Production possibilities curve

The production possibilities curve shows the various combination of X output and Y output that can be derived from the economy's input base. In our simple model above at the beginning in figure 5.5, this base is $O_F L$ units of labor and $O_F K$ units of capital. More specifically, it shows the maximum output of one good that can be produced under the assumption that the output of the other good is held fixed. For example; at point e_2 given quantity of X is II_x , the maximum quantity of Y that can be produced is III_y . Similarly, point e_3 of the contract curve of production shows that given III_x the maximum amount of Y the economy can produce is II_y .

Given the economy's input base and existing technology, it is impossible to attain a point like D in Figure 5.6 that is outside the production possibilities curve. On the other hand, a point inside the PPF corresponds to a point off the production contract curve indicates that the economy is not in general equilibrium of production, and is not utilizing its inputs of labor and capital most efficiently. It is possible to attain a point like E that is inside the production possibilities curve, but it would be inefficient to do so. Indeed, a point like E would correspond to a point like Z in Figure 5.6 (go to production part) an allocation of inputs that does not lie along the contract curve.

5.4.2 PRODUCTION EFFICIENCY AND EQUILIBRIUM

As long as production is efficient, input allocations lie along the contract curve for which the marginal rates of technical substitution are equal for each sector, production occurs at some point along the production possibilities curve. Once on the PPF, the output of either commodity can be increased only by reducing the output of the other. The amount of commodity Y that the economy must give up, at a particular point on the PPF so as to release just enough labor and capital to produce one additional unit of commodity X, is called the Marginal Rate of Transformation of X for Y (MRT_{xy}). This is given by the absolute value of the slope of the PPF at that point. It is also equal to the ratio of the marginal cost of X to the marginal cost of Y.

$$\text{That is, } = \frac{\Delta y}{\Delta x}$$

The MRPT measures the amount of Y that must be sacrificed in order to obtain an additional unit of X.

In perfect market the profit maximizing producer or firm equate the price of the commodity to the marginal cost production. i.e., $P = MC$ which implies

$$MRPT_{xy} = \frac{MC_x}{MC_y} = \frac{P_x}{P_y} \Rightarrow \boxed{MRPT_{xy} = \frac{P_x}{P_y}}$$

Given the commodity prices, general equilibrium of production can be attained on the PPC at which the slope of PPC equal to the ratio of the prices of the products. To further clarify with the help of graph assume that the market price of commodity define the slope of the line AB. The ratio OA/OB measures the ratio of the marginal cost of these two products. The general equilibrium of product mix is shown below in Figure 5.7. The country is in equilibrium at point T where the level production is X^* and Y^* .

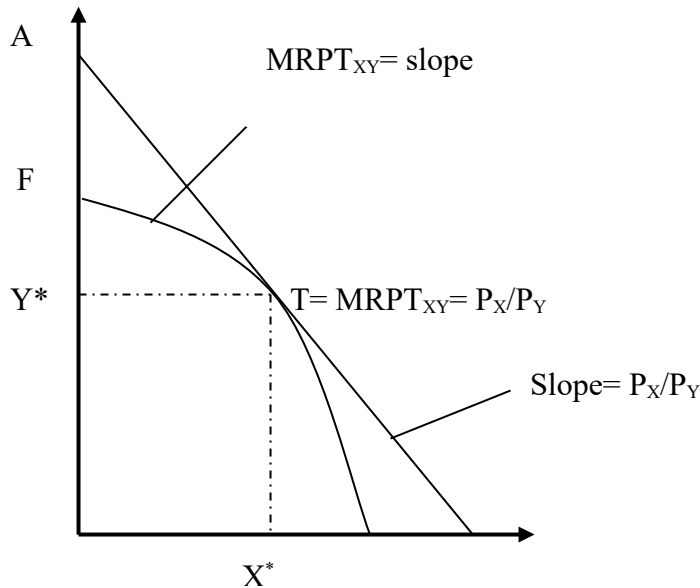


Figure 5.7: General equilibrium in product mix

5.5. PRODUCTION AND EXCHANGE

Dear readers, up until now we have been developing tools for measuring the attainment of general equilibrium in production and exchange respectively. This section brings the tools together so as to determine the simultaneous general equilibrium of production and exchange. That is, we can use the PPF and the contract curve for exchange to examine how our simple economy can reach simultaneously general equilibrium of production and exchange. We will analyze how the economy comes to equilibrium? For this purpose illustrations we combine our of exchange and production assumptions above together.

5.5 General Equilibrium in a Two-Factor, Two-Commodity, Two-Consumer (2x2x2) Economy

Assumptions of the 2x2x2 model

There are two factors of production (L and K) whose quantities are given exogenously.

These factors are perfectly divisible and homogeneous.

There is perfect competition in both commodity and factor markets. Consumers and firms face the same set of prices (P_x, P_y, w, r).

Figure 5.8: Production and exchange equilibrium

A. ALLOCATING CONSUMPTION BETWEEN INDIVIDUALS

If the isoquant map in each industry displayed in figure bb were applicable then we would know from the preceding section that the various combinations of X output and Y output that could be derived from this input base were given by the production possibilities curve PP' in Figure 5.6. This curve PP' reproduced in figure 5.8. suppose, for the moment, that we also knew the composition of output (i.e., the amount of X and Y) that should be produced in the economy. We could then insert an Edgeworth box diagram similar to that in Figure 5.5 as in exchange case into Figure 5.8. The upper-right corner of the box would simply be the point on the production possibilities curve corresponding to this predetermined composition of output.

To be more specific, suppose that we knew that the composition of output in the economy should be represented by point A', where the quantity of X produced would equal Q and the quantity of Y produced would be N. We could then draw an Edgeworth diagram for

consumption with Q as its width and N as its height. Since Q and N would be the total amount of X and Y to be distributed to A and B , respectively. This box diagram could be used to see how much of total output of each good would go to each consumer. Figure dd shows indifference curve for each consumer (T_1' , T_2' , and T_3 for A ; H_1' , H_2' , and H_3' for B) within the box defined by Q , A' , N , and the origin. It also shows the contract curve, C_1C_1' . Recall that this is the locus of points where A 's indifference curves are tangent to B 's.

We know that the distribution of output between the two consumers should be such that they should lie on the contract curve C_1C_1' . But, where? We show in the following section that, if the economy's output were allocated so that consumer satisfaction were maximized, then the marginal rate of product transformation (the magnitude of the slope of the production possibilities curve) should equal the marginal rate of substitution. If the economy's output were allocated to maximize consumer satisfaction, then the consumers should be at the point of the contract curve where the common slope of their indifference curves equals the slope of the production possibilities curve at A' .

Where on the contract curve would the common slope of their indifference curves equal the slope of the production possibilities curve at A' ? An examination of Figure dd shows that this condition would be fulfilled at point B on the contract curve, where A would receive F_1 units of X and M_1 units of Y while B would receive $Q-F_1$ units of X and $N-M_1$ units of Y . The slopes of the indifference curves at point B equal the slope of the production possibilities curve at point A' . How do we know? The tangent lines $B''B''$ and $A''A''$ are parallel. So, given the amount to be produced of each commodity, we have devised a way to determine the amount of each commodity that should be allocated to A and B .

B. ALLOCATING INPUTS BETWEEN COMMODITIES

We can also find the amount of labor and capital that should be allocated to the production of each commodity by consulting the Edgeworth box diagram in Figure 5.5. Recall that the contract curve drawn there underlies the production possibilities curve drawn in Figure dd. Recall, as well, that each point on the production possibilities curve corresponds to a point on the contract curve in Figure 5.5 that each point on the contract curve in Figure 5.5 corresponds

to a particular allocation of labor and capital between the production of the two commodities. For example, assume that point A' on the contract curve in Figure 5.5 corresponds to a particular allocation of labor and capital between the productions of the two commodities. For example, assume that point A' on the production possibilities curve corresponds to point U' on the contract curve in Figure 5.5. So, if we knew that the composition of output in the economy should be given by point A', then we would know that OxS units of labor should be devoted to the production of X and $OxL-OxS$ units of labor should be devoted to the production on Y. We would know that OxR units of capital should be devoted to the production of X, and that $OxL-OxS$ units of labor should be devoted to the production of Y. We would know that OxR units of capital should be devoted to the production of X, and that $OFK-OFR$ units of capital should be devoted to the production of Y. Finally, we would know that this allocation of inputs would guarantee that the marginal rates of technical substitution between capital and labor would be equal in the production of both X and Y.

C.THE MARGINAL RATE OF PRODUCT TRANSFORMATION, MARGINAL RATE OF SUBSTITUTION, AND CONSUMER SATISFACTION

We asserted in the preceding section that consumer satisfaction would not be maximized unless the marginal rate of product transformation between two goods is equal to the marginal rate of substitution between the two goods.

5.6. THREE CONDITIONS FOR ECONOMIC EFFICIENCY

The preceding sections have told a long story from which we can draw some general insight into the conditions that characterize an efficient allocation of resource-i.e, an allocation where it is impossible to improve somebody's welfare without causing harm to anybody else. These properties are called marginal conditions of Pareto optimality or Pareto efficiency. Indeed, it turns out that there are three such conditions to be fulfilled. If resources are to be allocated efficiently, simultaneous general equilibrium of production and exchange has to be reached. These require three conditions to be fulfilled. These are;

The marginal rates of substitution between any two goods must be the same for all individuals (otherwise, at least two individuals can get together for a mutually beneficial trade and thereby increase the utility of one or both people). That is, general equilibrium of exchange (distribution) for two individual is given by:

$$MRS_{XY}^A = MRS_{XY}^B$$

The marginal rates of technical substitution between any two inputs must be the same in the production of all goods (otherwise, two producers can get together to trade inputs and thereby increase the output of one or both products). General equilibrium in production is given by relation can be stated as

$$MRPT_{XY} = P_X/P_Y$$

The marginal rate of transformation between any two goods must equal the common marginal rate of substitution for all individuals (otherwise resources can be reallocated to increase the production of some goods at the expense of others to bring the product mix into line with consumer preference). Then the general equilibrium of the system as a whole requires the fulfillment of the third condition, i.e.

$$MRPT_{XY} = MRS_{XY}^A = MRS_{XY}^B \quad \text{or}$$

For perfect market case

$$MRS_{XY}^A = MRS_{XY}^B = P_X/P_Y$$

For instances, general equilibrium for the simple 2*2*2 model can be attained if the following three Pareto optimum (efficiency) conditions are satisfied;

The MRS_{XY} must be equal for both consumers (efficiency in exchange).

The $MRTS_{XY}$ must be equal for all firms (efficiency in production).

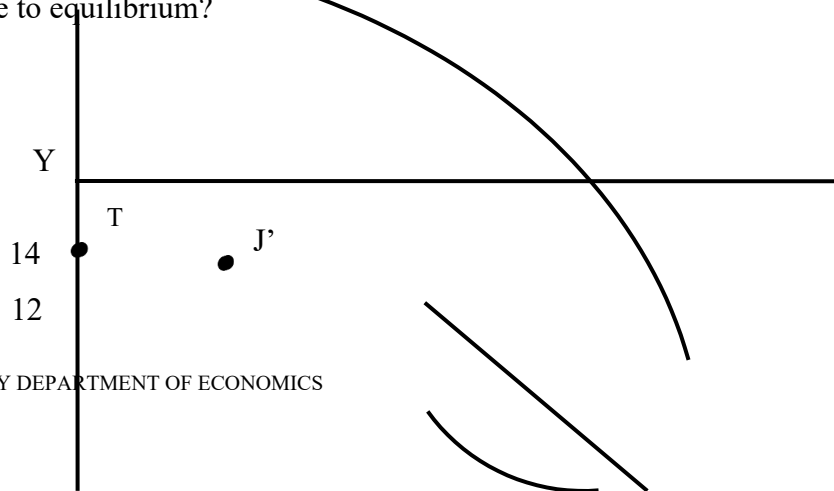
The MRS_{XY} and $MRPT_{XY}$ must be equal for two goods (efficiency in product mix)

5.4. Activity:

Explain marginal conditions of Pareto optimality in production?

Explain the general equilibrium of the simple 2*2*2 model?

Given the following simplified general equilibrium model in figure 5.9: below, explain how the system came to equilibrium?



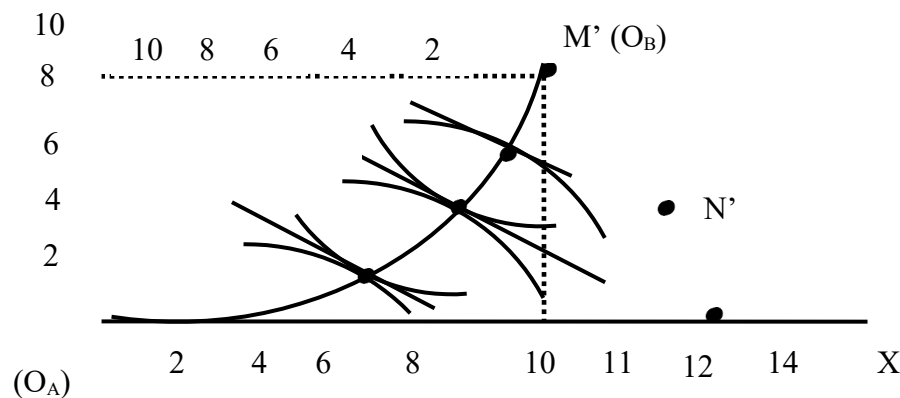


Figure 5.9; General equilibrium of production and exchange

5.7. SUMMARY

Previous chapters have been concerned with partial equilibrium analysis, in which it is assumed that changes in price can occur in one market without causing significant changes in prices in other markets. General equilibrium analyses account for the interrelationship among prices in various markets.

General equilibrium analysis has been used to help examine the conditions under which it is possible for equilibrium to occur simultaneously in all markets in a perfectly competitive economy. Modern work has established that a general equilibrium can be achieved in a perfectly competitive economy under a fairly wide set of conditions.

General equilibrium analysis provides a framework within which economists can study the relationships between prices and quantities for both commodities and inputs across the economy as a whole. Its purpose is to show what the equilibrium configuration of prices, outputs, and inputs will be in various markets given a certain set of consumer preferences, a set of production functions, and a set of input supply function.

The simplest of general equilibrium models involves three kinds of equations—one for products, one for inputs, and one that guarantee that the conditions of perfect competition are satisfied. Simplest model of all assumes that the supply of inputs is given (and independent of prices) and that the coefficients of production are fixed. It nonetheless illustrates the nature of general equilibrium models.

Expanding the scope of analysis beyond a single market make it possible to consider many interesting questions concerning the efficient allocation of resources across market. The edge worth box diagram can be employed to examine efficient allocation of commodity between

consumers. The contract curve summarizes those allocations. They are all characterized by equality in the marginal rates of substitution across all consumers.

The efficient allocation of inputs between industries also lies on a contract curve drawn in an Edgeworth box whose dimensions define the limits of input employment. This contract curve can be used to construct a production possibilities curve for an economy. Every point on the contract curve is characterized by equality of the marginal rates of technical substitution across all producers.

Even simple general equilibrium models can accommodate both production and exchange. Consumer satisfaction cannot be maximized unless the marginal rate of product transformation between (any) two goods is equal to the marginal rate of substitution between them for every consumer.

Perfectly competitive economies satisfy the three sets of conditions for economic efficiency. To the economic theorist, this is one of the basic arguments in favor of a perfectly competitive economy.

KEY TERMS

General equilibrium

Partial equilibrium analysis

Efficient allocation

Production

Exchange

Production possibility curve

Edgeworth box

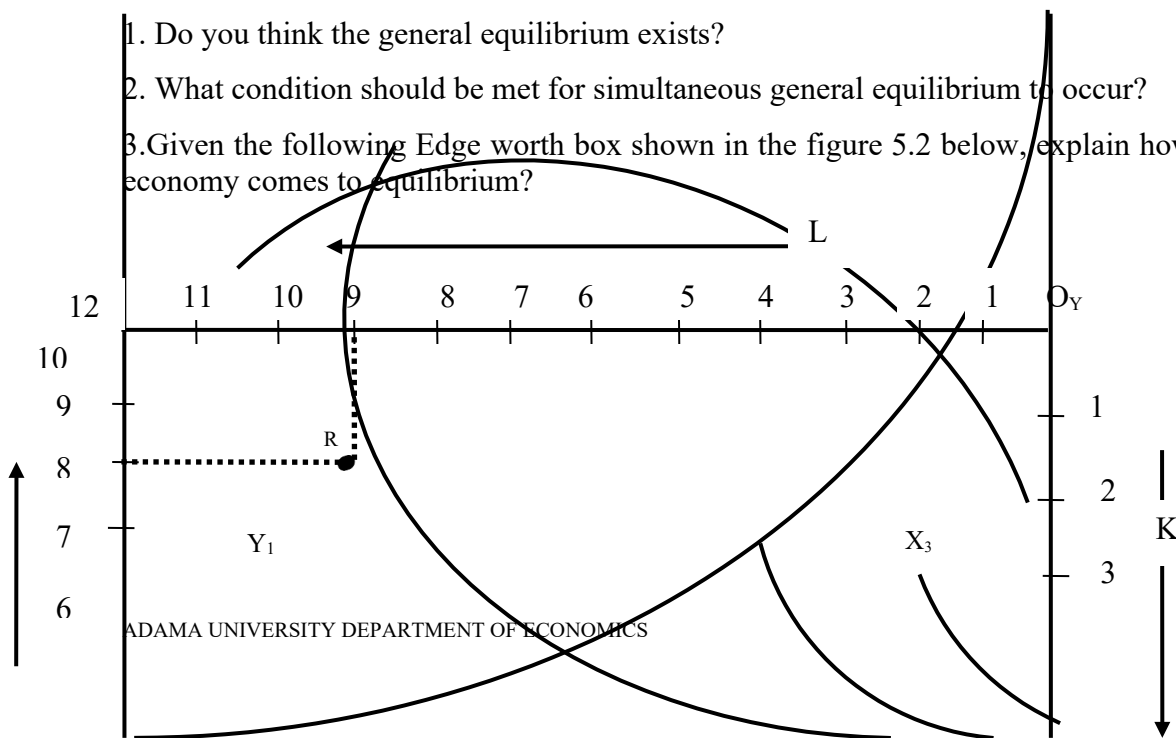
Consumer satisfaction

Pareto optimum conditions

Marginal Rate of Substitution

Review Questions

1. Do you think the general equilibrium exists?
2. What condition should be met for simultaneous general equilibrium to occur?
3. Given the following Edgeworth box shown in the figure 5.2 below, explain how the simple economy comes to equilibrium?



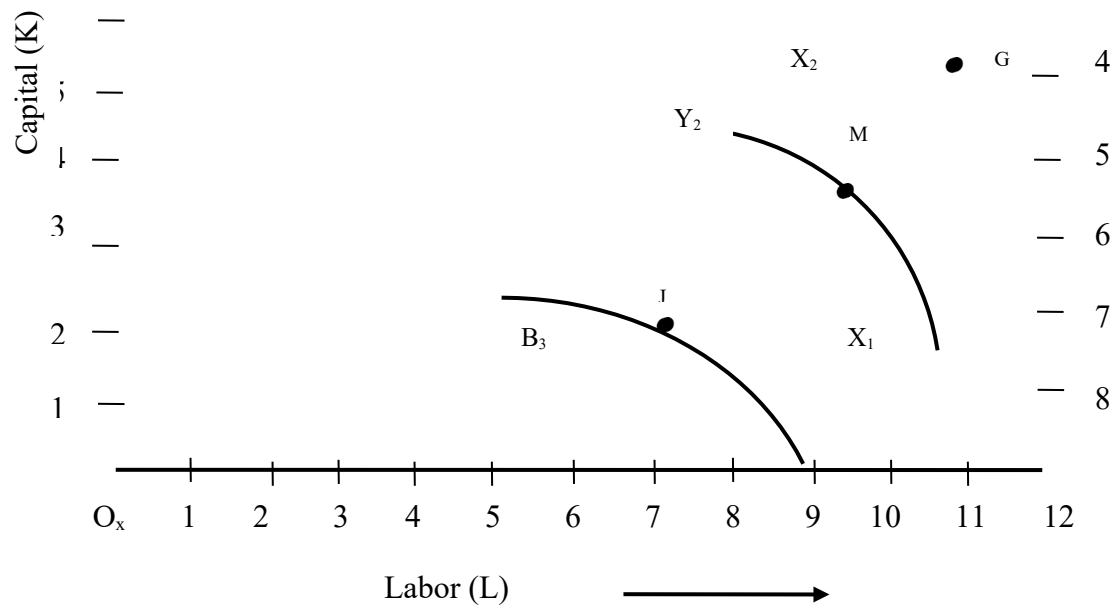


Figure 5.10 ; The Edgeworth box diagram for production

Suggested readings

Nicholson, w: **Microeconomic theory; basic principle and extension**. 5th edition

Salvatore, D(2003):**Microeconomics; theory and applications**.4th edition

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Varian, R. Hall (1999): **Intermediate microeconomics** 3rd edition

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Koutsoyiannis A. (1985): Modern microeconomics 2nd edition.

7.1. Welfare Economics: An Introduction

7.2. Aggregation of preferences

7.3. Utility Possibility Frontier and Grand Utility Possibility Frontier

Social Welfare Functions

7.5. Welfare Maximization

7.6 Individualistic social Welfare functions

7.7. Fair Allocations

7.8. Envy and equity

7.9. Summary

5.7 WELFARE ECONOMICS: AN INTRODUCTION

Welfare Economics studies the conditions under which the solution to the general equilibrium model is said to be optimal. It examines the conditions for economic *efficiency* in the production of output and the exchange of commodities and *equity* in the distribution of income. Note that the above definition points out that the maximization of society's well-being requires not only efficiency in production and exchange but also equity in the distribution of income.

The maximization of society's well being requires the optimal allocation of inputs among commodities and the optimal allocation of commodities (i.e. distribution of income) among consumers. Dear readers, the conditions for the optimal allocation of inputs among commodities and exchange of commodities among consumers have already been discussed. These are objective criteria devoid of ethical connotations or value judgments. In other word, it is impossible to objectively determine the optimal distribution of income. This necessarily requires interpersonal comparisons of utility and value judgment on the relative "deservingness" or merit of various members of society, and different people will inevitably have different options.

This unit wills presents criteria of social welfare. We will consider some four criteria like, classical welfare criteria, Neo – Classical welfare economics, Pareto welfare economics and post- Pareto contribution to welfare economics. It will also provide you with a brief account on different criteria of social welfare.

5.8 AGGREGATION OF PREFERENCES

Social welfare depends on the preferences of individuals. Let us return to our early discussion of consumer preferences. As usual, we will assume that these preferences are transitive. Originally, we thought of a consumer's preferences as being defined over his own bundle of goods, but now we want to expand on that concept and think of each consumer as having preferences over the entire allocation of goods among the consumers. Of course, this includes the possibility that the consumers might not care about what other people have, just as we had originally assumed.

Let us use the symbol x to denote a particular allocation or what every individual gets of every good. Then given two allocations, x and y , each individual i can say whether he or she prefers x to y . Given the preferences of all the agents, we would like to have a way to "aggregate" them into one social preference. That is, if we know how all the individuals rank various allocations, how one might go about bringing together the individual consumer's preferences to construct some kind of "social preferences"?

Two approaches that are of major interest are:

Majority voting approach

Rank – order voting approach

a) Majority voting approach (MVA)

Majority voting approach is one way to aggregate individual preferences. In this form of aggregation, allocation X is said to be preferred to allocation Y if a majority of the individuals prefer X to Y . For example, in an economy consisting of three individuals, if two of them prefer X to Y and one of them prefers Y to X , using majority voting approach we conclude that the society prefers X to Y .

However, there is a problem with this method in that it may not generate transitive social preference ordering. Suppose we have the following voting result.

Choice	Person A	Person B	Person C
1 st	X	Y	Z
		2 nd	Y
			Z
			X
3 rd	Z	X	Y

The ranking of the three individual is

Persons A and C prefer X to Y and person B prefers Y to X. Hence $X > Y$.

Persons A and B prefer Y to Z and person C prefers Z to Y. Hence $Y > Z$.

If this social preference ordering were transitive we could conclude that $X > Z$. However, from the table we know that persons B and C prefer Z to X implying that $Z > X$. Since the preferences are not transitive, there will be no “best” alternative from the set of alternatives (X, Y, Z). Which outcome society chooses will depend on the order in which the vote is taken.

That is, for example, if the three people vote first on:

X versus Y and then vote on the winner of this contest versus Z, the best alternative will be Z.

Z versus X and then vote on the winner of this contest versus Y, the best alternative will be Y.

Y versus Z and then vote on the winner of this contest versus X, the best alternative will be X.

Therefore, majority vote can be manipulated by changing the order in which things are voted so as to yield the desired outcome.

b) Rank–Order Voting Approach (RVA)

Here each person ranks the allocations according to his preferences and assigns a number that indicates its rank in his ordering. Consider the following table xx. For example, a 1 for the best alternative, 2 for the second best, and so on. Then we sum up the scores of each alternative across the people to determine an aggregate scored for each alternative and say that one outcome is socially preferred to another if it has a lower score.

Allocations	Person A	Person B	Sum of the ranks
X	1	2	3
Y	2	1	3

Since the sum of the ranks is equal for both alternatives, the society is indifferent between X and Y. Rank order voting has a limitation. It can be manipulated by introducing new

alternatives that change the final ranks of the relevant alternatives. For example, consider the following table yy.

Allocations	Person A		Person B		Sum of the ranks
X	1	3	4		
Y	2	1	3		
Z	3	2	5		

Note that the introduction of a new alternative (i.e., alternative Z) resulted in Y being preferred to X despite our earlier conclusion that the society is indifferent between X and Y when alternative Z was not introduced.

To summarize, both majority voting and rank–order voting have limitations in that their outcomes can be manipulated by astute agents. The question that normally follows is “Are there other social decision mechanisms, apart from majority voting and rank – order voting, that are immune to the limitations of these two voting approaches?” In other words, are there other social decision mechanisms that satisfy the following properties?

Given any set of complete, reflexive, and transitive individual preferences, the social decision mechanism should result in social preferences that satisfy the same properties.

If everybody prefers alternative X to alternative Y, the social preferences should rank X ahead of Y.

The preferences between X and Y should depend only on how people rank X versus Y, and not on how they rank other alternatives.

It can be quite difficult to find a mechanism that satisfies all the above properties. In fact, Kenneth Arrow has proved the following remarkable result which is known as Arrow’s Impossibility Theorem.

“If a social decision mechanism satisfies properties a, b and c, then it must be a dictatorship: all social rankings are the rankings one individuals.”

Arrow’s impossibility theorem shows that three very plausible and desirable features of a social decision mechanism are inconsistent with democracy. That is under democratic system, there is no perfect way to “aggregate” individual preferences to make one social preference. If we want to find a way to “aggregate” individual preferences to form social preferences, we will have to give up one of the properties of a social decision mechanism described in Arrow’s impossibility theorem.

Activity

What are the approaches of social decision mechanism?

Explain the difference between majority voting and rank ordering approach?

5.9 UTILITY POSSIBILITY FRONTIER/CURVE (UPF/UPC) AND GRAND UTILITY POSSIBILITY FRONTIER (GUPF)

The Utility Possibility Frontier (UPF here after) shows the various combinations of utilities received by individuals A and B (i.e., U_A and U_B) when our simple economy is in general equilibrium or Pareto optimum in exchange; it is the locus of maximum utility for one individual for any given level of utility for the other individual.

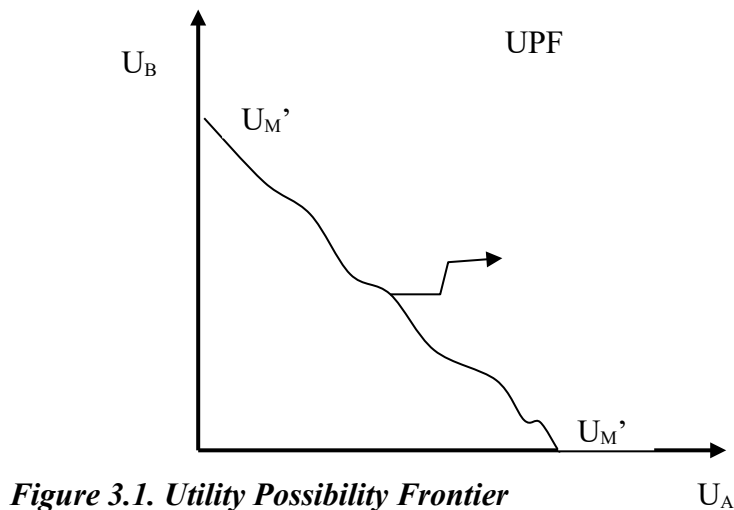


Figure 3.1. Utility Possibility Frontier

Note that UPF is associated with Pareto efficient allocations given equilibrium of production i.e., given a point on PPF where $MRPT_{X,Y} =$). This means we have another UPF for any other point on PPF. In general we can have as many UPFs as there are points on PPF. GUPF is the envelope of these UPFs associated with Pareto optimum points of production and exchange.

$U_M'U_M'$ is UPF drawn when the economy is in production equilibrium at point E' as shown above. $U_N'U_N'$ is another UPF drawn on the assumption that the economy is in equilibrium of

production at point H' on its PPF. By joining points E' , H' , and other points of equilibrium similarly obtained, we can derive GUPF which is curve $G E' H' G$ in the above diagram.

The GUPF indicates that no reorganization of the production – exchange process is possible that makes someone better off without, at the same time, making someone else worse off. Given GUPF of the above sort, in order to determine the Pareto optimum point in production and exchange at which social welfare is maximized, we need a social welfare function.

UPF $U_M' U_N'$ shows the various combinations of utilities received by individuals A and B (i.e., U_A and U_B) when the economy composed of individuals A and B is in general equilibrium or Pareto optimum in exchange. The frontier is obtained by mapping consumption contract (OC) in the Edge worth box of consumption from output or commodity space to utility space.

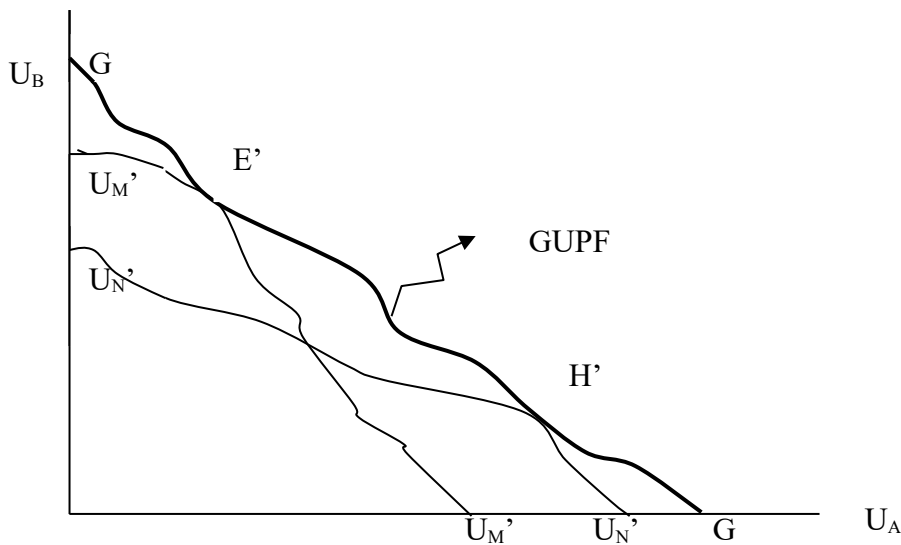


Figure xx. Grand Unity Possibility Frontier

5.10 SOCIAL WELFARE FUNCTIONS

If we were to drop any of the desired features of a social function described above, it would probably be property 3—that the social preference between two alternatives only depends on the ranking of those two alternatives. If we do that, certain kind of rank –order voting become possibilities.

Given the preferences of each individual I over the allocations, we can construct utility functions, $u_i(x)$, that summarize the individuals' value judgments: person i prefers x to y if and only if $u_i(x) > u_i(y)$. Of course, these are just like all utility functions—they can be scaled in any way that preserves the underlying preference ordering. There is no unique utility representation.

There is a name for this kind of aggregating functions; it is called a social welfare function. A social welfare function is just some function of the individual utility function: $W(u_1(x), \dots, u_n(x))$. It gives a way to rank different allocations that depends only on the individual preference, and it is an increasing function of each individual's utility.

There are different ways of formulating social welfare function from individuals' preferences, some which are:

By adding the individual utilities and use the resulting number as a kind of social welfare function. That is,

$$W(U_1(X) \dots U_n(X)) = \sum_{i=1}^n U_i(X)$$

This is sometimes known as *classical utilitarian or Benthamite welfare function*. In this form of social welfare function, we say allocation X is socially preferred to allocation Y if

$$W(U_1(X) \dots U_n(X)) > W(U_1(Y) \dots U_n(Y))$$

A slight generalization of the utilitarian welfare function is the weighted sum - of- utilities welfare function.

$$W(U_1(X) \dots U_n(X)) = \sum_{i=1}^n a_i U_i(X)$$

Where the weights $a_1, \dots, a_n$, are supposed to be numbers indicating how important each agent's is to the overall social welfare.

Another interesting welfare function is the minimax of Rawlsian social welfare function. Considering the welfare of the worse off agent as that affects the social welfare of an allocation. That is:

$$W(U_1(X), \dots, U_n(X)) = \min \{U_1(X), \dots, U_n(X)\}$$

This welfare function says that the social welfare of an allocation depends only on the welfare of the worst of agent –the person with the minimal utility.

Each of these is a possible way to compare individual utility functions. Each of them represents different ethical judgments about the comparison between different agents' welfares. About the only restriction that we will place on the structure of the welfare function at this point is that it be increasing in each consumer's utility.

5.11 WELFARE MAXIMIZATION

Once we have a welfare function we can examine the problem of welfare maximization. Let us use the notation x_i to indicate how much individual i has of good j , and suppose that there are n consumers and k goods. Then the allocation x consists of the list of how much each of the agents has of each of the good's.

If we have a total amount $X^1, \dots, X^k$ of goods $1, \dots, k$ to distribute among the consumers we can pose the welfare maximization problem.

$$\text{Max } W(u_1(x), \dots, u_n(x))$$

Such that

Thus we are trying to find the allocation that maximizes social welfare. What properties does such an allocation have?

We can illustrate this situation in Figure hh, where the set U indicates the set of possible utilities in the case of two individual. This set is known as the utility possibilities set. The boundary of this set- the utility possibilities frontier-is the set of utility levels associated with Pareto efficient allocation. If an allocation is on the boundary of the utility possibilities set, then there are other feasible allocations that yield higher utilities for both agents.

The “indifference curves” in this diagram are called is welfare curves since they depict those distributions of utility that have constant welfare. Isowelfare curves show those distributions

of utility that have constant welfare. The allocation that maximizes social welfare can then be obtained by bringing together our GUPF and the isowelfare curves.

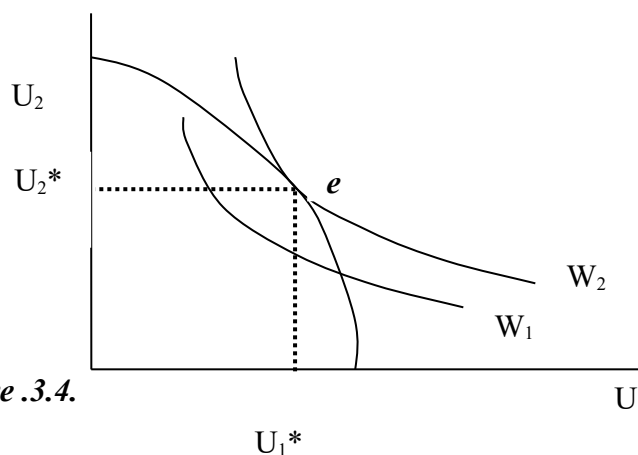


Figure 3.4.

In Figure 3.4 we have picked a Pareto efficient allocation and found a set of isowelfare curves for which it yields maximal welfare. As usual, the optimal point is characterized by a tangency condition. Social welfare is maximized when GUPF is tangent to the highest possible isowelfare curve. The economy maximizes social welfare when it achieves general equilibrium at an allocation which generates U_1^* and U_2^* levels of utility to consumer 1 and consumer 2 respectively. But for our purposes, the notable thing about this maximal welfare point is that it is Pareto efficient-it must occur on the boundary of the utility possibilities set.

5.12 INDIVIDUALISTIC SOCIAL WELFARE FUNCTIONS

Up until now we have been thinking of individual preferences as being defined over entire allocations rather than over each individual's bundle of goods. But, as we remarked earlier, individual might only care about their own bundles. In this case, we can use x_i to denote individual i 's consumption bundle, and let $u_i(x_i)$ be individual i 's utility level using some fixed representation of utility. Then a social welfare function will have the form

$$W=W(u_1(x_1), \dots, u_n(x_n)).$$

The welfare function is directly a function of the individuals' utility levels, but it is indirectly a function of the individual agents' consumption bundles. This special form of welfare function is known as an individualistic welfare function or a Bergson –Samuelson welfare function.

If each agent's utility depends only on his or her own consumption, then there are no consumption externalities. Thus we can have intimate relationship between Pareto efficient allocations and market equilibria: all competitive equilibria are Pareto efficient, and, under appropriate convexity assumptions, all Pareto efficient allocations are competitive equilibria.

7.7. FAIR ALLOCATIONS

The welfare function approach is a very general way to describe social welfare. But because it is so general it can be used to summarize the properties on many kinds of moral judgments. On the other hand, it isn't much use in deciding what kinds of ethical judgments might be reasonable once.

Another approach is to start with some specific moral judgment and then examine their implications for economic distribution. This is the approach taken in the study of fair allocations. We start with a definition of what might be considered a fair way divide a bundle of goods, and then use our understanding of economic analysis to investigate its implications.

Suppose that you were given some goods to divide fairly among n equally deserving people. How would you do it? It is probably safe to say that in this problem most people would divide the good equally among the n agents. Given that they are by hypothesis equally deserving, what else could you do?

What is appealing about this idea of equal division? One appealing feature is that it is *symmetric*. Each agent has the same bundle of goods; no agent prefers any other agent's bundle of goods to his or her own since they all have exactly the same thing.

Unfortunately, an equal division will not necessarily be Pareto efficient. If agents have different tastes they will generally desire to trade away from equal division. Let us suppose that this trade takes place and that it moves us to a Pareto efficient allocation.

The question arises: is this Pareto efficient allocation still fair in any sense? Does trade from equal division inherit any of the symmetry of the starting point?

The answer is: not necessarily. Consider the following example. We have three people, A, B, and C. A and B have the same tastes, and C has different tastes. We start from an equal division and suppose that A and C get together and trade. Then they will typically both be made better off. Now B, who didn't have the opportunity to trade with C, will envy A—that is, he would prefer A's bundle to his own. Even though A and B started with the same allocation, A was luckier in her trading, and this destroyed the symmetry of the original allocation.

This means that arbitrary trading from an equal division will not necessarily preserve the symmetry of the starting point of equal division. We might well ask if there is any allocation that preserves this symmetry. Is there any way to get an allocation that is both Pareto efficient and equitable at the same time?

5.13 ENVY AND EQUITY

Let us now formalize some of these ideas. What do we mean by “symmetric” or “equitable” anyway? One possible set of definitions is as follows:

We say an allocation is equitable if no agent prefers any other agent's bundle of goods to his or her own. If some agent i does prefer some other agent j 's bundle of goods we say that i envies j . Finally, if an allocation is both equitable and Pareto efficient, we will say that it is a fair allocation. Instead of just any arbitrary way to trade, if we use the special mechanism of the competitive market, a trade away from equal division will result in a Pareto efficient allocation. It is also impossible for A to envy B in these circumstances. A competitive equilibrium from equal division must be a fair allocation. Thus the market mechanism will preserve certain kinds of equity: if the original allocation is equally divided, the final allocation must be fair.

Activity

When do we say the allocation is equitable?

7.9. Summary

1. Arrow's Impossibility Theorem shows that there is no ideal way to aggregate individual preference into social preference.
2. Nevertheless, economists often use welfare function of one sort or another to represent distributional judgments about allocations.
3. As long as the welfare function is increasing in each individual's utility, a welfare maximum will be Pareto efficient . Furthermore, every Pareto efficient allocation can be thought of as maximizing sine welfare function.
4. The idea of fair allocation provides an alternative way to make distributional judgments, This idea emphasizes the idea of symmetric treatment.
5. Even when the initial allocation is symmetric, arbitrary methods or trade will not Necessarily produce a fair allocation. However m it turnout that the market mechanism will provide a fair allocation.

KEY TERMS

Aggregation of preferences	Arrow Impossibility Theorem
Envy and Equity Arrow's	Grand Unity Possibility Frontier
Majority voting approach (MV	Benthamite
Welfare function	Efficiency
Fair Allocations	Rank–Order Voting Approach
Social Decision Mechanism	<i>Utility Possibility Frontier</i>
Welfare economics	Welfare Maximization
Social Welfare functions	

CHAPTER SIX

MARKET FAILURE AND GOVERNMENT POLICY

6.1 INTRODUCTION

6.2. EXTERNALITY

6.2.1 TYPES OF EXTERNALITIES

6.2.2. INEFFICIENCY OF MARKET WITH EXTERNALITIES

6.2.3 REMEDIAL MEASURES OF EXTERNALITIES

6.3. COMMON PROPERTY AND ITS TRAGEDY

6.3.1. TYPES OF PROPERTY

COMMON PROPERTY AND ITS TRAGEDY

TRAGEDY OF COMMONS

6.3.2.2 SOLVING THE COMMONS PROBLEM

6.4. PUBLIC GOODS

6.4.1. INEFFICIENCY OF PUBLIC GOODS UNDER MARKET PROVISIONS

6.4.2. DEALING WITH PROBLEM PUBLIC GOOD

6.5. ECONOMICS OF INFORMATION

6.5.1.INFORMATION AS ECONOMIC COMMODITY

6.5.2.Information problems: Asymmetric information

6.5.3.MEASURES TO REDUCE INFORMATION ASYMMETRY

SUMMARY

CHAPTER SIX

MARKET FAILURE AND GOVERNMENT POLICY

6.1 INRTODUCTION

In the previous chapters, we have seen the function of the market and its outcome. The interactions among economic agents (consumers, firms, and government agencies) in market through invisible hand of Adam smith influence their behavior. Market through its incentives in the form of prices, profits, and cost differentials helps economic agent allocate resources efficiently in the society's best interest. For instances, price reductions can change consumers buying habits(increases the consumption and hence utility) where as an increase in prices form high profit margin for the firm that encourages productions of desired goods and services... etc.

But, there are quite a lot of areas where the market outcomes are not as such efficient for the economy. That is, the invisible hand pushes in such a way that individual decisions do not lead to socially desirable outcomes. Economists call this situation a market failure where the market outcomes lead to inefficient resources allocation. There are three common sources of market failure:

A. Externalities

Public goods

Imperfect information

Any time the market fails government interventions are among the possible remedial measures.

In this and the following this chapter we will see the sources of market failure, their consequences and remedial measures in detail. Further more, we will look different forms of intervention as a remedial. To this end, the objectives of the chapter are:

To enable the students with sources of market failures

To acquaint the student with the consequences of market failure

To see the possible remedial measure and public choice problems

Objectives:

Accordingly, at the end of this, the students will be able to:

Explain different forms of externalities.

Show the social marginal cost is different from the private marginal cost when there is externality.

Explain the optimal search rule.

Reason out why and how resource inefficiency in consumption and production lead to overall economic inefficiency.

Mention different forms of interventions

6.2. EXTERNALITY

"There's so much pollution in the air now that if it were not for our lungs there'd be no place be place to put it all"
Robert Orben

Have you ever heard of, faced with or see;

An evil entrepreneur who dumps deadly toxins in children's play grounds,

A firm whose production process lets off fumes that harms its neighbors,

Global warming as a result of the advanced countries industrial emissionetc.

These are some example of externalities. Then what is externality?. What do you think of pollution?

Economics in part has answer to these and other related questions. Every one wants a cheaper and safer environment; activities free of any effect on the other's decisions that lead to self-maximization of objective function and that of the society. However, many activities generate certain effect on people not directly in actions. Those effects are generally unintended and not taken in to account by the doer of the activities. To be more specific when two parties (People, group whatever type of economic agent) pursue their own interest and do some activity, their action may produces unintended effects or impacts on people not directly engaged in the activity (or third party). Economists call the effects on the third party not taken in to account by the doer of the action **externality**. The above are some example of externalities that affects the third party not involved in the actions.

An externality occurs when a person's well being or a firm's production capability is directly affected by the actions of others consumers or firms rather than indirectly through changes in prices as compared to the action determined by the market system. It is the external effect on the third party not involved in the actions. A person smoking in the middle of meetings, for instance, affects those peoples in the meeting who may not have interest for smoking.

6.2.1 TYPES OF EXTERNALITIES

Externality can be of two types based the effects that it has on other economic agents. It may either help or harm others. Accordingly, it is of two type:

Positive externality

Negative externality.

Positive externality: it is a benefit of an activity received by peoples other than those who pursue the activity. For example, honey bees where someone earns his living as keeper of honey bee and his neighbors on all sides grow apples. Because bees pollinate apples trees as they forage for nectars. The more hives he keeps, the larger the harvest will be in the surrounding orchards. Honey bee keeper generates external benefits to the other parties. By installing attractive shrubs and outdoor sculpture around its plants, a firm provides externalities to its neighbors. The person (party) that creates it is not paid for for these benefits (external).

Negative externality: The externalities harm someone. You are harmed if your neighbor keeps you awake by screaming at each other late at night. It is the negative effect of an activity that falls on people other than those who pursue the activity. The activity involves cost on those not directly involved. These negative externalities generate extra cost on the third party for which they have not compensated for the effects. One can mention quite a lot of this type of externalities, but the most common negative externality arise due to different type of pollution. For instance, air pollution and acid rain as a result of burning high sulfur fuels (coals) in industrial plants and electric utilities that destroys vegetation and forests.

Note that: A single action may confer positive externalities on some people and negative externalities on others. The smell of pipe smoke pleases some people and annoys others. Some people think that their wind chimes please their neighbors, whereas anyone with an ounce of sense would realize that those chimes make us want to strangle them. It was reported that efforts to clean up the air in Los -Angeles, while helping people breath more easily, caused radiation levels to increase far more rapidly than if air had remained dirty.

6.1. Activity:

Do you think education have any externality? Suppose someone educate herself /himself, how do he or she benefits himself and his society?

On the other hand, we have said that all economic activities of economic agents are related to production and consumption activities based on their respective objective function in previous chapters. We discussed the conditions for objective function optimization by implicitly assuming the actions of economic agent have no external effect on the others. However, some activities have spillover effect or neighborhood effects. These are two common economic activities or decisions of economic agent that are sources of externalities. In other word, externalities are by products of consumption and production that may be benefits or harm other peoples.

I. CONSUMPTION EXTERNALITIES

It rises if one consumer does not care directly about another agent's production or consumption. These are of two types according to their effect:

A. External economies of consumption (positive externality) that occurs when an action taken by a consumer, rather than a producers results in an uncompensated benefits to others. For examples, pleasure that one get from observing one's beautiful flower garden.

B. External diseconomies of consumptions (negative externality) that occur when action taken by a consumer results in an uncompensated cost to others. For examples, my neighbor playing loud music at 3 hour in the morning or in the even when the neighbor go to sleeping or person next to me in a restaurant smoking a cheep cigarette.

II. PRODUCTION EXTERNALITIES

These types of externality arise when the production possibilities of one firm are influenced by the choice of other firms. Like that of consumption it is either harm or benefit the third party.

A. External economies of production occur when firm's production may benefits the other not directly involved in the actions. For example, training workers who eventually go to work for other firms who do not pay training costs but lead to increased output.

B. External diseconomies of production that arise when the action taken by a consumer results in an uncompensated costs to others. For instance, a firm on the upstream might pollute many parties such as fisherman and irrigation along downstream used by dumping out waste materials.

6.2.Activity:

What type of externality are vaccinations? (Not that helps not only the person vaccinated but also the entire neighbor hood that the person lives in by preventing the spread of contagious diseases)

Suppose certain textile industry is constructed in Ethiopia and it increase the employments, create demand for input producers. What type of externality is it?

6.2.2. INEFFICIENCY OF MARKET WITH EXTERNALITIES

We have discussed the conditions for efficiency in consumption and production and overall economic efficiency under Pareto-optimality. The conditions involved marginal rates of substitution (MRS) and marginal rates of transformation (MRT). The conditions were derived on the assumption that production costs are born only by the producers of the product and the utility derived from consumption enjoyed only by the purchasers. There is no difference between private and social benefits or between private and social costs as all the relevant costs and benefits of an activity accrue directly to the person who carries it out the activity.

In many instances where there is externality (positive or negative externality of production or consumption) these assumption do not hold. Individuals follow their self interest that may not produce the best allocation of resources. Because individuals consider only their own costs and benefits and tend to engage much in activities that generate negative external costs to the society and too little in activities with positive externalities to the society and vice versa. Hence, externalities create divergences between private and social costs and benefits that change economic patterns of production and consumption.

Hence, if externalities exist, our earlier model breaks down for two reasons:

Prices that plays crucial role in determining the equilibrium, losses its significance as they do not reflect all costs and benefits

The Pareto-optimality conditions no longer work as prices provide misleading signals for optimum allocation of resources.

The famous economist A.C. Pigou was the first to present this in systematic way in his book entitled "Wealth and Welfare" in 1912. He argued that in the presence of externalities, even if we have perfect competition we do no achieve Pareto- efficiency. The social benefit or cost is a combination of private and external benefits or costs. We will use the following notation to denote these costs and benefits

MPC = marginal private cost

Similarly

MPB = marginal private benefit

MEC = marginal external cost

MEB = Marginal external benefit

MSC = Marginal social cost

MSB = Marginal social benefit

Hence; $MSB = MPB + MEB$

Hence; $MSC = MPC + MEC$

The overall economic efficiency requires that $MSC = MSB$ for each product. The reason is that as long as $MSB > MSC$, production should be expanded because additional benefit exceeds additional cost. Similarly, if $MSB < MSC$, then production should be decreased since only private cost and benefit is covered by the producers of externalities, the economy will not reach economic efficiency. For economic efficiency consumers and producers must weigh the full social benefits of consumption or production. The effects of externalities on resources allocation can be portrayed graphically under each case that we have seen above.

EXTERNALITIES IN CONSUMPTIONS

Recall that under utility chapter we have assumed that the utility (satisfaction) of one consumer is independent of a consumption pattern of the other. The economic efficiency in consumption is attained when marginal rate of substitution of commodities must be the same for any of two consumers. If we assume there are no externalities in consumption the marginal social cost and marginal private costs are equal, and the competitive supply curves reflect the common marginal cost. Similarly, marginal social benefit and marginal private benefits are equal, and the demand curve reflects the benefits.

However, in case of externality the two graphs differ. In the case of consumption, we have explained above that there are two forms/types of consumption externalities (negative and positive). If we assume there are no externalities in consumption is marginal social cost and marginal private costs are equal, and the competitive supply curves reflect the common marginal cost. However, on the demand side, the demand curve reflects only the marginal private benefit (MPB) in which the economic agent does not take in to account the externalities. Hence the marginal social benefit (MSB) differs from MPB.

External benefits = social benefits - private benefits or

We will look at the impact of the difference between MPB and MSB on resource allocation under two scenarios.

NEGATIVE EXTERNALITY OF CONSUMPTION.

Recall that the marginal private benefit is the demand curve for the firm and the marginal cost is the supply curve of the firm as well as the society. In this case the marginal social benefit (MSB) is below that of marginal private benefit (MPB) as can be seen from the figure 6.1-A below. The reason is that the economic agent consume something that result in lower benefit to the society than to the private as he/she pursue his self-interest. There is an over production of the commodity as the individual is not take in to account all the cost and benefit. Accordingly, the amount of goods and services an economic agent want to consume is at point where $MPB = MPC$ which is point Q_1 at price is P_1 is the market equilibrium where as the optimal quantity for the society is Q_0 (where $MSB = MSC$) with the corresponding C_0 .

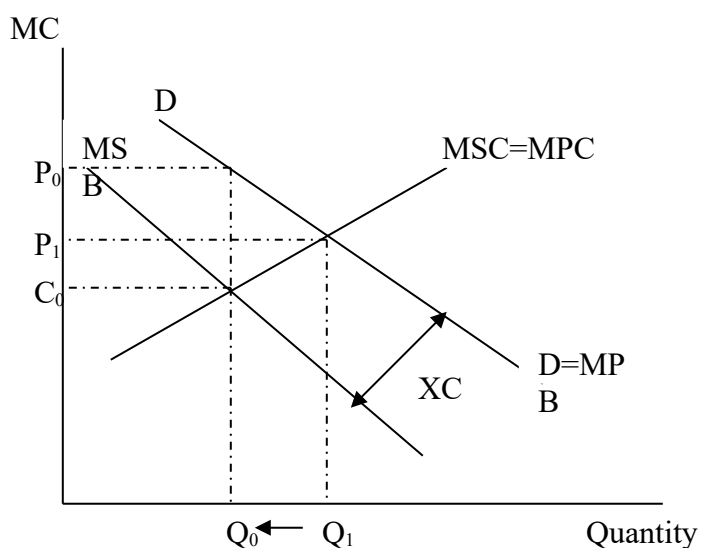


Figure 6.1-A: Negative externality in consumption

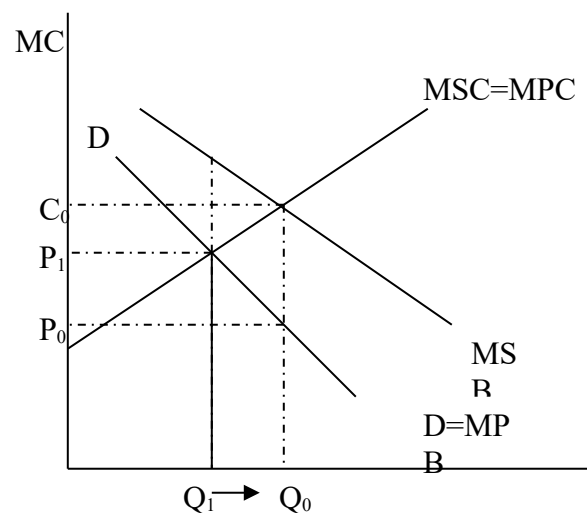


Figure 6.1-B: Positive externality in consumption

At the socially optimum point the $MSB = MPB + MEB$ that is $P_0 = C_0 + C_0 P_0$ over the entire private demand curve; the external social cost is XC at all level of production. Hence it is not at the socially optimal level of production as it leads to overproduction on the part of private.

B.POSITIVE EXTERNALITIES OF CONSUMPTION

Again, the MSC (equal to MPC) curve is the supply curve. The demand curve D is the marginal private benefit that is different from that of the society's as shown by figure 6.1-B above. Since there are external benefits, $MSB > MPB$, and the MSB curve was above the demand curve. The socially optimal quantity is given by Q_0 where $MSB = MSC$. However, for the private economic agent point of view the equilibrium attained when Q_1 is produced with corresponding prices is P_1 which is smaller than the socially optimal level. Hence, there is underproduction as compared to the socially optimal level. The socially optimal, Q_0 is the output level at which $MSC = MSB$ is obtained by adding the external benefit, XB, to every value along private demand curve. That is, to produce at sociality optimum (Q_0) the consumer need to be compensated (subsidized) by the amount equal to $C_0 - P_0$. (This will be covered under legal remedies topic).

Thus, in both cases private equilibrium under condition externalities fails to exhaust all possible gains from consumptions in such a way that the societies reach its equilibrium (Socially optimal quantity).

EXTERNALITIES IN PRODCUTIONS

Under ideal circumstances producer's private cost or benefit will encompasses all the attendant social costs and social benefits. Hence, the production decision will be consistent with our social welfare. Unfortunately, this happy identity does not always exist as social cost differ from private costs by certain amount termed as external social costs.

$$\text{External cost} = \text{social costs} - \text{private costs}.$$

In this case the price signal confronting producers is flawed by not conveying the full (social) cost of scarce resources, the market encourages excess pollution and we end up with non optimal mix of output and wrong production process. Let us consider first the case of negative production externalities and then positive production externalities.

A.NEGATIVE EXTERNALITY IN PRODUCTION

Figure 6.2(A) illustrate the case of negative externality in production. Assume there are no externality in consumption, the demand curve D shows the marginal private and social benefits ($MPB=MSB$) as identical. The competitive supply curve, however, reflects only the marginal private costs(MPB) but the MSC curve lie above the competitive curve as it includes marginal external costs to every volume along private MC by “ XC .” The optimal output is Q_0 with a price P_0 . At this level all possibilities from exchange is exhausted. But the competitive market, if it left alone, will produce Q_1 , with a price of P_1 . Thus, there is a tendency to over produce by XC amount.

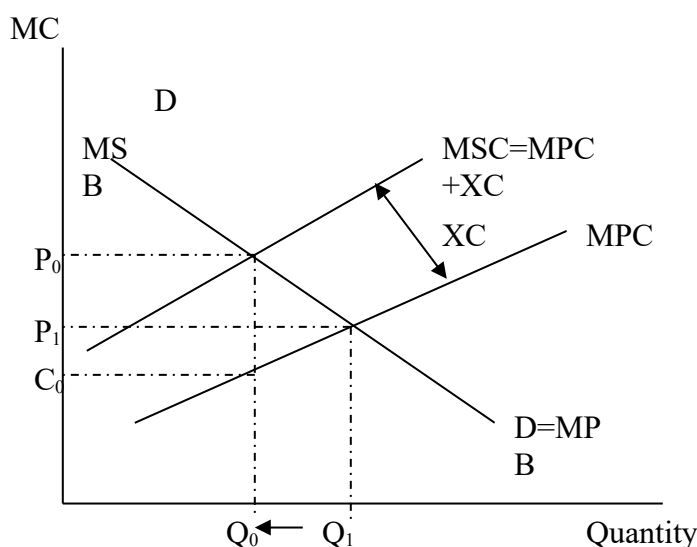


Figure 6.2-A: Negative externality in production

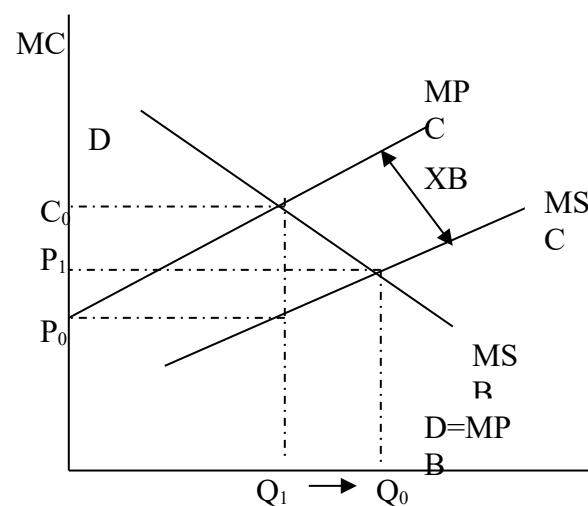


Figure 6.2-B: Positive externality in production

B. POSITIVE EXTERNALITIES IN PRODUCTION

This case is illustrated in Figure 6.2 –B above. The demand curve represent identical the marginal social and private benefit curve, but supply curve differs in that the MSC curve is below the MPC curve ($MSC < MPC$). The socially optimal level of output is given by the intersection of the demand curve with the MSC curve as shown by point ‘e’ with Q_0 output and P_0 price. However, if left alone the competitive market will produce Q_1 with price P_1

where the demand curve intersect the MPC curve, Thus, too little will be produced from the social point of view. At output level Q_0 , producers received a price of P_0 , but their marginal cost is C_0 the difference between the two is the amount BX is the external benefit generated from the action of economic agent in production.

In general, no matter whether externalities are positive or negative they distort the allocation of resources in otherwise inefficient markets. When externalities are present, the individual pursuit of self interest will not result in the largest possible economic surplus for the society.

6.2.3 REMEDIAL MEASURES OF EXTERNALITIES

We have seen that externalities lead to inefficient allocation of resources. What are the measures to be taken to reduce the inefficiencies generated by externalities? What are government policies or concerned bodies' action to reduce its effect?

Under ideal conditions a firm or an economic agent that generate external activities must include external costs or benefits in its calculations of private gains and costs. The internalization can be through private price tags placed on external costs (or benefits) so that private and social costs (or benefits) coincide. Other possible way is through quantity of product adjustment in which one can increase or decrease depending on the type of externalities. For instances, pollution can be reduced either by reducing the activity of the polluting agents or paying the cost of pollution abatement. What is the optimal level of externality? What is the optimal level of pollution abatement?

The optimization of externality is just like the production and consumption issues we have discussed so far employ the concept of marginality analysis. In dealing with any form of externalities the cost and benefits matters and optimal level of production depends on the

marginal cost and benefits of producing additional good. It is until the marginal external social benefit of externality is equal to the marginal external social cost.

This can be illustrated with our pollution examples. Figure 6.3 shows the marginal social costs and marginal social benefits of different amount of pollution abatement – the number of gallons of water purified through filtration and how it is determined. The marginal social cost of each successive unit of abatement increases as more and more water is purified.

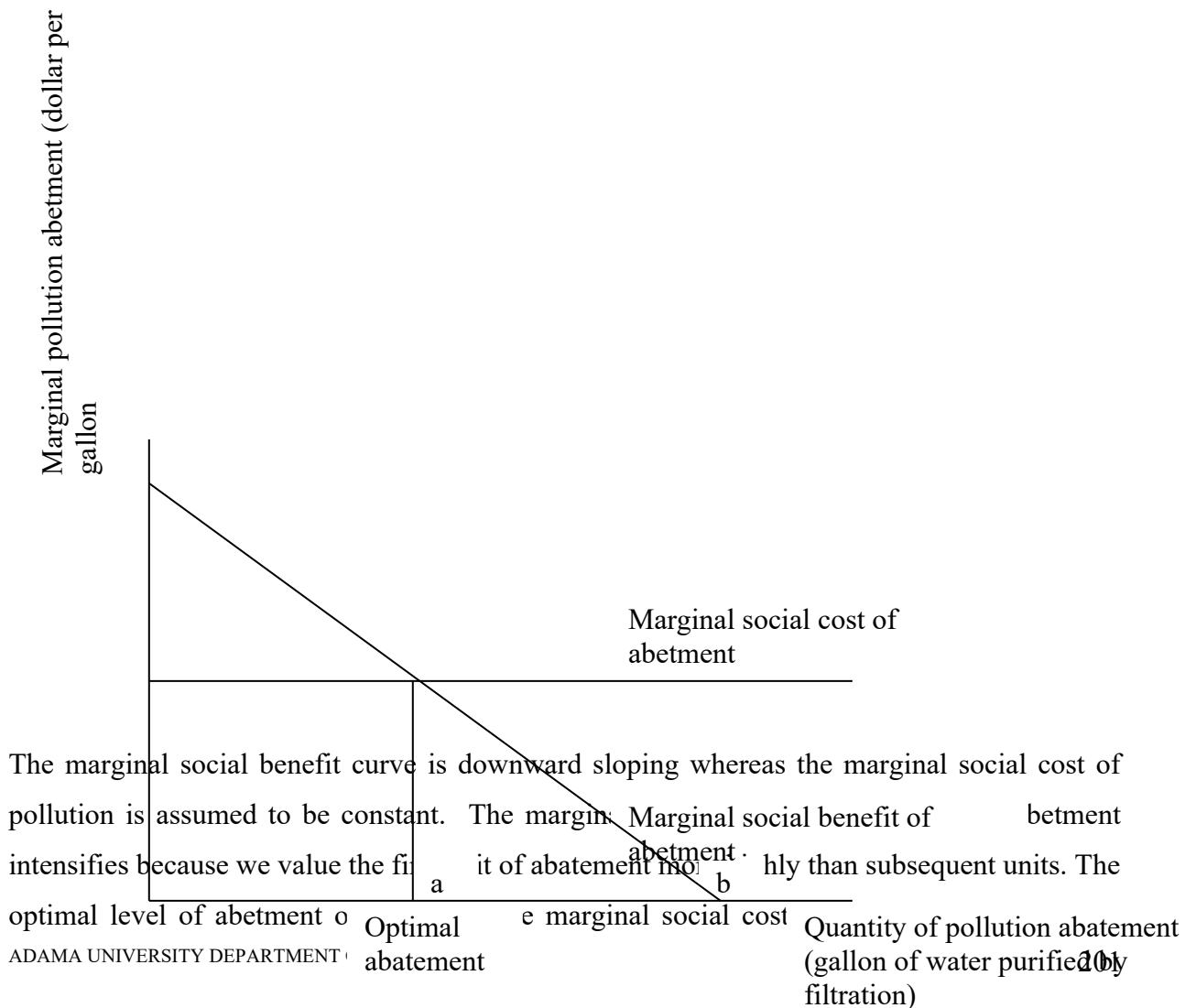


Figure 6.3 : Optimal pollution

equals its marginal social benefits. If the abatement is undertaken beyond this optimal level, the extra benefits fall short of the extra opportunity costs. On the contrary, if the abatement is undertaken below this optimal level, the extra costs fall short of the extra opportunity benefits.

From our analysis one may think whether the optimal level of pollution should be greater than zero. The optimal externality is greater than zero does not mean any positive level of negative externality (pollution) is good. It is merely to recognize that society has an interest in cleaning up the environment, but only up to the certain point. If you aim for the total elimination of pollution we would have to devote more and more scarce resources to abatement, until the marginal social benefits are zero. Hence, the last dollars spent on pollution abatement will yield zero benefits and will have a high marginal cost.

The idea can be seen from dirt in an apartment even if you spent the whole day, every day, vacuuming your apartment, there would be some dirt that left in it. You probably tolerate substantially more than the minimal amount of dirt. Further more, the society will incur cost that exceed the benefit by trying to make it zero. However, the efficient level of pollution would be zero if the pollutant were so damaging that the marginal cost of even the first unit released into the environment exceeded the marginal cost of not allowing releases.

The above optimum level of externality can be attained through internalization of the cost and benefits as we indicated above. The internalization can be accomplished by the

Redefinition of property rights

Voluntary agreements

C. Government action

A. REDEFINITIONS OF PROPERTY RIGHTS

Poor definition of property right is a prominent source of externalities. Property rights are social arrangements that govern the ownership, use, and disposal of resources, goods and services in modern society. It is a legally established title that is enforceable in the courts. If we do not know who owns the property right to the river, there is no one to limit its use and

resources will likely be exploited and abused. For instances, fishing business will be over fishing ocean water, factors will pollute more...etc.

However, if private property right can be clearly defined, the private owner will make sure that the resource is efficiently used as (the assigning property rights) give an exclusive privileged to use assets. For instances, by owning this test book (module), you have a property right to read it and to stop others from reading or taking it. If you had a property right that assured you of the right to be free from noise pollution, you could get the courts to stop your neighbor to play the music. The owner of hunting land can set fees high enough to prevent games from being depleted. The assignment of may not matter on individual preferences.

B.VOLUNTARY AGREEMENT

Voluntary agreements are another means of internalizing externalities. When property rights are well defined, voluntary agreements can internalize external costs. The proposition that voluntary agreements can solve externality problems was first introduced by R.H. Coase and termed as Coase theorem. He profoundly changed the way economists, legal scholars, politicians, philosophers and others think about externalities and the legal and social institutions that have evolved to deal with them. The Coase theorem states that externalities (costs/benefits) can be internalized by negotiation among the affected parties as long as the cost of negotiating is very small or null and the numbers of parties are small.

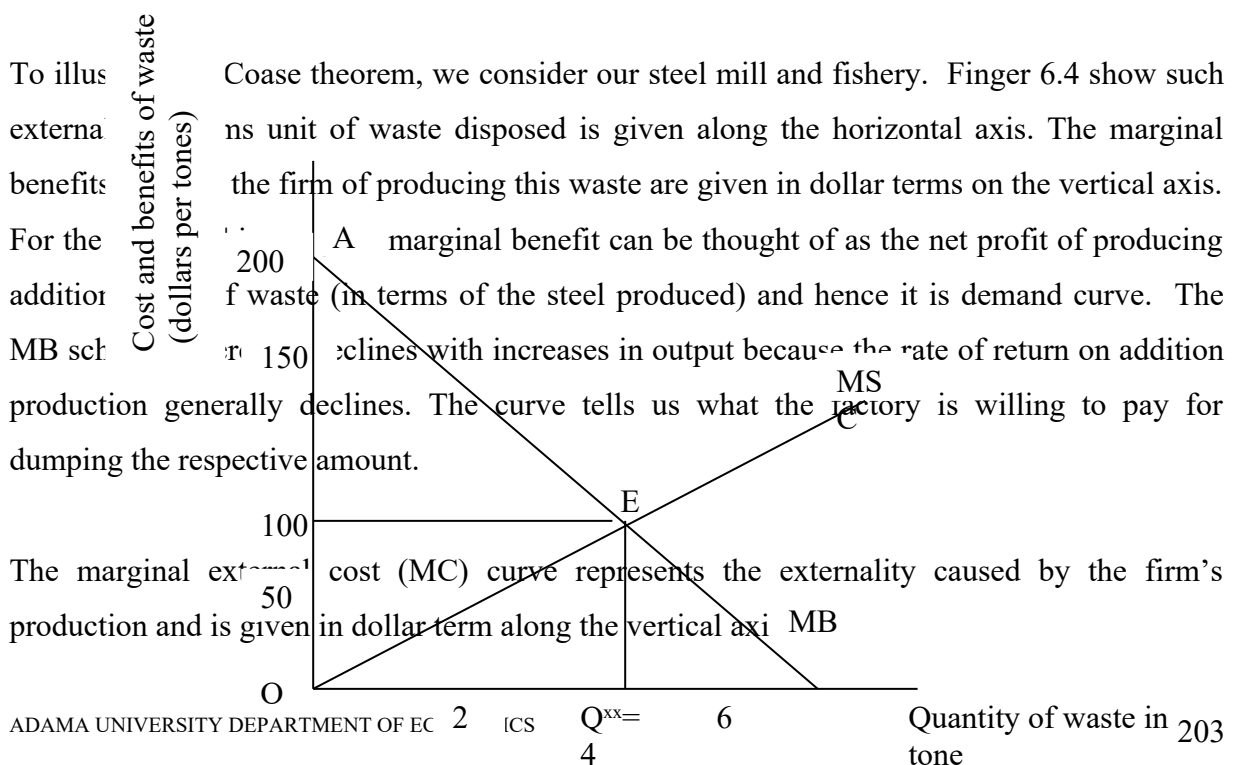


Figure 6.4: optimal waste based on voluntary

Now we apply Coase's analysis. As an experiment, we give the ownership right to fishery. Assuming that bargaining cost is nothing, we can predict what will happen with the help of figure. For the amount of waste up to, we observe that $MB > MSC$. The firm's profit on additional units of waste are greater than the additional pollution cost born by the neighborhood (fisher) would be willing to sell the right of using the water to produce these units. Since the total cost of the externality caused by this output is represented by the area under the MSC curve, a bargain can be truckle between the firm and the fishery to produce by agreement on how to divide the surplus marginal benefits AEO. The factor can pay the fishery and hence both can gain from voluntary agreement up to $= 4$ waste disposal.

Beyond $MSC > MB$, a bargain to produce these units cannot be reached between the firm and fishery as firm would not be willing to bid enough to obtain the right to produce these units hence, the fishery would not sell the right. So Q, where $MB = MSC$ is the equilibrium outcome when the neighborhood is given the transferable right to use of river.

Suppose that a judge decided instead to award the firm the right to use the air as it choose. What happens? We go through the same analysis, but now we see that the fishery must pay the firm to produce less waste. For up to units of output (waste) $MB > MC$, implying the fishery cannot offer a large sum of money to induce the firm to reduce it. But beyond Q, firm would be willing to accept any offer but the fishery is not willing to make it as it costs him more. It is worth while for firm to reduce pollution up to where $MSC = MB$. The result shows that no matter who

has the legal right to use, the amount of pollution is the same. This is the central insight offered by the Coase theorem.

C. GOVERNMENT ACTIONS

Many real world externality problems can be analyzed with Coase theorem, as long as the bargaining and transaction costs are low. Nevertheless, the parties may not be able to bargain successfully for at least three important reasons. First, transaction cost is not very small, if the numbers of plants are numerous, it is difficult to organize and reach on the solution. Imagine the transaction costs if 50 factories and 10 fishers as compared to the one we have seen so far. Second, if firms engage in strategic bargaining behavior, an agreement may not be reached. Third, if either side lacks information about the costs or benefits of reducing pollution, a non efficient outcome may occur. It is difficult to know how much to offer the other party and to reach an agreement if you do not know how the polluting activity affects the other party. For these reasons, Coasian bargaining is unlikely to occur in relatively few situations. So the cost of defining and enforcing a system of ownership can be so high that some system could be adopted.

Under these condition government intervention is one of the option for solution. The government can cope with the problems in two ways /approaches.

A. Regulatory approach

B. Incentive based approach

Regulatory approach

The government requires externality producing agent (pollution agent) to limit the level of externality to a certain level. Usually the limits are set to the optimal level based on the marginal or the cost -benefit calculation analysis. There are two major approach through which the government can cope with externalities problems under these . These are:

I.Limiting the level of externalities

Incentive based approach

I. Limiting the level of externalities

Under this case the government limits the amount of externality generated by each economic agent. Each factory, for instances, would have a limit on the amount of pollutant allowed to discharge. Let's substantiate our analysis with our specific examples mention above using graph. Figure: 6.5 below show the total social cost of each level of discharge of an industry's wastes. The more, the industry cut down on the amount of waste it discharges, the higher are its costs of pollution control at each low of discharge of the industry's wastes.

Specifically, the efficient level of the pollution in the industry is R. Because increasing pollution from a level lower then R would improve social welfare. Discharging one more unit of pollution would increase the cost of pollution, but it would reduce the cost of pollution control by more. Hence, the optimal level of pollution is R.

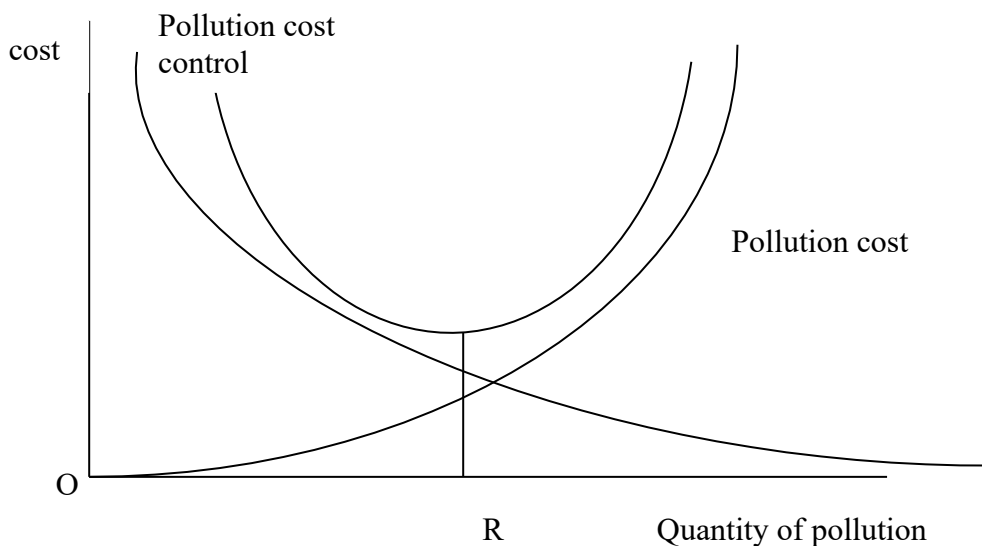


Figure 6.5: optimal pollution discharge

In the United state, various state, locals, and federal agencies are responsible for pollution control. The agencies derive its legislatives authority from a number of Congressional Acts, including the Clear Air Act, the Water Pollution Control Acts, the National Environmental policy....etc. In enforcing Federal Environmental laws, the Environmental Protection Agency (EPA) has usually followed the regulatory approach. It specifies standard for wastes discharge with respect to air, water, and noises pollutions. It issues permits setting ceiling on the amount of pollution discharged.

6.3.Activity:

Is there such type of approach in our country? If there is what do you think the policies?

II. Incentive based approach

This is the case where governments uses economic incentives or penalties (such tax or subsidies) to encourage polluting agents to restrict emissions to efficient levels.

Some of these are:

Emission charges

Tax and subsidies

Emission charges

The government (the regulatory agency established by the government) sets the emission charges, which are in effect, a price per unit of pollution. The more pollution a firm creates, the more it pays in emission charges. To work out the emission charge that achieve efficiency, the regulator must determine the marginal social cost and marginal social benefit of pollution. The optimum level is where price per unit of pollution must be set to make the marginal social cost of the pollution equal to its marginal social benefit.

In practice, it is hard to determine the marginal benefit of pollution, and people who are best informed about the marginal benefit, the polluters, have an incentive to mislead the regulators. As a result, if a pollution charge is used, the most likely outcome is for the price to be set too low.

Government taxes: Efficient fee

Effluent fee is certain money that a polluter must pay to the government for discharging waste. The tax system is a mechanism through which the amount of external costs (the deviation between private and society) is imposed on private that create externality. Taxes provide

incentives to producers or consumers to cut back on activities provide that create external cost. An appropriate tax with externality will force the business firm to take in to account the costs imposed on others. Accordingly, reduces the amount of production to the efficient level where marginal social costs equal marginal social benefits.

6.3. COMMON PROPERTY AND ITS TRAGEDY

6.3.1. TYPES OF PROPERTY

There are different types of goods and services that can be produced and consumed by economic agents. Some of these goods and services are termed as private goods, common goods, collective goods and public goods. They are classified as such based on their property: varying degree of exclusion and rivalry. Rivalry means that only one person can consume the goods: the good is used up in consumption-it is depletable. If an individual consumes a particular quantity of a good, such as apples, these same apples are no longer available for the others to consume. Exclusion means that others can be prevented from consuming the good. Only the person who owns the candy bar may eat it. For instances, when you eat a cheeseburger, it is no longer available for any one else. More over, people can be easily prevented from consuming cheeseburgers that the individual do not pay for.

The classification scheme defined by the non rival and non excludable property is summarized on table XY. These are explained below in four classifications with relevant examples. The columns of the table indicate the extent to which one person's consumption of a good fills to diminish its availability for others. Goods in the right column are non-rivals and those in the left columns are not. Likewise, goods on the top are highly non excludable as compared to the

bottom once. The two hybrid categories are common goods which are rival but non excludable and collective goods which are excludable but nonrival.

The first type of good is private good. Private goods have the properties of the rivalry and exclusion. Private goods are on the upper - left column. Second, there are resources with rivalry but without exclusion. These are most of the called common good or common property resources. Common goods (pure) is one for which non payers can not easily be excluded and for which each unit consumed by one person means one less unit available for the others. In an open access ocean with fishery, any one fish (no exclusion), but once a fish is caught, no one else can catch it (rivalry).

Table 6.1: Private, public and hybrid goods non rival

	Exclusion	No exclusion
Rivalry	<u>Private goods</u> - wheat - candy bar	<u>Commons goods</u> - fishery - hunting - highway
No Rivalry	<u>Collective goods</u> - Pay - per - view TV - Club goods (concert, tennis club)	<u>Public goods</u> -national defense -aerial spraying pesticide, -clean air

The third type of good is collective goods. They are goods with strong non-exclusive and non-rivalry. Club goods (concert, tennis club) are typical example of these types. For instances, club-goods, security guards prevent people who don't have a ticket from entering a concert hall. Until the concert hall is filled, the cost (marginal) of providing the concert to one extra person is zero. Adding another person creates congestion or other externalities that harm concert goers once the concert hall is filled. Similarly, allowing more people to join a swim club does not inflict extra costs until members start getting in each other's way. These are goods with both private (exclusion) and public property (non- rivalry). These are collective goods .That is goods or services that at least some degree is non rival but excludable (exclude non payers). They are provided sometimes by government, sometimes by private companies

The fourth types of resources are public goods .Public goods are those goods or services that are in varying degrees, non-rivalry and non –excludable. The Good that are highly non-excludable and non -rival are often called pure public goods. First, some public goods are exclusive but lack rivalry in consumption. The consumption by one person does not diminish its availability for others. That is, once the good is provided for someone other can also consume it at no extra cost (the marginal cost of providing the good to the additional consumer is zero). And it is technically difficult or extremely costly to exclude non payers from consuming it. For example, the national defense, Law enforcement, fire and police protection, and flood control, the Ethiopian Radio broadcasting... etc. The fact that it is a public good does not necessarily mean that government ought to provide it. Some times, private companies can find it profitable way of producing goods with both characteristics.

Some goods even changes characteristics over time and some vary back and forth. Consider, for example, an uncrowned bridge. If you crossed an uncrowned bridge, it would not interface with my crossing, so the use of this bridge could be labeled "no rival" .It is, however, not a non exclusive good because it is perfectly feasible to charge a fee for crossing the bridge and to prevent people who do not pay from crossing it. At the peak times during the day, it is not even non rival; because your crossing during a congested time could easily interfere with my crossing. Markets for public goods exist only if non purchasers can be excluded from consuming it.

Other goods lack rivalry or exclusion or both. Furthermore, many goods differs in the degree to which the have non -rivalry or non- exclusive and hence their name. Many goods are hybrids, with properties of both private and public goods not all non-trivial goods are non-exclusive, and not all non exclusive goods are non rivals. Most of them are exclusive but lack rivalry in consumption.

6.4. Activity

Discusses the above type of property resources with examples that you think in your environments?

What type of is fishing in lake Zeway if you assume every body has free access to resources?

COMMON PROPERTY AND ITS TRAGEDY

6.3.2.1. TRAGEDY OF COMMONS

So far we have examined externalities that arise as by product of a production or consumption activities mainly related to effect on individual economic agent be it producer or consumer (third party). Other important externalities arise with common property- resources to which everyone has free access. Unlike private property, for which the owner can exclude others from using the property right, common property is not subject to such exclusions. This principle applies both to the renewable resources, such as fish, and non renewable resources such as oil. Some other examples of common property resources are common pools, Internet, road, & fisheries.

Common pool: common parking, petroleum and other fluid and gases are often extracted from a common pool. Those individual using the resources (common pool) compete to remove the substances most rapidly there by going ownership of the good. This competition creates externality by lowering fluid and other resources through opposing pressure. Free common parking where every one can freely enter and enjoy urban parks such as a central park in city. Parks with fee entry often become crowded, an outcome that reduces every one's enjoyment. The most common resources with this property in developing countries like ours are: hunting, grazing land on which the economic agent over utilize the resources.

The Internet: There is overcrowding on the Internet for a flat fee that does not vary with the number of hours of use. The Addis Ababa University where students are using for many hours can be cited.

Roads: if you own a car, you have a property right to drive that car, but you lack a property right to the highway on which you drive, you cannot exclude others from deriving on the highway and must share it with them. Each driver, however, claims temporary property right in a portion of the highway by occupying it. Competition for space on the highway leads for congestions a negative externality), which slow every drivers speed and hence, lot of emission of carbon.

Fisheries: many fisheries men have common access to ocean such that everyone can fish and no one has a property right to a fish until it is caught. Each fisherman wants to land a fish before others do so to gain property right to that fish. Each fisherman has a negligible impact on the total stock of fish, but the cumulative efforts of thousands of fishermen result in serious depletion. This tendency to overfish has diminished many fish populations dramatically as the rate at which fish are being born is lower than the rate at which they are being caught. This is one of the typical problems with Ethiopian lakes (example Zeway lakes).

In each of these cases no one owns the resource, no one has any incentive to invest in it in a productive way or take the opportunity cost of using it into account. Individuals neglect their effects on the resource's productivity and hence exploit it. Because people will tend to use it until its marginal benefit is zero. This property of the goods makes the good a rival good and also non-excludable that always results in **tragedy of the commons**. These problems, and others similar to it, are known as Tragedy of commons. There is a tendency for a resource that has no price to be used until its marginal benefit falls to zero.

The essential cause of the tragedy of the commons is that one person's use of commonly held property imposes an extra cost on others by making the property less valuable. The outcomes of economic interaction with the tragedy of commons are inefficiency in resource allocation. We will pose this problem in the original content of a common grazing land, although there are many other possible illustrations. Consider a peasant economy in which the villagers graze their cows on the private and common fields under different scenarios.

We will see allocation of resources under private ownership and common grazing land. First, assume the grazing land is owned by private (private ownership) who rears cows with the main intention of dairy production. Suppose that it costs 'a' dollars to buy a cow. How many cows should the person decide to graze? How much milk a cow produces will depend on how many other cows are grazing on the grazing land. We will let $f(c)$ be the value of the milk produced if there are 'c' cows grazed on the common. Thus the value of the milk per cow is just the average product, $f(c)/C$. What is the number of cows that maximizes the total wealth? In order to maximize the total amount of wealth, we set up the following problem:

$$\text{Max } f(c) - ac$$

Our condition for maximization of production will occur when the marginal product of cow equals its cost, a , $MP(C) = a$ which is the competitive equilibrium. Hence, the common grazing ground were owned by someone who could restrict access to it, the solution it would be optimum. Each villager has a choice of a cow or not grazing one, and it will be profitable to graze a cow as long as the output generated by the cow is greater than the cost of a cow.

If there are c cows are currently being grazed so the current output per cow is $f(c)/C$. To add a cow the villagers look at the total output $f(c+1)$, and the total cow number of cow $(k+1)$. Thus, total average revenue per cow is $f(c+1)/(k+1)$. He compares revenue with cost of the cow, a . If $f(c+1)/(k+1) > a$, it is profitable to add the cow since the value of the output exceeds the cost. Then each village chooses the point.

On the contrary, the grazing land is owned communally and the cost is the same as that of the above. Everybody is free to enter (non exclusion) as long as it is profitable to graze a cow on the common field, villagers will purchase cows. They will stop adding cows to the common only when the profits have been driven to zero.

6.5. Activity

Is there any common property in your surrounding? If so, what are they? How do individuals are using the resources?

6.3.2.2 SOLVING THE COMMONS PROBLEM

There are two approaches to ameliorate common problems. These are;

- A. Direct government regulation through taxation or restriction of access
- B. Property right

A. REGULATION

Overuse of common resources occurs because individuals do not bear the full social cost. However, by applying taxes or fees equal to the external harm that each individual impose on

others, a government forces each persons to internalize the externality. Some of practical experience that can be cited as example may be; government often charges an entrance fee to a Art or Museum. Another example is related to congestion. San Francisco- Oakland Bay Bridge in California among the most congested bridge in the world. It is about 8.5 mile long and out of 275,000 vehicles that drive across the bridge in a day 100, 000 cross the bridge in one direction in the morning. But the bridge can handle only 10,000 commuters per hours in one direction without congestion. The bridge's morning traffic jam is costly in terms of time, money and pollution. Time lost is approximately 20 minute crawling across a cross the bridge at low speed. Gasoline is wasted, and tail pipe pollution is 250% greater under congested condition than when cars drive at the speed limit.

The optimal level of toll is estimated to be \$ 3.6 at peak hour. Although ,politician does not imposed it because of its welfare effect on the poor in that it help wealthier commuter at the expense of harming poorer, starting from 1980, \$ 1 toll was collected at all hour in one direction. Imposing a lighter toll to cross the bridge during rush hour would reduce congestion by discouraging some current drivers.

Alternatively the government can restrict access to the commons. One typical approach is to grant access of a first - come - first - served basis on which people who arrive early get access. In this regard as The New York Times reported overusing has decimated the stocks of cod, hand dock and flounder that have lack sustained. As result The New England Fisheries Management Council is attempting to alleviate the problem by banning new entry to the industry requiring fishermen to limit their days at sea and increasing the mesh size of their nets.

B. Property right

The other solution may be assigning property right to private that removes the incentives to overuse the resources. That is converting the common access property to the private property. In many developing countries over the past century have the common agricultural land has been broken down in to smaller private farms. As a result the overexploitation problems have been reduced.

In sum, we have seen that if every thing that people care about is owned by someone who can control its use and, in particular, can exclude others from overusing it, a Pareto efficient outcome could be reached. However, if a government set a tax or fee that is less than the marginal externality harm, it reduces but does not eliminate the externality problem.

6.6. Activity

1. How do the societies solve the problem? How do you relate the societal solution with what we have learnt?
2. Assume there is a common pool of water resources called 'Birka' in a certain village and the cost per m³ is 50 centes. What happens to the resource when there is no restriction to access beyond the limit? If the above amount of charge is imposed on an individual, what do you think happens to the utilization of the resources?

6.4. PUBLIC GOODS

In the previous topic we have seen production good consumption externalities and tragedy of commons. As an example, smokers, fishers and chemical factory, problem of grazing common land, and the way to deal with them such as assigning property right, voluntary agreement and so on.

Unfortunately, not all externalities are only limited to them and can be handled in that manner. As long as the number of economic agents increases (more than two economic agents involved), things become much more difficult. Suppose, instead of one factory (steel mill) and one fishery, the number of factories producing chemicals increases to about 50 or 100 and fishers 20 or 50. Then the externalities that result as the number of producers increases become common for the country, a certain portion of the society. They are particularly troublesome kinds of externality, for the decentralized market solution and costly even for non-market solution that we have discussed so far with the other forms of externalities. This is an example of a good with the characteristics of public good.

Having said these, it is appropriate to ask questions related to public goods.

Some of these are:

What are public goods?

How do we identify public goods from private goods?

What are the optimality conditions for the provision of public goods?

Who is responsible for the supply of public good and how?

As it is explained above public goods are those goods or services that are in varying degrees, non-rivalry and non –excludable. Consider the case of national defense. A specific citizen's interest in the nation can be protection by the nation's military without reducing the protection that it offered to any other Ethiopian citizens. Further more, once a country has created military establishment, all of its citizens enjoy its protection at the sometime. Since there is no practical way of excluding citizens from its protection, national defense is non exclusive good.

6.4.1.INEFFICIENCY OF PUBLIC GOODS UNDER MARKET PROVISIONS

Markets for the public goods exist only if non purchasers can be excluded from consuming them. Thus markets do not exist for nonexclusive public goods. The market tend to produce too little of an exclusive public good because of the lack of rivalry. In the absence of rivalry, the marginal cost of providing public good to one extra person is zero. Firms have no incentive to produce at a zero prices, if firms set a price above zero, consumers buy too little of this public.

This is a complicated issue whose answer is best explored initially with in the context of partial equilibrium analysis. However, we can illustrate with simple diagram.

We have constantly argued that the competitive market will provide optimal quantity of private goods because output will be expanded just to the point where $D=S$.And the demand curve represents the social benefits of additional units MSB and the supply curve reflect the marginal social cost of production MSC.

Suppose, to this and for simplicity, that there were only two consumers, the Ababe family and Bekele family. Let D_A in figure 6.6 be the Abebe's family demand curve for some goods, D_B be the Bekel's demand curve for the same good. The supply curve, S , is also noted. If we assume the good in question is private good we follow an ordinary efficiency criteria of

competitive market. The optimum quantity of a private good is at point when market demand for the product equals the market supply for the product.

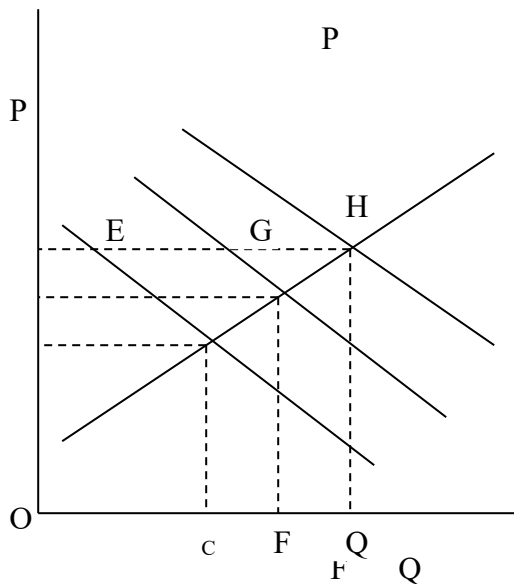


Figure 6.6. A. private good

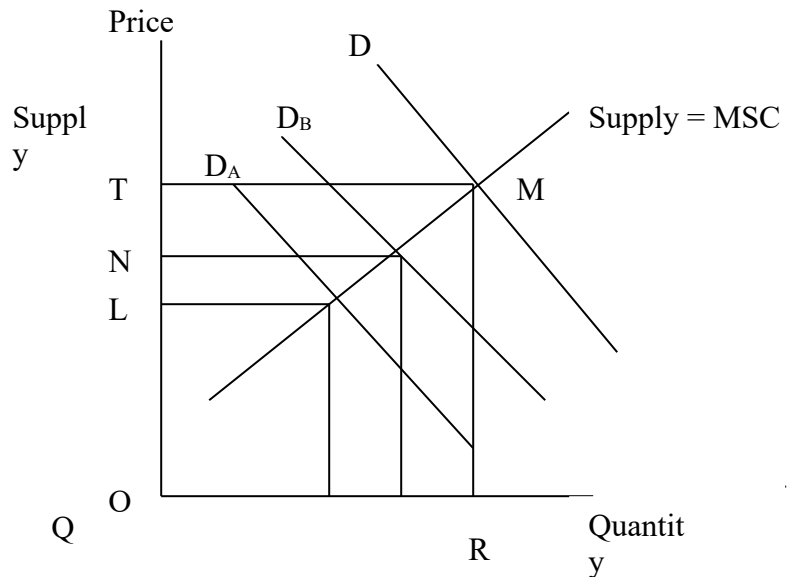


Figure 6.6.B. Public goods

We would then derive the market demand curve D by summing horizontally the demand curves of the two consumers. Furthermore, the social marginal benefit of a private good is the same as the marginal benefit to the individual who consumes that good. Hence, the demand curve represents the full social benefits of additional units MSB and the supply curve reflects the marginal social cost of production MSC . The efficient output could be Q , where market demand intersects the market supply curve. At this point the marginal benefit that each consumer would obtain from an extra unit of the good would equal its marginal cost.

If on the other hand, the good were public goods, though the efficiency /optimality criteria are the same, there is modification of the supply/demand model used. The key to understanding the difference is to recognize that once a pure public good is supplied by one individual, it is simultaneously supplied to all, and whereas a private good is supplied only to individual who purchased it. In contrast with the case of private good, where the social marginal benefit of a

public good is the sum of the marginal benefit to each person who consumes the good, it is marginal social benefit of one individual. Because a public good lacks rivalry, many people can get pleasure from the same unit of output. As a consequence, the social demand curve or willingness to pay curve for a public good is a vertical sum of the demand curves of each individual.

In this case, the market demand curve would be obtained by summing the individual demand curves vertically as a given amount of public good can be consumed by both of our consumers (in our example) at the same time. The combined prices that indicate willingness paid by the two consumers for the provision of good should be the sum of the prices that each would pay individually.

Thus, aggregate demand curve for product X, (D) is obtained by vertical summation of D_A and D_B . Given the market supply curve (S_X) for product X, the optimal amount of X is R unit per period at point M. To see this point more precisely, recall the marginal benefit of each consumer from the good is reflected by the individual demand curve. If output is R, then the marginal social benefit from an extra unit of output would be vertical sum of L (the marginal benefit to Abebe) and N (the marginal benefit to Bekel); that is $L+N = T$. It follows that R must be efficient output, since marginal social benefits at R (OT) equals the marginal social cost at R (RM).

The above analysis reminds us that economic efficiency for a private good requires that each consumer's marginal benefit equals marginal cost. As shown in panel A. By contrast, panel B, shows that economic efficiency for public good requires that the sum of marginal benefits of all consumers must equal marginal cost.

$$MB_A + MB_B = MC$$

Thus, note once again the marginal principle is at work. The problem is that in general, less than the optimal amount of public good will be supplied under perfect competition, and this prevents the attainment of maximum efficiency and Pareto-optimum.

When public goods output should be expanded to the point where the $MSB=MSC$, but the market demand curve which is derived by vertical summation of individual demand curves is no longer reflective of the MSB. In addition, even if people could be selectively

excluded from consuming it, the non rivalry in consumption means that it is inefficient to exclude any one. Because it costs nothing to provide the good to each additional consumer after the first social welfare is maximized by giving away the good free. This implies that once a good is produced, the marginal cost of provision of the good to any additional individual is zero. These create an incentive for the economic agent not to pay for the good that lead to **free rider problem**. That is many peoples are unwilling to pay for their share of public good. They try to get others to pay for it, so that they get benefit from the actions of the other without paying. In this case, individuals know that it is possible to get it free once as someone also pays for its provision, this is similar but not identical to the prisoner's dilemma we examined so far.

Let us construct a numerical example on TV problems where two roommates (payer-A and Player –B) try to decide whether or not to purchase a TV. Given the small size of third apartment, the TV will necessary go the living room and both roommates will be able to watch it and there is no way to exclude the other from watching. Thus it characterizes public good. Suppose that each person has a wealth of \$500, that each person values the TV at \$100 and the cost of TV \$ 150. Since the sum of the **reservation price** (the money they are watching to pay individual to acquire) exceeds the cost, it's Pareto efficient to buy the TV. Consider the decision of one roommate, player if he buys the TV, he gets benefits of \$100 and pays a cost of \$150, leaving him the net benefits of \$50. However, if player- A buys the TV, player B gets to watch it for free, which give him benefit of \$ 100. Player B's interest is to free ride: to watch the TV but not contributing. The free riding game payoff matrix is depicted in table xx.

		Player B	
		(-50,-50)	(-50,100)
Player-A	(100 -50)	(0, 0)	

The same logic hold if player' B buy the TV. The dominant strategy equilibrium is for neither player to buy the TV. This is similar to prisoner's dilemma, but not exactly the same. In prisoner's dilemma, the strategy that maximizes the sum of player's utilities is for each player's to make the same choice. Here the strategy that maximizes the sum of the utilities is for just one of the players to buy the TV.

If player -A buy the TV and both players watch it, we can construct a Pareto improvement simply by having player- B make a "side payment" to player- A. For example, if player- B gives player- A \$51, then both players will be made better off when player -A buy the TV. In fact, this is probably what happens in practice; each player would contribute some fraction of the cost of the TV.

Thus, a free rider problem arises because each consumer believes that the public good will be provide any way, whether he or she contribute to its payment or not. The problem is that as many individuals behave this way, less than optimal amount is produced. Given all these complications, it is perhaps not surprising to learn that the market mechanism is generally ill -equipped to provide public goods in the right amounts. The markets operate on the principle that those who do not pay for a good cannot consume it, but it is impossible to prevent people from consuming a public good if they do not pay for it.

6.6.Activity

Suppose certain Microsoft computer software program specifically ‘Microsoft office’ without protection by the company of not using the package. Do you think these resources have the public good characteristics? If so why and how?

6.4.2.DEALING WITH PROBLEM PUBLIC GOOD

We have seen that market mechanism of dealing with Public lead to inefficient allocation of the pubic goods. There are different mechanisms to handle the problem depending upon the situation at hand. Remedial measures are more or less similar with what we discussed under externalities and tragedy of commons and hence one can apply those means of solving the problem. However, among the mechanism the following four are the most commonly ways of dong with the problem. These are:

- A Social pressure
- B. Mergers
- C. Compulsion and
- D. Privatization

Social pressure is the action through which the local people forces the person to act in appropriate manner. Sometime, especially when the group is small, social pressure eliminate free riding. Social pressure results in at least minimal provision of some public goods. Such pressure can cause most firms at a using public to contribute "voluntarily" to a fund for protections and optimal provisions.

A direct way to eliminate free riding is by merging firms and thereby internalizing the positive externality. Privatization (exclusion) that restricted the access to the resources

Compulsion, some outside entity such as government may dictate a solution. The governments intervene in the function of the market through a series of regulation. Some examples are:

- antitrust laws aimed at preserving competition in the economy
- minimum wage laws
- agricultural product prices support program
- motor vehicle pollution control act
- a host of regulation on consumer protections
- regulation regarding licensing of occupations

Further more, the government affect the allocation resources through imposing taxes reasonable for compensation as well as retarding affect them .to pay "taxes" that are assessed through tenants "votes". If the majority votes to hire guards, all must share the costs or government may impose certain amount of taxes.

6.7.Activity

What mechanism of solutions for the problems of public goods we have seen under activity above in your surrounding?

6.5. ECONOMICS OF INFORMATION

Information is vitally importance to firms, consumers, and input owners. Up until now we have generally assumed that economic agents have complete information about the matters that influence their economic choices or decision –full information. For instance, we have said that when a firm is making production decisions is absolutely certain about the prices it will be paying for the inputs and the prices it will be selling the output. Under this situation the market allocation lead to efficient association of resources.

However, in real world, the information on certain issues fluctuate overtime and is not perfect, some parties to a given transaction might have more or better information than the others. Hence, market few is to achieve efficient allocation of economic resources. There are two major problems: uncertainty and imperfect information. These problems form an area called “Economics of information”. In information economics information itself is a commodity, which economic agents can acquire only at some cost. Information provided by those in business for instances speculators, retailers, agents, stockbrokers, and others are valuable commodity for the other parties.

The general objective of this chapter is to introduce the concept of information economics to see the problem related to in formations, and the mechanism to reduce information problems.

Accordingly, after completing this chapter, you will be able to;

Explain the causes of information problems.

Mention the remedial measures.

Describe how information costs affect the contracts between parties to a business contract.

Understand how opportunistic behavior affects business contracts works?

6.5.1.INFORMATION AS ECONOMIC COMMODITY AND OPTIMAL SEARCH RULE

Many peoples such as speculators, intermediaries, and others in the information business are often unpopular. Most of the time peoples such as production workers, views them as more middleman which are parasites on the effects of others who do the real work. Sometimes farmers complain about the speculators who buy form them when the price is low and resale later at a higher price. Consumers complain about rising grocery prices, which they attribute to greedy grocery’s.

But on closer examination, we can see that the economic role of sales agents is essentially the same as that of production workers. These agents add genuine economic value by increasing the extent to which goods and services find their way to the consumers who value them most. For instances, one of the most common problems consumer confront is the need to choose among different versions of a product with many complex feature that they do not fully understand. The middlemen assist them in giving information about the quality. They supported the fact that many people go to great lengths to avoid paying for the services of sales agents. Many manufacturers cater to them by offering consumer a chance to buy direct and side step the middleman's commission

These economic agents also assist manufactures or producers in their sale product. Hence, the information provided by those in the "information business" (speculators, realtors, agents, intermediaries, stock brokers and others) is a valuable commodity.

Because of the above reasons and others, which will be mentioned in detail shortly, economic agents spent a huge quantity of scarce resources on economics of information. People and businesses voluntarily pay billions of dollars to acquire information on prices, locations and quantities. This information includes data on the prices, quantity and quality of good, location, and product performance.

In contrast with models of perfectly competitive, monopoly and other market structure where information is free, in real world most of the time it is scarce. If it were not, we would not need, the Wall Street Journal, CNN, The Reporters, the Ethiopian Herald, Ethiopian TV and Radio...etc. And we would not need to shop around for bargains or spend time looking for jobs. These and other information are distributed over the population in bits and pieces and every market participant have no equal assess to them.

Because of its scarcity, information acquiring typically does have its costs. Information gathering is an activity like any other. People value all the time, effort, money or financial costs and related benefits of gathering information. That is, the cost benefit principle provides a strong conceptual framework for looking at the way to optimize the information cost. These information costs include the cost of telephoning, shopping, checking credentials, inspecting goods, monitoring the honesty of workers or customers, placing Ads and reading Ads... etc. For instances, consumer incurs search cost while shopping, reading or consulting experts in

order to acquire pricing and quality information. This information is costly, as we have limited capacity to acquire, process, store, and retrieve facts and figures about the required issue even though some technological change such as Internet has reduced it.

Information cost has substantial effect on real-world markets. It is one of the crucial costs that affect the transaction costs. The buyers and sellers must first find each other and they agree on the prices and other terms of the contract. Knowledge of existences and location of willing buyer is valuable information to sellers just as knowledge of a willing seller is valuable information to the buyers.

Price is the central variable for the market adjustment and has signal implication in market for goods and services. With regard to information we have said that it is scarce and valuable commodity. By similar analogy the price system of information economizes on the commodity certain (costs) by allowing people to pay for the search information. Many models discussed above we have seen so far assume perfect information where its cost is zero. The fact that economic models ignore information cost does not make this model useless. They give insight to the force-generating trend in prices and quantities occur along period where information limits not to be important. Now we relaxed our assumption in that information is not free. It is appropriate to ask and understand how markets work in short run. That is, how economic agent optimizes information when it includes cost.

We know that in a perfect competitive market all buyers pay the same price for the same product. In many real world markets, however, the prices of even these homogenous goods differ from store to store, and shops to shops. Consumers or buyers feel difference in prices and want to find explain why homogenous product sell for different prices in different location as the cost of going to the cheapest store may out weigh the advantages of the lower prices.

Because of these rational economic agents (buyer) use the optimal search rule to optimize his transaction. The optimal – search rule states that people will continue to acquire economic information as long as the marginal benefits of gathering information exceed the marginal cost. That is, the buyer must balances the expected gain from further search against the cost of further search or search for lower price until the expected marginal benefit of additional search equals the marginal cost of search.

Example, when a person decides to buy a new car, the more information that person has on the prices and on technical qualities of various automobiles, the better the eventual choice is likely to be. But it is costly to drive all over town to the various dealers, take time off from work or from leisure activities to compare prices, acquire and master technical information on new car. Because of these, prospective buyer must draw the line at the point where the marginal benefits from acquiring information equal with marginal cost.

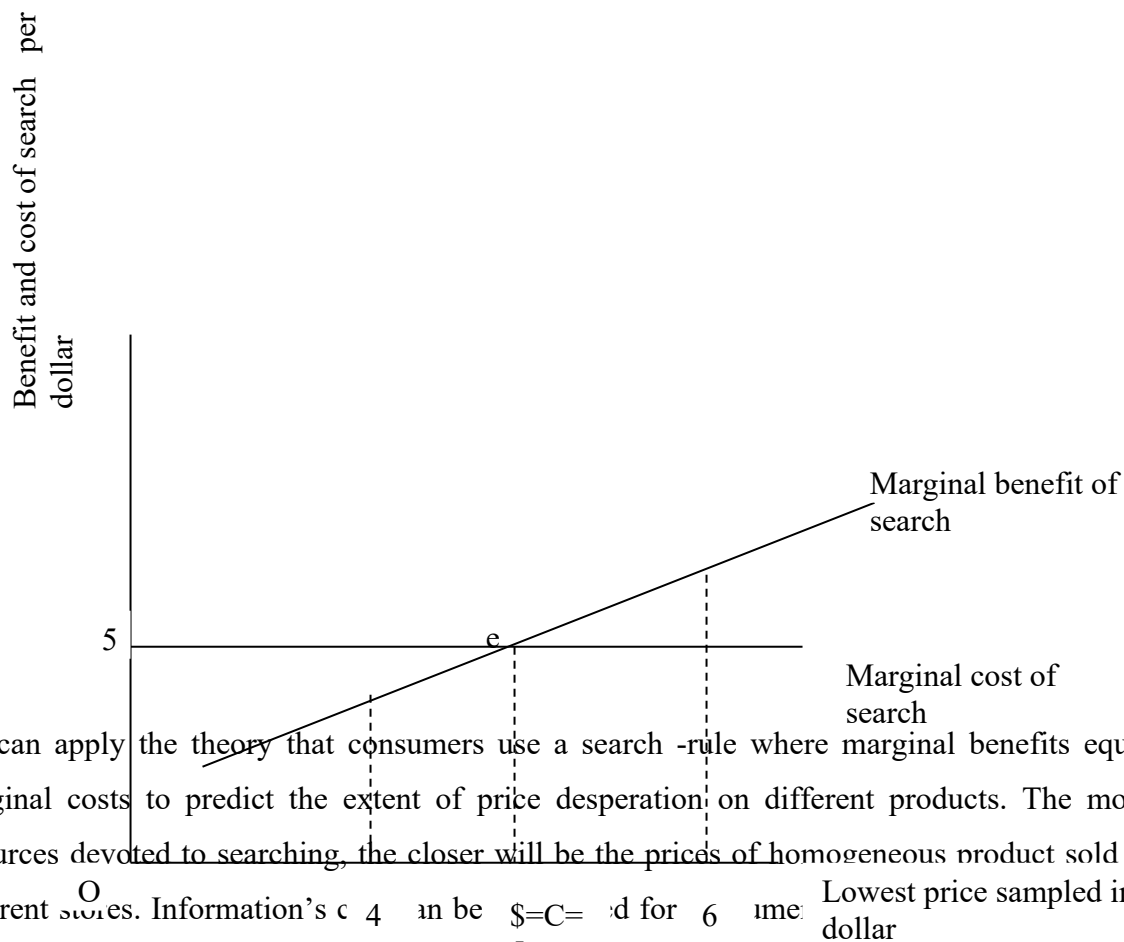
This optimal search rule can be illustrated using diagram. Suppose a consumer has just moved to a new town and is looking for the best place to buy a particular product. The consumer might visit several stores to collect varied price information. Thus, after some comparison shopping, the consumer will have a sample of the various prices charged say the product is BM car.

Let vertical axis measures the benefits and costs of search per visit. Horizontal axis measures the lowest known price (s) that the consumer has collected through search. Your marginal cost of search is \$C (\$5 for case) per dealer visited and is shown by horizontal line in the figure. The costs include time cost and the amount spent on transportation and advice. Your expected marginal benefit of visiting one more dealer depends on the lowest price that you have found. The marginal benefits differ with prices. The lower the price you have already found, the lower is your expected marginal benefit of visiting one more dealer, as shown by the curve

Until point 'e' where the price at which the marginal cost of search equal the marginal benefit of search (the consumer's reservation price), the buyer marginal benefit is greater than the cost and hence they search for the lowest price possible. The reservation prices are the highest price at which the consumer will buy a good. Consumer will to search for lowest prices if the lowest prices so far found exceeds the reservation prices but will stop searching and buy it if the lowest price found is less than or equal to the reservation prices.

The reservation price is \$5 in figure xx. If the lowest price sampled is \$6, the consumer should still search because the marginal benefit of search exceeds the marginal cost of search. If the lowest price is below \$5 say \$4 the Consumer will purchase the good because the marginal

cost of search exceeds the marginal benefit of search. The lowest price sampled is \$5, because \$5 is the highest price the consumer will pay for the product.



We can apply the theory that consumers use a search rule where marginal benefits equal marginal costs to predict the extent of price desperation on different products. The more resources devoted to searching, the closer will be the prices of homogeneous product sold at different stores. Information's cost can be \$=C= d for 6 time. Lowest price sampled in dollar the consumers union, which tests products and sell the results in monthly magazine. Furthermore, gov't Figure 6.7; Optimal-Search Rule by setting minimum standard and carrying out inspections to ensure that standard is being observed. Advertising and other information sources can reduce the cost. It can reduce price for two reasons, firstly, it can increase competition by telling potential buyer about alternative sources of supply. Second, it enables firms to increase their output and reap economics of scale.

6.8Activity:

Do you think house rent or sale brokers in your surrounding has any economic role for the transaction activity undertaken by the two major parties: buyer and seller? How and what do they do?

Suppose you want to buy a tape recorder as a competitive market where there is a difference in cost of the product in different shops. How do you decide to buy the good using the optimal-search rule criteria?

6.5.2. Information problems: Asymmetric information

So far we have looked at situations in which information is available to everyone and can be obtained with an expenditure of resources. But not all situations are like this. For example, someone might have private information that may affect many economic transactions as what is available to one person is too costly for anyone else to obtain. Typical examples may be your knowledge about your driving. You know much more than your auto insurance company does about how carefully and defensively you drive. Another is your knowledge about your work effort. You know far more than your employer about how hard you work.

Economic dealings between economic agents (individuals or organizations) are governed by contract. One party to a transaction may know a material fact that the other party does not know about the good. For example, the seller knows the quality of his product whereas the buyer does not. When a good is purchased, the seller explicitly or implicitly guarantees that the good will work according to an expected performance standard. The more informed party may exploit the less informed party.

Economists use the term asymmetric information to describe situations in which buyers and sellers are not equally well informed about the characteristics of products or services. In these situations, sellers are typically much better informed than buyers, but sometimes the reverse will be true. This asymmetric information affects the behavior of market participants and explains a wide variety of phenomena in modern economy that lead to opportunistic behaviors which in turn lead to market failure, destroying many desirable properties of competitive markets.

Used Cars: An example of asymmetric information

The classic example of asymmetric information is the market for used cars-lemons. Why is this not a surprise? Clearly, the seller of a used car generally knows a great deal more about its performance and deficiencies than the potential buyer. This asymmetry of information will influence how the market works. How? Suppose that all new cars are either good or defective and that buyers find out which after they have purchased a car. If the owner then turns around and offers the car for sale, then a potential buyer of what is now a used car will not be able to determine (before buying it) whether it is good or defective, but the seller will know.

Under these circumstances, the equilibrium price of a used car would be less than the price of a new car. Note first that good and bad used cars should both sell at the same prices because the potential buyer cannot tell the difference prior to the sale. But this common price must be less than the price of a new car. If the prices of a used car were not less than the price of a new one, there would be no demand for used cars; people would buy a new car, and hence determine whether it is defective, and (if it proves to be defective) sell it and buy another new car.

“Lemons” (defective cars) would likely constitute a portion of the used cars offered for sale. Owners of good used cars could easily find the equilibrium price of a used car to be so low that they would not be motivated to offer many of their cars for sale. On the other hand, owners of defective used cars would not feel similarly restrained. And this insight would make potential buyers even more inclined to be wary of entering the market for used cars and to offer relatively low prices for used cars if they did.

The buyer of a used car would be willing to pay more than the equilibrium price if he or she were sure of getting a good one and the seller of a good used car would be delighted to agree to such a transaction. But the asymmetry of information across the market and the fact that only the seller knows whether the used car is good or defective makes it difficult for these sorts of transactions to occur. Faced with this situation, sellers of used cars try in various ways to signal buyers that their cars are good. They cite relevant facts concerning their cars. They encourage the buyer to have his or her experts inspect the cars before purchase.

6.9. Activity

Is there something similar to the used car problem do you know? If so what is that, and how did it happen?

We have said that economic dealing between individuals is governed by contract either implicitly or explicitly. When both parties to a contract (transaction) have no equally information, transaction is not efficient as information advantage leads to problems of opportunism, where by the informed party or person benefit at the expense of the person with less information. The opportunistic behavior or asymmetric information creates two problems:

Moral Hazard

Adverse selection

A. Moral hazard problem

The moral hazard problem occurs when one party to a contract engages in opportunistic behavior after the contract is made. A Moral hazard problem exists when one of the parties to contract has an incentive to alter his or her behavior after the contract is made at the expenses of the second party to get benefit. It arises as it is too costly for the second party to obtain information about the first party's post contractual behavior and hidden action. Stating differently it is opportunism characterized by an informed person's taking advantage of a less informed person through an unobserved action.

Some examples are

If Mr.-X hires Mr.-Y as a sales person and pays him a fixed wage regardless of his sales. Mr. – Y has an incentive to put in the least possible effort, benefiting himself and lowering profits of Mr. as the pay is not based on the volume of sales.

An insured people find to take unobserved action (engage in risky behavior or take lesser precaution) that increase the probability of large claims against insurance companies. Individual is will spend less on preventive health care and thus the probability of getting in to risk rises

But if a consumer can purchase car insurance on theft, then the cost inflicted on the individual of having his car stolen is less as he/she can report to the companies and get insured if the car is stolen. Similar analogy, applies to car damage/ collusion. Although it is unlikely that insured driver would deliberately have collusion, such an insurance policy might give the

deriver an incentive to alter deriving behavior. The deriver might be less cautious in parking lots and might generally drive less defensively than normal.

Such activities as failure to take care and reckless behavior reduce output or increase accidents, which show why market failures and hence harm society. These are moral hazard in which individuals have an incentive to do opposite to get advantage at the expense of contracting parties. The Moral hazard occurs in all economic activities where economic agents do not bear the full consequences of their action. In such instances the economic agents maximizes their own utility to determinant of others.

As opposed to competitive market where the transaction is determined by the condition that demand equals supply the marginal willingness to pay equals the marginal willingness to sell here is no smooth transaction up to the equilibrium in case of more hazard, hence no market equilibrium.

In our insurance case, for instances, has the property that each consumer would like to buy more insurance, and the insurance companies would be willing to provide more insurance if the consumers continued to take the same amount of care. But this trade won't occur because if the consumers were able to purchase more insurance they would rational choose to the less care.

Despite the pejorative name, not all moral hazard, are entirely harmful. For instances, pregnant women with health insurance make more prenatal doctor visits. Although the extra cost from these visits is a moral hazard to insurance companies, society benefits from healthier mothers or babies.

6.10 Activity

Someone, whose car is insured, may take less care to prevent from being stolen parking or/and damage while deriving as compared someone with no insurance. Why? What kind of problem is it? Is there any similar problem in you surrounding?

B. ADVERSE SELECTION

The adverse selection is other problem which arises prior to the making of the contract transactions between parties. It is the tendency for people to enter in to agreements in which they can use their private information to their own advantage and to the disadvantage of the less informed party. It occurs as the cost of obtaining the relevant information makes it difficult to determine whether the real is a good one or a bad one.

Some example may be:

In a shop or supermarket contract between the business man say Mr. X and sales person Mr. Y, if Mr. X offers sales people a fixed wage, she will attract lazy sales people, hard working sales people will prefer not to work for Mr. X because they can earn more by working for someone who pays by results. The fixed wage contract adversely selects those with private information (knowledge about their work habits) who can use that information/ knowledge to their own advantage and to disadvantage of the other party.

Suppose there are many producer of umbrella in which some produce high - quality product at with P_H prices and the others produces low- quality product at P_L . If the market is competitive, it has an incentive to produce only low-quality umbrellas as each producer has negligible effect on others. The Low-quality umbrella production has destroyed the market, for both

qualities of good. That is, low quality item crowded out high quality items because of the high cost of acquiring information.

These adverse selection problems are more prevalent in insurance industry. Insurance companies face a different world; for instance good and bad car or drivers. If we assume Mr. G a good drivers low rate accident and Mr. F bad driver with high rate accident, unless the companies evaluate when they derive, it can not differentiate between two drivers. Due to high cost of gathering information, the insurance company sells automobile insurances to both party at the same rate, even through individuals with in the group often differ sharply in terms of their likelihood of filing claims. Thus, buying insurance is attractive to those individual with the highest likelihood of accidents them good driver.

Similar problem arises with health insurances in which case those people with high incidences of health problems tend to buy health insurances at same rate with that of people with low incidences. The high risk people are better off because they can purchase insurances at rates that are lower than the actual risk they face and the lower people can purchase insurances.

Adverse selection forces insurance companies to raise their premium, which makes buying insurance even less attractive to low-risk individuals, which raises still further the average risk of those who remain insured. In some cases only those individuals faced with extreme risk may continue to find insurance an attractive purchase. Adverse selection create a market failure by reducing the size of market or dominating it, thereby preventing desirable transactions insurance companies have to charge higher rates for insurance due to adverse selection or choose not to offer insurance at all.

6.11.Activity

Suppose there are two individuals Abebe with high level health problem and Kebede with low level of health problem and each person knows himself but the others do not. Furthermore, they have access to health insurances at the same rate payments \$R. How does each of these behave? What type of problems may arise?

6.5.3.MEASURES TO REDUCE INFORMATION ASYMMETRY

Up until now we have seen that asymmetric information result in inefficiencies due to moral hazard as well as adverse selection problems. How do these problems can be solved without shifting risk to people who hate bearing it or contracts at least reach a good compromise between the two goals. Insurance companies, for instances, face trade-off between reducing moral hazards and the risk of insurance buyers as the company has a pools risk. It acts as through it is risk neutral. Similar trade-off may arise under adverse selection. Hence, how to solve these problems?

There are two main methods of solving adverse selection and moral hazard. These are:

- A. restrictions opportunistic behavior or controlling opportunistic behavior and
- B. equalizing information

A.CONTROLLING OPPORTUNISTIC BEHAVIOR

Adverse selection can be prevented if informed have no choice. For example, a government can avoid /reduce asymmetric information by providing insurance to everyone or by mandating that everyone buy insurance. Many states require that every driver carry auto insurances there by reduce adverse selection that would arise from having a disproportionate number of bad drivers buy insurances. Similarly, firms often provide health insurance to all employees as a benefit rather than paying a higher wage and letting employees decides whether to buy such insurance on their own. By doing so, firm reduce adverse selection problems for their insurance carrier. Both healthy and unhealthy will be covered as a result firms can buy medical insurance for their workers at a lower cost per person than workers could obtain on their own. Insurance companies have responded to this problem by imposing limits on the amount to be insured. Particular policies many, for example, pay for no more than 8 days in the hospital for a person undergoing a gall bladder operation.

Further more, many insurance policies have a feature of **co-insurances** in which the companies cover some part so that the policyholder is obliged to pay the balance health insurance policies. Often have a 20% co-insurance rate, which implies that the insurance company pays 80% and policy holders pays 20% of any bill. Someone with a \$500 medical bill would then pay 0.20×500 or \$100. This 560 – insurance is to give the policyholder more incentive to hold down medical bills and to take pervasive care.

Insurance companies also respond to moral hazard by inserting **deductibles** in to the policies that they write. With a deductible, a policy maker has to bear part of the insured loss up to some limit. For example, a medical insurance policy may stipulate that the patient must pay the first \$200, of medical expenses that he or she incurs..

In some market it is possible to avoid from consumer ignorance through laws might provide protection against being selling lemons. Consumers might screen by collecting the information themselves, the government or others third party might supply reliable information. For instance, product liability laws protect consumers from being stuck with non functional or dangerous products. More over, many states super courts have concluded that the products are sold with on implicit understanding that they will safely perform their intended function. If they do not, consumers can sue the sellers even in the absence of product liability laws.

6.12.Activity

What are similar policies or remedial measures in our country about the insurance industries related to car insurances and health insurances?

B. EQUALIZATION INFORMATION

Either informed or uninformed parties can eliminate information problems through:

Screening

Signaling

Screening is an action taken by a uninformed person to determine the information possessed by informed people. A buyer may test – drive (screen) several used cars to determine which one starts and handles the best. When the originally uninformed people obtain better information, they may refuse to sign a contract or insists on changes in contract clauses or on the prices of a good. Insurance companies, for example, try to reduce adverse selection problems by learning the health history of their potential customers; information on age and general health, individuals dangerous habit (drinking or smoking)...etc.

Consumer can avoid the lemons problems if they can obtain reliable information about quality. Consumers can buy information from experts or infer product quality from seller's reputation as it is easy to establish reputation in a market in which the same consumers and firms trade

regularly. In some market, consumers learn of a firm's reputation from other consumers or from observation. If the mechanic can reliably determine whether the car is lemon, the information asymmetry is limited.

Signaling is an action taken by an informed person to send information to a less informed persons. A firm may send a signal such as widely distributing a favorable report on its product by an independent testing agency to try to convince buyers that its product is high quality. In some markets government agencies or non-profit organization such as customers unions also provide consumers with information.

Firm and individuals both engage in an activity called 'market signaling' to convey information about the quality of the products or services that they are offering. Companies usually, for example, are interviewing and giving exam to select relatively better workers from the labor market. Firm usually use education as an important indicators or signal of person's productivity when they are make hiring decisions as the more educated people, the more the productivity of the worker.

Similarly, the producers signal product quality through different mechanism such as issuing guarantee and warranties for customers implying their product is of high quality. The high quality product is also signaled through high prices for their goods. Further more, advertising and other market information give valuable result in this regard even though is persuade individual to buy a product or services.

Moreover, if an outside organization to provide believable information, it must convince consumers in that it is trustworthy. Some nonprofit organization, such as consumers groups, and non-profit organization publish expert comparisons of brands. To the degree that this information is credible, it may reduce adverse selection by enabling consumers to avoid buying low-quality goods. In some case there is also standardization and certification in which the government, consumer groups, Industry group and other provides information based on standard: a metric or scale for evaluating quality of a particular product. Consumers learn of a brand's quality through certification: a report that a particular meets or exceeds a given standard levels.

6.13.Activity

Explain why more is spent on the advertising of deodorants than on the advertising of farm of machinery?

Do you think the certification and standardization of product especially food items by The Ethiopian Quality and Standard Authority helps consumer in selecting relatively good product? How?

SUMMARY

There are quit a lot of areas in which the market can not allocate of resources efficiently. The presence of externality is one of the main reasons. It arises when the action of an economic agent the other party not involved in the action and not comp sated for that. There are two major type of externality based on the sources.

These are consumption and production externality which may be either help or harm to the third agent. These actions can be resolved thought internalization of externality by the party generated the action. There are three major ways of internalizing: redefinition of property right, voluntary agreement, and government action.

Externality occurs when a consumer's well-being or a firms production capability are directly affected by the actions of other consumers or firms rather than indirectly affect through change in prices.

Externalities are mainly by product of consumption or production which either harm or help the third party not involved in the action. The existences of externality lead to market failure as those who produce something do not include the full cost of their actions.

Common property is other sources of externalities where the resources to which everyone has free access. The common property resources are overexploited as everyone wants get benefit before his fellowmen makes his or her private property.

Public goods are goods for which everyone must consume ones the good is supplied. The private market economy does not do well at providing public goods. If no one can be excluded from consumption there is no way to charge a price for the good and, therefore no incentives for private entrepreneurs to produces and sell as private companies would have obvious difficulty recovering their cost of production.

Many people might be willing to pay enough to cover the cost of producing the good, but as it is non excludable, the company cannot easily charge for it. In the provision of public good leads to free rider problems. Because, there are many people sharing the cost of providing the public good, each individual feels that with drawing his and her financial support will practically go unnoticed by others and will have little or no effect on the provision of the good. Market provisions of public goods lead under optimal supply of the resources. Economic agents have constantly faces with problem of hidden information that result in moral hazard and adverse selection. The problems of moral hazard arise because those who have information advantages have incentives to change their behavior.

The equilibrium out comes in these markets where hidden information problem there appear to be inefficient, but one has to be careful in making such a claim. The equilibrium in this case will always be inefficient relative to the equilibrium with full information.

There are two main methods of solving adverse selection and moral hazard. These are: restrictions opportunistic behavior or controlling opportunistic behavior and equalizing information. In the first case the government or the concerned body restrict this behavior by law or use different mechanisms such as standardization, certifications...etc. In relation to the second either informed or uniformed parties can eliminate information problems through screening and signaling

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KEY TERMS

Externalities	Effluent fee	Subsidies	Regulatory approach
Pollution	Production externalities	Consumption externalities	
Coase theorem	marginal private cost and benefits	pollution	
Government intervention	marginal social cost and benefits	positive externalities	
Production externalities	consumption externalities	Negative externalities	
Public goods	tragedy of commons	private property	
Socially optimum level of production	free rider problem		
Adverse selection	Economics of information	Optimal –search rule	
Information asymmetry	Moral hazard	Screening	Signaling
Principal -Agent problem	Lemon's	deductibles	co-insurances

REVIEW QUESTIONS REVIEW QUESTIONS

What is externality?

What are the remedial measures for externalities?

Why zero pollution not the best solution for society? Can there be too little pollution? Why or why not?

Analyze the following extract. Is garbage a positive or negative externality? Why is market solution practical here?

Garbage in Philadelphia is undesirable of course. Since the turn of the century, however, hog farmers in New Jersey have been feeding wastes 3\$ million a year and reduce its garbage mound by allowing New Jersey farmers to pick up leftover food scraps for their porcine recyclers. The city pays \$1.9 million to the New Jersey pig farmers for picking up the waste each year, which is about \$79 a ton. Otherwise, the city would have to pay \$125 a ton for curbside recycling of the same food waste.

What type of externality do greenhouse effects have on the environment?

Explain why it is Pareto- inefficient to exclude anyone from consuming a pure public good once it is produced?

When the consumption of a good involves external benefit, can subsidizing the producers bring about the socially optimal output? Why or why not?

Why is free-rider problem more likely to occur when a large number of people are involved?

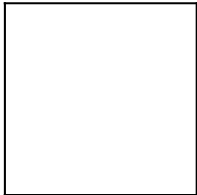
Why is zero pollution not the best solution for society? Can there be too little pollution? Why or why not?

Assuming there are two firms in the economy: chemical factory that emit waste on the upper stream and fishermen on the downstream of the same river, show that the market mechanism lead to inefficient outcome?

What are the remedial measures for externalities in production or consumption?

Explaining the significances of the 'free rider, problem? explain the meaning of the following terms and their implications for economic policy:

adverse selection



l hazard

ling

ning

What is the moral hazard problem in health insurances? Suppose that once a person pays for health insurance, all health care is free. How do the long waiting lines at doctors' offices mitigate the problem of moral hazard?

How the adverse selection problem is be affected by Internet?

If the search cost in a market is zero and the market is competitively organized what predictions can you make about prices in this market?

What is optimal search rule says about the utilization of resources?

Do intermediaries have any role to play in economic activities in market economy? What are their activities?

Suggested readings

Nicholson, w: **Microeconomic theory; basic principle and extension**. 5th edition

Salvatore, D(2003):**Microeconomics; theory and applications**.4th edition

Mansfield, E. and Yohe, G (2000):**Microeconomics, theory and applications**.10th edition

Varian, R. Hall (1999): **Intermediate microeconomics** 3rd edition

Pindyck, R.S. and Rubinfeld, D.L. (2003): **Microeconomics**. 6th edition

Koutsoyiannis A. (1985): Modern microeconomics 2nd edition.